



OneATS

User manual

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TAKAOKA TOKO CO., LTD.



OneATS user manual

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Symbol	Description
CIM	Common Information Model
DMS	Distributed Management System
DCOM	Distributed Component Object Model
FEP	Front End Processing
HMI	Human Machine Interface
I/O	Input/Output
IED	Intelligent Electronic Device
LAN	Local Area Network
OPC UA	OPC Unified Architecture
RTU	Remote Terminal Unit
SCADA	Supervisory Control and Data Acquisition
SSL	Secure Socker Layer
TCP	Transmission Control Protocol
TLS	Transport Layer Security
XML	Extensible Markup Language

Text color:

 Comment

 Notes

 Warnings

Chapter	Content
A.	System overview
B.	Database
C.	OneATS DataEditor – Project Management
D.	OneATS FEP
E.	OneATS DataServer
F.	OneATS Grid Studio
G.	OneATS SmartHIS
H.	OneMasterTool



OneATS User Manual

- OneATS Grid Studio

17/Feb/2025

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SOFTWARE UPDATE VERSION INFORMATION

Software Version	Date of Issue	Change Description
V4.1.7291	24/Jun/2024	
V4.1.7299	07/Jul/2024	- Layer panel menu: + Clone: copy all shapes in this layer to new layer. + Show Only This: will show & active only this layer. + Show All - Increase max screen-size to 20k - System variable: _playbackState (0: running, 1: pause, 2: stopped) - Add _loggedin system variable (Login state) - Change minimum zoom percent from 10% to 5%, zoom wheel step from 20% to 5% - Enable change background color of DataGrid, Tabular controls - Add Location, Identifier columns on Output Window - Search control by double click message on Output Window - Search control by double click on Variables Configuration dialog - Ctrl + Alt + Arow keys: move shape by 100 pixels - Update Ctrl + U: copy reference source id - Symbol Template: config keep-size & keep-rotate, skip auto-save, allow override import symbol (Note: file format change, need to backup project!) - Eventview query like message on both History and Realtime mode. - Sub select Reference Shapes - Config Undetermine state color - Tracing from terminal: Upstream, Down Stream, Ground - Search control on Alarm Subscription dialog - Export screens keep folder tree - Add close-button on Screen system menu - Script NetworkColorSetting() show network color setting dialog - Script EnableColoring() show Enable Coloring config dialog - Fix bugs: + Draw dash-line for line-width == 2 + Project Setting: disable maximum, minimum buttons + DataGrid OnSelectRow() event with header and no-row + Busbar terminal port size update when change busbar width + Connectivity & control arrow connection draw
V4.1.7356	06/Jan/2025	- Search Control on Viewer (Show list of controls sastified search pattern by name or caption, can locate control on screen by Double Click) - Screen search toolbar on Project Pane (Show list of screen sastified search pattern by name): - Modeless shape configuration dialog (Show config dialog of new shape without closing the config dialog) - Miscelllaneous

		+ Limit pen per chart to 20 + Run Profile menu + Fit Frame to Canvas menu + Diagnostic Log level config on Viewer (Menu > Tools > Diagnostic Log) + Hotkey show Layer Manager Dialog on Viewer (F4) + Add App minor version to Setup + Retain drawing control (exit by ESC or right-click) + Symbol Template Paned device type grouping + Auto calculate chart legend column when zoom screen + Auto-gen AClname name (Connectivity1Name-Connectivity2Name) + Add suffix _Copy to copied shapes + Add Search Control to Layer list dialog - Screen search toolbar: + Switching mode (Home/Search) + Hotkey: Ctrl + Q activate search mode - Update script: ExecuteFile(filePath) add optional param argument. - Update lock shape: disable change font, color, ... - Update lock layer: lock all shapes in that layer

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F. OneATS Grid Studio

OneATS Grid Studio is a software module that designs and operates the HMI interface. In the OneATS Platform system, there are two main components:

- OneATS Grid Designer: HMI interface design module.
- OneATS Grid Viewer: HMI operating module (runtime).

Configuration information from OneATS Grid Designer and OneATS DataEditor is compiled, to become run-time data. OneATS Grid Viewer will download run-time data and operate the project.

The database on OneATS Base Layer (including Data Model, Grid, Authorization Data) is fully synchronized between system configuration tools (OneATS DataEditor & OneATS Designer), users just need to configure in one location and use anywhere.

OneATS Platform ensures the data is configured and operated according to the user's authority. Users are only allowed to operate on the permitted data area, so the data is protected as much as possible.

Run-time data is configuration data that has been optimized for operation and interaction with users. OneATS Grid Studio applies incremental compilation techniques to reduce maximum compiling time and improve the performance.

F.1. Overview of OneATS Grid Studio

F.1.1. Architecture

The diagram illustrates the common OneATS Grid Studio deployment architecture in control centers:

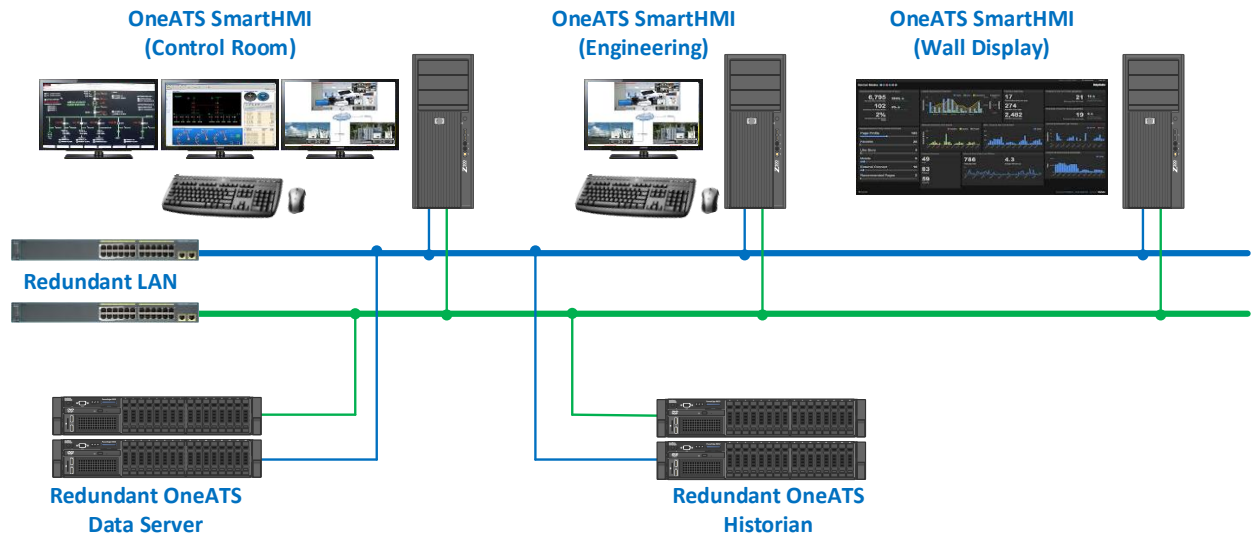


Figure F-1. OneATS Grid Studio architecture in control centers

F.1.2. Communication

F.1.2.1 Technical description

OneATS Grid Viewer collects real-time data by OneATS DataServer and queries historical data with OneATS Historian according to OPC UA (IEC-62541).

Communication supports fail-over and reconnect mechanisms, ensuring that the system always operates stably throughout the operation.

OPC UA connections may use SSL/TLS coding standards and digital certificates to increase channel security.

Users can monitor the connection status of OneATS Grid Viewer and OneATS DataServer, and OneATS Historian through the system variables provided.

F.1.2.2 Technical specification

- OPC UA(IEC-62541).
- Support fail-over and switch-back in Redundancy mode.
- Automatically reconnect and restore the connection session.
- Support SSL/TLS connection and certificate-based authentication.
- Write data and call asynchronous API.

- Configure session time-outs, connection time-outs, and other connection properties.

F.1.3. Data flow

The diagram describes the data flow for the OneATS Grid Studio:

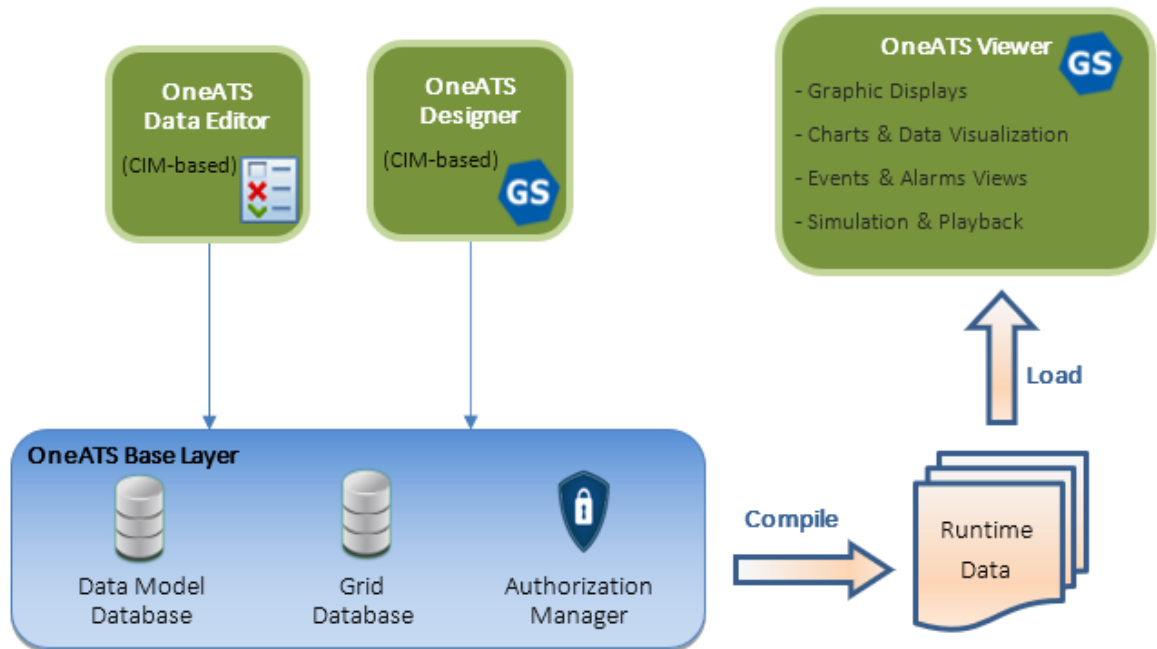


Figure F-2. Data flow for the OneATS Grid Studio

Configuration information from OneATS Grid Designer and OneATS DataEditor is compiled, to become run-time data. OneATS Grid Viewer will download run-time data and operate the project.

The database on OneATS Base Layer (including Data Model, Grid, Authorization Data) is fully synchronized between system configuration tools (OneATS DataEditor & OneATS Designer), users just need to configure in one location and use anywhere.

OneATS Platform ensures the data is configured and operated according to the user's authority. Users are only allowed to operate on the permitted data area, so the data is protected as much as possible.

Run-time data is configuration data that has been optimized for operation and interaction with users. OneATS Grid Studio applies incremental compilation techniques reduce maximum compiling time, improve the performance.

F.2. Installation

F.2.1. Install OneATS Grid Studio

Run the latest version of **Grid StudioStudio.exe** installation file. The installation interface appears, select **Next** to continue:

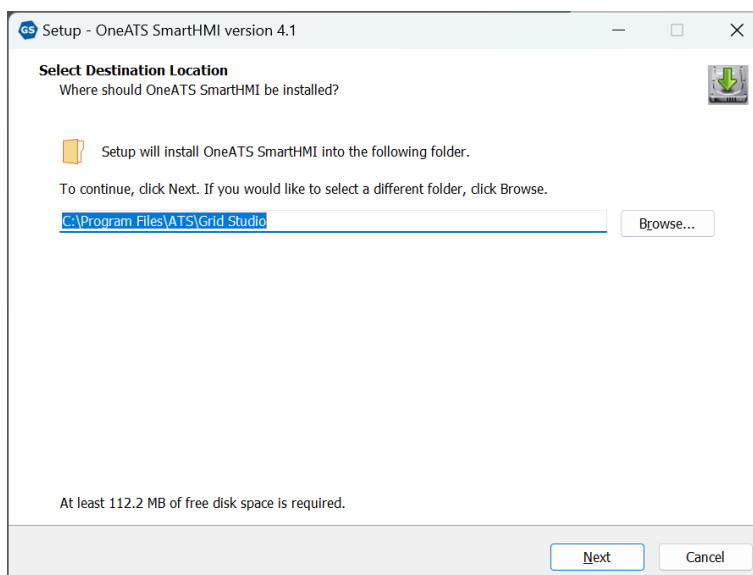


Figure F-3. Installation of One ATS Grid Studio

- Select **Next**:

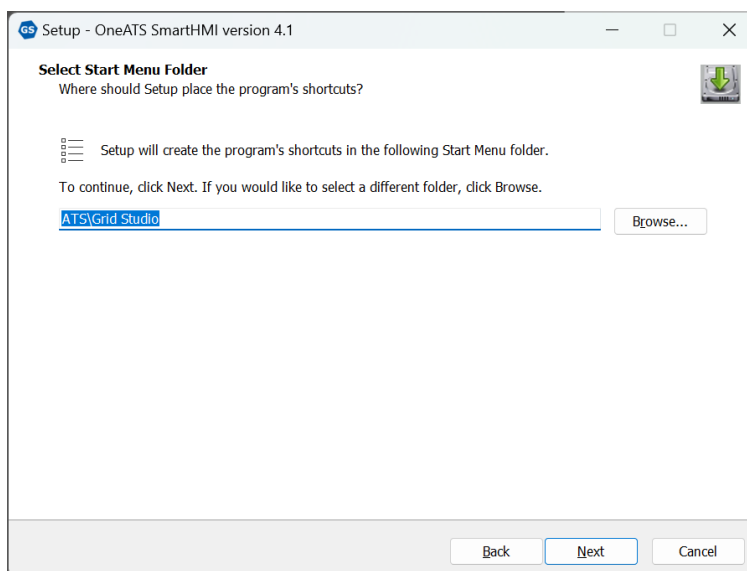


Figure F-4. Installation of One ATS Grid Studio

- Select **Next** to continue:

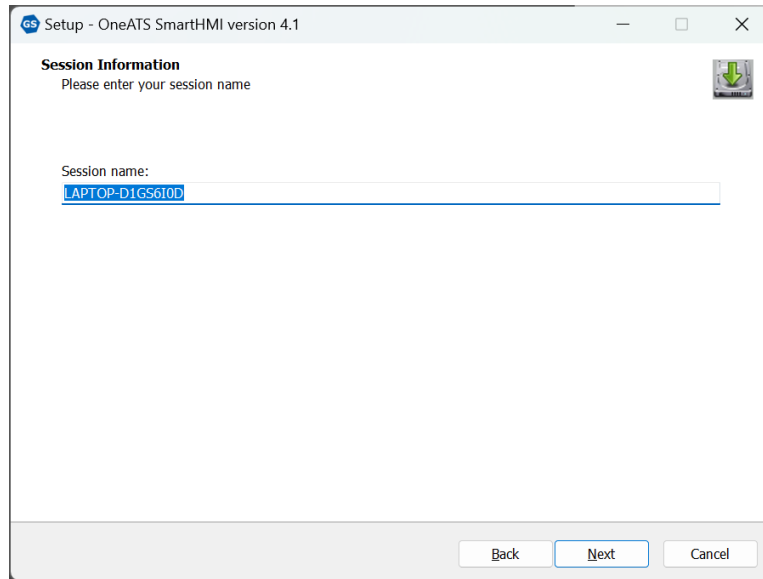


Figure F-5. Installation of One ATS Grid Studio

Select **Install**:

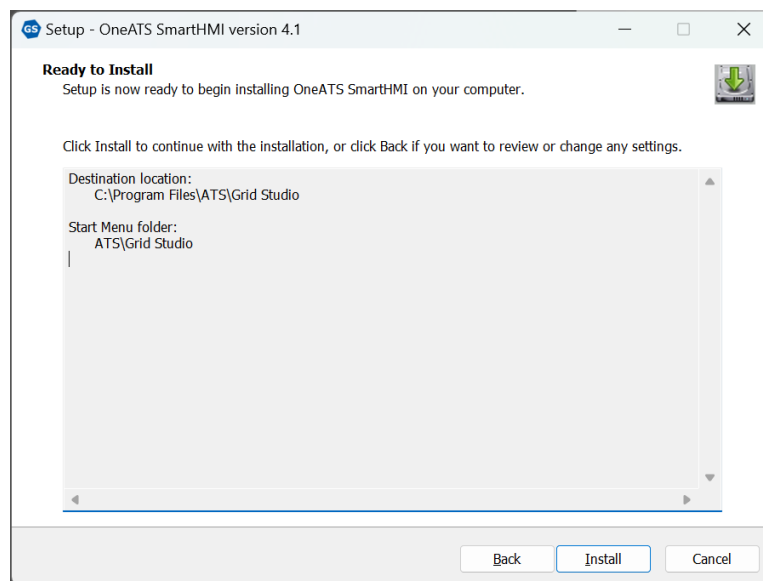


Figure F-6. Installation of One ATS Grid Studio

Select **Finish** to finish the installation process:

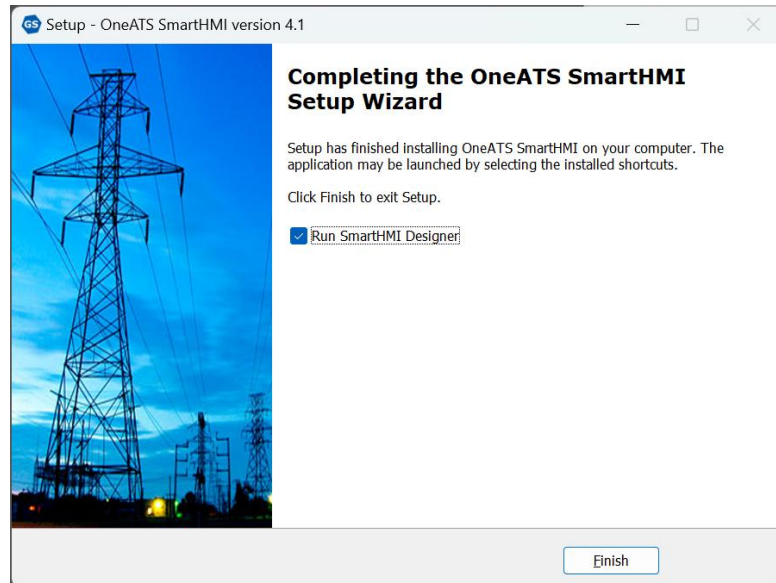


Figure F-7. End of installation of One ATS Grid Studio

F.2.2. Install and check license

Refer to ChapterH.OneMasterTool.

F.2.3. Install Git environment

Not Available

F.3. Design HMI by OneATS Grid Designer

F.3.1. Design tools

F.3.1.1 Menu bar

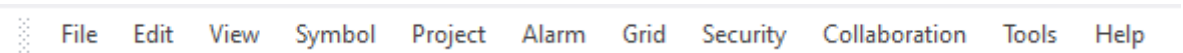


Figure F-8. Menu bar

File	Open/close project tools
Edit	Tools to edit copy/paste/...
View	Hide toolbars, tool windows
Symbol	Symbol Template tools
Project	Project tools: configure, compile, run Viewer, ...
Alarm	Alarm configurations
Grid	Network object tools
Security	Security tools, users
Collaboration	Manager Version Project (<i>Unavailable</i>)
Tools	Add-on configurations
Help	Information about the application: version, license, ...

Allows using keyboard shortcuts to open the menu by pressing **Alt+[Key]**.

F.3.1.2 Basic toolbar



Figure F-9. Basic toolbar

	Create a new project
	Open a project
	Save the project, save the changes

	Cut
	Copy
	Paste
	Show project pane window
	Show properties window
	Show Control View
	Show Symbol Template
	Print
	Help

F.3.1.3 Project toolbar



Figure F-10. Project toolbar

	Undo
	Redo
	Compile project
	Run runtime project
	Zoom in
	Zoom out
	Pan / Allows to move to areas in the screen by holding left mouse click
	Rectangle Zoom on objects in an area limited by the selected rectangle
	Select / Selection of objects

F.3.1.4 Control Utilities Toolbar



Figure F-11. Control Utilities Toolbar

	Group objects
	Ungroup objects
	Lock /Object cannot change size, font, color, location... by mouse operation
	Unlock / Allows to change size, font, color, location... of the object by mouse operation
	Transfer objects located above all other objects
	Transfer objects under all other objects
	Move the object that is on top of the object directly in front of it
	Move the object below the object directly behind it
	Rotate objects 90° counterclockwise
	Rotate the object 90° clockwise
	Take axial symmetry
	Take horizontal axis symmetry
	Left align objects, the last selected objects are selected as landmarks
	Center objects vertically
	Right align objects, last selected object is selected as landmark
	Align on objects, last selected object is selected as landmark
	Center objects horizontally

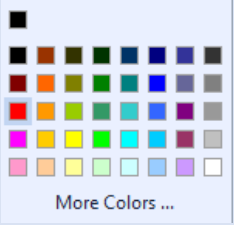
	Align under objects, the last selected object is selected as a landmark
	Objects are the same horizontal size of the last selected object
	Objects are the same vertical size of the last selected object
	Objects have the same horizontal and vertical size of the last selected object
	Objects are evenly spaced vertically
	Objects are evenly spaced horizontally

F.3.1.5 Font Toolbar



Figure F-12. Font toolbar

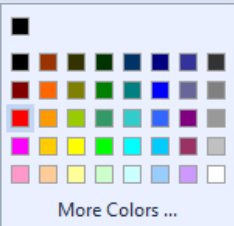
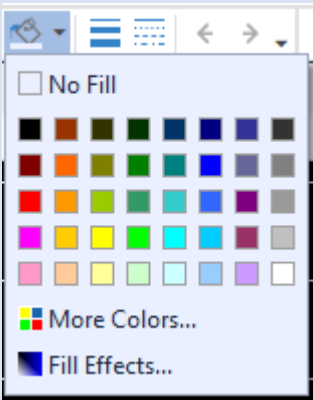
	Select text font
	Select text size
	Make your text bold
	Italicize your text
	Underline your text
	Left Alignment
	Center Alignment
	Right Alignment
	Top Alignment
	Middle Alignment
	Bottom Alignment

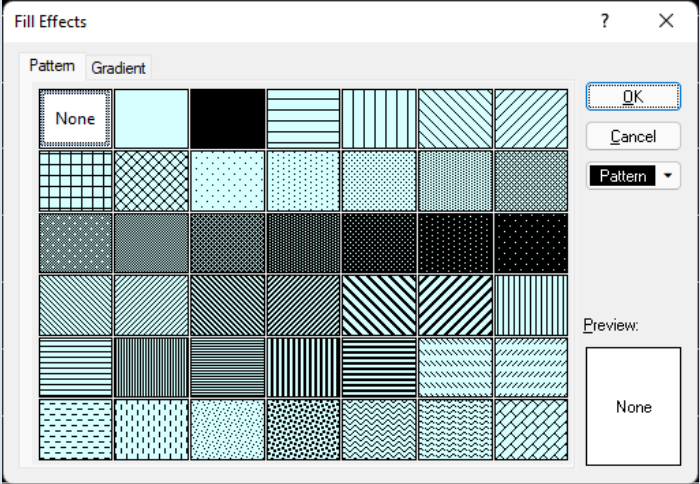
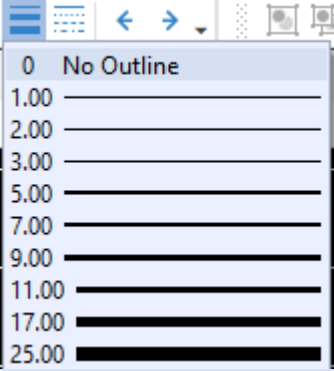
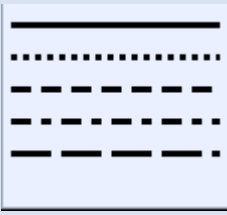
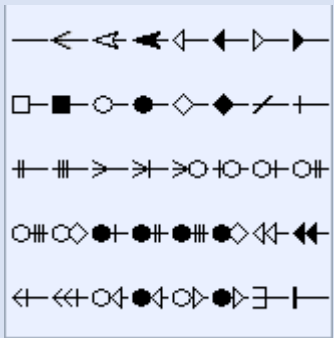
	Increase font size
	Reduce font size
	Font color  Click More Colors to adjust the color

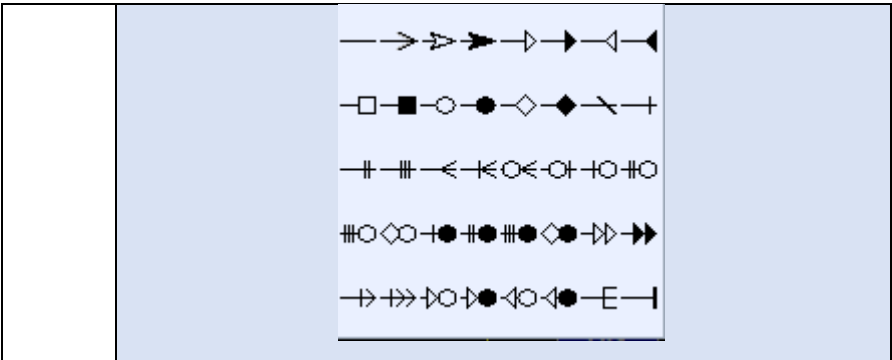
F.3.1.6 Line Toolbar



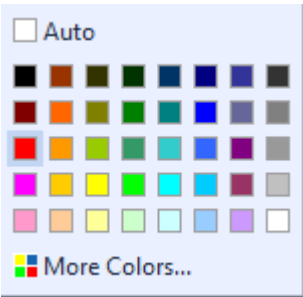
Figure F-13. Line Toolbar

	Choose a color for the line  Click More Colors to adjust the color
	Choose a color for the objects  Click More Colors to adjust the color

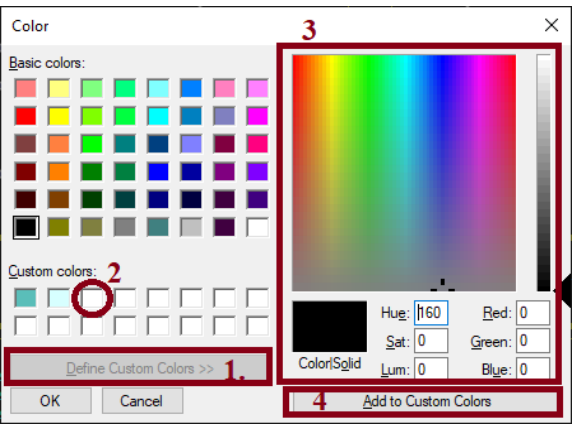
	Click Fill Effect to adjust color effect 
	Selection of line width 
	Selection of line style 
	Select shape style of the line's starting point 
	Select shape style of the line's ending point



Choose color Text Color / Filled Color / Line Color



- Click More Colors to adjust the color



F.3.1.7 Status Bar



Figure F-14. Status Bar

(X=1022, Y=791)	Current coordinates of the mouse pointer
Grid	Turn on/off grid mode
Snap	Turn on/off snap mode

96% -  +	Zoom rate
	Connection status of DataModel

F.3.1.8 Based Controls Toolbar – basic graphic objects



Figure F-15. Basic graphic object

	Line object
	The object has multiple serial lines.
	Insert an image object
	Edit Box object
	Button
	Label
	Ellipse
	Circle
	Rectangle
	Square
	Round rectangle
	Polygon
	Check Box
	List Box
	Radio Button
	Combo Box
	DateTime

F.3.1.9 Advanced Controls Toolbar



Figure F-16. Advanced graphic objects

	Event View
	Trend View
	Data Grid View
	Relay Event View
	Chart
	Tabular View
	Insert ActiveX

F.3.1.10 Built-in Control Toolbar – (Electrical) Network



Figure F-17. Network objects

	Connectivity Node
	Busbar
	AC line

F.3.1.11 Template Controls Toolbar



Figure F-18. Template Controls

	Control Arrow
	Control Led

F.3.1.12 Project Pane

Manage project components

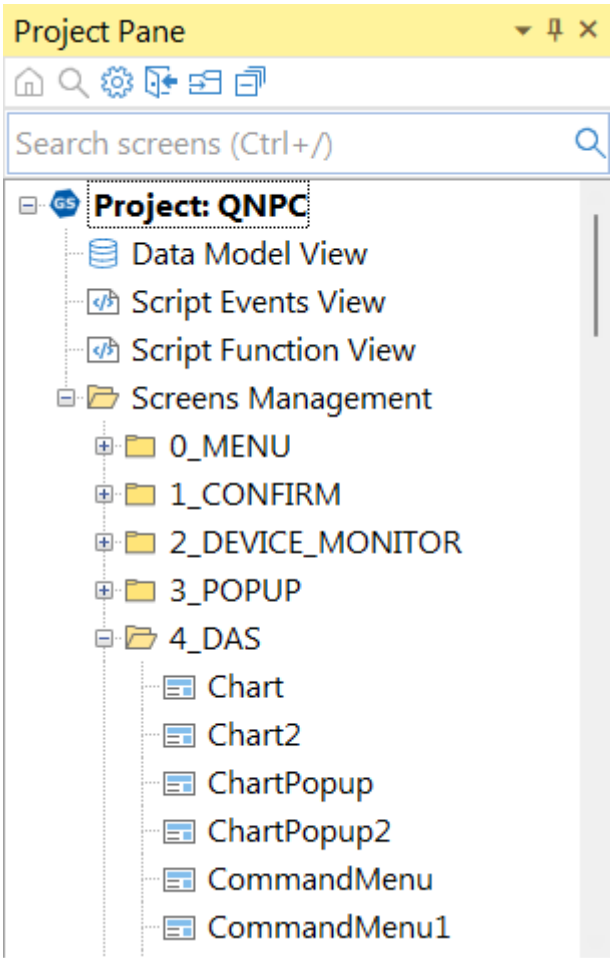


Figure F-19. Project Pane

	Show screens in default mode
	Show screens in search mode
	Show Project Properties Reference to F.3.3.7
	Database Log-out
	Database Log-in
	Open Project folder
	Collapse all (screen tree)

Search screens (Ctrl+/)	Enter the name of screen to search
 Project: T500HATINH	Project Name
 Data Model View	Display the Data Model management window Reference to section F.3.5
 Script Events View	Displays the Event Script management window
 Script Function View	Display the Function Script Management Window
 Screens Management	Default parent screen folder

F.3.1.13 Properties Window

Properties of controls based:

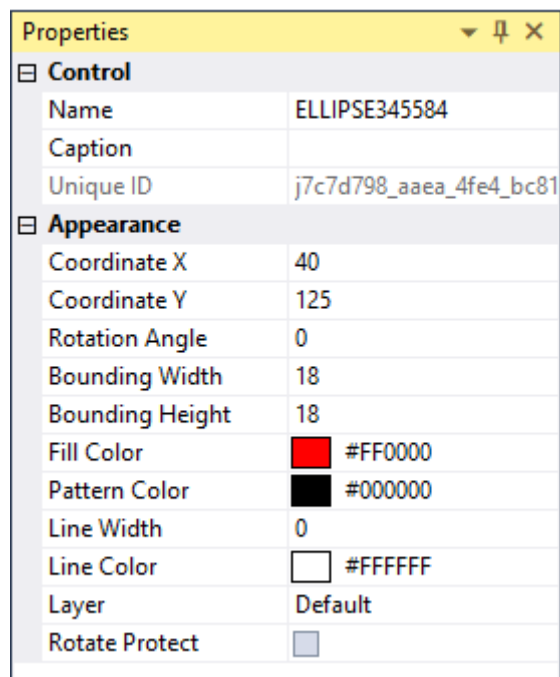


Figure F-20. Properties window

Control	
Name	Object name

Unique ID	The object ID, automatically managed by the application
Appearance	
Coordinate X, Y	Coordinates of the top left point of the object. The landmark (0,0) is the top left corner point of the screen containing the object
Rotation Angle	Rotation angle compared to the original drawing (mouse) operation
Bounding Width, Height	Width and Height of object
Fill color	Background color / fill color
Pattern color	Fill effect
Line color	Color of contour
Layer	Set the visibility layer for the object
Rotation Protect	Allows object rotation
Word Wrap (Text)	Allows text to automatically wrap to the next line when the content exceeds the width of the display area
Style (Button)	Select style of the button (Classic, Vista or Image)

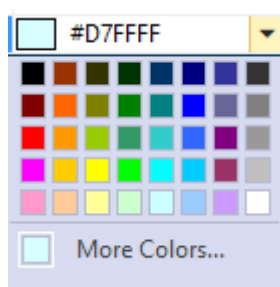
Properties of screen:

Screen	
Name	Screen name
Path	The screen file path in the project folder
Background	Background color of screen
Control count	Number of controls on the screen
Refresh Rate (ms)	Refresh rate of screen

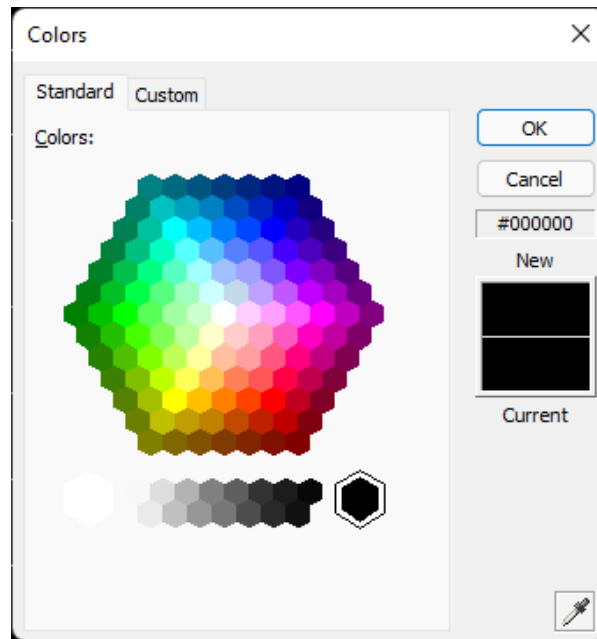
Resizable Screen	Allow screen to change the size (Can only be set when the screen is in Restore mode)
Zoomable Screen	Allow screen to zoom in/out in Viewer
Frame Properties	
X Coordinate Y Coordinate	Coordinates of the top left point of the screen. (Can only change when the screen is in Restore mode)
Canvas Width Canvas Height	The size of the workspace area where content or objects will be drawn or displayed.
Frame Width Frame Height	The size of screen frame (Can only change when the screen is in Restore mode and Resizable Screen is selected)
Show Title	Show the title bar of screen (Can only set when the screen is in Restore mode)
Screen Events	
On Open	The script executed when the screen is opened
While Running	The script executed continuously with a specified frequency (timer) while the screen is displayed.
On Close	The script executed when the screen is closed.

Color picker or Fill Color / Pattern Color / Line Color

- Choose color from the button 



- Choose **More Colors** to open wide color range palette



- Use Pick-up tool  tool to pick a color by selecting directly into any point.

F.3.1.14 Control View Window

- The tool helps to search for objects by object names or object ID

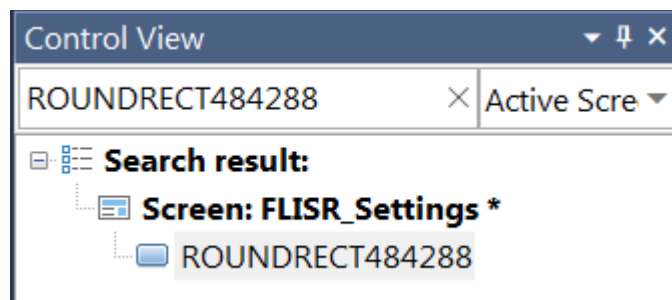
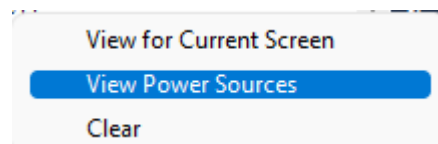


Figure F-21. Control View window

- Right-Click inside the Control View window to filter the search result.



View for Current Screen	Show only controls on the active screen
View Power Sources	Displays only Network objects are configured the “Properties Source” properties.
Clear	Clear the search result

- Filter/ search objects by name/ ID with Search tool (Using Ctrl + Q to activate search mode).

- Right-Click on the object → Menu Right-Click object calibration tools.
Reference to – Menu Right-Click graphic objects.

F.3.1.15 Symbol Library Window

Manage all symbols of project.

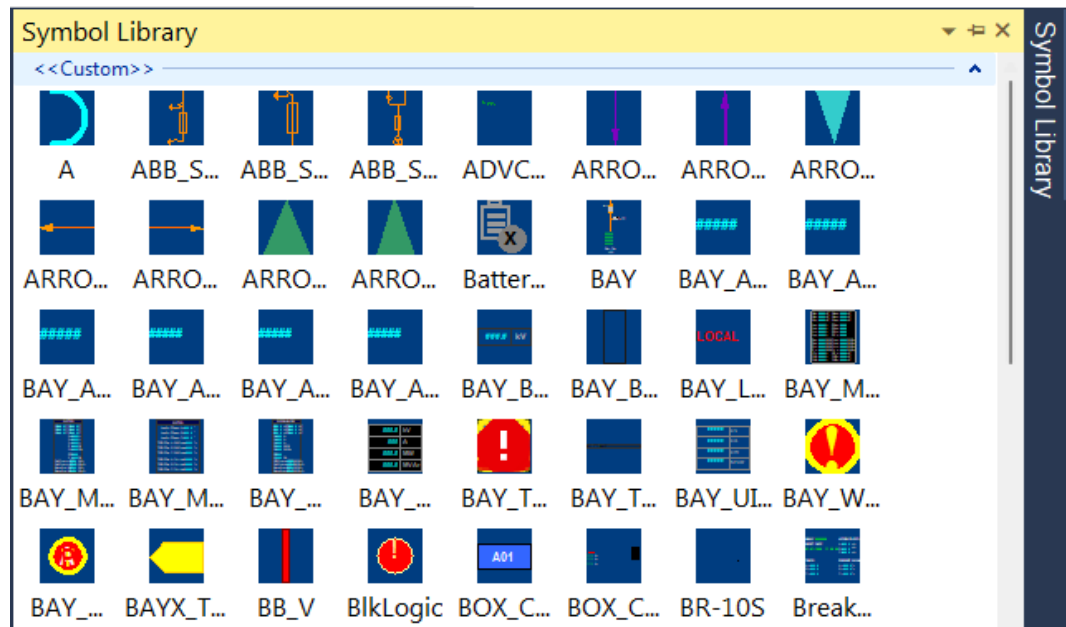


Figure F-22. Symbol Library Window

F.3.1.16 Layer Window

Manage layers of a screen.

Layers				
Visible	Name	Zoom Visible (%)	Active	Lock
☒	Default	[:]	☐	☐
☐	LF	[:]	☐	☐
☐	SHC	[:]	☐	☐
☒	DF	[:]	☒	☐
☒	CAMLO	[:]	☐	☐
☒	DAKRONG	[:]	☐	☐
☒	DONGHA	[:]	☐	☐
☒	GIOLINH	[:]	☐	☐
☒	HAILANG	[:]	☐	☐
☒	KHESANH	[:]	☐	☐
☒	THANHCO	[:]	☐	☐
☒	TRIEUPHONG	[:]	☐	☐
☒	VINHLINH	[:]	☐	☐
☐	Hide	[:]	☐	☐
☐	Line	[:]	☐	☐

Figure F-23. Layer Window

F.3.1.17 Output View Window

To provide information, warnings, errors during project compilation.

Output View		
All		
Error		
Warning		
All		
Warning	7SL82 CPM	eafe968c_21a5_4650_bcl0a_7bdf4b7f1d60
Warning	7SL82 CPM	e621a94e_2cc9_4b5e_94a2_572a5dd50888
Warning	7SL82 CPM	a41d59e8_f29d_42e2_a2a9_de782f8f4de4
Warning	7SL82 CPM	g842bc7f_d33e_4903_ad90_c065c26004a4
Warning	7SL82 CPM	g0e6753d_b065_4ec3_8331_3b560394c703
Warning	7SL82 CPM	ea7182f9_c7e7_44f8_8282_9eeeee0b1dba
Information	Project Manager	
Information	Project Manager	
Information	Project Manager	
Information	Project Manager	

Figure F-24. Output View Window

F.3.1.18 Profile Window

Manage profiles. Reference to section F.3.20

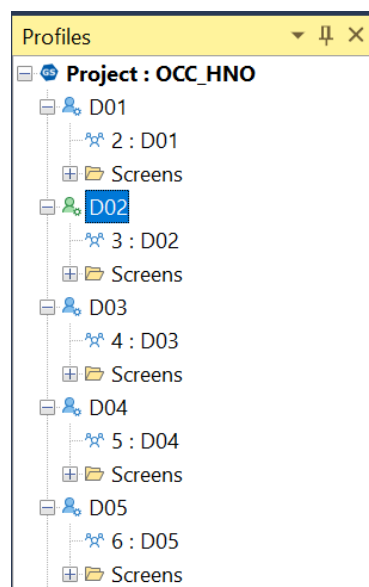


Figure F-25. Profile Window

F.3.2. Configure common parameters for the application

On the Menubar, select **Tool/Option** to open the window that allows you to set default configurations for the software.

F.3.2.1 General Tab

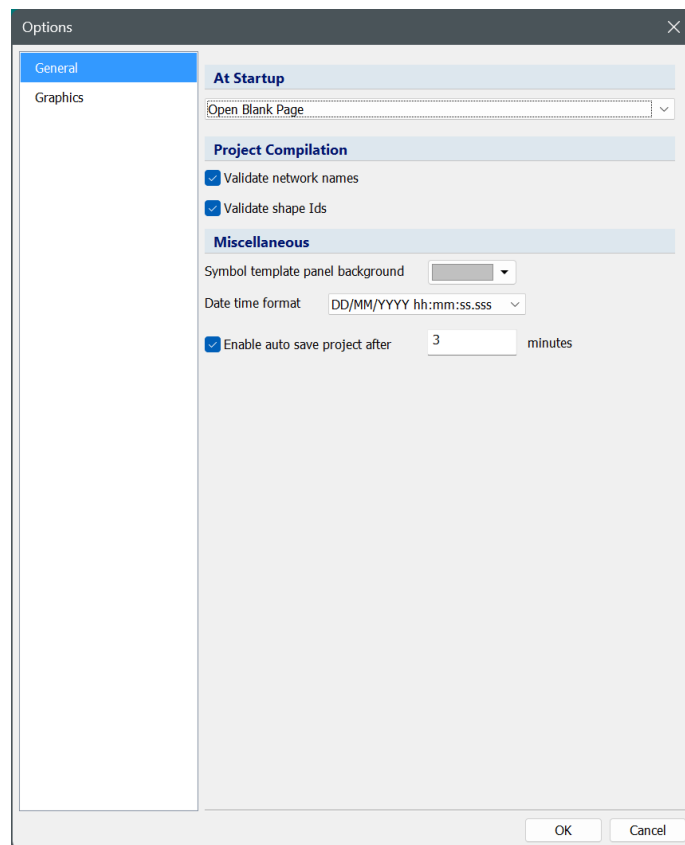


Figure F-26. Options for General

At startup: Options to open the project when running the application (Designer):

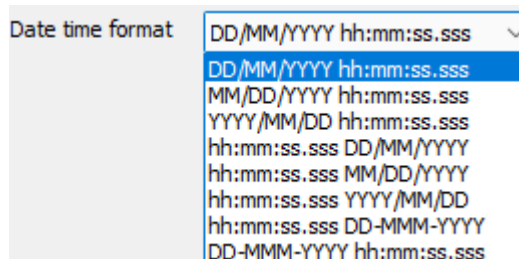
- **Load Last Loaded Project:** open the last opened project
- **Open Blank Page:** do not open any project

Project Compilation: Information compilation:

- **Validate network names:** validate network objects name
- **Validate shape Ids:** Validate the IDs of graphic objects

Miscellaneous:

- **Symbol Template Panel Background Color:** allows you to choose a background color for **Symbol Template Library** window.
- **Date time format:** allows you to choose to display different date and time formats. (Controls like Datetime, EventView, TimeFormat function, Chart, ... will display in the same format)



- **Enable auto save project after:** Automatically save project after a period of time.

F.3.2.2 Graphic

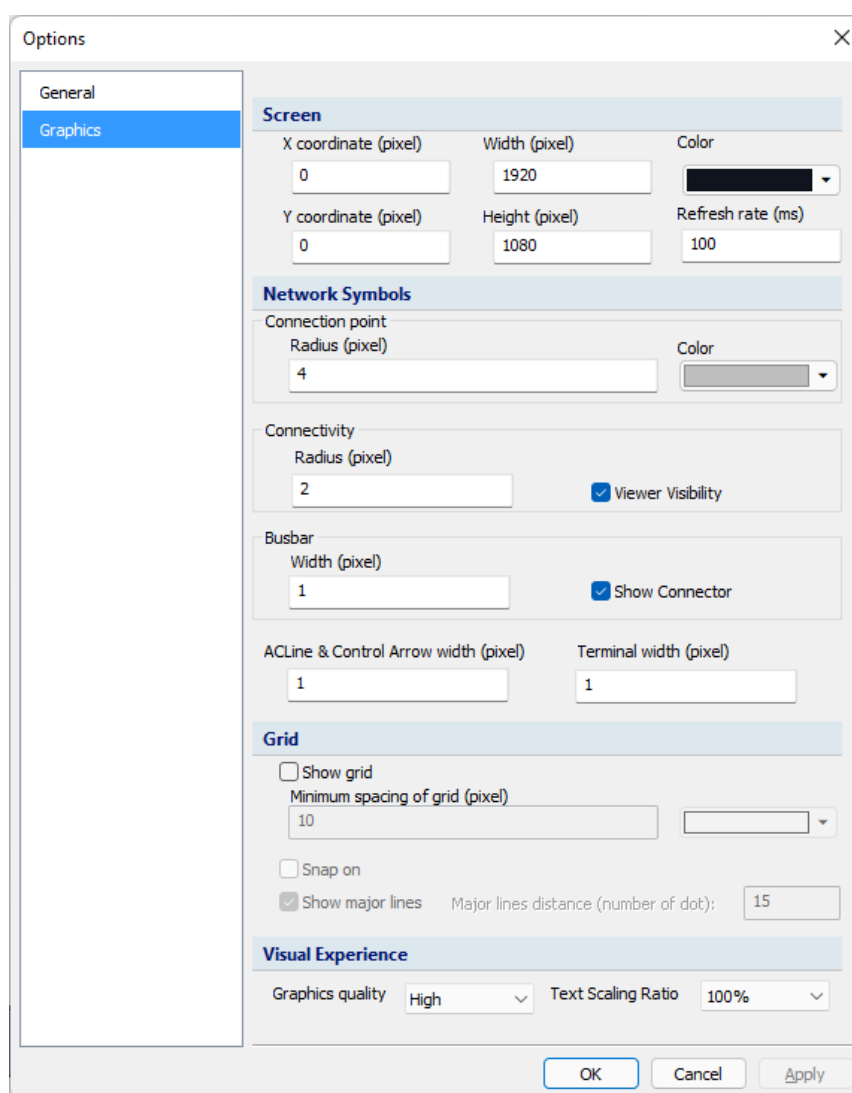


Figure F-27. Options for configuring the project's graphics

Screen: contains settings related to the screen when it is created, including:

- **X coordinate, Y coordinate:** Coordinates in pixels.
- **Width, Height:** Dimensions width, height.
- **Color:** Screen background color.

- **Refresh rate:** Graphics update frequency.

Network Symbol: configure graphical display parameters for the *Electrical Network* objects:

- **Connection point:** Set size and color.
- **Connectivity:** choose the size, the option to hide / show on the Viewer.
- **Busbar:** configure width, option to hide/show Connectors.
- **ACLine & Control Arrow width (pixel):** Width option for ACLine, Control Arrow, Terminal objects.
- **Terminal width (pixel):** The width of terminal

Grid: Configure grid mode:

- **Show grid :** Allow to show or hide the grid on screen
- **Minimum spacing of grid (pixel):** Adjust distance between the grid nodes.
- ▼ to choose the color for grid nodes.
- **Snap on:** Enable/disable automatic snap.
- **Show major lines:** Turn on/off the main trunk lines.
- **Major lines distance (number of dot) :** The spacing between major grid lines

Visual Experience: graphic options.

- **Graphics quality:** graphic quality option.
- **Text Scaling Ratio:** display text according to the specified zoom ratio. When transferring projects from different machines, maybe "text scaling" configurations on different Windows, resulting in text not being displayed as expected (too large or too small), then it will be necessary to adjust the ratio. This ratio on Grid Studio for proper display of text.

F.3.3. Project management

F.3.3.1 Create new project

From the menubar choose **File → New Project:**

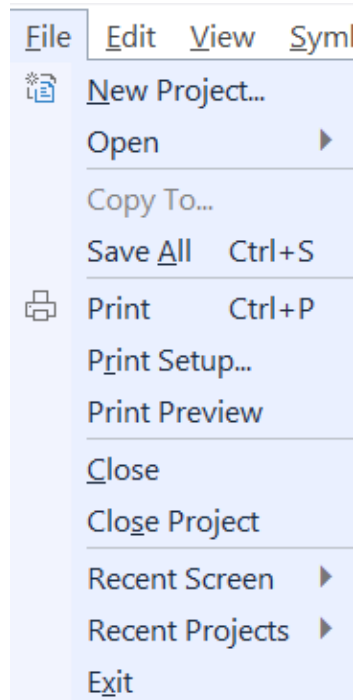


Figure F-28. Menu File

A dialog box appears:

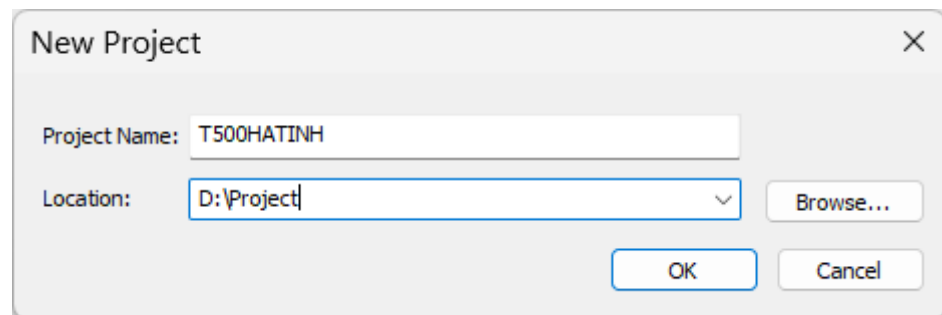


Figure F-29. The dialog box creates new project.

Project Name	Project name
Location	Location to save the project. Click Browse to choose location.

F.3.3.2 Open Project

From the menubar choose **File** → **Open** → **Project** (Ctrl+O). Go to the existing project folder, select filename .hdp, select **Open**.

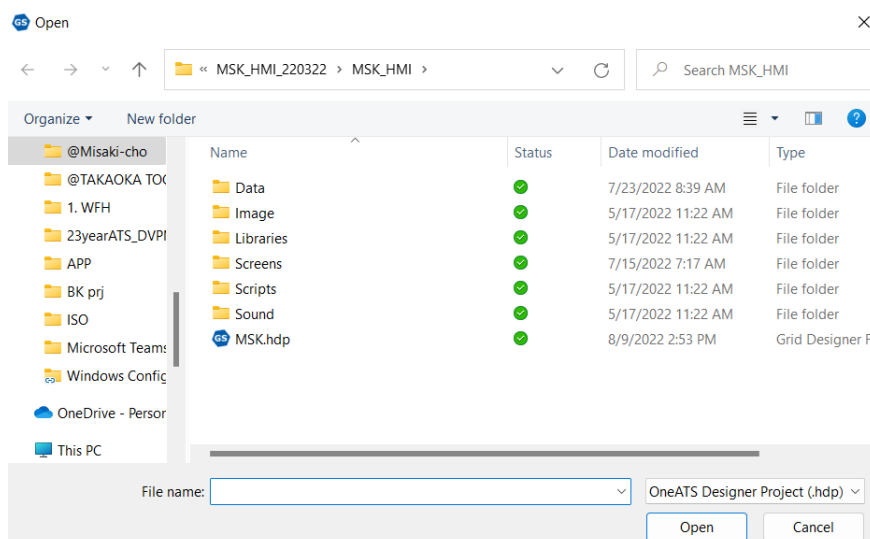


Figure F-30. The location dialog opens the project

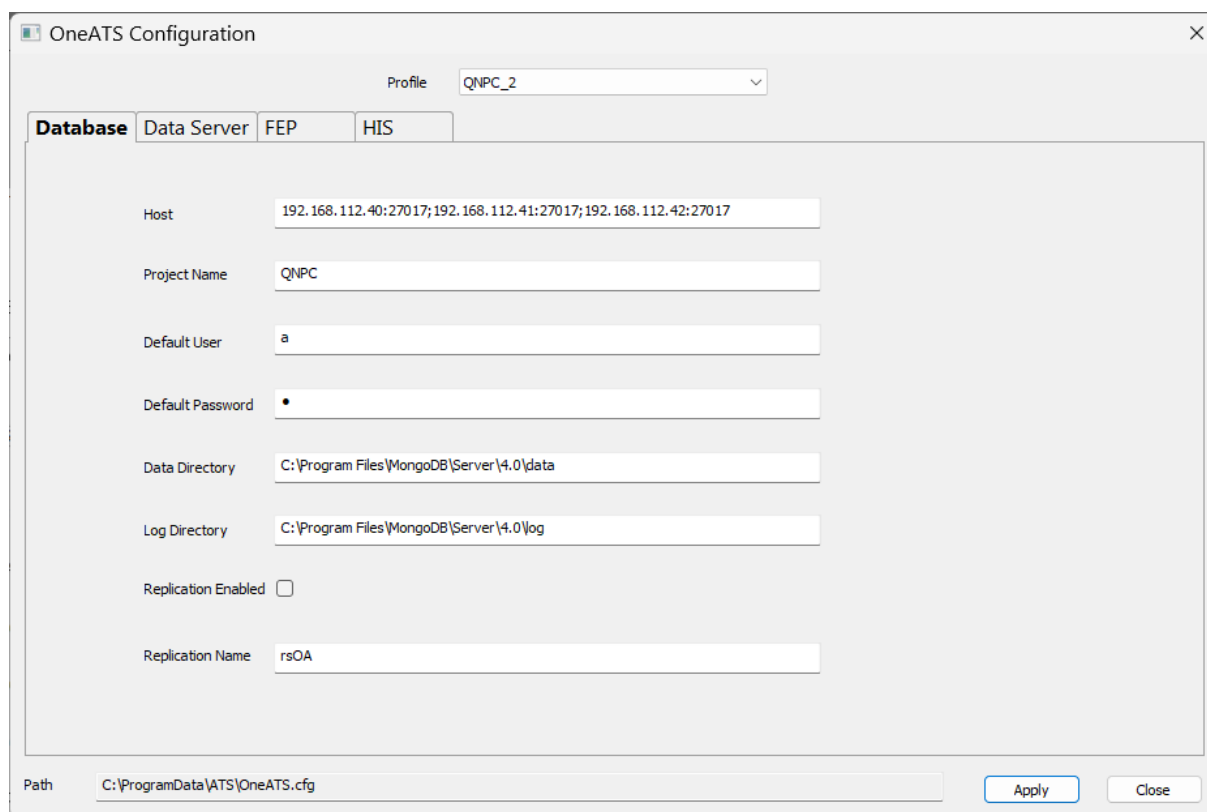
If the project has been logged in before, logging into the project is as simple as entering the information in the **Database Login** window and selecting **OK**.

Users can open recently opened projects by selecting **File → Recent Project** from the menu >> selecting the project they want to open.

In case you open the project with other Database address settings, the software will notify "Cannot Connect to Database", users need to pay attention to edit appropriate information **Host** (Address/Port), **Database** (Project Name) in the **OneATS Master Tool** application. In the common case of different Database address due to project conversion from personal engineer computer to workstation computer, at the initial login dialog select "Skip".

F.3.3.3 Login Project

For first-time login, use **OneATS Master Tool** software to configure **Host** and **Project Name** according to the database configuration (IP, Port, Project Name) of the system before logging in.



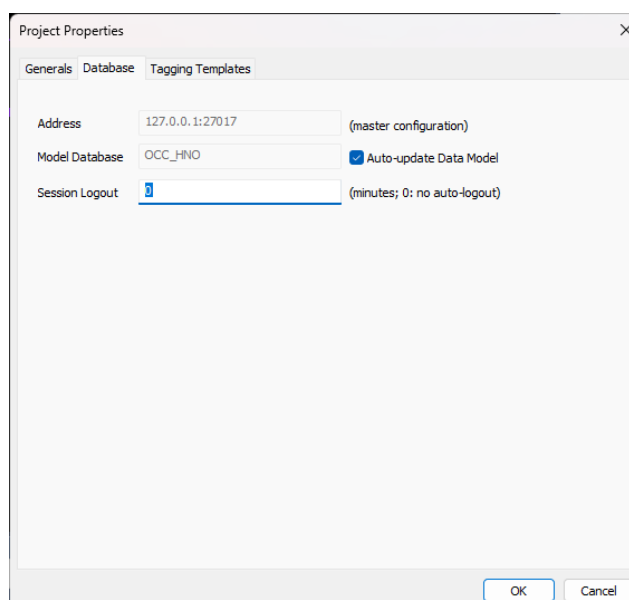
The OneATS Configuration window shows the 'Database' tab selected. The 'Profile' dropdown is set to 'QNPC_2'. The configuration fields are as follows:

Field	Value
Host	192.168.112.40:27017;192.168.112.41:27017;192.168.112.42:27017
Project Name	QNPC
Default User	a
Default Password	•
Data Directory	C:\Program Files\MongoDB\Server\4.0\data
Log Directory	C:\Program Files\MongoDB\Server\4.0\log
Replication Enabled	☐
Replication Name	rsOA

The Path field at the bottom is set to 'C:\ProgramData\ATS\OneATS.cfg'. The 'Apply' and 'Close' buttons are visible at the bottom right.

Figure F-31. OneATS Master Tool Configuration window

After configuring the **Host** and **Project Name**, open the HMI project, select **Security** → **Settings**, and check that the configuration information is matched.



The Project Properties window shows the 'Security' tab selected. The configuration fields are as follows:

Field	Value	Notes
Address	127.0.0.1:27017	(master configuration)
Model Database	OCC_HNO	☒ Auto-update Data Model
Session Logout	0	(minutes; 0: no auto-logout)

The 'OK' and 'Cancel' buttons are visible at the bottom right.

Figure F-32. Security window

Click **Security** → **Login** to open the database login window. Enter **User name** and **Password**.

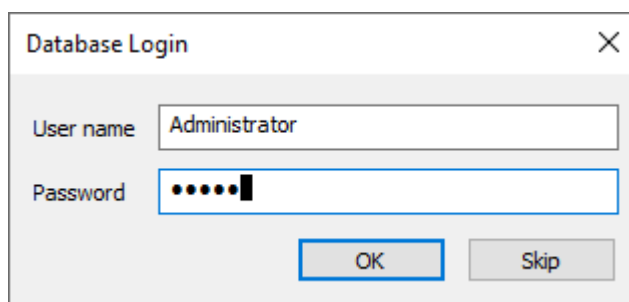


Figure F-33. Database login window

F.3.3.4 Close Project

Choose the **File** → **Close Project** menu to close current project.

F.3.3.5 Backup project

- From the menu choose Project/Backup Project:

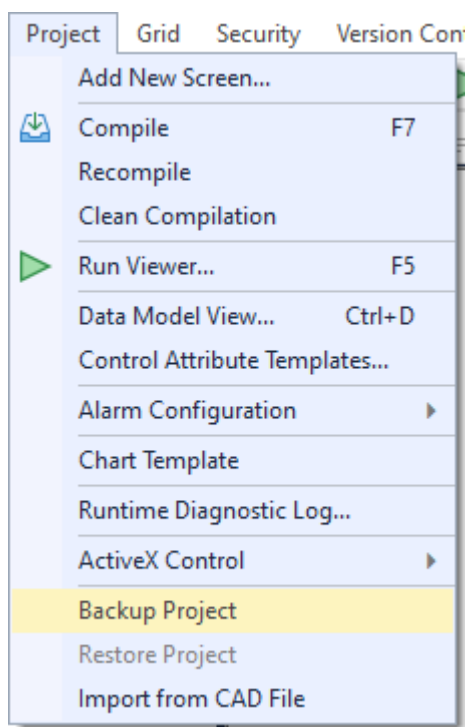


Figure F-34. Backup Project

- Select a location to store the backup project.

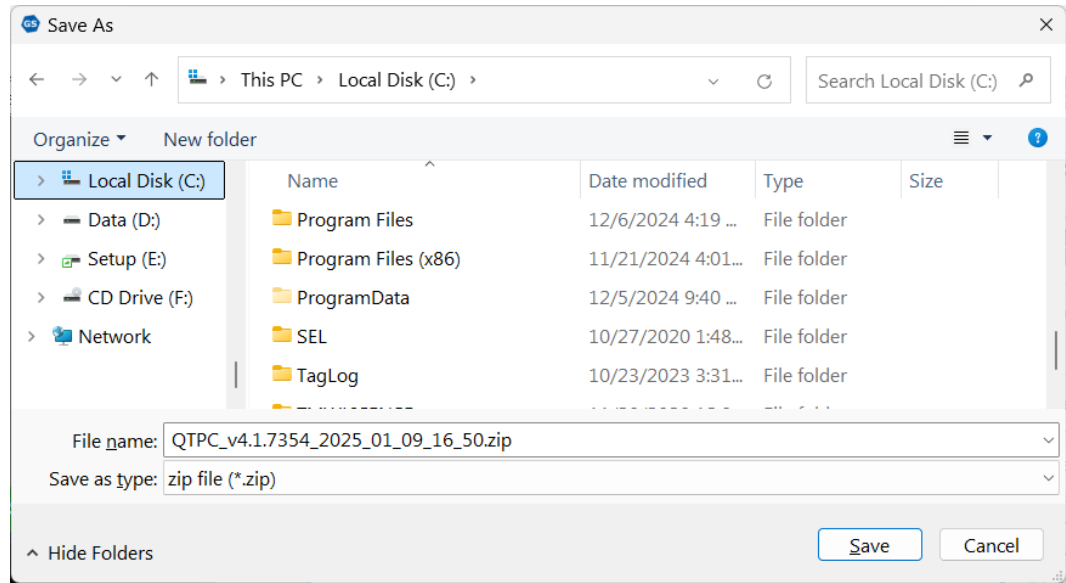


Figure F-35. Select a location to backup

- The project is backed up as a file
[project_name]_[version]_[timestamp].zip in the selected folder.

F.3.3.6 Restore project

Unavailable.

F.3.3.7 Project properties

Project Properties table includes the common properties of the project.

From **Project Pane**, Right-Click → **Properties** ⚙️ :

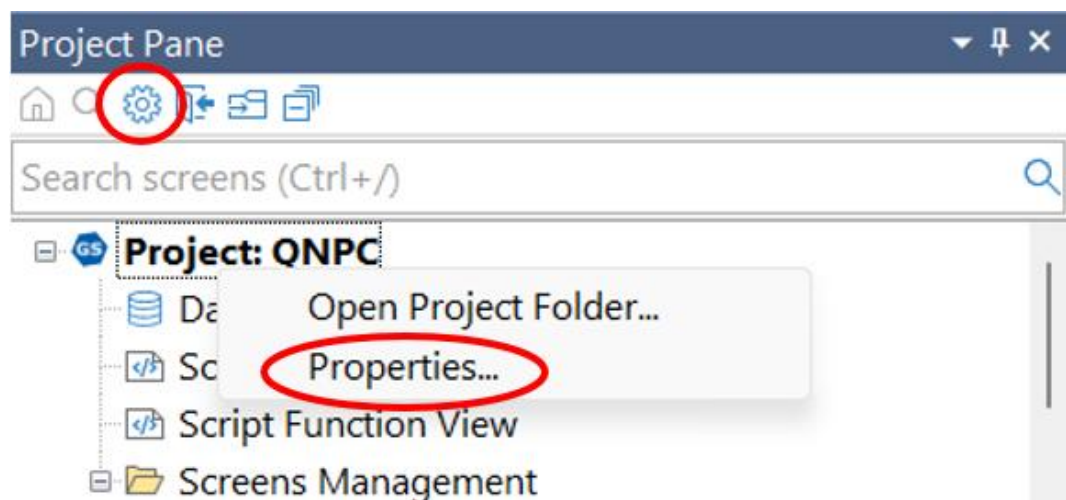


Figure F-36. Menu opens project properties

F.3.3.7.1 Generals Tab: Viewer properties that run the Viewer application

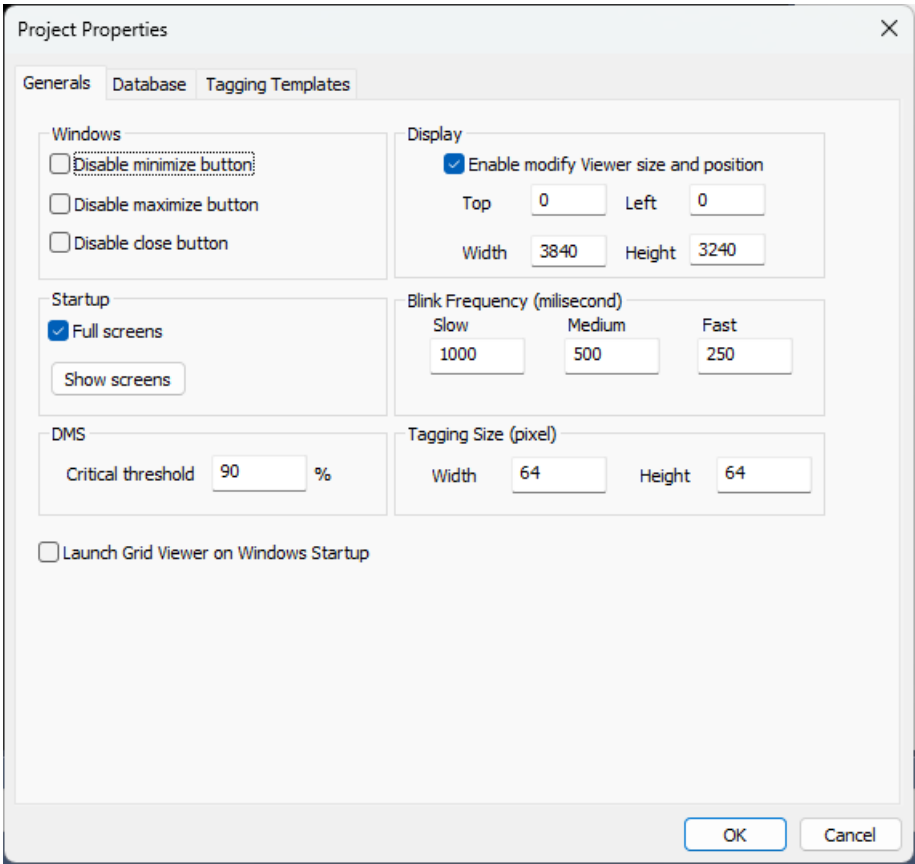
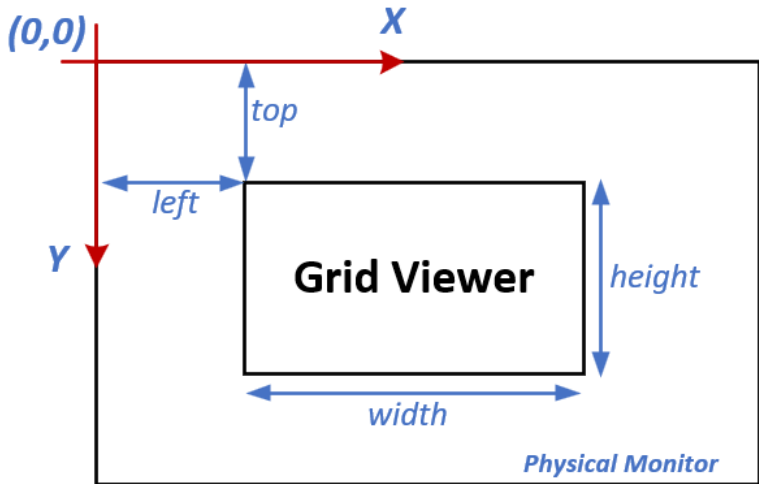
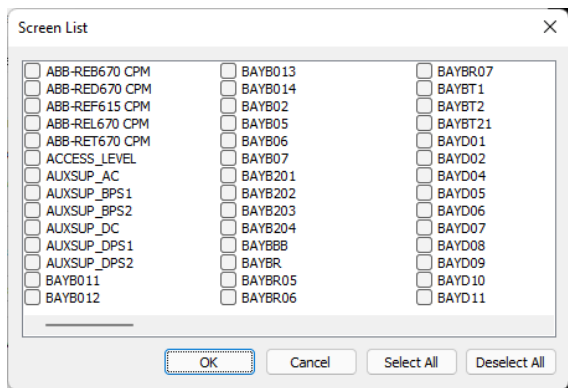
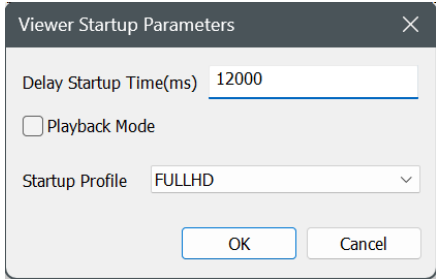


Figure F-37. Project Properties – General Tab

Properties	Description	
Windows	Hide / show min/max/close button of Grid Viewer application	
	Disable minimize button	Hide minimize button 
	Disable maximize button	Hide maximize button  / 
	Disable Close button	Hide Close button 
Display	Display coordinate, application window size Grid Viewer Display.	
	Enable modify Viewer size and position	Allows resizing, moving the Grid Viewer application window frame.

	Top, Left	Grid Viewer coordinates in the physical screen coordinate.
	Width, Height	Size of Grid Viewer application window.
Coordinates, display size, in units of Pixel (0,0): upper left corner of Grid Viewer Window 		
Startup	Select the screens to display when starting the Viewer	
	Full Screen	Grid Viewer will run with maximize mode and hide toolbars
	Show Screens	Select screens to be displayed when Grid Viewer starts. 
Blink Frequency (milisecond)	Set the frequency for the blink properties of graphic objects.	
DMS	Critical Threshold (%) – P limit threshod (%) using for warning operator in PVCurves calculation	

Tagging Size (pixel)	Configure the size for the Tagging icon.
Launch Viewer on Windows Startup	Setting registry to have Grid Viewer run from Start-up. Manage startups by Window Task Manager / Startup  - Delay Starup Time(ms): Starting the Viewer after a Delay - Playback Mode: Start the viewer with Playback mode - Startup Profile: Launch the Viewer according to the selected profile

F.3.3.7.2 Tagging Templates Tab

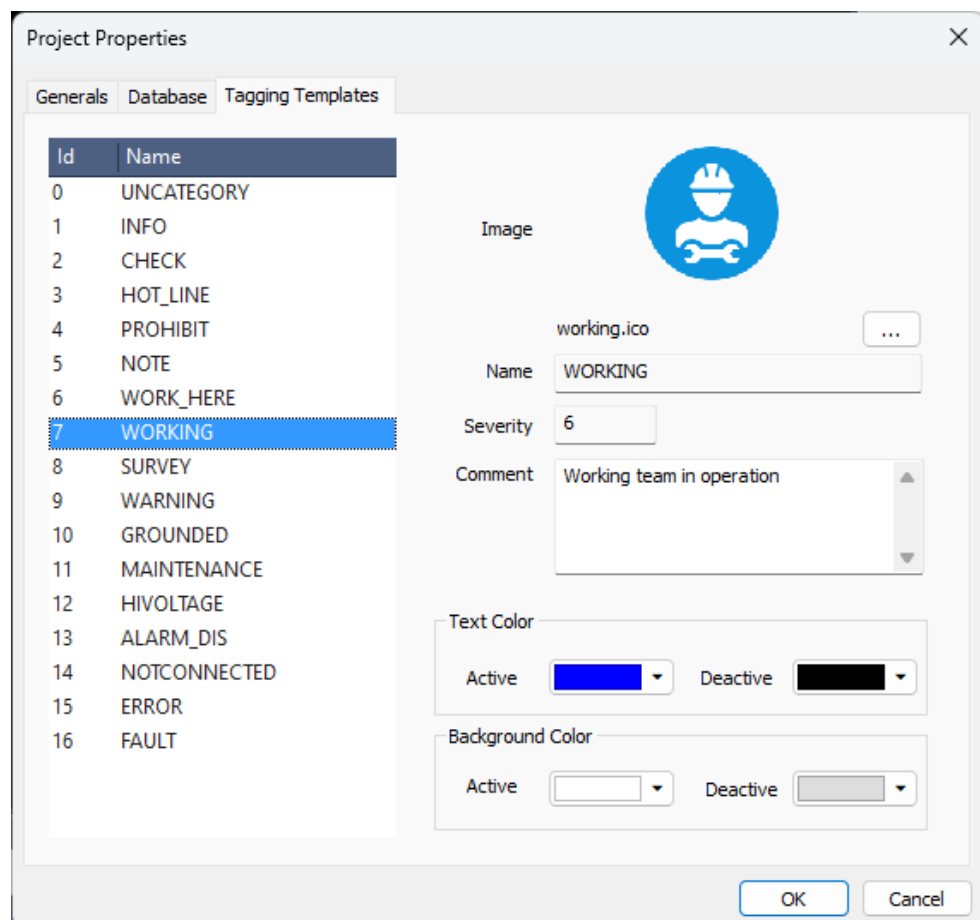


Figure F-38. Project Properties –Tagging Templates Tab

Property	Description
Id	Order Number
Name	Tagging Name.
Image	icon of the corresponding tagging, click the icon  to change the icon (request file ".ico").
Severity	The severity of the tagging corresponds.
Comment	The description of Tagging Icon
Text Color	Text color of activated and deactivated tagging
Background Color	Background color of activated and deactivated tagging's row

F.3.3.7.3 Security Tab

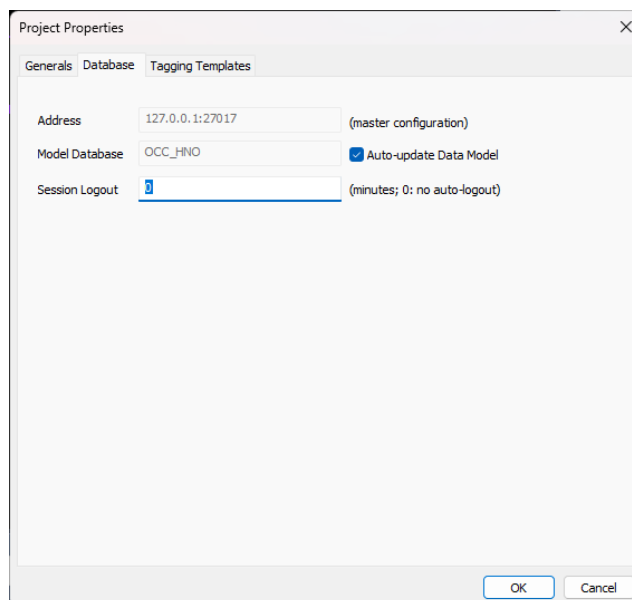


Figure F-39. Security Tab

Properties	Description
Address	IP address, TCP port of (MongoDB) Database is the location where Data Model and project configuration are stored.

Model Database	Name of Project Database (MongoDB Collection)
Auto-update	Automatically update when Data Model has new changes
Session Logout	Time to automatically terminate a user's session on Grid Viewer Note: = 0 (zero) means not automatically log out.

F.3.4. Compile and debug in Designer environments

The program compiles screen file of the form **screen_name.scn** into the form **screen_name.scnv**. Compilation progress, warnings, error message can be viewed in the **Output View** window. Warnings, error message in this window are related to the compilation progress.

Type	Location	Identifier	Description
Warning	SEL421 CPM	dcb909fb_d002_4096_b7b8_68a5ae80d308	has part outside screen, position TL(789, 15), RB(475, 834)
Information	Project Manager		Compiling E:\OneDrive - ATS\JSC\OCC\HANOI\HMI\OCC_HNO_v4.1.7170_2023_10_24_15_00\OCC_HNO\Screens\3_STATIONS\8_RELAY\RELAY CPM\SEL451 CPM.scn
Warning	SEL451 CPM	c3065c60_46a2_447d_88e6_f2057b49dfce	has part outside screen, position TL(510, 5), RB(990, 880)
Information	Project Manager		Compiling E:\OneDrive - ATS\JSC\OCC\HANOI\HMI\OCC_HNO_v4.1.7170_2023_10_24_15_00\OCC_HNO\Screens\3_STATIONS\8_RELAY\RELAY CPM\SEL487B CPM.scn
Warning	SEL487B CPM	1e18d12b_24e9_4df1_9deb_94160568d8ae	has part outside screen, position TL(5, 5), RB(961, 770)
Warning	SEL487B CPM	a5f3099_5752_4c06_a1d7_53700c3ff424	has part outside screen, position TL(5, 511), RB(961, 506)
Warning	SEL487B CPM	fad2325d_8c31_4d99_9543_b17970d1683e	has part outside screen, position TL(720, 15), RB(675, 765)
Warning	SEL487B CPM	e17c503e_d1b7_4f1a_a5a6_6832b2193a82	has part outside screen, position TL(720, 690), RB(950, 765)
Information	Project Manager		Compiling E:\OneDrive - ATS\JSC\OCC\HANOI\HMI\OCC_HNO_v4.1.7170_2023_10_24_15_00\OCC_HNO\Screens\3_STATIONS\8_RELAY\RELAY CPM\SEL487E CPM.scn
Warning	SEL487E CPM	h49106aa_3605_4126_b138_12887278b60f	has part outside screen, position TL(625, 5), RB(990, 935)
Warning	SEL487E CPM	dcb909fb_d002_4096_b7b8_68a5ae80d308	has part outside screen, position TL(884, 15), RB(475, 929)
Information	Project Manager		Compiling E:\OneDrive - ATS\JSC\OCC\HANOI\HMI\OCC_HNO_v4.1.7170_2023_10_24_15_00\OCC_HNO\Screens\3_STATIONS\8_RELAY\RELAY CPM\SIEMENS\75J62 CPM.scn
Warning	75J62 CPM	b7f7b1c80_2631_4538_ae5fd_736c5ad1be5	has part outside screen, position TL(5, 5), RB(489, 635)
Information	Project Manager		Compiling E:\OneDrive - ATS\JSC\OCC\HANOI\HMI\OCC_HNO_v4.1.7170_2023_10_24_15_00\OCC_HNO\Screens\3_STATIONS\8_RELAY\RELAY CPM\SIEMENS\75L82 CPM.scn
Warning	75L82 CPM	eafe968c_21a5_4650_bca7_7bdf4b7f1d60	has part outside screen, position TL(1, 2), RB(482, 661)
Warning	75L82 CPM	e621a94e_2cc9_4b5e_94a2_572a5dd50888	has part outside screen, position TL(31, 7), RB(477, 656)
Warning	75L82 CPM	a41d59e8_f29d_42e2_a2a9_de782f9f4de4	has part outside screen, position TL(534, 75), RB(160, 639)
Warning	75L82 CPM	g842be7f_d33e_4903_a490_c065c26004a4	has part outside screen, position TL(534, 168), RB(201, 639)
Warning	75L82 CPM	g0e6753d_b065_4ec3_8331_3b560394c703	has part outside screen, position TL(566, 248), RB(457, 646)
Warning	75L82 CPM	ea7182f9_c7e7_44f8_8282_9aeae0b1dba	has part outside screen, position TL(566, 214), RB(249, 646)
Information	Project Manager		Compiling E:\OneDrive - ATS\JSC\OCC\HANOI\HMI\OCC_HNO_v4.1.7170_2023_10_24_15_00\OCC_HNO\Screens\3_STATIONS\9_BUSBAR\TEAM D02\BAYDBB_E1_40.scn
Warning	BAYDBB_E1_40	i554cbb6_332e_492b_8729_095e46b3d8f4	is duplicated network name
Information	Project Manager		Compiling E:\OneDrive - ATS\JSC\OCC\HANOI\HMI\OCC_HNO_v4.1.7170_2023_10_24_15_00\OCC_HNO\Screens\3_STATIONS\9_BUSBAR\TEAM D02\BAYEBB_E1_40.scn
Warning	BAYEBB_E1_40	77780dbd_6aaa_4e5d_b968_a126c19f29f2	is duplicated network name
Warning	BAYEBB_E1_40	h4334f12_30c6_49f0_990b_d369cf6916b6	is duplicated network name
Warning	BAYEBB_E1_40	j4b01712_cf06_4214_a288_fedc8f809478	is duplicated network name
Warning	BAYEBB_E1_40	b831f923_1bdc_4d4e_830c_abe4c05ba17	is duplicated network name
Information	Project Manager		Compiling E:\OneDrive - ATS\JSC\OCC\HANOI\HMI\OCC_HNO_v4.1.7170_2023_10_24_15_00\OCC_HNO\Screens\3_STATIONS\xxxx.scn
Information	Project Manager		----- Compilation completed. 0 error(s), 1046 warning(s) -----

Figure F-40. Output View window

From the menu bar, choose Project, drop-down menu displays:

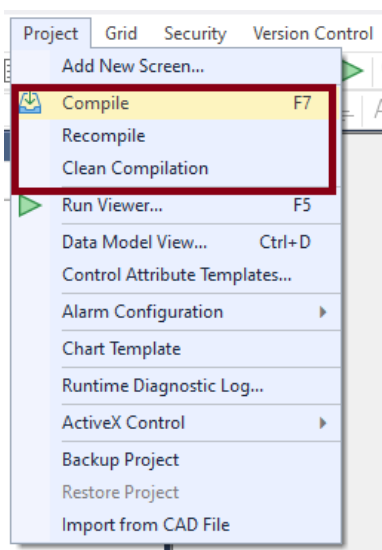


Figure F-41. Compile Project Menu

- **Compile:** start compiling. (or use F7 key)
- **Recompile:** clear the previously compiled file, then start the compilation process.
- **Clean Compile:** clear the compiled file.

From the Output Viewer, you can double-click to select the object with the corresponding Unique ID on the Event line.

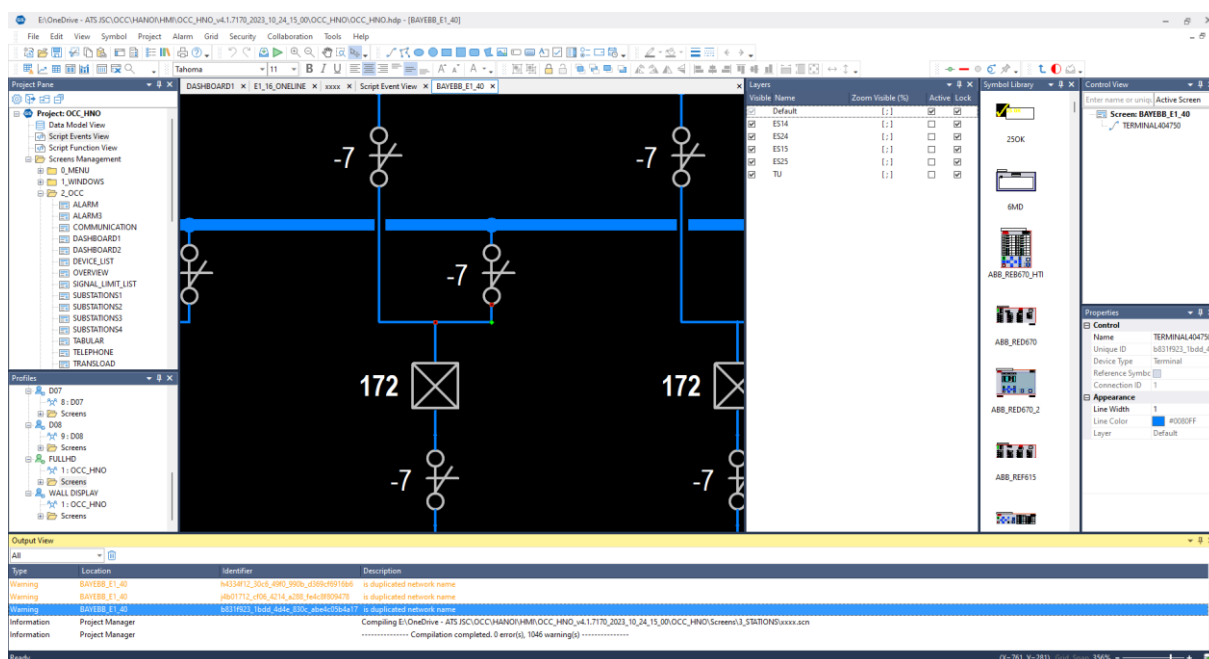


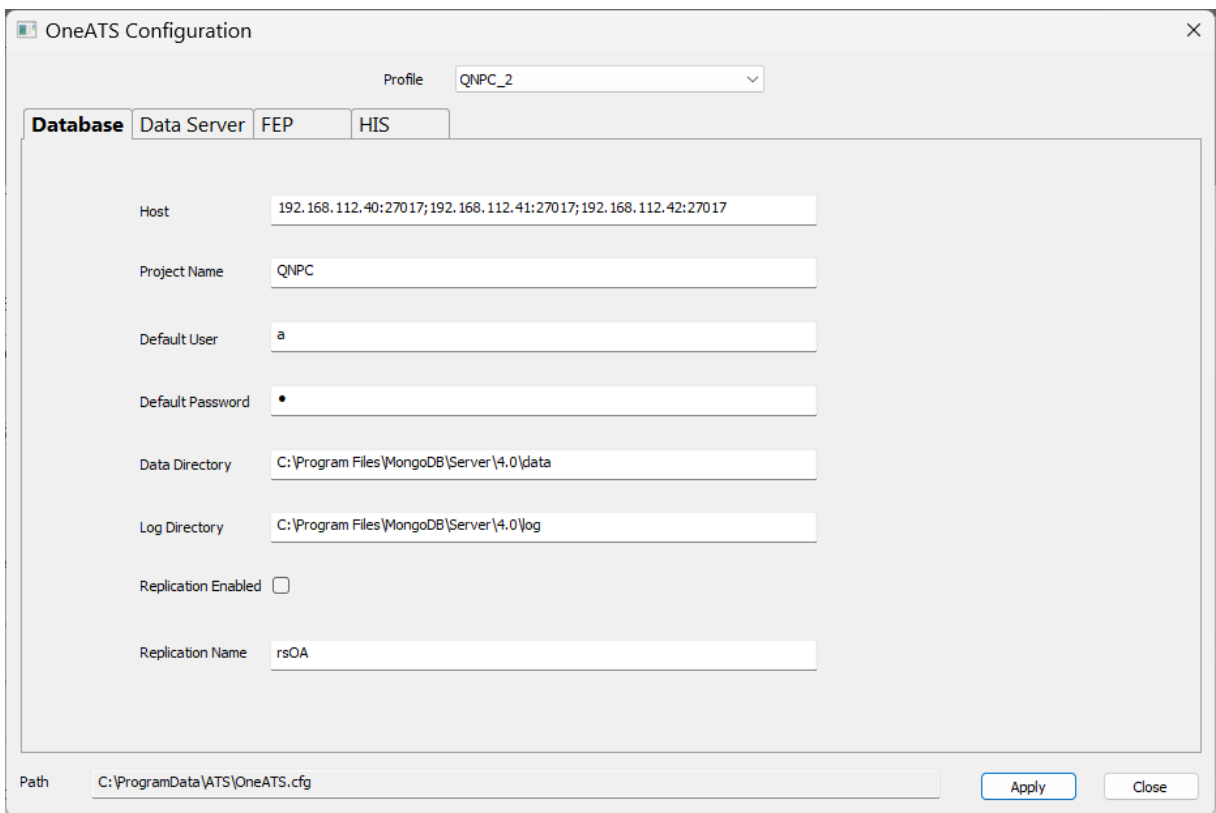
Figure F-42. Select Object from Output View

F.3.5. Data management/ Data model

OneATS Grid Studio manages information data by **tags**. Each data point is a Tag. Each Tag contains many various fields of information, including **Keyname**. **Keyname** is technical identifier name of a **Tag**. **Keyname** has a unique value and distinguishes between Tags.

F.3.5.1 Manage data connection to Database

Connect to the Database configured on OneMasterTool software. Connections to the Database/DataServer/HIS Server are configured on the Data Model, which will be updated in Designer when performing "Update Data Model" (Reference at F.3.5.3.2)



The image shows the 'OneATS Configuration' window with the 'Database' tab selected. The 'Profile' dropdown is set to 'QNPC_2'. The configuration fields are as follows:

Field	Value
Host	192.168.112.40:27017;192.168.112.41:27017;192.168.112.42:27017
Project Name	QNPC
Default User	a
Default Password	•
Data Directory	C:\Program Files\MongoDB\Server\4.0\data
Log Directory	C:\Program Files\MongoDB\Server\4.0\log
Replication Enabled	☐
Replication Name	rsOA

At the bottom, the 'Path' field is set to 'C:\ProgramData\ATS\OneATS.cfg'. There are 'Apply' and 'Close' buttons at the bottom right.

Figure F-43. OneATS Master Tool Configuration

F.3.5.2 Tag Classification

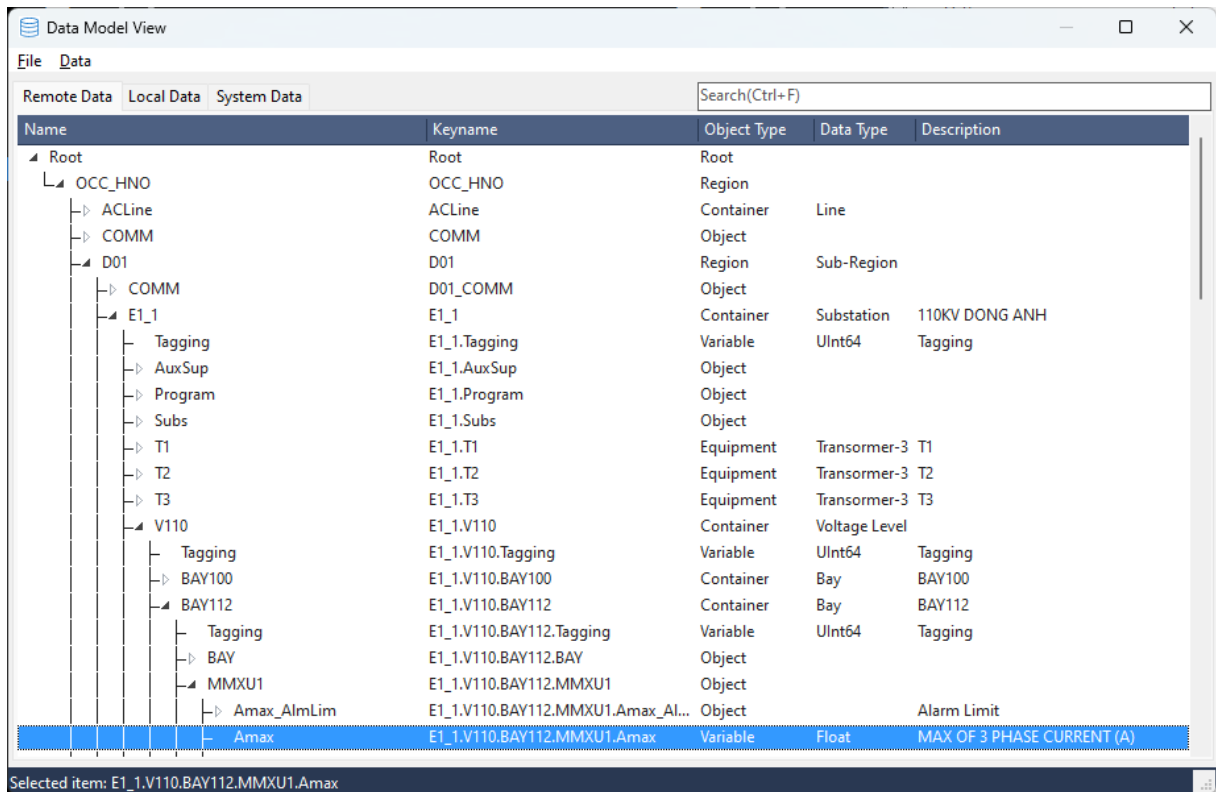


Figure F-44. Data Model Windows

Tag type	Description
Remote Data	The data model is structured by tags, which are used in all OneATS platform's modules by the imported/downloaded data model.
System Data	System tags are built-in inside the OneATS Grid Studio application. These tags are used internally in the OneATS Grid Studio module and have a read-only attribute.
Local (Application) Data - Variable Type	Tags are created and scoped for use within the OneATS Grid Studio module.
Local (Application) Data - Reference Type	Reference is a special type of Tag/Object whose data type is Reference Object/Reference Variable that belongs to the Local Data scope. A reference carries the property of the

	object/variable it refers to. There are two types of references: ❖ Reference Object: refers to an object. ❖ Reference Variable: refers to a variable.
Local Script Variables	Variables are declared and scoped for use within the script.

F.3.5.3 Remote Data in Data Model

F.3.5.3.1 Remote Data List Management

The **Data Model View** is the place to manage all tags for application. To open the window, from the menu bar of Designer selects the **Project/Data Model View** or press the keyboard shortcut **Ctrl + D**.

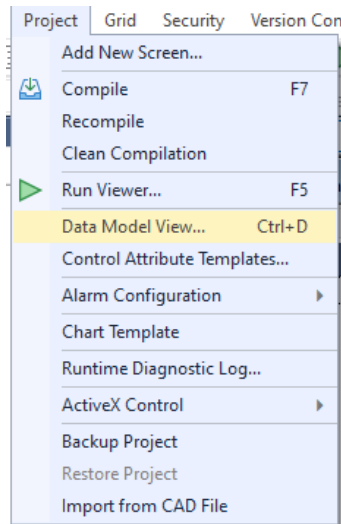


Figure F-45. Data Model View

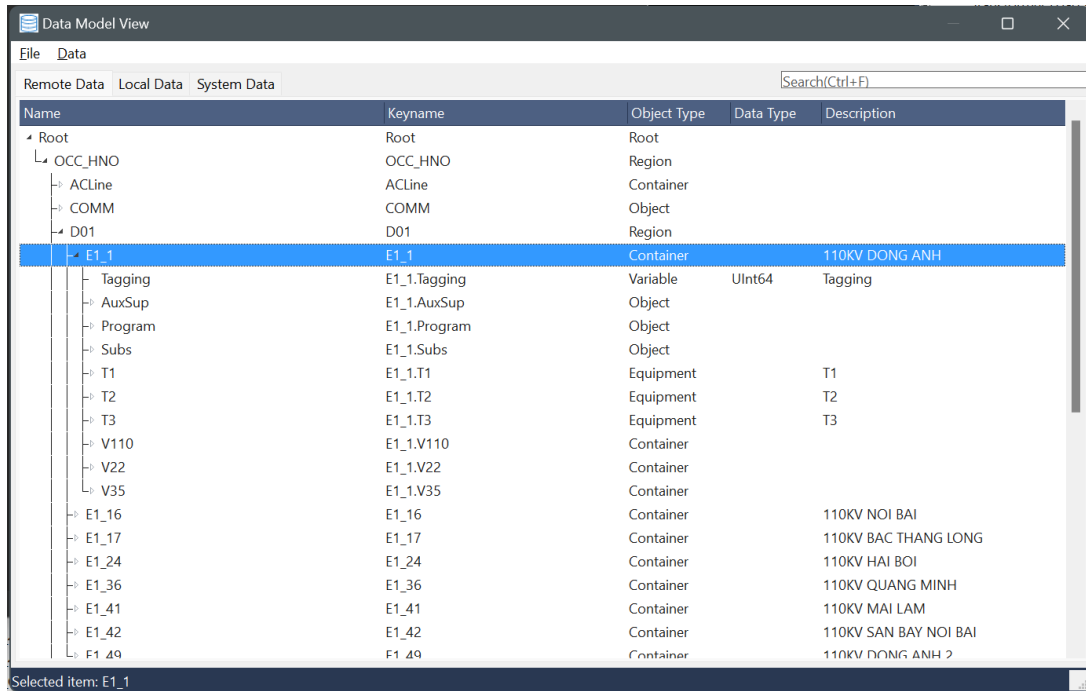


Figure F-46. Data Model View

F.3.5.3.2 Update Remote Data

When there is a change in data from the Data Model, the user needs to update the Remote Data. This update ensures that the data from Remote Data is always uniform with the data in the Database. The process of updating data for Remote Data is called Update Data Model.

To update Remote Data, Grid Designer needs to log in and connect to the Database (Mongo DB).

Open the Data Model View window, choose Data Model >>Update Data Model.

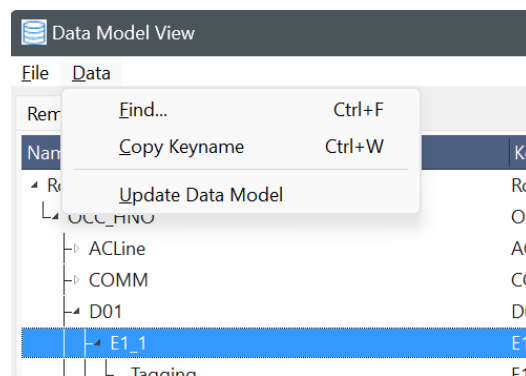


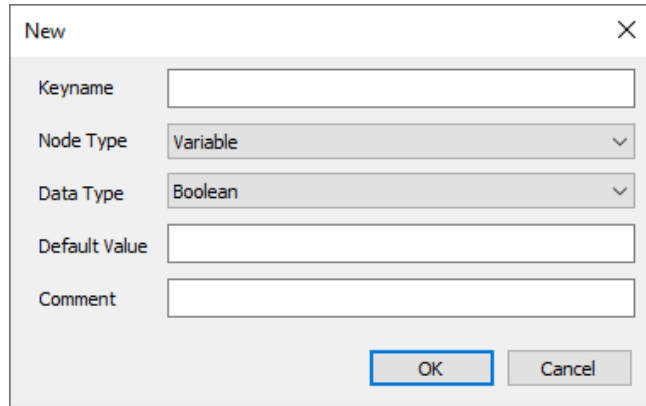
Figure F-47. Update Data Model

Select the **Remote Data** tab in the **Data Model View** window. Here, the model structure is displayed in the form of a tree. The user can use this structure to check that the data has been updated. Each tag has data fields **Keyname**, **object type**, **data type**, **description**, ...

F.3.5.4 Local Data – Variable Type

F.3.5.4.1 Adding a Local Variable tag

Open the Data Model View window → Local Data tab → Right Click → New Local Data



The dialog box titled 'New' contains the following fields and controls:

- Keyname:** A text input field.
- Node Type:** A dropdown menu with 'Variable' selected.
- Data Type:** A dropdown menu with 'Boolean' selected.
- Default Value:** A text input field.
- Comment:** A text input field.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom right.

Figure F-48. New Local Data

Keyname	Data tag name Refer to naming rules at F.5.2.1
Node Type	Variable
Data Type	Data type Data type reference at F.5.2.2
Default value	Default value
Comment	Description

F.3.5.4.2 Editing a Local tag

Open the Data Model View window, choose Local Data tab.

Double Click on the Tag to be edited (or right-click and select Edit). The Edit window appears, users can edit information such as Data Type, Default value, comment.

Click OK to save the changes.

F.3.5.4.3 Delete a Local tag

Open the Data Model View window, choose Local Data tab.

Right Click on the tag you want to delete, select Delete. The Confirm Delete window appears.

Click Yes.

F.3.5.5 Local Data – Reference Type

F.3.5.5.1 Add a new Reference tag

Open the Data Model View window, select Local Data tab.

Right Click and select New Local Data. The New window appears allowing users to fill in the following information:

Keyname	Data tag name Refer to naming rules at F.5.2.1
Node Type	Reference Object or Reference Variable
Data Type	Data type - apply with Reference Variable Data type reference at F.5.2.2
Comment	Description

Click **OK**.

F.3.5.5.2 Editing Reference tag

Open the Data Model View window, select the Local Data tab.

Right Click the tag you want to edit, select Edit. The Edit window appears, users can edit information such as Data Type, comments.

Click OK to save the changes.

F.3.5.5.3 Delete Reference tag

Open the Data Model View window, select the Local Data tab.

Right Click to Tag you want to delete, select Delete. The Confirm Delete window appears.

Click Yes.

Deleting multiple local tag at once is not supported. Local tag is stored in **ProjectName_lo.xml** file in the folder **Data** of the project. In case we need to adjust multiple or inherit from another project we can edit this file.

F.3.5.6 System Data

System Data names start with **_**. List of system tags:

System Tag	Data type	Description	Note
_mode	Int16	Data connection method, return result is 1/0 1 = SCADA / Realtime mode 0 = PLAYBACK / Stop Realtime mode	
_time	Int64	The system time is the number of seconds since 00:00:00 1/1/1970 UTC. In Realtime/SCADA mode: _time is the current time – operating system time – of the Grid Viewer. In Historical/Playback mode: _time is the time to retrieve past data	In the expression, use the TimeFormat function to convert to String
_userName	String	Display the logged-in user.	
_accessLevel	Int16	Level of permission granted to a user Value range: 0 – 999	Usual design conventions: The higher the value, the more authority
_loggedin	Boolean	Logged in State	
_queryHisState	Boolean	Display value 1 while the system is in the progress of querying historical data. Otherwise, display 0 when system stops to query.	
_profileName	String	Profile name	

_historyServerStatus	Int16	Show connection status to His Server • 1 = Connected 0 = Disconnected	
_dataServerStatus	Int	Shows connection status to DataServer • 1 = Connected • 0 = Disconnected	
_enableCalculation	Int16	Enable Calculation (for DMS version) • 0 = Disable • 1 = Enable	
_enableUpdateModelEngine	Int16	Enable to Update Engine (for DMS version) • 0 = Disable • 1 = Enable	
_temporaryGDState	Int16	Temporary Ground Disconnector State (for DMS version)	
_playbackState	Int16	Playback current state • 0 = playing • 1 = pause • 2 = stopped	

F.3.6. Alarm and Event

Alarm is an abnormal condition that occurs that requires operator attention. These conditions can be single-state or multi-state. For example, a circuit breaker with only closed state or not is a single-state condition; while the oil level of a transformer can have many states such as low, high, high high. Alarm is generated when the value of data exceeds a predetermined threshold.

Events are events that are recorded for analysis and monitoring, the operator does not need to react to events. The generated event may or may not have conditions attached. For example, the transition from alarm condition to normal state is conditional; the user's system login action is not conditional. Thus, all types of Alarms can be considered as Conditional Events.

In OneATS system, **Alarm** and **Event** are generated from SCADA Server – Alarm Server module. OneATS Grid Studio is Client, receives Alarm/Event Messages from DataServer and presents in audio, graphic (via Data Object/Variables alarm properties) or tabular form with **Event View** object.

F.3.6.1 Alarm Subscription

Alarm Subscription allows OneATS Grid Studio, as a Client, to choose to receive alarms/events from the Alarm Server (DataServer) appropriate to the Client's needs.

Select **Menu bar** → **Alarm** → **Alarm Subscription**: choose **Alarm Subscription Nodes** then press **Right click** → choose **Subscribe**:

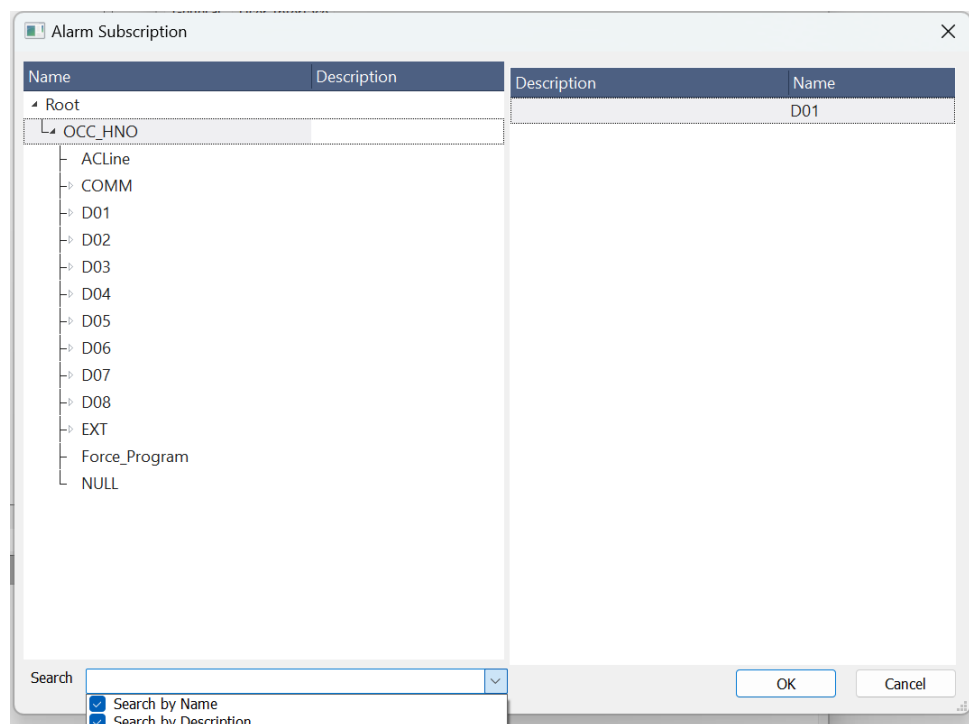


Figure F-49. Alarm Subscription configuration window

Use the Search Box to filter nodes by Name or Description

To Unsubscribe a node or all nodes, Right Click on subscribed node, select Delete or Delete All → OK.

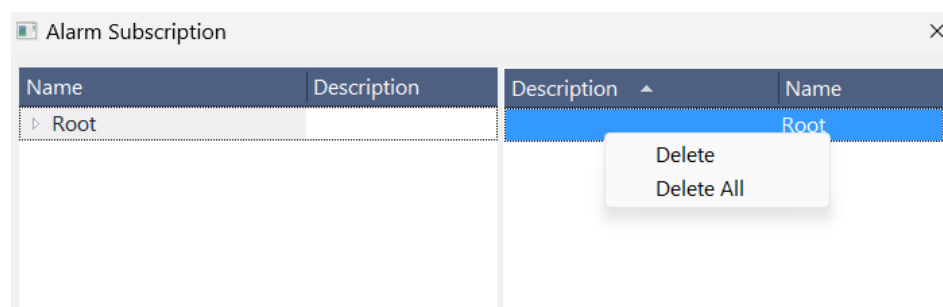


Figure F-50. Alarm Subscription - Unsubscribe Nodes

F.3.6.2 Alarm by sound

Grid Studio provides a sound alert feature when an alarm occurs.

At DataServer, each Variable has an Alarm sound property defined.

At Grid Studio, the list of Alarm sounds (name) (List of Alarm Sound) is synchronized with DataServer (through Update Data Model activity). The operation of configuring the Alarm Sound at Grid Studio is the activity of attaching the Alarm sound (name) to a specific sound file depending on the needs of the Client.

From the **Menu bar** → **Alarm** → **Alarm Sounds**.

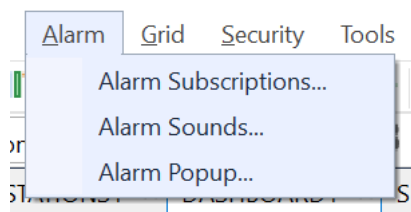


Figure F-51. Alarm Menu

Alarm Sound Configuration window appears.

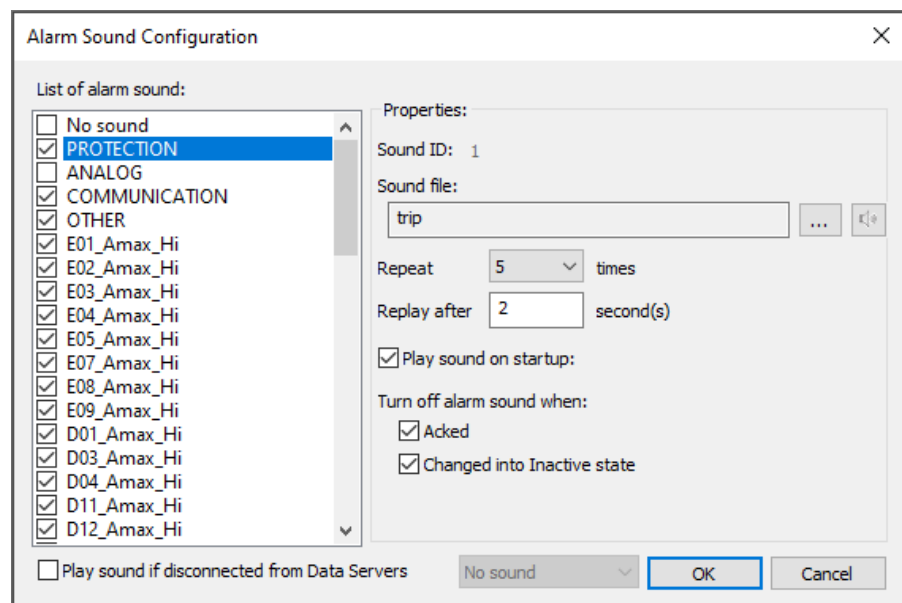


Figure F-52. Configuring alarm sound

Properties	Description
List of alarm sound	List of alarm sounds. Each alarm sound is specified by a Sound ID.

	To configure an alarm sound, check the corresponding box on the left in the list.
Play sound if disconnected from DataServers	Play alarm sound when connection to DataServer is lost. The sound file is selected from the list of alarm sounds.
Attributes corresponding to each alarm sound	
Sound file	Audio file, format *.wav Click  to open Window Explorer window and select the audio. Click  to listen to audio file with Window Media application.
Repeat	Number of times the audio file is played when the alarm occurs
Replay after	Time interval between playing audio files. Unit: second
Play sound on startup	Play sound with alarms appearing right after Viewer starts (first data connection)
Turn off alarm sound if Alarm Acked	Stop sounding when alarm is confirmed
Turn off alarm sound if Alarm changed into Inactive state	Stop sounding when the alarm is an inactive state.

F.3.6.3 Alarm by View List

To display alarms & events as a list, Grid Studio provides the Event View graphical object. For details, please refer to F.3.14.2

F.3.6.4 Alarm by Graphic Blinking

To display alarm as flashing graphics, Grid Studio provides the “**Blink when Acknowledgment state change**” function in the Master Properties of the graphical objects. Refer to F.3.10.4.7

F.3.6.5 Alarm Popup

Support Alarm Popup (4.1.7068 onwards)

From the **Menu bar** → **Alarm** → **Alarm Popup**.

Alarm Popup Configuration window appears.

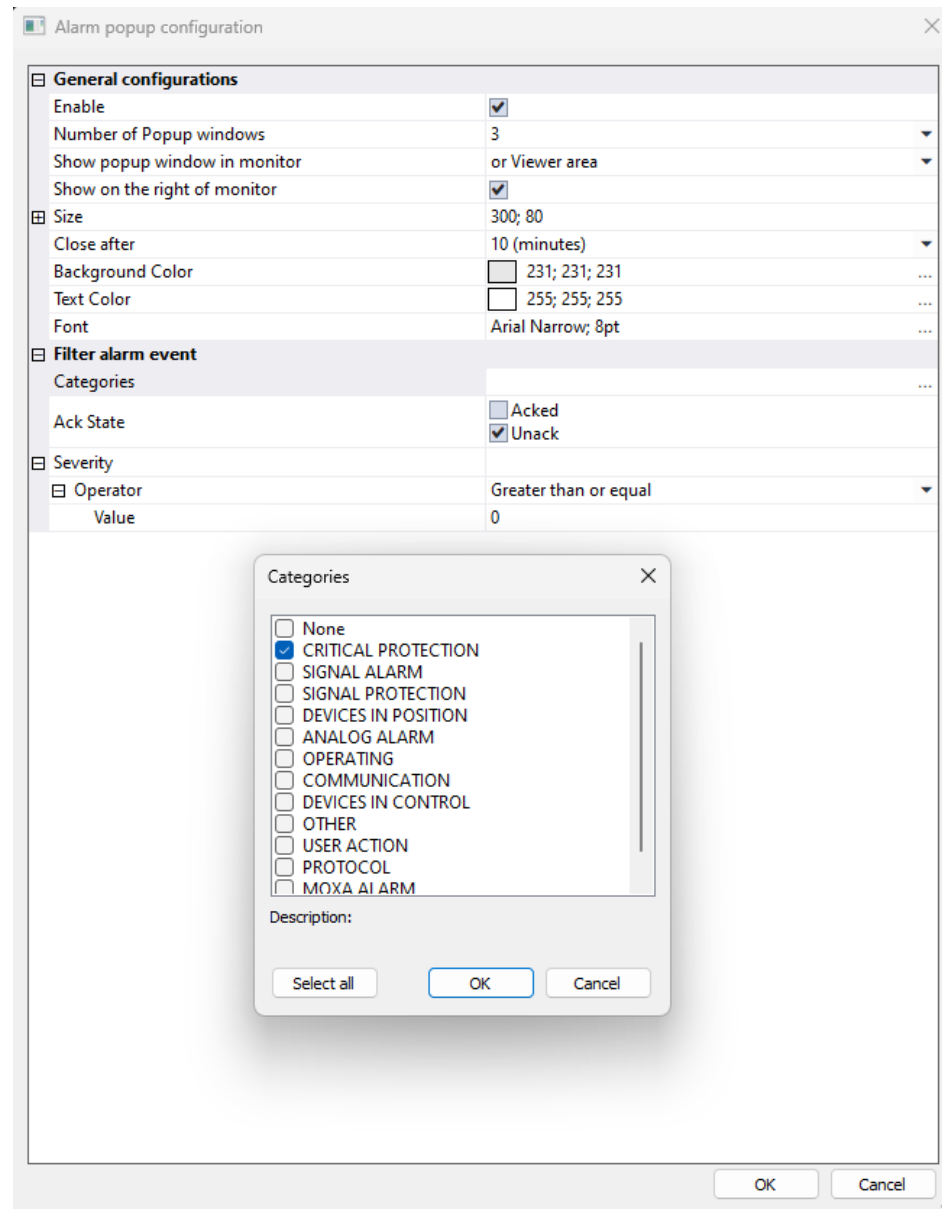


Figure F-53. Configuring alarm popup

Properties	Description
General Configurations	• Enable: Enable Alarm Popup. • Number of popup windows: Maximum number of popups displayed (1~10).

	• Show popup windows in monitor: Display position at the physical screen (1: 1- Main Display; or Viewer area). • Show on the right of monitor: Displays the Popup at the bottom and right of the physical screen, otherwise it will be displayed at the bottom left.
Size	Size of Popup
Close after	Close Popup after: • 5, 10, 30 (minutes). • User Action: When a user impacts Ack or click close popup.
Background Color	Background Colour Popup.
Text Color	Text color.
Font	Font, Size.
Filter Alarm Event	• Categories: Filter by Categories • Ack State: Filter by Ack status (Acked/UnAck) • Severity: Filter by severity.

F.3.7. Template of static properties (CAT)

CAT (Control Attribute Template) – Template Static Properties (Font, Color, ...) allows to set a common template of static properties for graphic objects. Graphics objects that use the same CAT template have the same static properties. Updating a CAT change automatically changes the static properties of all objects that use that CAT. The design using CAT makes it simple and convenient to change the static properties of a series of graphic objects with the same static properties.

F.3.7.1 Properties of CAT

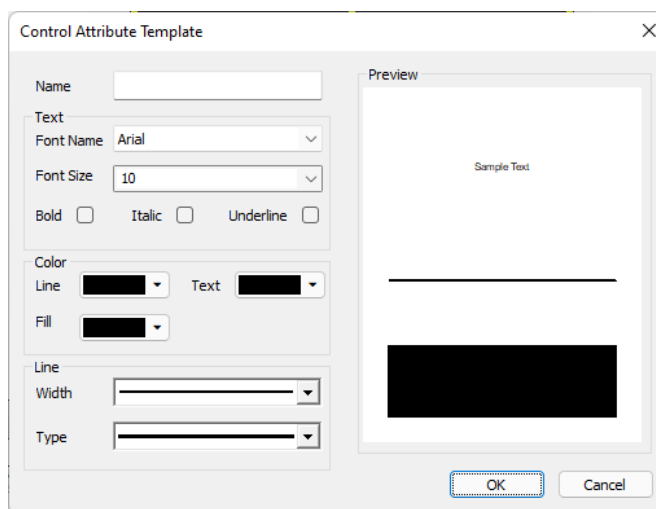


Figure F-54: Properties of CAT

Properties	Description
Name	Name of CAT template
Text Font Configuration	
Font Name	Font name
Font Size	Text size
Bold	Bold style
Italic	Italic style
Underline	Underline style
Color Configuration	
Line	Line color
Text	Text color
Fill	Background color
Line Configuration	
Width	Line width
Type	Line style (solid net, dashed line, ...)

F.3.7.2 Managing static properties templates– CAT

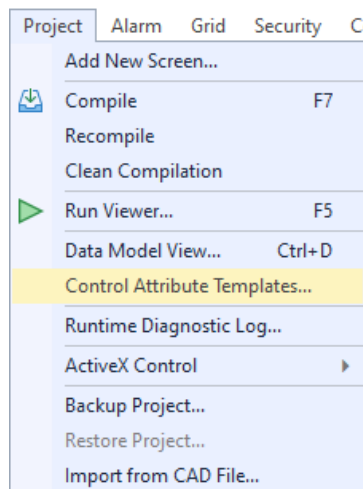


Figure F-55: CAT in Project Menu

- Click Menu Project → Control Attribute Templates. The CAT management window appears.

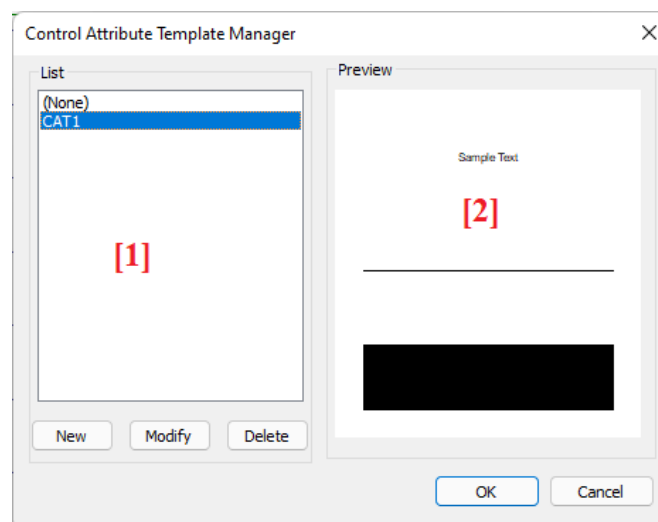


Figure F-56: The CAT management window

- [1]: list of available CATs
- [2]: preview (illustration) for CAT design

F.3.7.3 CAT list editing operations

F.3.7.3.1 Add new CAT

- Click New button to create new CAT
- Configuring properties for CAT

F.3.7.3.2 Edit existing CAT

- Choose CAT
- Click Modify button

- Edit selected CAT

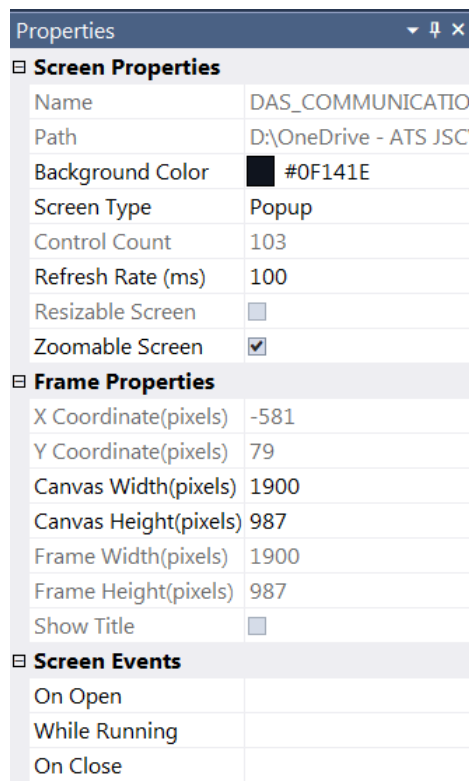
F.3.7.3.3 Delete existing CAT

- Choose CAT
- Click Delete button

F.3.8. Screen Management

Screen is a place to contain graphic objects that perform specific functions and tasks according to the designer's purpose.

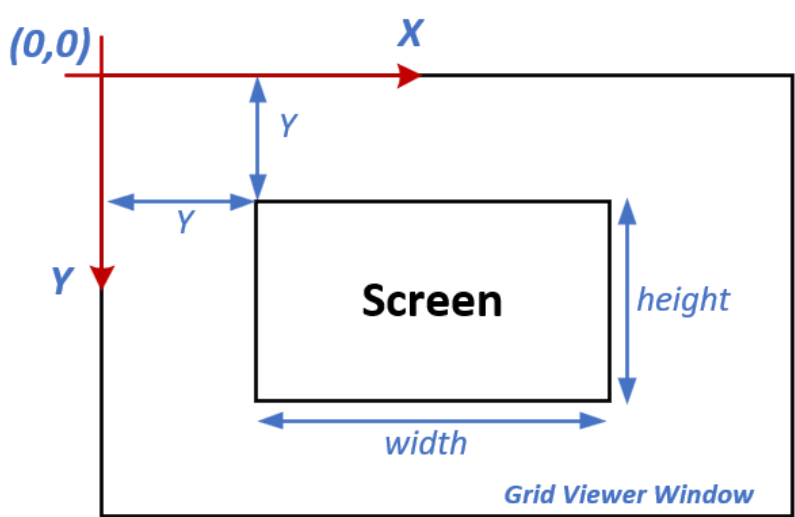
F.3.8.1 Screen properties



Properties	
Screen Properties	
Name	DAS_COMMUNICATION
Path	D:\OneDrive - ATS JSC\
Background Color	#0F141E
Screen Type	Popup
Control Count	103
Refresh Rate (ms)	100
Resizable Screen	☐
Zoomable Screen	☒
Frame Properties	
X Coordinate(pixels)	-581
Y Coordinate(pixels)	79
Canvas Width(pixels)	1900
Canvas Height(pixels)	987
Frame Width(pixels)	1900
Frame Height(pixels)	987
Show Title	☐
Screen Events	
On Open	
While Running	
On Close	

Figure F-57: Screen Properties

Properties	Description
Screen Properties	
Name	Name of Screen, this name is unique and distinguishes between Screens.
Path	Screen file path

Background Color	Screen Background
Screen Type	Screen types such as Popup/Overlay/Menu
Refresh Rate	Graphics objects update frequency
Resizable	Option to allow resizing Screen
Zoomable	Option to allow zooming in Viewer environment
Frame Properties	
X, Y Coordinate	Display coordinates, in units of Pixel (0,0): upper left corner of Grid Viewer Window 
Canvas Width, Height	Canvas size
Frame Width, Height	Screen size
Show Title	Title display option

Screen Type	Description
Popup	Always appear above all other Screens

Overlay	Appears above the currently open Overlay Screen. When a Screen Overlay closes, the Screen immediately behind it appears.
Menu	Always appear above all Overlay Screens

F.3.8.2 Create a new Screen

From Project Pane, select **Screen Manager** → **Add** → **New Folder** to create new folder containing Screen:

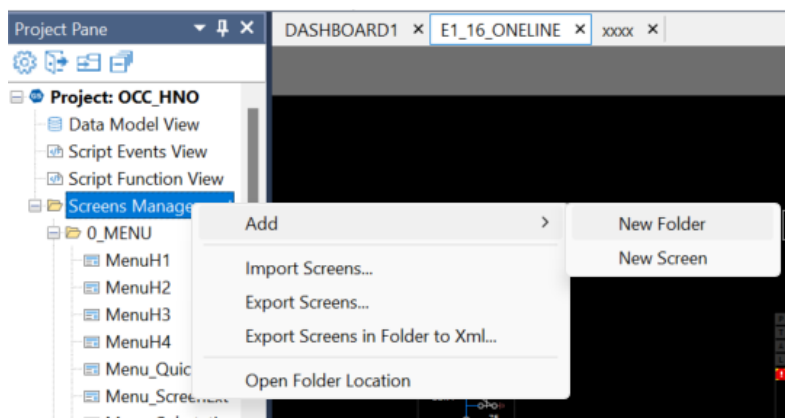


Figure F-58. Menu create new screen

From the new folder, right-click and select to create new screen located in that folder:

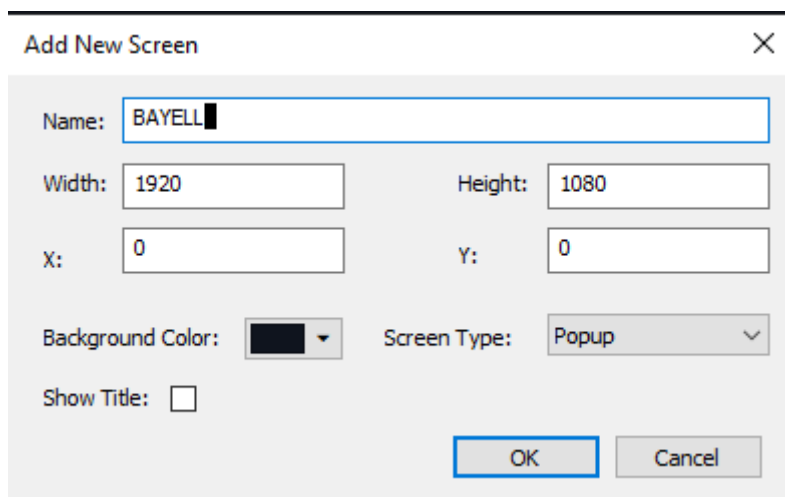


Figure F-59. Add new screen window

F.3.8.3 Open Screen

From the screen you want to open on the Project Pane, right-click and select Open Screen or double click:

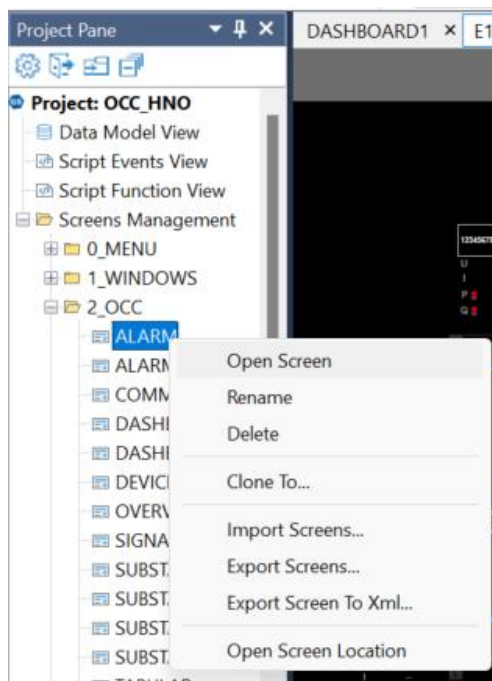


Figure F-60. Menu open screen

F.3.8.4 Rename Screen

From the screen you want to rename on **Project Pane**, right-click and select **Rename** or press **F2**.

F.3.8.5 Save Screen

To save changes on the screen, select icon **Save Document** on the standard toolbar or select **File → Save All (Ctrl+S)** to save the changes on all existing open screens.

F.3.8.6 Close Screen

In the screen window, right-click to select **Close** or press **⌘** to close screen:

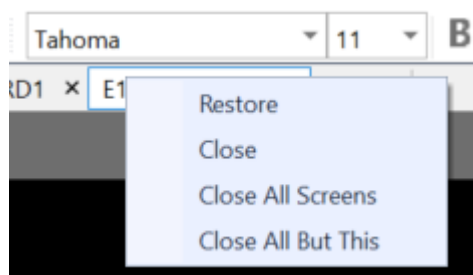


Figure F-61. Close screen menu

F.3.8.7 Delete Screen

Right-click on the screen to be deleted on the **Project Pane** and select **Delete** to delete the screen:

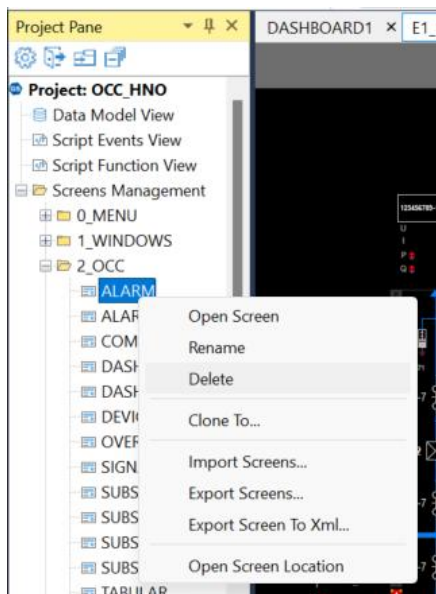


Figure F-62. Delete screen menu

F.3.8.8 Clone Screen

Right-click on the screen you need to copy and select **Clone To ...**

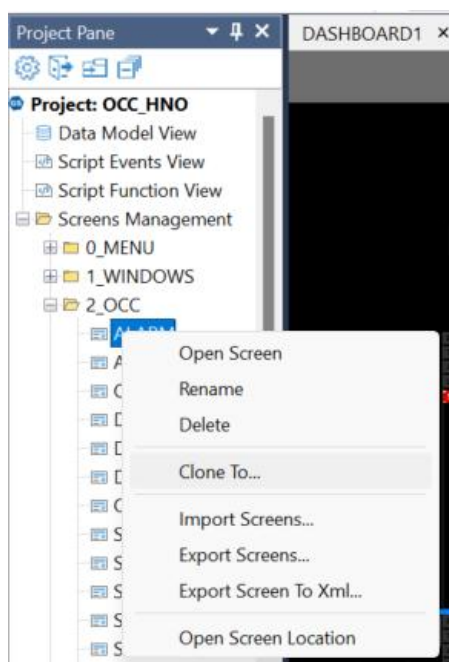


Figure F-63. Clone screen menu

Fill in screen name and press **OK**:

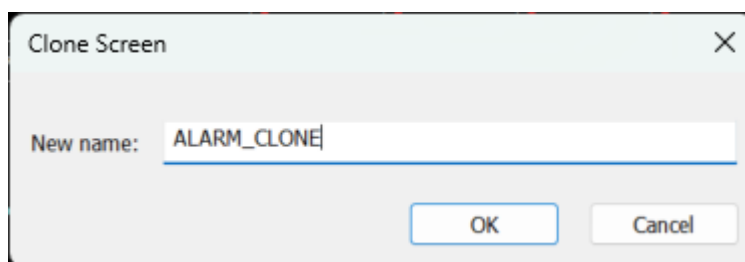


Figure F-64. Dialog box sets the new screen name

F.3.8.9 Import/Export Screen

F.3.8.9.1 Export screen

- Right-Click Project Pane/Screen → **Export Screens...**

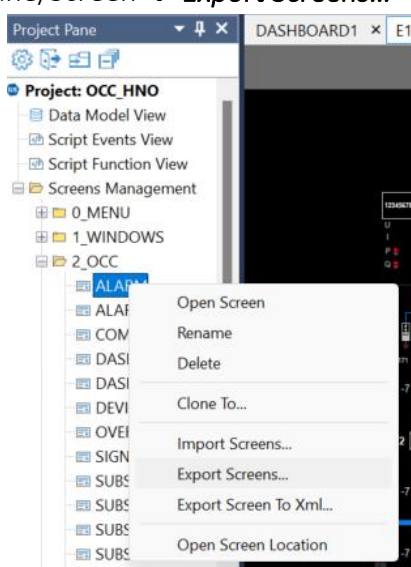


Figure F-65. Export screens

- Select a storage location.

F.3.8.9.2 Import screen

- Right-Click Project Pane/Screen → **Import Screens...**

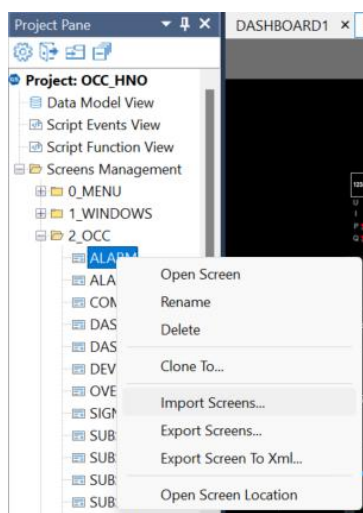


Figure F-66. Import screens

- Select a storage location.

F.3.8.9.3 Export screen to Xml

- Right-Click Project Pane/Screen → **Export Screens to Xml...**

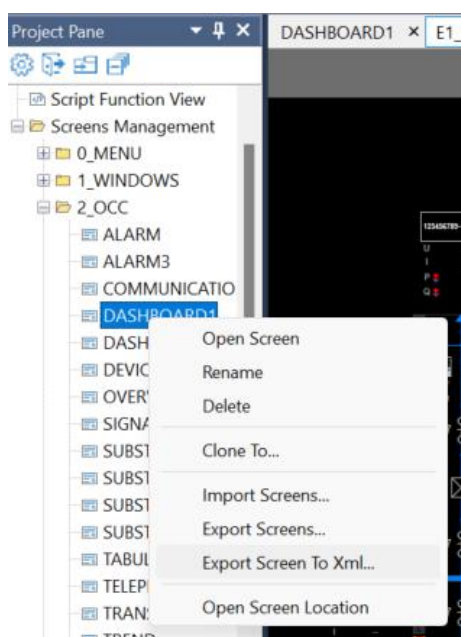


Figure F-67. Export Screens to Xml

- Select a storage location.

F.3.8.10 Import CAD file

To import AutoCAD files in .dwg, .dxf format to create screen:

- Tab Project → **Import from CAD file**

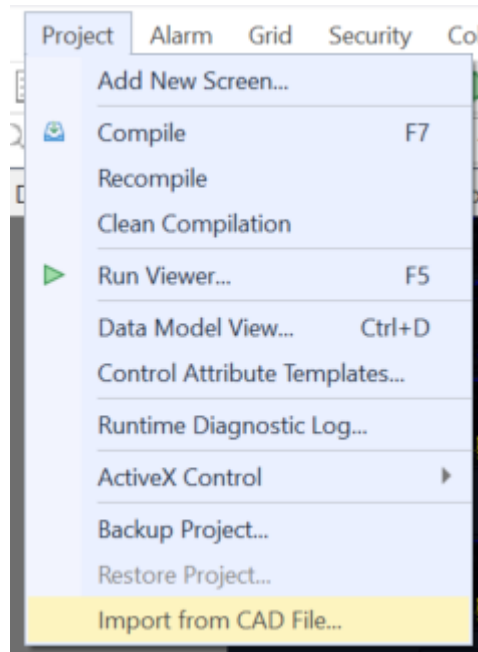


Figure F-68. Import CAD file

F.3.8.11 Right-Click menu on the screen

Right-Click on an empty area on the screen

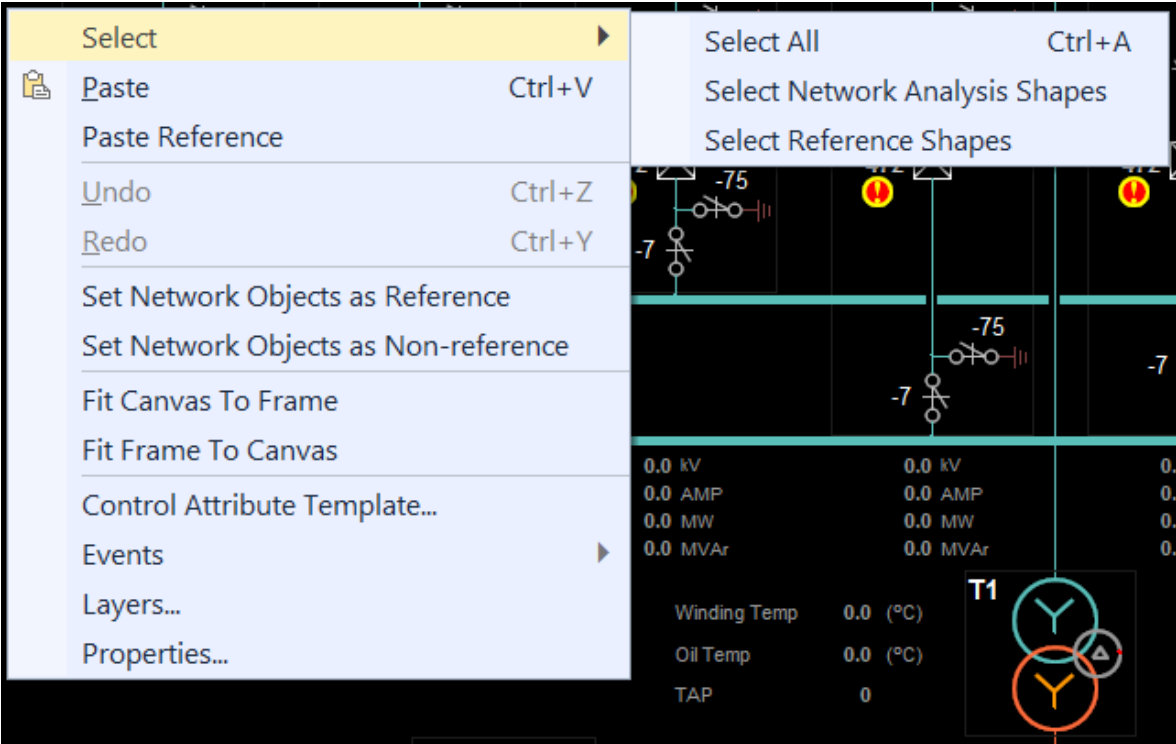


Figure F-69. Right-Click menu on the screen

Function	Description
Select	Select All (Ctrl+A): Select all objects in screen.

	Select Network Analysis Shapes: <i>Unavailable.</i> Select Reference Shapes: Select all reference shape on screen
Paste	Reference F.3.16.5
Paste Reference	Reference F.3.16.6
Set Network Objects as Reference	Convert all electrical network objects to reference. Reference F.3.12.2
Set Network Objects as Non-reference	Convert all electrical network objects to non-reference Reference F.3.12.2
Fit Canvas To Frame	Adjust the size or position of the workspace (canvas) to fit perfectly within the frame
Fit Frame To Canvas	Adjust the size or position of the frame to fit perfectly within the workspace (canvas)
Control Attribute Template	Reference F.3.7.2
Events	Scripts based on screen events. Reference F.3.17.4
Layer	Manage layers in screen. Reference F.3.9.1
Properties	Open Screen Properties pane Reference F.3.8.1

F.3.8.12 Open Screen Location

Right-click on the screen you need to open folder location and select **Open Screen Location**

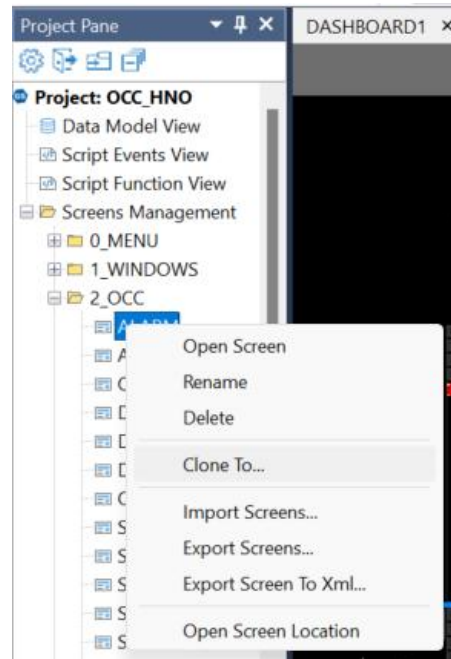


Figure F-70. Open screen Location

F.3.9. Layer

Layer is a layer that displays objects.

In one screen, many different objects can be grouped into one layer.

Each screen allows creating multiple layers, can display one or more different layers at the same time.

Layer is a handy feature that allows displaying multiple layers of information in the same screen by enabling/disabling the appropriate layers.

Each screen always has a Layer Default.

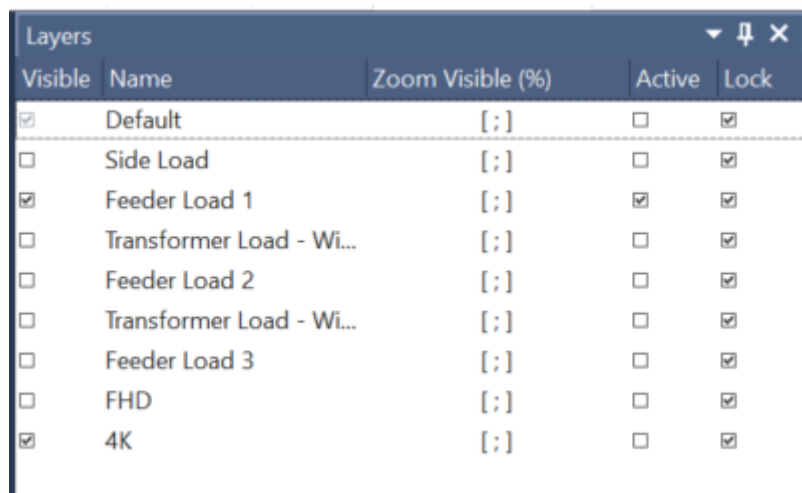
The application supports scripts that control layers. Reference F.5.7.4.11 – Layer management scripts

F.3.9.1 Layer management

Right Click on the design area in Screen → **Layers** or View Menu→**Layers**

Properties	Description
Visible	Layer display options. When a layer is hidden, all objects belonging to that layer are also hidden. This property takes effect in Grid Designer & at startup Grid Viewer
Name	Name of the layer.

Zoom Visible (%)	Zoom strip but still layer is still displayed. Example: [50;120] → When Zoom goes to 121% or 49%, the layer will be hidden. [:] : default – The layer is always visible regardless of the zoom level
Active	Indicates which layer is active. When a Layer is in the active state, all the new objects we create belong to this layer
Lock	If selected, all objects in that layer are locked (user cannot change object properties (size, font, color, location...)) If not selected, all objects in that layer are unlocked (user can change object properties)



Visible	Name	Zoom Visible (%)	Active	Lock
☒	Default	[:]	☐	☒
☐	Side Load	[:]	☐	☒
☒	Feeder Load 1	[:]	☒	☒
☐	Transformer Load - Wi...	[:]	☐	☒
☐	Feeder Load 2	[:]	☐	☒
☐	Transformer Load - Wi...	[:]	☐	☒
☐	Feeder Load 3	[:]	☐	☒
☐	FHD	[:]	☐	☒
☒	4K	[:]	☐	☒

Figure F-71. Layer management window

F.3.9.2 Add, rename, delete layer

F.3.9.2.1 Add new Layer

Right Click on the Layer Manager window → New and fill name for Layer.

F.3.9.2.2 Clone Layer

Right Click on the Layer → select Clone to Clone Layer.

F.3.9.2.3 Delete Layer

Right Click on the Layer → select Delete to delete Layer.

F.3.9.2.4 Edit Layer name

Right Click on the Layer → select Edit to edit Layer name.

F.3.9.2.5 Show Only a Layer

Right Click on the Layer → select Show Only This to show Layer.

F.3.9.2.6 Show All Layers

Right Click on the Layer Manager window → select Show All to show all Layers.

F.3.10. Basic graphic objects/ basic control



Figure F-72. Basic graphic objects/ basic control

Graphic properties of basic object include design graphic properties – Designer Properties – and graphic properties/ interaction properties in the Viewer – Master Properties environment.

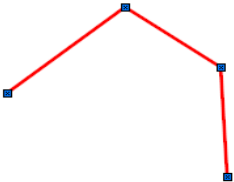
Designer Properties are described in F.3.10.3

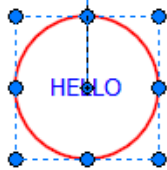
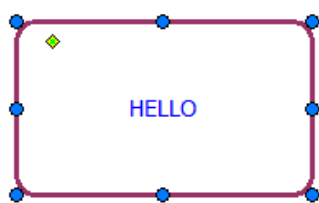
Master Properties are described F.3.10.4

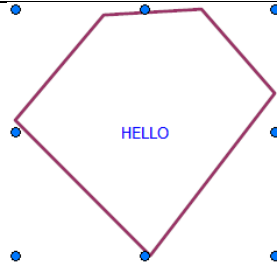
General graphic display principle: the graphical display properties of an object will prioritize Master Properties and then Designer Properties.

F.3.10.1 List of basic graphics objects

Basic graphics objects are drawn from the basic graphics objects toolbar. Refer to section F.3.1.8.

Object	Symbol	Description
Line		The line object is used to draw straight line  * Hold Shift while drawing to create a perpendicular line
Polyline		Polyline object used to draw many continuous lines: 

		* Hold Shift while drawing to create a perpendicular line
Ellipse		The <i>ellipse</i> objects use to draw ellipses.  * Hold Shift while drawing an ellipse to create a circle
Circle		The <i>Circle</i> objects use to draw circles. 
Rectangle		The <i>Rectangle</i> objects use to draw rectangles:  * Hold Shift while drawing an ellipse to create a square
Square		The <i>Square</i> draw squares: 
Round Rectangle		The <i>Round Rectangle</i> object draws a rectangle with rounded corners  * Hold Shift while drawing Round Rectangle to create a rounded square
Polygon		The <i>Polygon</i> object helps to draw polygons:

		
Picture		Picture objects allow you to insert an image file into the screen. To insert an image, click the Picture icon on the toolbar containing basic object, find to image file you want to insert and press Open button.  Supported image formats: PNG, JPEG, TIF, ICO, BMP, GIF, PCX, MNG, WMF.
Edit box		Allows displaying and importing data. The value of the analog/string tag is the input value 
Button		Button object allows the creation button. 
Label		Label object allows displaying data information. 

F.3.10.2 Design support with right-click menu for object

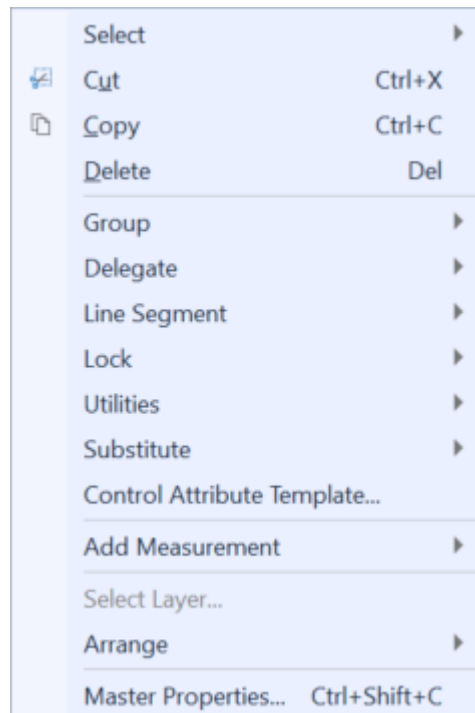


Figure F-73. Right-click menu on the object

Tools	Shortcut keys	Description
Select		Select All (Ctrl+A): Select all objects in screen. Select Network Analysis Shapes: Unavailable. Select Reference Shapes: Select all reference shape on screen
Cut	Ctrl + X	Cut selected objects
Copy	Ctrl + C	Copy selected objects
Delete	Del	Delete objects
Paste	Ctrl + V	Paste without getting reference properties objects is copied or cut previously. Refer to section F.3.16.5 for reference properties definition.
Copy Network ID	Ctrl+U	Copy object Unique ID <i>(Only valid for Network Object)</i>

Apply Network ID	Ctrl+T	Paste Unique ID to Reference Symbol Object (Only valid for Network Object)
Group / Ungroup		Group / ungroup selected objects Refer to section F.3.16.8
Delegate – Make/Break		Group/ungroup selected objects with the same Master properties Refer to section F.3.16.9
Line Segment		Select Connectivity 1: Select the first connectivity node that the line connects to Select Connectivity 2: Select the second connectivity node that the line connects to
Lock / Unlock		Lock/Unlock position and size of selected objects
Utilities → Revert Image		Restore to the original image before changes (Only valid for image objects)
Substitute → Variables		List of variables in the Master Properties of selected objects
Substitute → Text		List of text in selected objects
Control Attribute Template		Apply the Designer Properties design properties according to the pre-made template (CAT). Refer to section F.3.6
Select Layer		Select Layer for Object (Require Unlock Layer Before)
Add Measurement		Unavailable.
Edit Symbol Template		Edit Template with instance objects of Template Refer to section F.3.15

Arrange		Arrange the object order within a layer Refer to section F.3.1.4
Master Properties	Ctrl + Shift + C	Configuring dynamic graphic properties – Master Properties. Refer to section F.3.10.4

F.3.10.3 Static graphics properties - Designer Properties

- The design parameters of the coordinates, Bounding Width-Height, Word Wrap line color, and background color of the object are set in the **Properties** window. Refer to section F.3.1.13
- Text design parameters (fonts, colors ...) are set through the **Font toolbar**. Refer to section F.3.1.5
- The design parameters of the line style are set through the **Line toolbar**. Refer to section F.3.1.6
- Set the **Fill Effect** on the **Line toolbar**. Adjust the auxiliary color in the fill effect with the **Pattern Color** parameter in the Properties window.
- Alignment of objects, rotations, and symmetry is established with the help of the **Control Utilities toolbar**. Refer to section F.3.1.4
- Templating static attributes (CAT) with **Right-Click Menu / Control Attribute Template**. Refer to section F.3.7

F.3.10.4 Dynamic graphics properties - Master Properties

Master Properties provides dynamic graphic properties and mouse interaction properties of the object according to the **Tag / Expression value** and the **action script**.

Double Click on the object or select the object and press **Ctrl + Shift + C** to open the **Master Properties** window.

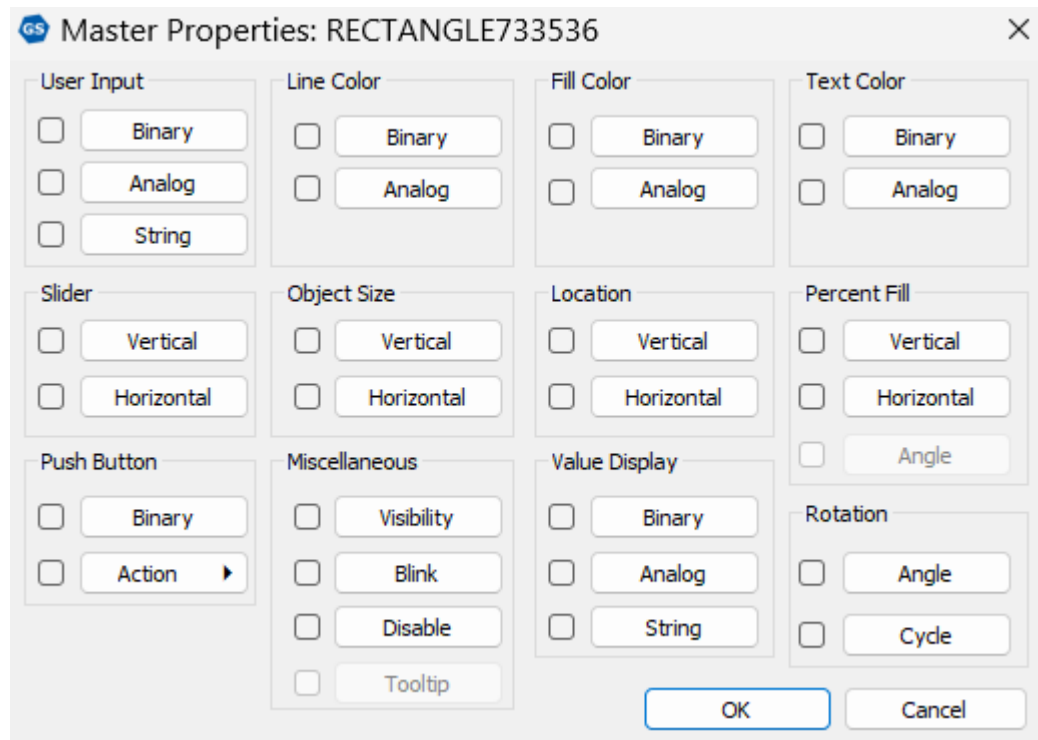


Figure F-74. Master Properties Window

F.3.10.4.1 List of Master Properties:

Master Properties	
Attribute group	Description
- Line Color - Fill Color - Text Color - Object Size - Location - Percent Fill - Value Display - Miscellaneous - Rotation	Representing data information in the form of graphical attributes
- User Input - Slider - Push Button	Mouse interaction with the object changes the data values

Master Properties – Value Display group	
Data display	
Binary	Display Boolean data (true/false)

Analog	Displaying numeric data
String	Display data as character string

Master Properties –Miscellaneous group	
Visibility	Hide / Show Objects
Blink	Blinking object
Disable	Prohibit/Allow mouse operations on the object
Tool Tip	Unavailable

Master Properties – Line Color / Fill Color / Text Color groups	
Color Display	
Line Color	Custom change the object's line color according to the Expression value.
Fill Color	Custom change the color inside the object's outline according to the Expression value.
Text Color	Custom change the color of the text element of the object according to the Expression value.
Binary	Display Line Color / Fill Color / Text Color in 2 colors corresponding to Boolean Expression value (true/false).
Analog	Display Line Color / Fill Color / Text Color in 10 colors corresponding to Analog Expression value.

Master Properties – Percent Fill group	
Color the object as a percentage	
Vertical	Fill scale object color (%) vertically

Horizontal	Fill scale object color (%) horizontally
Angle	Fill ellipse/round object color (%) according to rotation angle

Master Properties – Object Size group Resize object by percentage (%) depending on the result of Expression.	
Vertical	Resize vertically
Horizontal	Resize horizontally

Master Properties – Location group Move the object display coordinates according to the Expression value	
Vertical	Vertically shift object display coordinates according to expression value.
Horizontal	Horizontally shifts the displayed coordinates of the object according to the expression value.

Master Properties – Rotation group Rotate object by Expression value or fixed frequency	
Angle	Rotate the object with rotation according to the value of expression.
Cycle	Rotate the object with the rotation speed according to the value of expression

Master Properties – User Input Display the data input window when there is a Left Click event on the object	
Binary	Enter Boolean data (true/false)

Analog	Enter numeric data
String	Input string data

Master Properties – Slider Enter data according to the object position that changes by moving the object.	
Vertical	Enter data according to the Click event on the object and drag the mouse vertically
Horizontal	Enter data according to the Click event on the object and drag the mouse horizontally.

Master Properties – Push Button Interaction by operation mouse Click on the object.	
Binary	Change the Boolean Tag value.
Action - Left Click	Run the corresponding script for the Left Click + Release event.
Action - While Left	Run the script corresponding to the Left Click + Hold an event with the set Timer frequency.
Action - Left Down	Run the script corresponding to the Left Click event.
Action - Left Double Click	Run the corresponding script for the Left Double Click event.
Action - Right Click	Run the corresponding script for the Right Click + Release event.
Action - While Right	Run the script corresponding to the Right Click + Hold an event with the set Timer frequency.

Action - Right Down	Run the script corresponding to the Right Click event
Action - Right Double Click	Run the corresponding script for the Right Double Click event.
Action - Mouse Move	Run the script corresponding to the mouse move On Object event.

F.3.10.4.2 Syntax of Expression

Expression is the basic configuration information for the Master Properties dynamic data properties.

Expression support **Int64** (4.1.6932 onwards)

Support operators and functions:

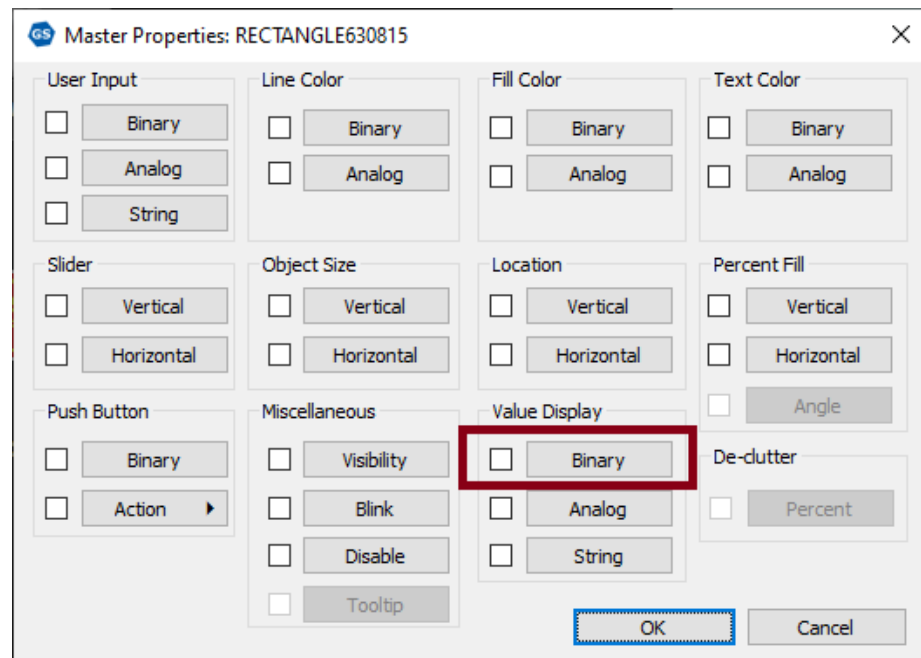
Operators	Description	Sample	Example
+ - * /	Addition, subtraction, multiplication, division		
%	Modulo		
^	Exponential		
!	Negative		
//	String Concatenation		"aa"// "b"
&& and or	Boolean operators: and, or		
& << >>	Bitwise operator		
< > >= <= == !=	Comparison spells		
?	Condition: if, else	Condition ? (true return) : (false return)	a == 1 ? "Close" : "Open"

Function	Description	Sample	Example
CheckBit	Check bit	CheckBit(n, k)	CheckBit(5, 1)
max	Returns the largest value in the range.	max(...)	max(5, 1, 2)
min	Returns the smallest value in the range	min(...)	min(5, 1, 2)
TimeFormat	Returns the string corresponding to the time parameter. Parameter passed in	TimeFormat(time)	TimeFormat(_time)

	type time_t – is the number of seconds since 00:00 1/1/1970 UTC Display type (DD/MM/YYYY) depends on project configuration Menu Tool → Options/(Miscellaneous) Date Time (format)		
strlen	Returns string length	strlen(string)	strlen("aaaa")
strfind	Finds a string within a string, returns the substring index	strfind(string, substring)	strfind("aaba", "b")
TaggingTopIndex (4.1.6932 onwards)	Generate a tagging index of the highest priority displayed in Tagging (1~16)	TaggingTopIndex(tagging_var)	TaggingTopIndex(E01.XCBR1.Tagging)
TaggingCount (4.1.6932 onwards)	Get the number of tags currently attached that appear in the Tagging variable	TaggingCount(tagging_var)	TaggingCount (E01.Tagging)

F.3.10.4.3 Configuring Value Display Binary

Open the **Master Properties** window of object.



Click **Binary** in the **Value Display** section. The **Value Display Binary** window appears.

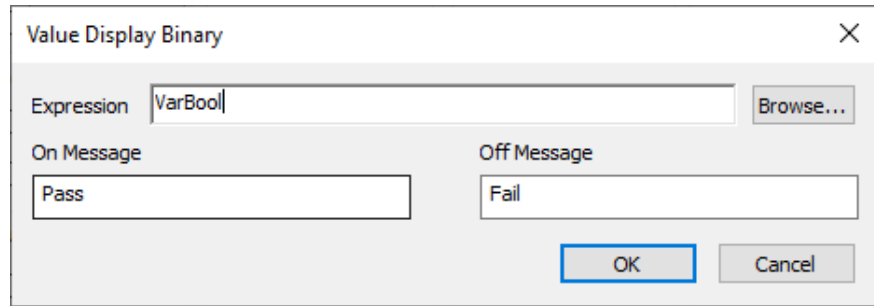


Figure F-75. Value Display Binary Configuration

Master Properties – Value Display – Binary	
Expression	Keyname of a boolean tag or expression with a boolean result
On Message / Off Message	The content displayed when the expression has a value of true / false.
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

Click **OK** to close **Value Display Binary** window.

Click **OK** to save.

F.3.10.4.4 Configuring Value Display Analog

Open the object's **Master Properties** window.

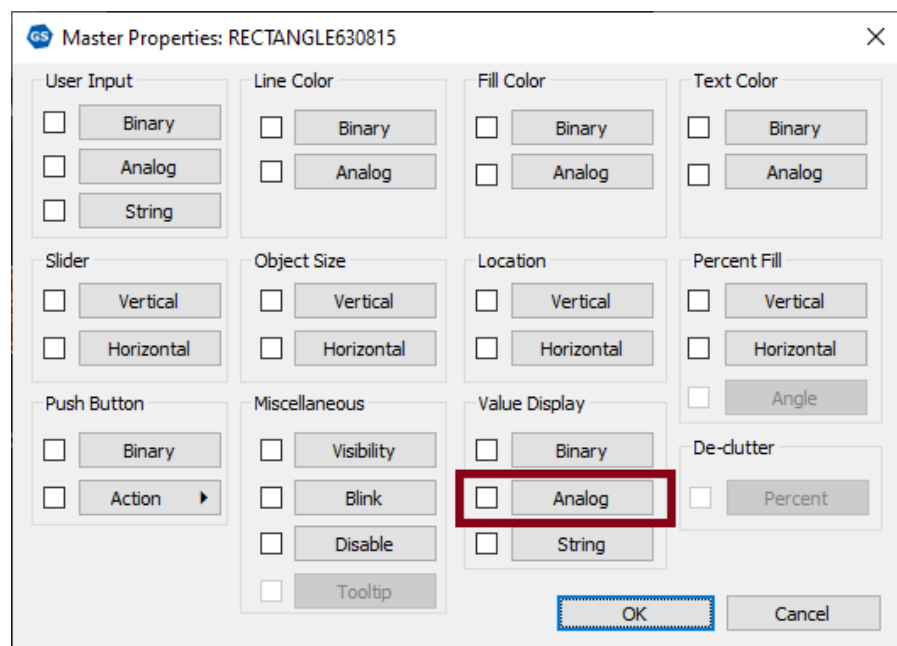


Figure F-76. Value Display - Analog

Click **Analog** in the **Value Display** section. The **Value Display Analog** window appears.



Figure F-77. Value Display Analog Configuration Window

Master Properties – Value Display – Analog	
Expression	Keyname of an analog tag (int, real) or expression whose result is a number.
Decimal Places	The number of decimals after the comma you want to display
Browse	Support tag selection. Open the Data Model View window. Select tag from list → right click → Copy Key Name → Close Data Model View → Paste to Expression

Click **OK** to close **Value Display Analog** window.

Click **OK** to save.

F.3.10.4.5 Configuring Value Display String

Open the object's **Master Properties** window.

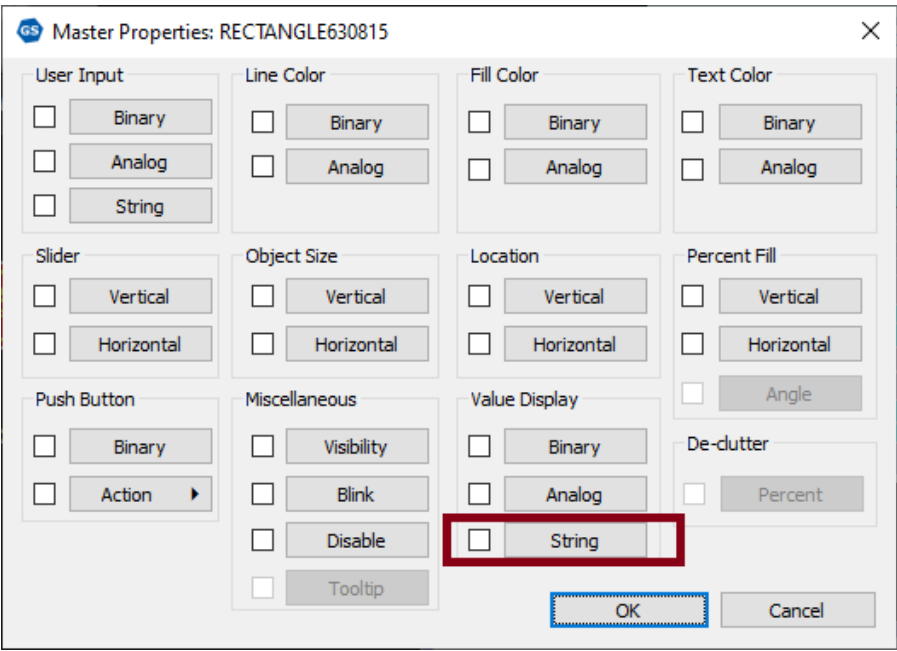


Figure F-78. Master Properties Value Display - String

Click **String** in the **Value Display** section. The **Value Display String** window appears.

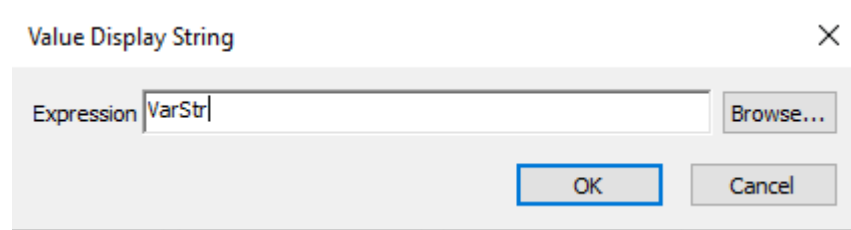


Figure F-79. Value Display String Configuration Window

Master Properties – Value Display – String	
Expression	Keyname of a tag of type string or DateTime or expression that returns string
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

Click **OK** to close the **Value Display String** window.

Click **OK** to save.

F.3.10.4.6 Configuring Miscellaneous – Visibility

Open the object's **Master Properties** window.

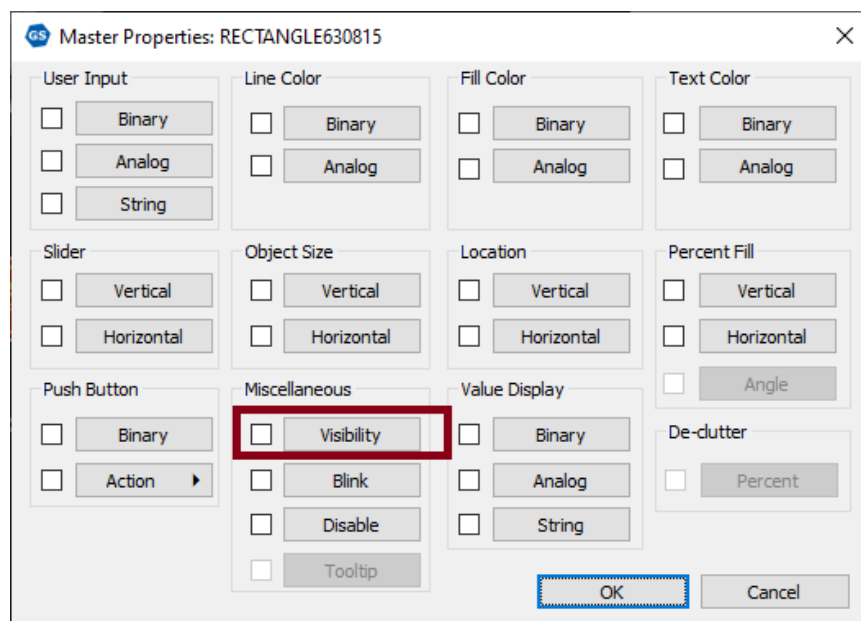


Figure F-80. Master Properties Miscellaneous – Visibility

Click **Visibility** in the **Miscellaneous** section. The **Miscellaneous Visibility** window appears.

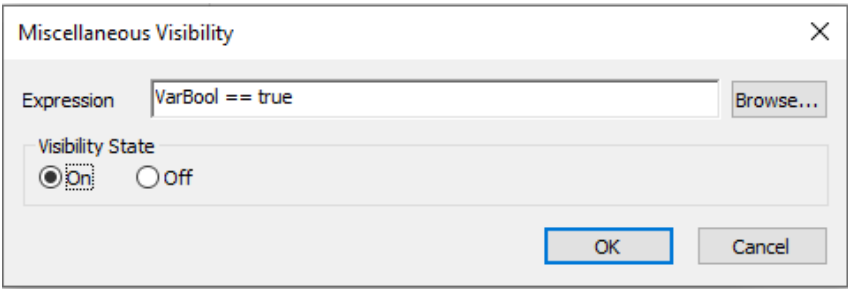


Figure F-81. Miscellaneous Visibility Configuration window

Master Properties – Miscellaneous – Visibility	
Expression	The keyname of a boolean tag or expression with a boolean result.
Visibility State - On	Object is visible if Expression = TRUE, hidden if Expression = FALSE
Visibility State - Off	Object is visible if Expression = FALSE, hidden if Expression = TRUE.
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

Click **OK** to close the **Miscellaneous Visibility** window.

Click **OK** to save.

Graphical objects displayed under the **Visibility State configuration** are initialized when the expression fails (4.1.6946 onwards)

F.3.10.4.7 Configuring Miscellaneous – Blink

Open the object's **Master Properties** window.

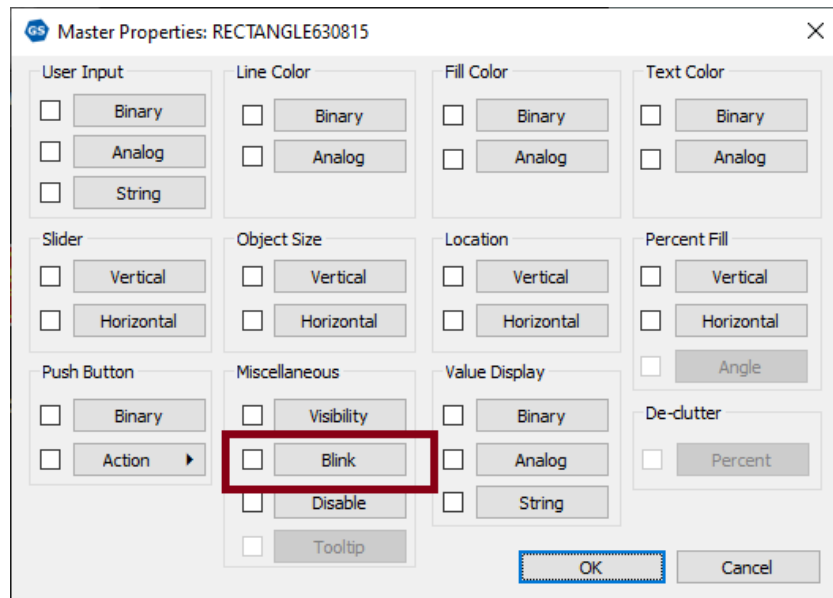


Figure F-82. Master Properties Miscellaneous – Blink

Click **Blink** in the **Miscellaneous** section. The **Miscellaneous Blink** window appears.

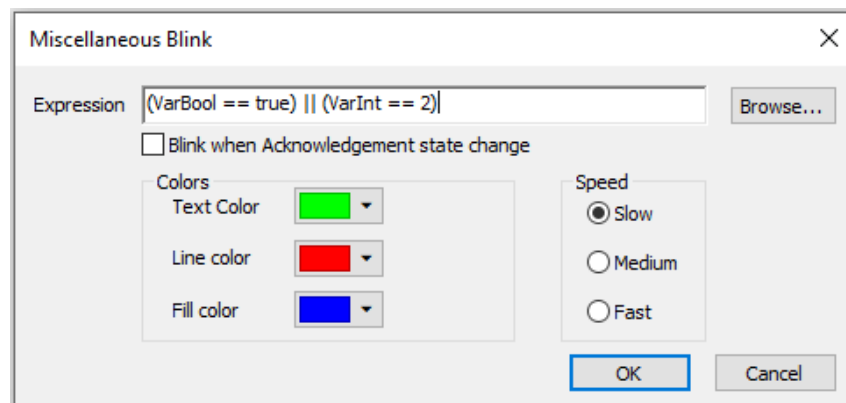
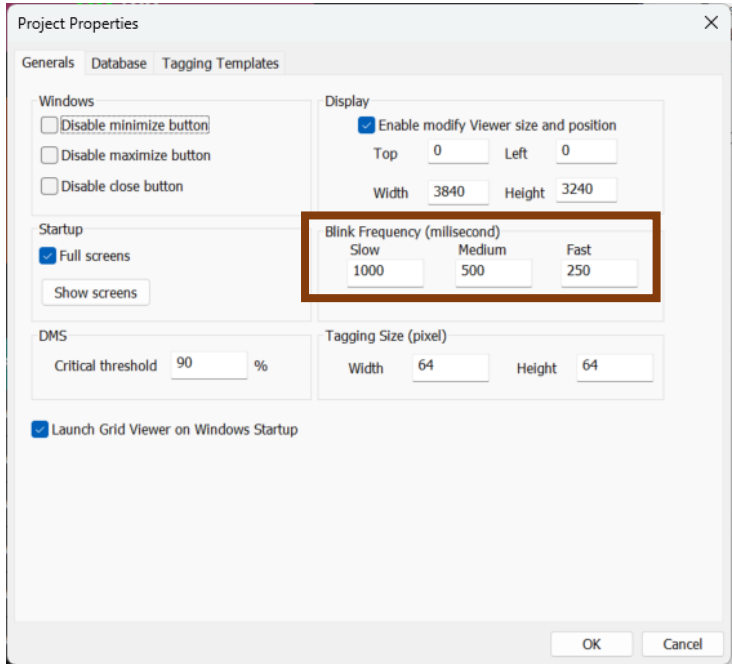


Figure F-83. Miscellaneous Blink Configuration Window

Master Properties – Miscellaneous – Blink	
Expression	The keyname of a boolean tag or expression with a boolean result.
Blink when Acknowledgement state change	When the ack state of the tag in the expression changes, a "blink" condition will occur.
Color – Text Color	When Expression = TRUE, the object will blink (change continuously) the color of the text between 2 colors: Miscellaneous – Blink/Text Color and Properties / Text Color (design color in Designer environment)

Color – Line Color	When Expression = TRUE, the object will blink (continuously change) the color of the Line between 2 colors: Miscellaneous – Blink/Line Color and Properties / Line Color (design color in Designer environment)
Color – Fill Color	When Expression = TRUE, the object will blink (change continuously) the color of the Fill between 2 colors: Miscellaneous – Blink/Fill Color and Properties / Fill Color (design color in Designer environment)
Speed	Level of color conversion speed. There are 3 levels: • Slow • Medium • Fast Actual speed for each level is configured in Project Properties 
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

Click **OK** to close **Miscellaneous Blink** window.

Click **OK** to save.

F.3.10.4.8 Configuring Miscellaneous – Disable

Open the object's **Master Properties** window.

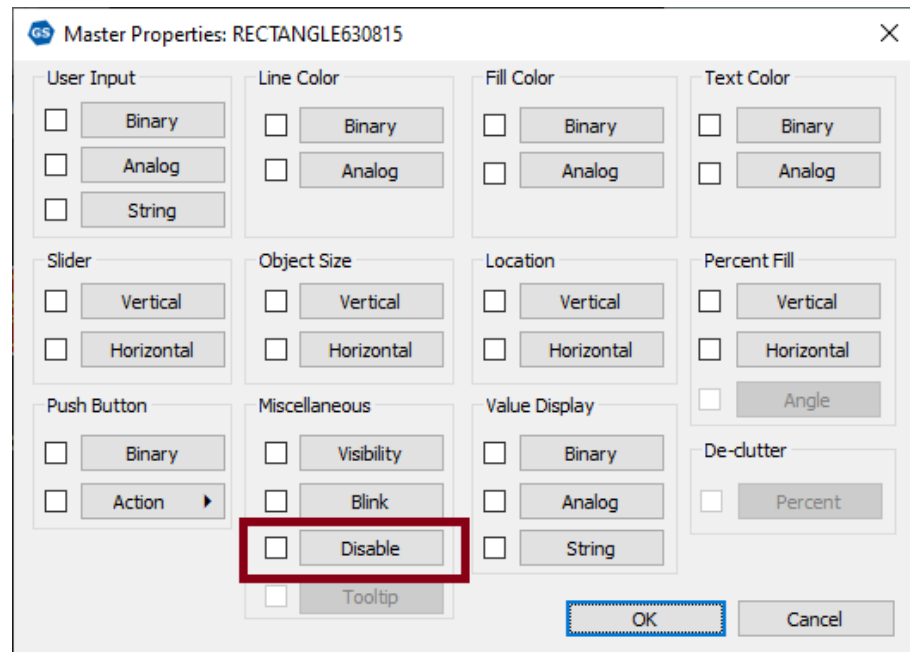


Figure F-84. Master Properties Miscellaneous – Disable

Click **Disable** in the **Miscellaneous** section. The **Miscellaneous disabled** window appears.

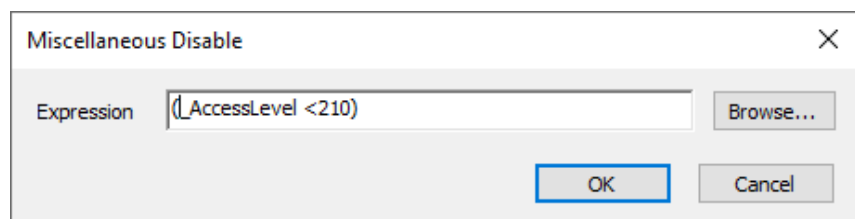


Figure F-85. Miscellaneous Disable Configuration Window

Master Properties – Miscellaneous – Disable	
Expression	The keyname of a boolean tag or expression with a boolean result. When Expression = TRUE, mouse actions with the object (defined in the interactive property groups – User Input, Push Button, Slider, ...) will be disabled.
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

Click **OK** to close the **Miscellaneous disabled** window.

Click **OK** to save.

F.3.10.4.9 Configuring Line Color/ Fill Color / Text Color – Binary

Open the object's **Master Properties** window.

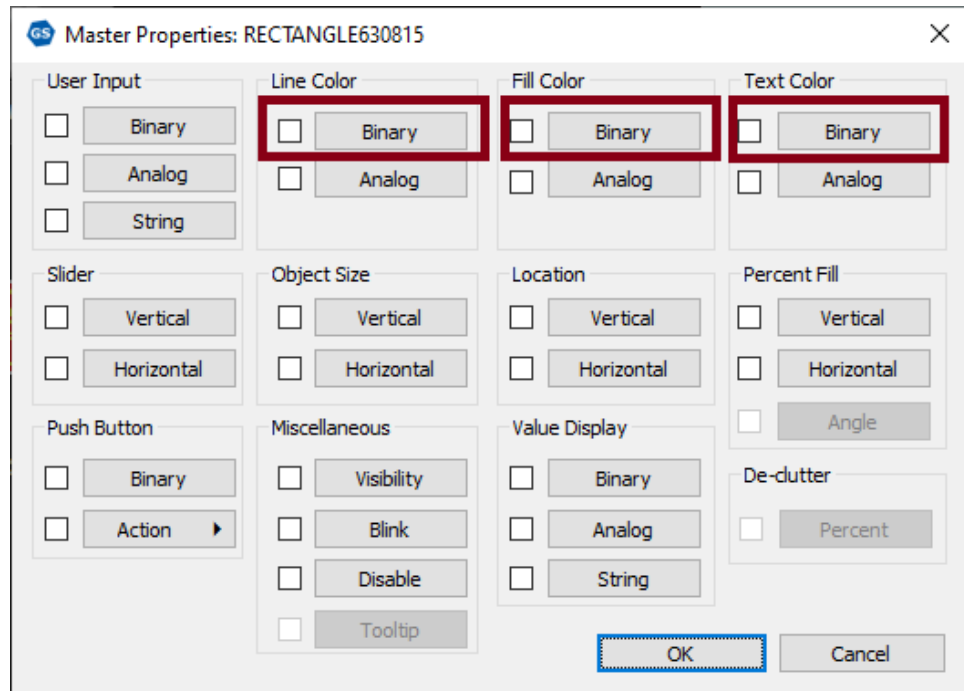


Figure F-86. Master Properties Line Color/ Fill Color / Text Color – Binary

Click **Binary** in the **Line Color / Fill Color / Text Color** section. The **Line Color / Fill Color / Text Color - Binary** window appears.

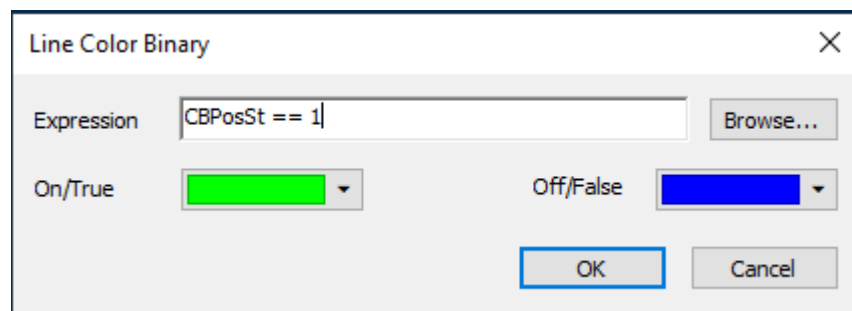


Figure F-87. Line Color Binary Configuration Window

Master Properties – Line Color / Fill Color / Text Color – Binary	
Expression	The keyname of a boolean tag or expression with a boolean result.
On/True	Color when Expression = TRUE.

Off/False	Color when Expression = FALSE.
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

Click **OK** to close the **Line Color / Fill Color / Text Color Binary** window.

Click **OK** to save.

F.3.10.4.10 Configuring Line Color/ Fill Color / Text Color - Analog

Open the object's **Master Properties** window.

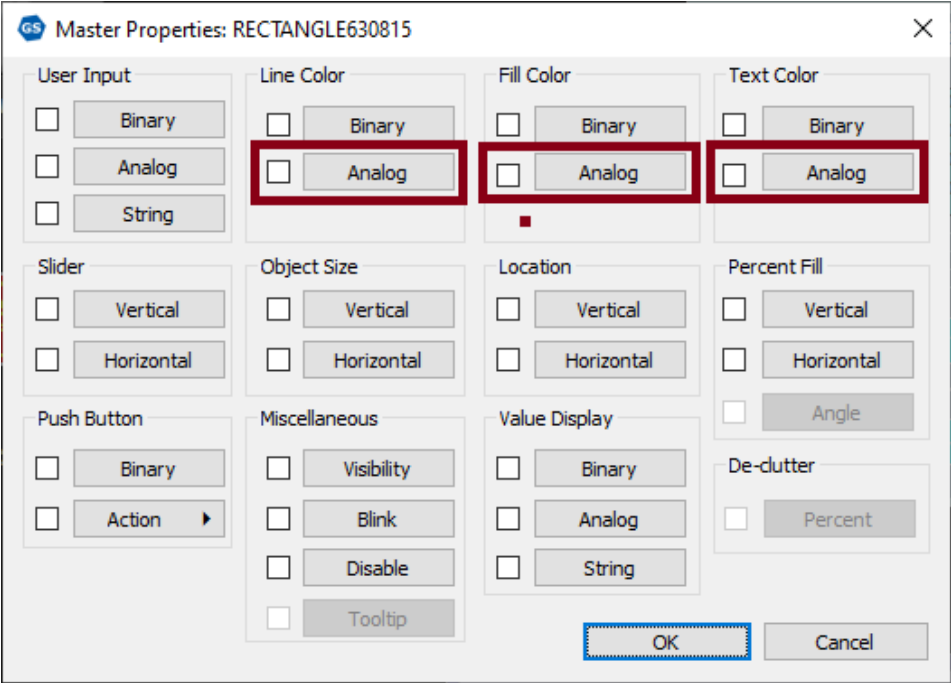


Figure F-88. Master Properties Line Color/ Fill Color / Text Color – Analog

Click **Analog** in the **Line Color** section. The **Line Color / Fill Color / Text Color - Analog** window appears.

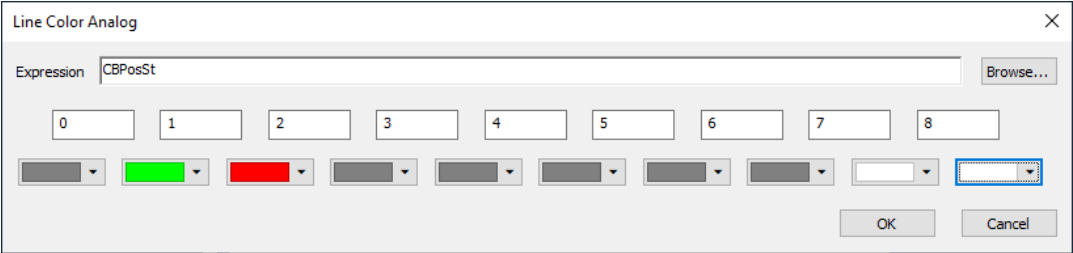
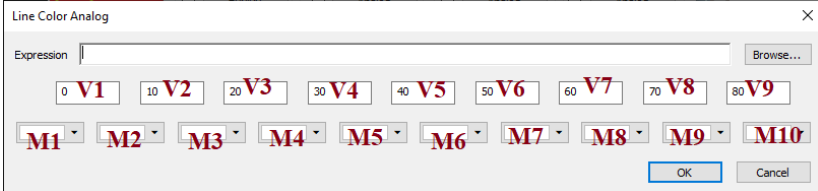


Figure F-89. Line Color Analog Configuration Window

Master Properties – Line Color / Fill Color / Text Color – Analog

Expression	Keyname of a numeric tag (integer, real number) or expression with numeric results.																								
List 09 values & List of 10 colors	Colors displayed according to the configured color/value table.  Table of values: <table> <thead> <tr> <th>Value</th><th>Color</th></tr> </thead> <tbody> <tr> <td>$-\infty < \text{Expression} \leq V1$</td><td>M1</td></tr> <tr> <td>$V1 < \text{Expression} \leq V2$</td><td>M2</td></tr> <tr> <td>$V2 < \text{Expression} \leq V3$</td><td>M3</td></tr> <tr> <td>$V3 < \text{Expression} \leq V4$</td><td>M4</td></tr> <tr> <td>$V4 < \text{Expression} \leq V5$</td><td>M5</td></tr> <tr> <td>$V5 < \text{Expression} \leq V6$</td><td>M6</td></tr> <tr> <td>$V6 < \text{Expression} \leq V7$</td><td>M7</td></tr> <tr> <td>$V7 < \text{Expression} \leq V8$</td><td>M8</td></tr> <tr> <td>$V8 < \text{Expression} \leq V9$</td><td>M9</td></tr> <tr> <td>$V9 < \text{Expression} \leq +\infty$</td><td>M10</td></tr> <tr> <td></td><td></td></tr> </tbody> </table>	Value	Color	$-\infty < \text{Expression} \leq V1$	M1	$V1 < \text{Expression} \leq V2$	M2	$V2 < \text{Expression} \leq V3$	M3	$V3 < \text{Expression} \leq V4$	M4	$V4 < \text{Expression} \leq V5$	M5	$V5 < \text{Expression} \leq V6$	M6	$V6 < \text{Expression} \leq V7$	M7	$V7 < \text{Expression} \leq V8$	M8	$V8 < \text{Expression} \leq V9$	M9	$V9 < \text{Expression} \leq +\infty$	M10		
Value	Color																								
$-\infty < \text{Expression} \leq V1$	M1																								
$V1 < \text{Expression} \leq V2$	M2																								
$V2 < \text{Expression} \leq V3$	M3																								
$V3 < \text{Expression} \leq V4$	M4																								
$V4 < \text{Expression} \leq V5$	M5																								
$V5 < \text{Expression} \leq V6$	M6																								
$V6 < \text{Expression} \leq V7$	M7																								
$V7 < \text{Expression} \leq V8$	M8																								
$V8 < \text{Expression} \leq V9$	M9																								
$V9 < \text{Expression} \leq +\infty$	M10																								
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression																								

Click **OK** to close the **Line Color Analog** window.

Click **OK** to save.

F.3.10.4.11 Configuring Percent Fill – Vertical

Open the object's **Master Properties** window.

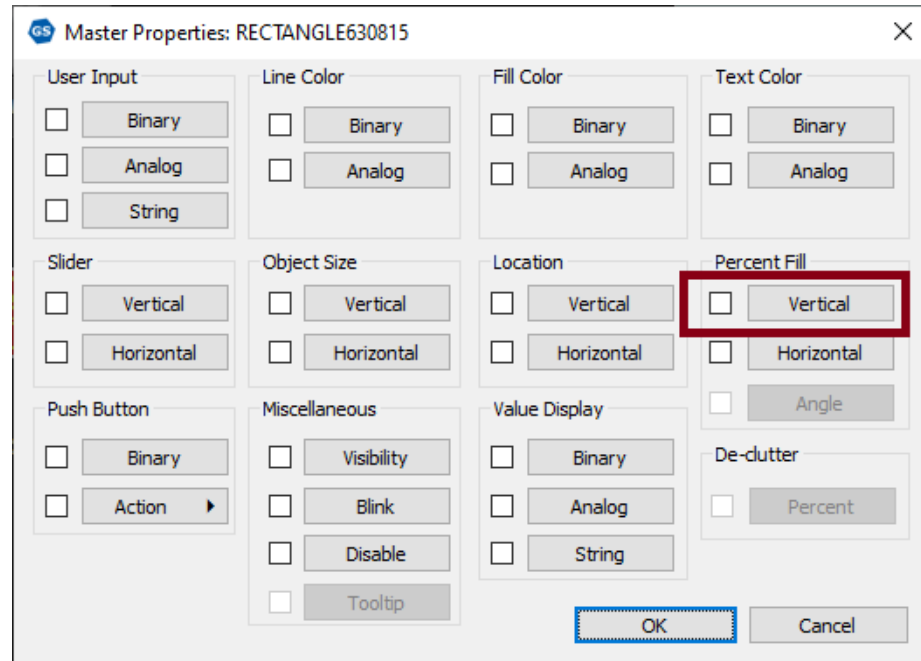


Figure F-90. Master Properties Percent Fill – Vertical

Click **Vertical** in the **Percent Fill** section. The **Percent Fill – Vertical** window appears.

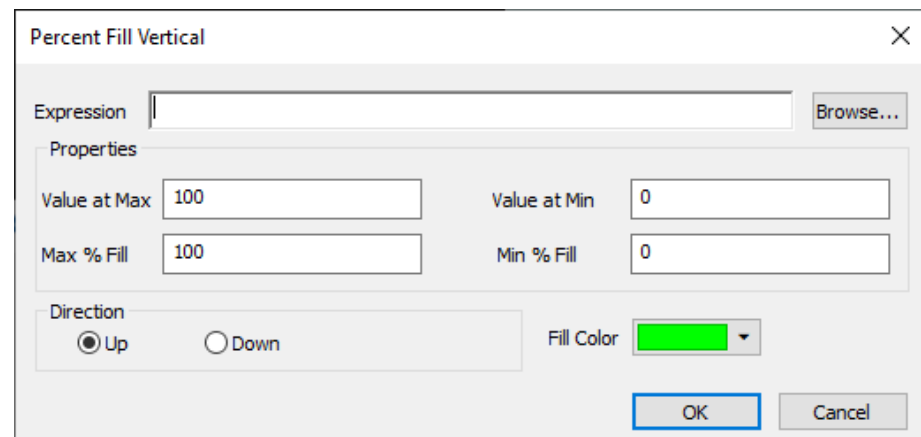


Figure F-91. Percent Fill Vertical Configuration Window

Master Properties – Percent Fill – Vertical	
Expression	Keyname of an analog tag (integer, real number) or expression with analog results.
Value at Max Value at Min	Expression's Max-Min range is used to color. Expression values outside the max-min range will be converted to max-min values in the % formula.
Max % Fill Min % Fill	The % range of colored values corresponds to the max-min range of the expression.

	Max ≤ 100%. Min ≥ 0%
Direction - Up	Color direction from bottom to top.
Direction - Down	Color direction from top to bottom.
Fill Color	Colors are poured in percentages. The background color is the color designed in the Designer.
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

The % fill color value on the Grid Viewer is converted from the Expression value:

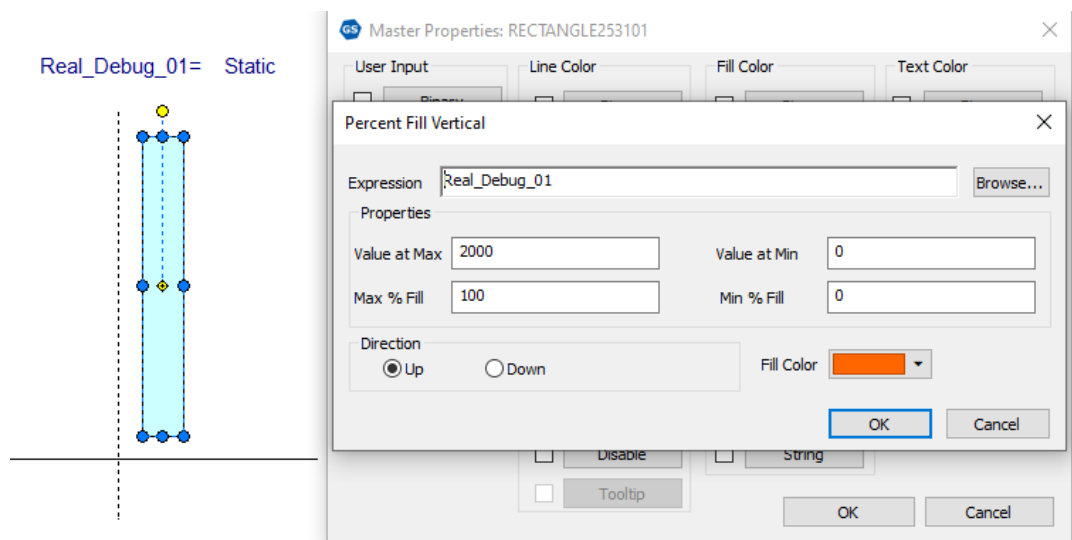
$$\frac{\% \text{Fill} - \% \text{Min}}{\% \text{Max} - \% \text{Min}} = \frac{\text{Expression} - \text{Value at Min}}{\text{Value at Max} - \text{Value at Min}}$$

Click **OK** to close the **Percent Fill – Vertical** window.

Click **OK** to save.

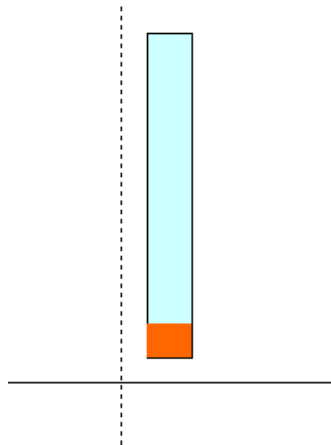
Illustration example:

- Design on the Grid Designer



- Effect on Grid Viewer

Real_Debug_01= 200.000



F.3.10.4.12 Configuring Percent Fill – Horizontal

Open the object's **Master Properties** window.

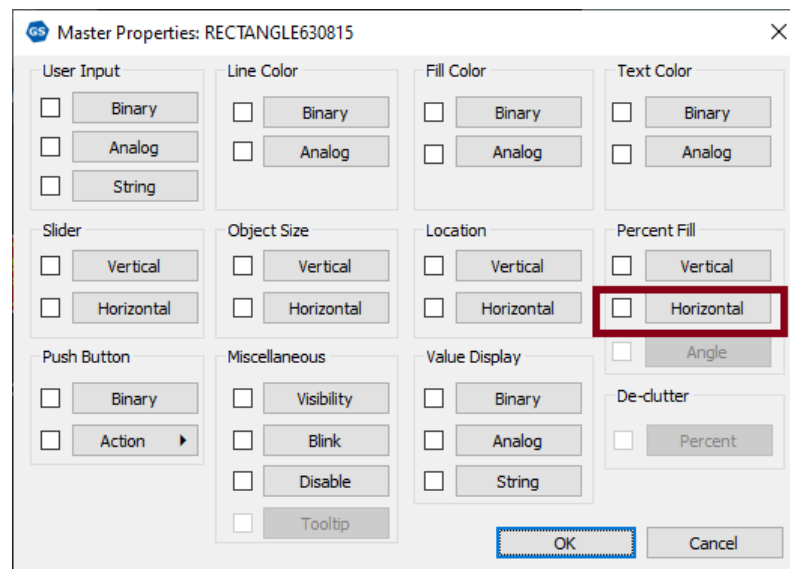


Figure F-92. Master Properties Percent Fill – Horizontal

Click **Horizontal** in the **Percent Fill** section. The **Percent Fill – Horizontal** window appears.

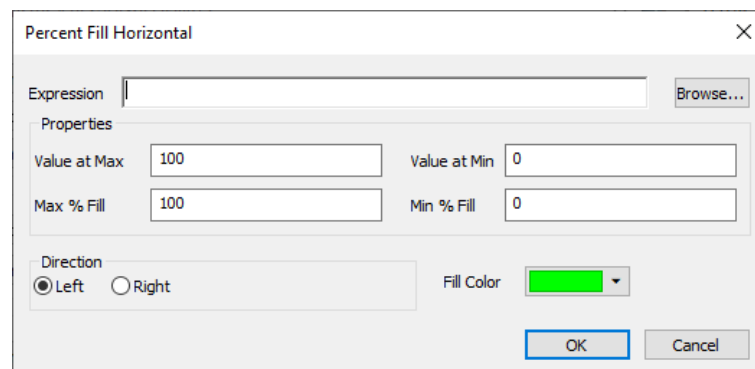


Figure F-93. Percent Fill Horizontal Configuration Window

Master Properties – Percent Fill – Horizontal	
Expression	Keyname of an analog tag (integer, real number) or expression with analog results.
Value at Max Value at Min	Expression's Max-Min range is used to color. Expression values outside the max-min range will be converted to max-min values in the % formula.
Max % Fill Min % Fill	The % range of colored values corresponds to the max-min range of the expression. Max ≤ 100%. Min ≥ 0%
Direction - Left	Color direction from left to right.
Direction - Right	Color direction from right to left.
Fill Color	Colors are poured in percentages. The background color is the color designed in the Designer.
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

The % fill color value on the Grid Viewer is converted from the Expression value:

$$\frac{\% \text{Fill} - \% \text{Min}}{\% \text{Max} - \% \text{Min}} = \frac{\text{Expression} - \text{Value at Min}}{\text{Value at Max} - \text{Value at Min}}$$

Click **OK** to close the **Percent Fill – Horizontal** window.

Click **OK** to save.

Illustration example:

- Design on the Grid Designer

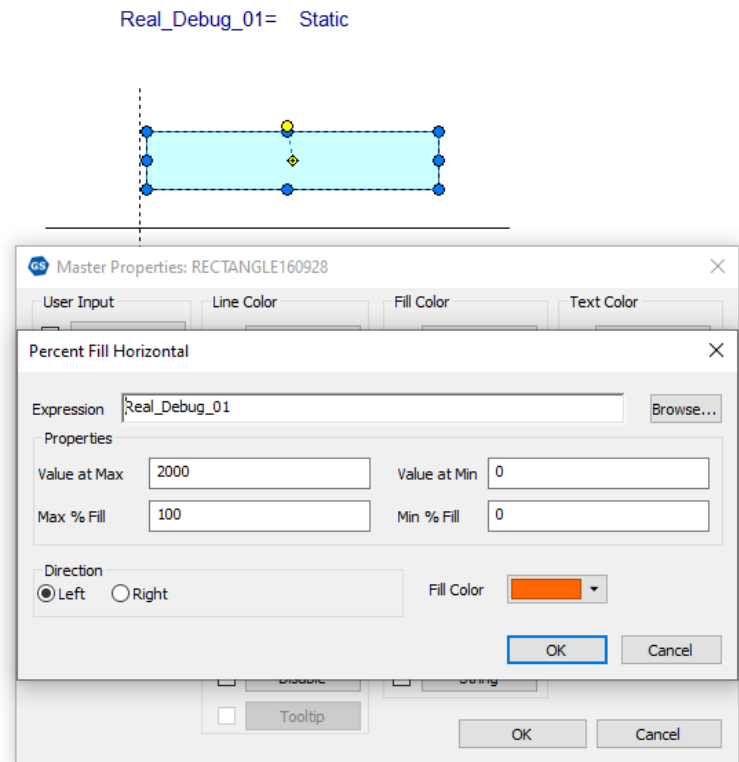
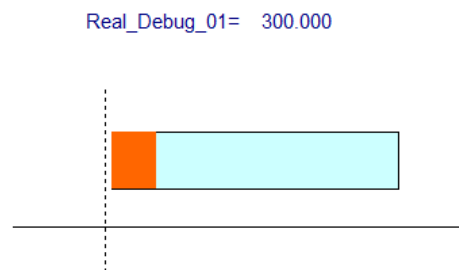


Figure F-94. Example of Percent Fill Horizontal Configuration

- Effect on the Grid Viewer



F.3.10.4.13 Configuring Percent Fill – Angle

Open the **Master Properties** window of ellipse / Round object.

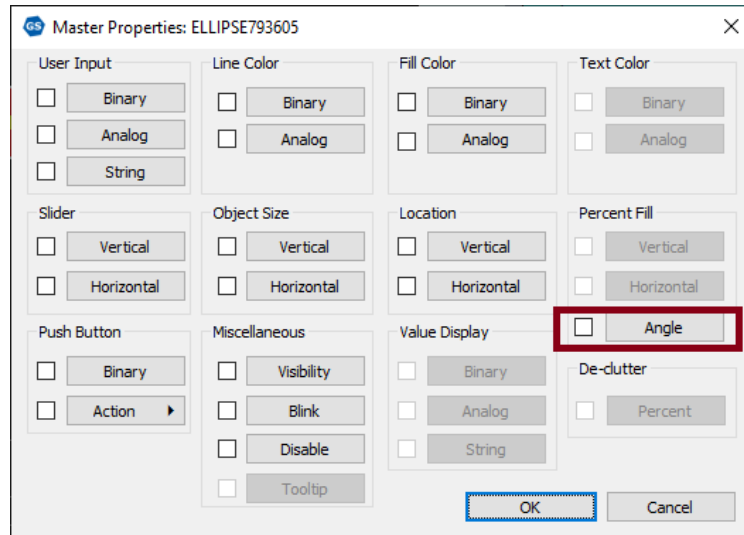


Figure F-95. Master Properties Percent Fill – Angle

Click **Angle** in the **Percent Fill** section. The **Percent Fill – Angle** window appears.

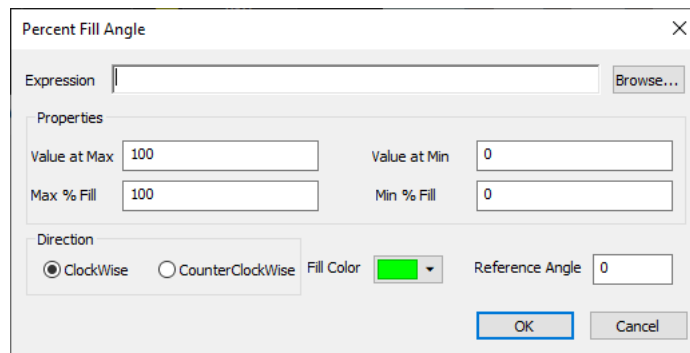
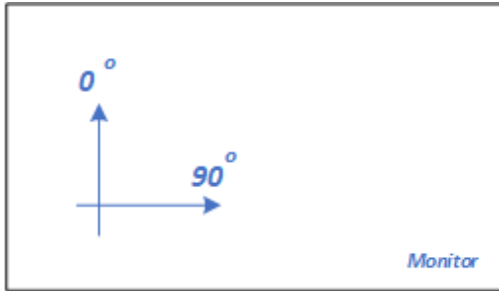


Figure F-96. Percent Fill Angle Configuration Window

Master Properties – Percent Fill – Angle	
Expression	Keyname of an analog tag (integer, real number) or expression with analog results.
Value at Max Value at Min	Expression's Max-Min range is used to color. Expression values outside the max-min range will be converted to max-min values in the % formula.
Max % Fill Min % Fill	The % range of colored values corresponds to the max-min range of the expression. Max ≤ 100%. Min ≥ 0%
Direction ClockWise	– Color direction is clockwise.

Direction – Counter ClockWise	Color direction is counter-clockwise.
Fill Color	Colors are poured in percentages. The background color is the color designed in the Designer.
Reference Angle	The absolute angle corresponds to the Min % level. Where the 0° angle convention as described: 
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

The % fill color value on the Grid Viewer is converted from the Expression value:

$$\frac{\% \text{Fill} - \% \text{Min}}{\% \text{Max} - \% \text{Min}} = \frac{\text{Expression} - \text{Value at Min}}{\text{Value at Max} - \text{Value at Min}}$$

Click **OK** to close the **Percent Fill – Angle** window.

Click **OK** to save.

Illustration example:

- Design on the Grid Designer

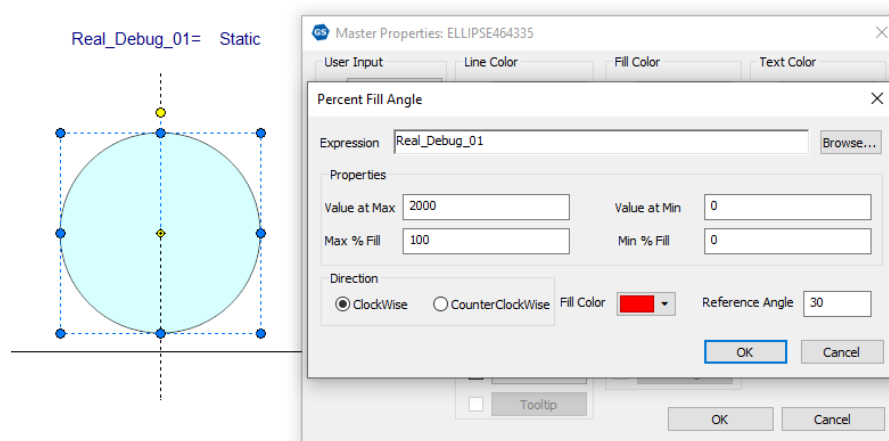
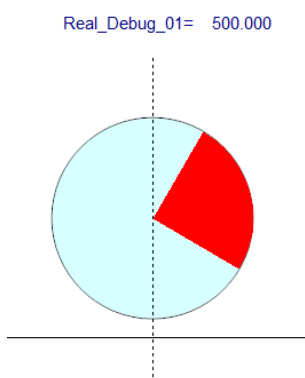


Figure F-97. Example of Percent Fill Angle Configuration

- Effect on the Grid Viewer



F.3.10.4.14 Configuring Object Size – Vertical

Open the object's **Master Properties** window.

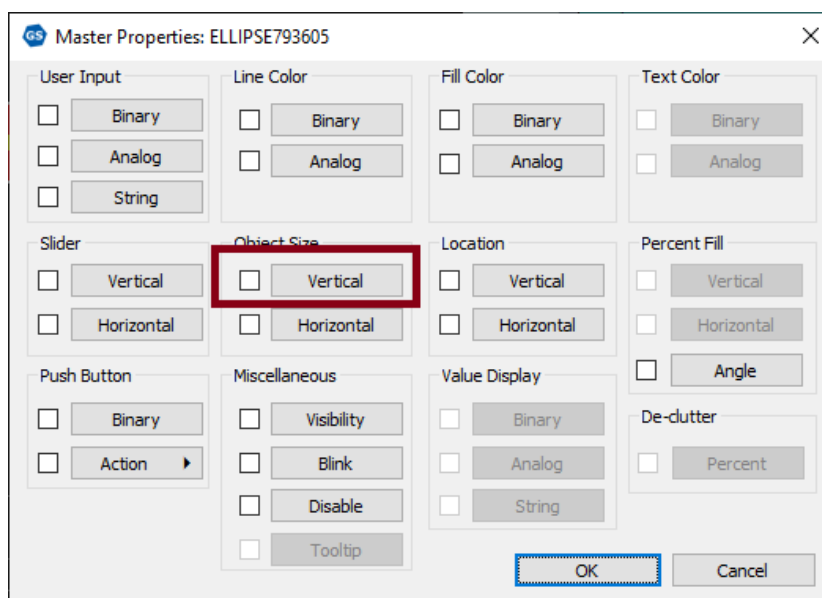


Figure F-98. Master Properties Object Size – Vertical

Click **Vertical** on the **Object Size** section. The **Object Size Height** window appears.

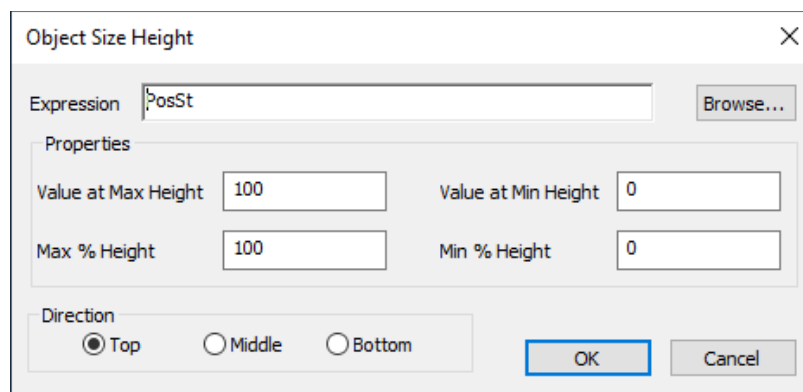


Figure F-99. Object Size Height Configuration Window

Master Properties – Object Size – Vertical	
Expression	Keyname of an analog tag (integer, real number) or expression with analog results.
Value at Max Height Value at Min Height	Expression's Max-Min range is used to resize. Expression values outside the max-min range will be converted to max-min values in the % formula.
Max % Height Min % Height	The % range of vertical dimensions corresponds to the max-min range of the expression. The 100% landmark is the height of the object designed on the Grid Designer. Max ≤ 100%. Min ≥ 0%
Direction - Top	Origin points to scale object vertically - at the top point.
Direction - Middle	Origin points to scale object vertically - at the center point.
Direction - Bottom	Origin points to scale object vertically - at the bottom point.
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

The height % value on Grid Viewer compared to the design value on Grid Designer is converted from the Expression value:

$$\frac{\% \text{Height} - \% \text{Min}}{\% \text{Max} - \% \text{Min}} = \frac{\text{Expression} - \text{Value at Min}}{\text{Value at Max} - \text{Value at Min}}$$

Click **OK** to close the **Object Size Height** window.

Click **OK** to save.

Illustration example:

- Design on the Grid Designer

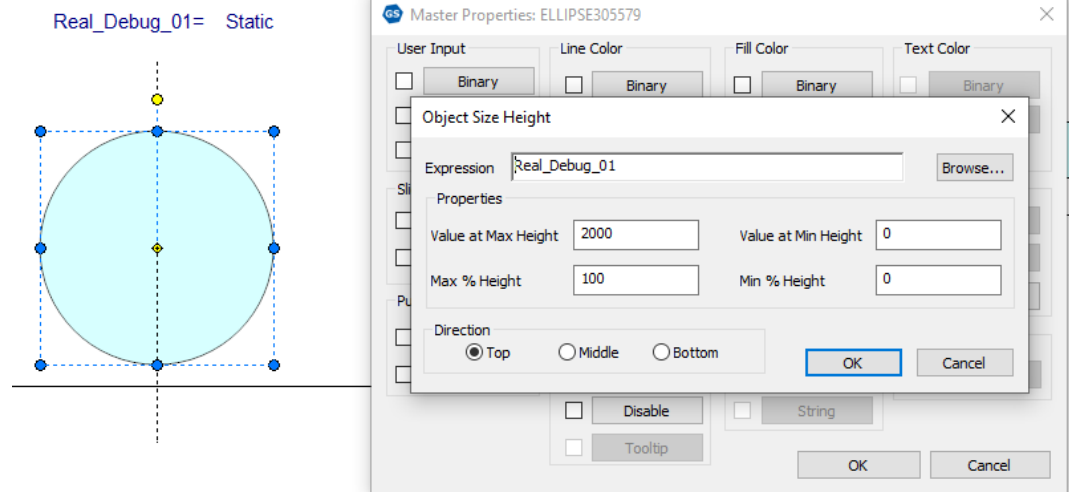
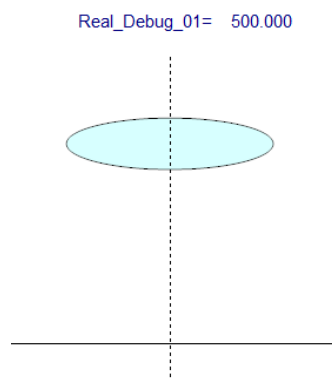


Figure F-100. Example of Object Size Height Configuration

- Effect on the Grid Viewer



F.3.10.4.15 Configuring Object Size – Horizontal

Open the object's **Master Properties** window.

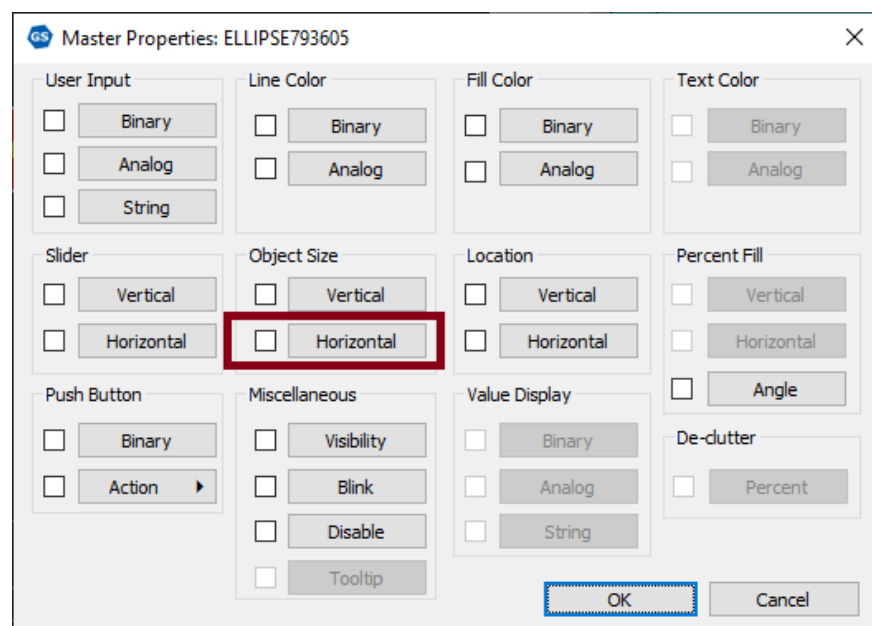


Figure F-101. Master Properties Object Size – Horizontal

Click **Horizontal** in the **Object Size** section. The **Object Size Width** window appears.

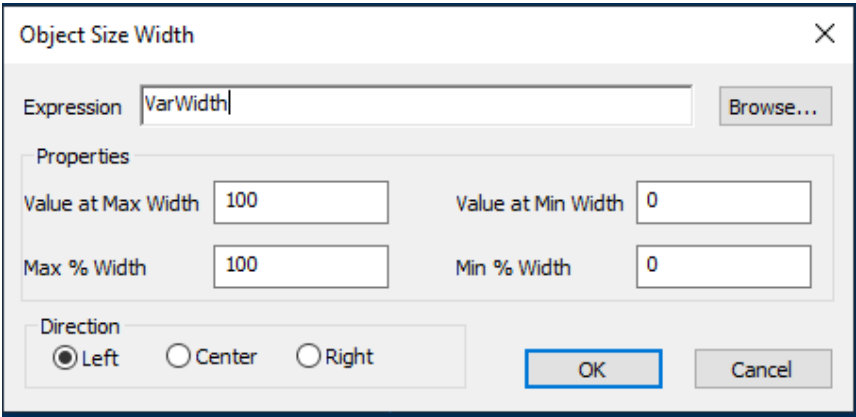


Figure F-102. Object Size Width Configuration

Master Properties – Object Size – Horizontal	
Expression	Keyname of an analog type of tag (integer, real number) or expression with similar results.
Value at Max Width	Expression's Max-Min range is used to resize.
Value at Min Width	Expression values outside the max-min range will be converted to max-min values in the % formula.
Max % Height	The range % of the horizontal dimension corresponds to the max-min range of expression The 100% landmark is the horizontal of the object designed on the Grid Designer. Max ≤ 100%. Min ≥ 0%
Min % Height	
Direction - Left	Origin points to scale object horizontally - at the leftmost point.
Direction – Center	Origin points to scale object horizontally - at the center point.
Direction - Right	Origin points to scale object horizontally - at the rightmost point.
Browse	Support tag selection.

	Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression
--	---

The horizontal % value on the Grid Viewer compared to the design value on the Grid Designer is converted from the Expression value:

$$\frac{\%Width - \%Min}{\%Max - \%Min} = \frac{\text{Expression} - \text{Value at Min}}{\text{Value at Max} - \text{Value at Min}}$$

Click **OK** to close the **Object Size Width** window.

Click **OK** to save.

Illustration example:

- Design on Grid Designer

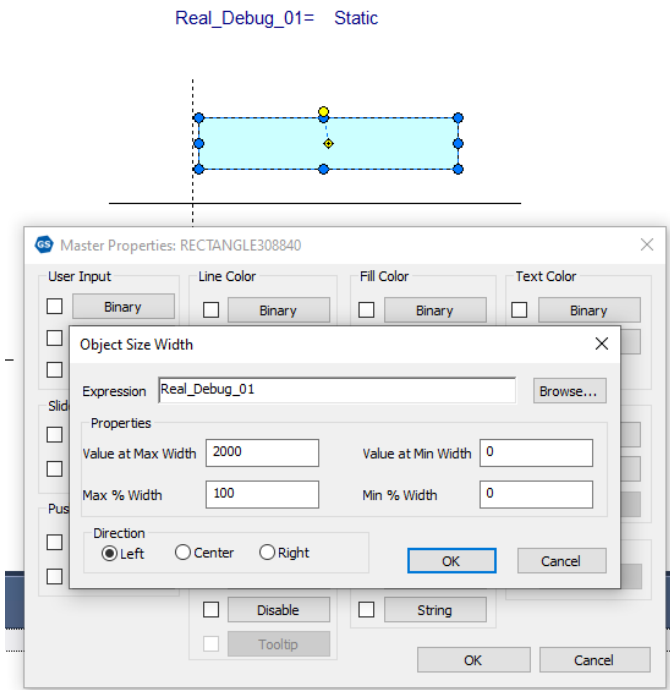
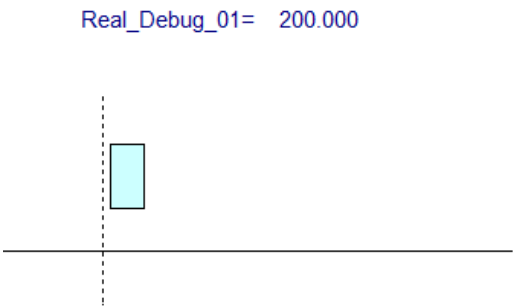


Figure F-103. Example of Object Size Width Configuration

- Effect on Grid Viewer



F.3.10.4.16 Configuring Location – Vertical

Open the object's **Master Properties** window.

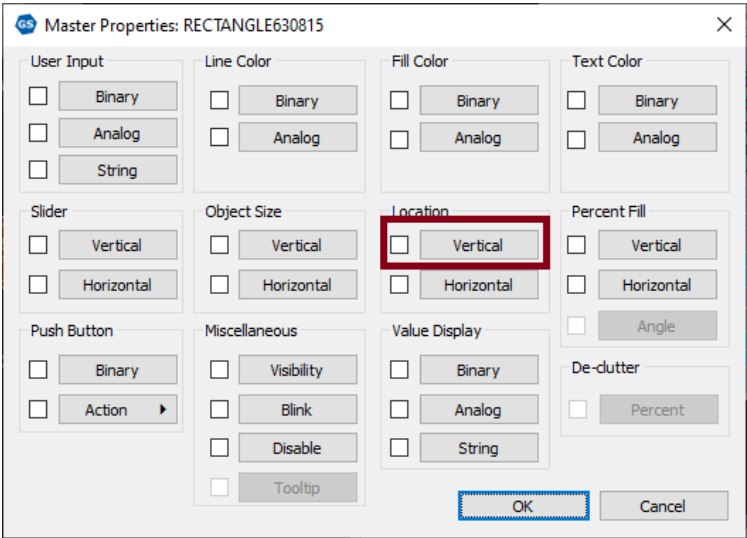


Figure F-104. Master Properties Location – Vertical

Click **Vertical** in the **Location** section. The **Location Vertical** window appears.

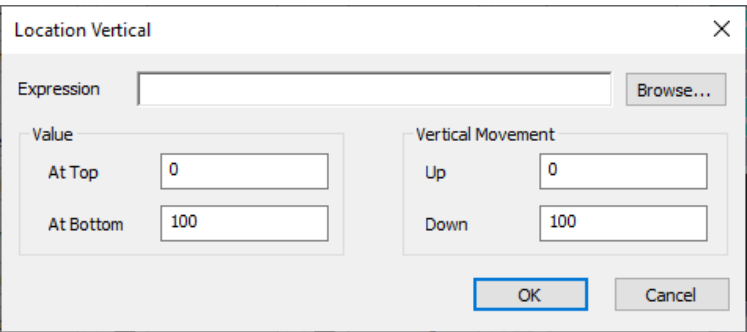


Figure F-105. Location Vertical Configuration Window

Master Properties – Location – Vertical	
Expression	Keyname of an analog tag (integer, real number) or expression with analog results.
Value at Top Value at Bottom	Expression's range of values corresponds to the positions at the top (at Top) and at the bottom (at Bottom). Expression values outside the top-bottom range will be converted to top-bottom values in the coordinate formula.
Vertical Movement Up - Down	Shift the coordinates along with the Y-axis corresponding to the range at Top - at Bottom of the expression.

Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression
--------	---

The Y-axis coordinate value on the Grid Viewer is calculated by the formula:

$$Y.Viewer = Y.Designer + \frac{\text{Expression} - \text{at Top}}{\text{at Bottom} - \text{at Top}} \cdot (\text{Down} + \text{Up}) - \text{Up}$$

Click **OK** to close the **Location Vertical** window.

Click **OK** to save.

Illustration example:

- Design on Grid Designer

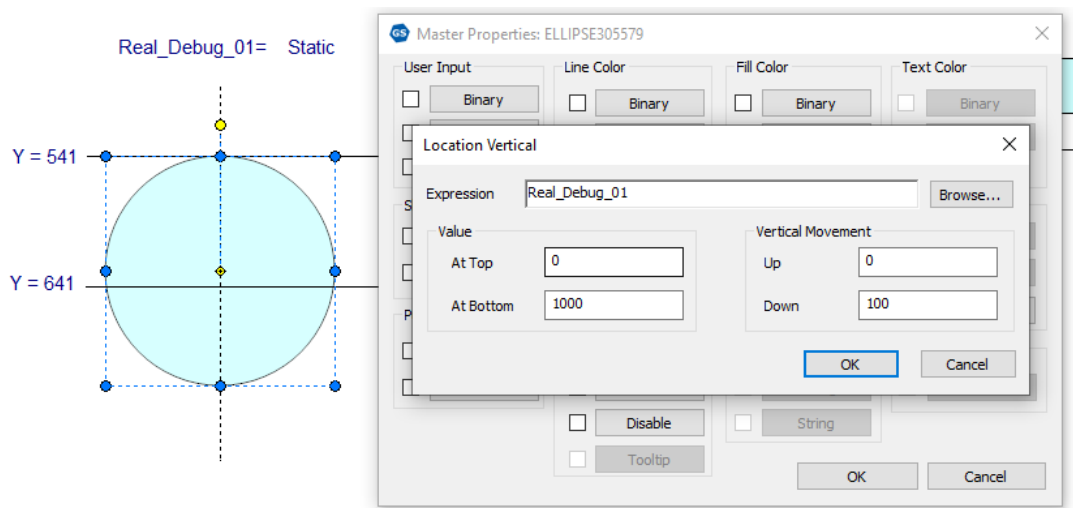
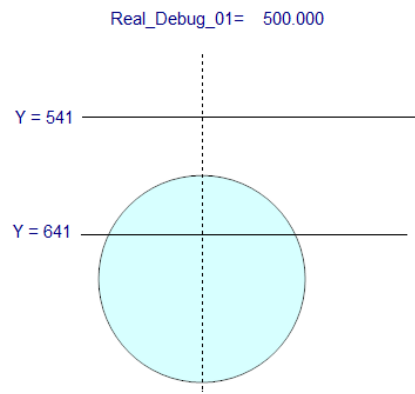


Figure F-106. Example of Location Vertical Configuration

- Effect on Grid Viewer



F.3.10.4.17 Configuring Location – Horizontal

Open the object's **Master Properties** window.

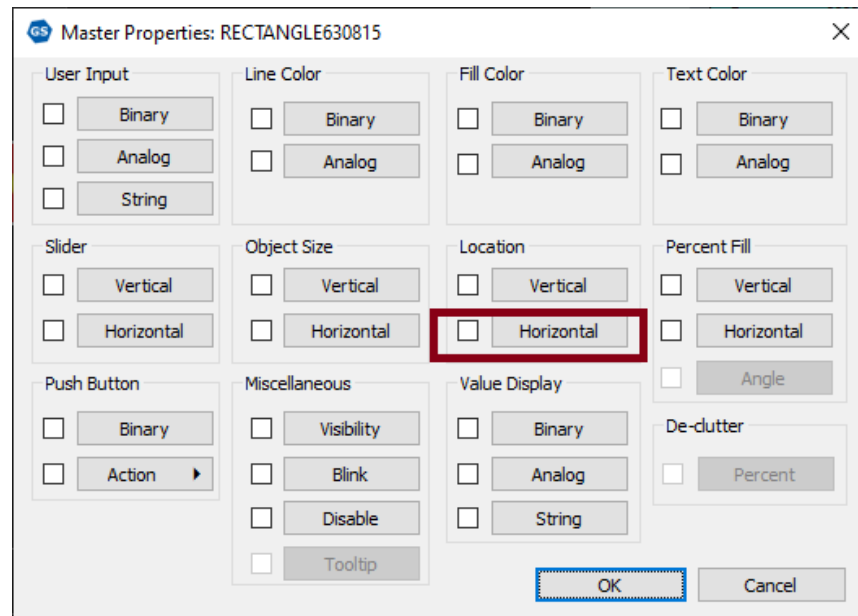


Figure F-107. Master Properties Location – Horizontal

Click **Horizontal** in the **Location** section. The **Location Horizontal** window appears.

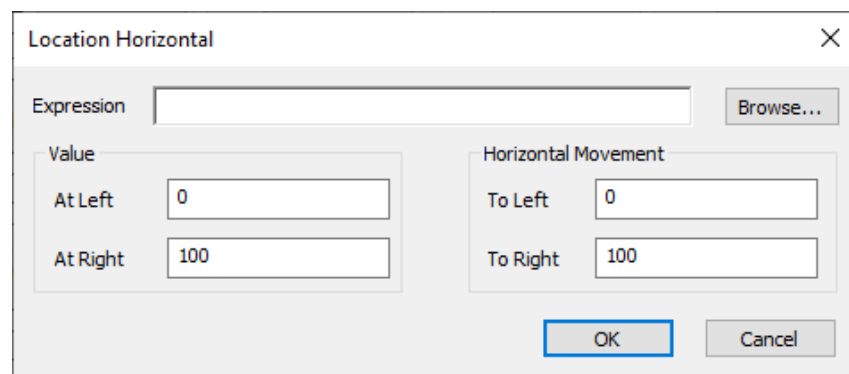


Figure F-108. Location Horizontal Configuration Window

Master Properties – Location – Horizontal	
Expression	Keyname of an analog tag (integer, real number) or expression with analog results.
Value at Left Value at Right	Expression's range of values corresponds to the positions on the leftmost (at Left) and the rightmost (at Right) Expression values out of range at Left - at Right will be converted to at Left - at Right values in the coordinate formula.

Vertical Movement To Left – To Right	Shift the coordinates horizontally on the X axis corresponding to the range at Left - at Right of the expression.
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

The coordinate value in the X axis on the Grid Viewer is calculated by the formula:

$$X.Viewer = X.Designer + \frac{\text{Expression} - \text{at Left}}{\text{at Right} - \text{at Left}} \cdot (\text{to Right} + \text{to Left}) - \text{to Left}$$

Click **OK** to close the **Location Horizontal** window.

Click **OK** to save.

Illustration example:

- Design on Grid Designer

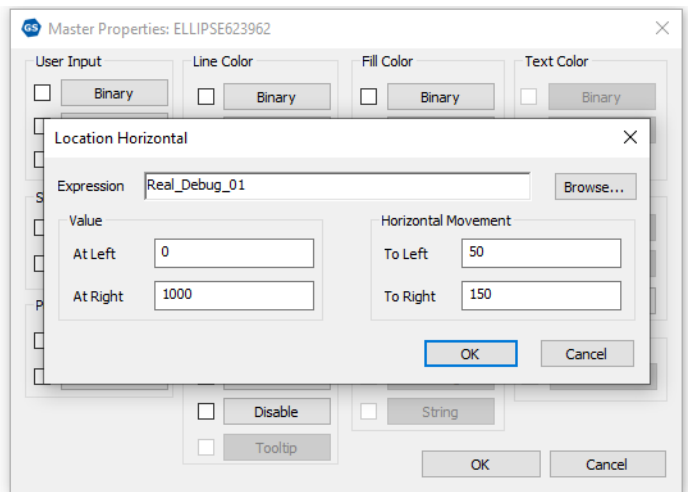
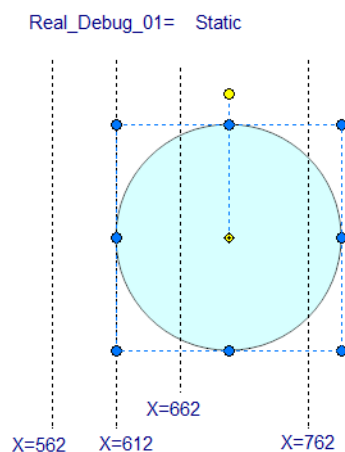
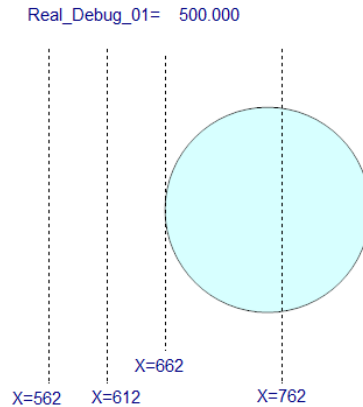


Figure F-109. Example of Location Horizontal Configuration

- Effect on Grid Viewer



F.3.10.4.18 Configuring Rotation – Angle

Open the object's **Master Properties** window.

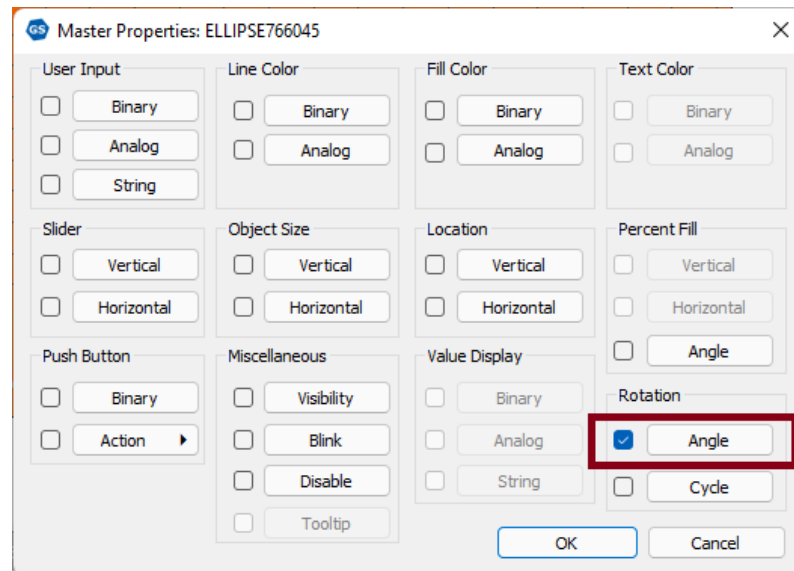


Figure F-110. Master Properties Rotation – Angle

Click **Angle** in the **Rotation** section. The **Rotation – Angle** window appears.

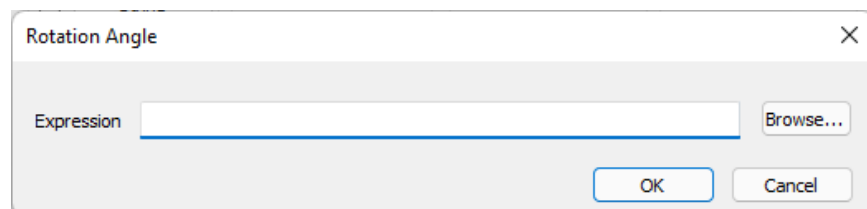


Figure F-111. Rotation Angle Configuration Window

Master Properties – Rotation Angle	
Expression	The object's rotation angle - Unit Degree

	- Keyname of an analog tag (integer, real number) or expression with analog results.
--	--

Click **OK** to close the **Rotation – Angle** window.

Click **OK** to save.

Illustration example:

- Design on Grid Designer: 45-degree rotation.

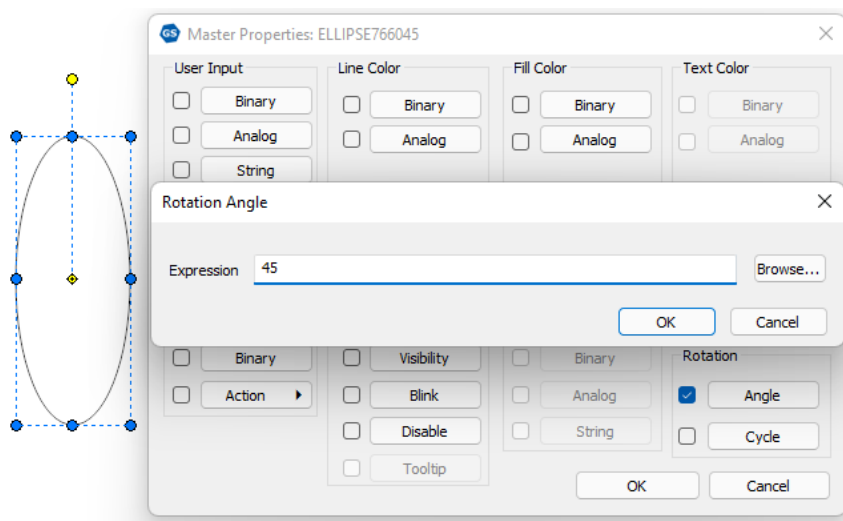
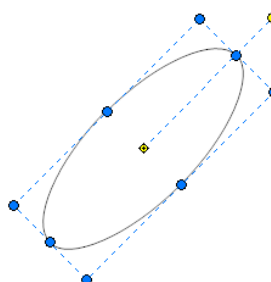


Figure F-112. Example of Rotation Angle Configuration

- Effect on Grid Viewer



F.3.10.4.19 Configuring Rotation – Cycle

Open the object's **Master Properties** window.

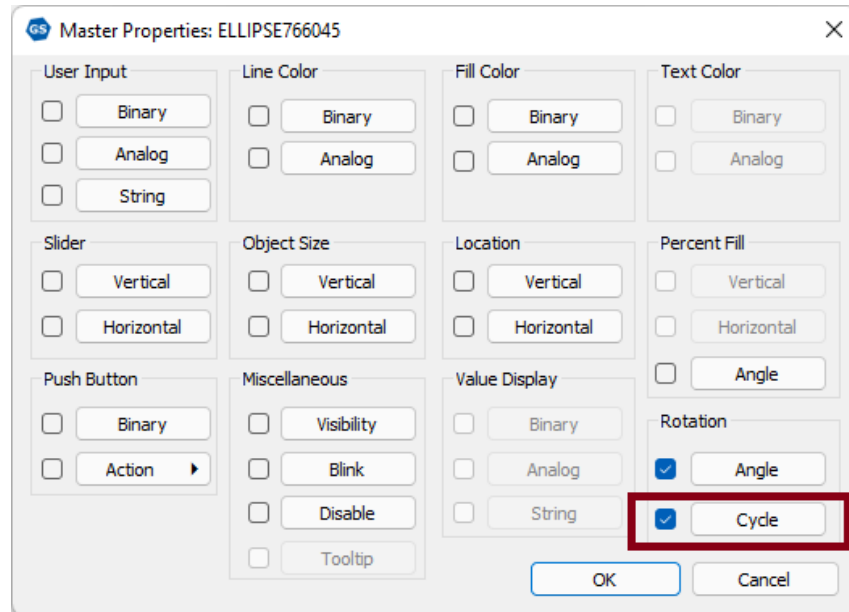


Figure F-113. Master Properties Rotation – Cycle

Click **Cycle** in the **Rotation** section. The **Rotation – Cycle** window appears.

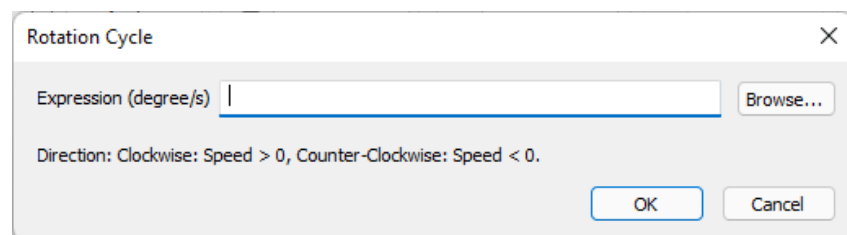


Figure F-114. Rotation Cycle Configuration Window

Master Properties – Rotation Cycle	
Expression	The object's speed rotation - Unit Degree/s - Direction: ▪ Clockwise: speed >0 ▪ Counter-Clockwise: speed <0 - Keyname of an analog tag (integer, real number) or expression with analog results.

Click **OK** to close the **Rotation – Cycle** window.

Click **OK** to save.

F.3.10.4.20 Configuring User Input – Binary

Open the object's **Master Properties** window.

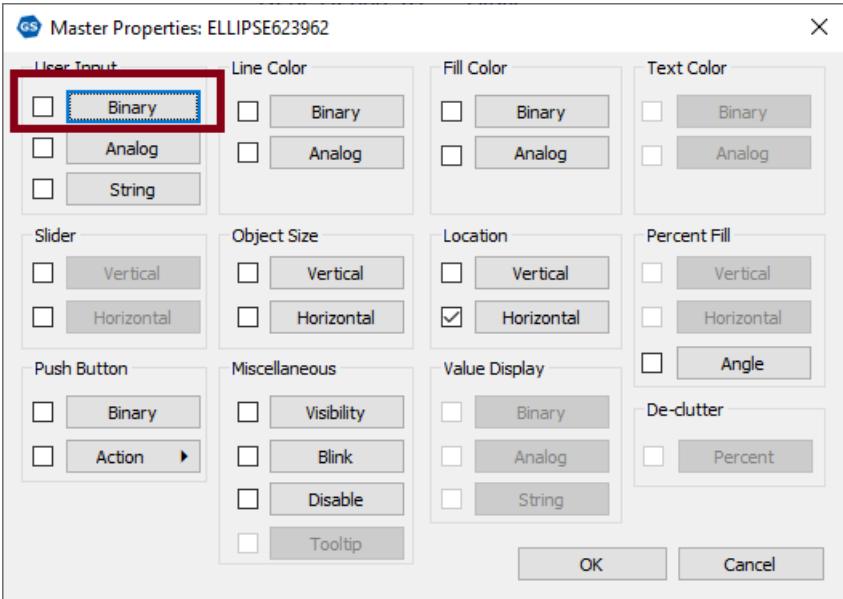


Figure F-115. Master Properties User Input – Binary

Click **Binary** in the **User Input** section. The **User Input Binary** window appears.

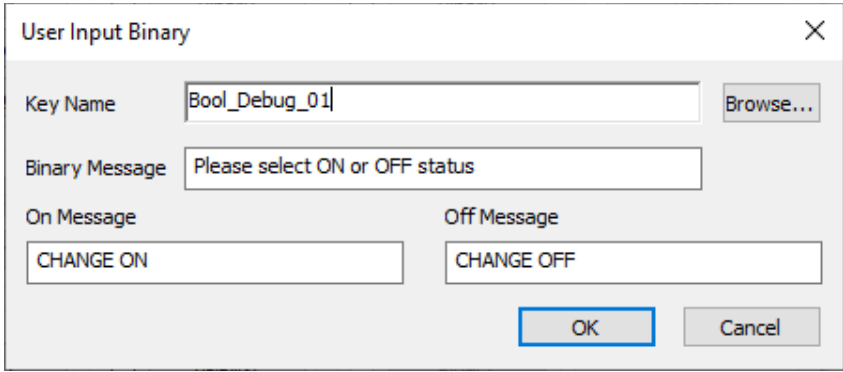


Figure F-116. User Input Binary Configuration Window

Master Properties – User Input – Binary	
Key Name	Keyname of a boolean tag.
Binary Message	Notice in the data input window.
On Message	Text on button set tag value = true
Off Message	Text on button set tag value = false
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

Click **OK** to close the **User Input Binary** window.

Click **OK** to save.

Illustration example:

- Design on Grid Designer

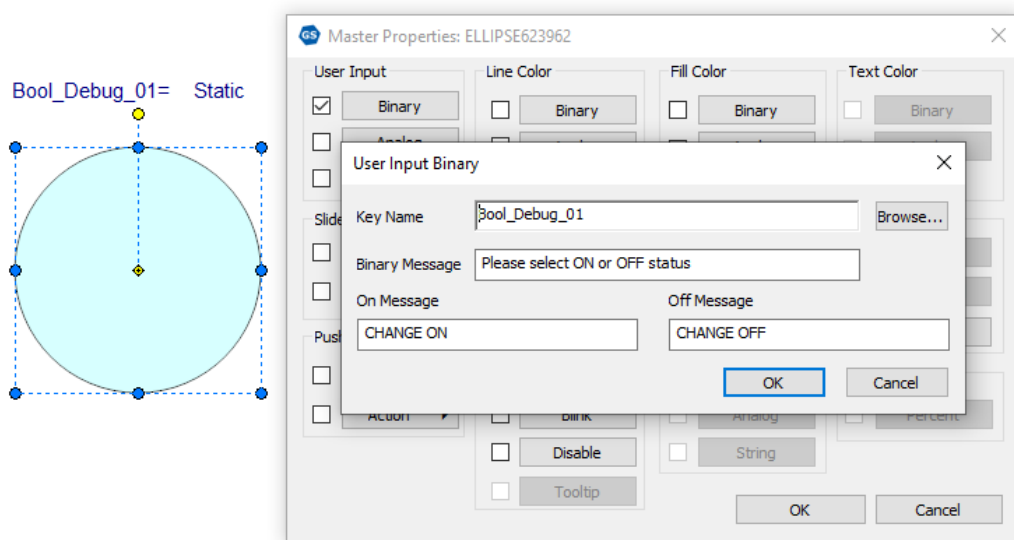
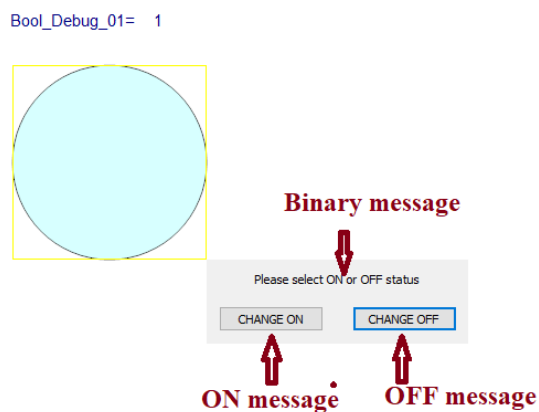


Figure F-117. Example of User Input Binary Configuration

- Effect on Grid Viewer when click to object.



F.3.10.4.21 Configuring User Input – Analog

Open the object's **Master Properties** window.

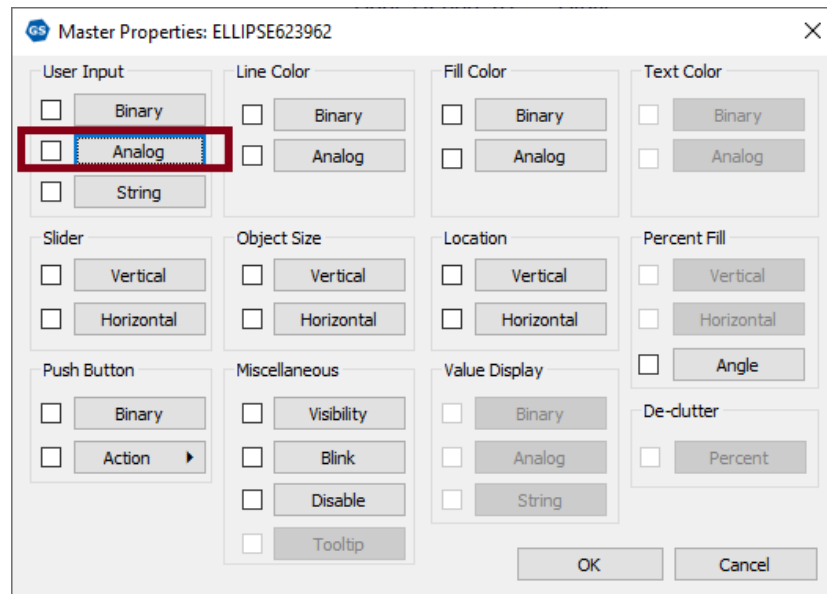


Figure F-118. Master Properties User Input – Analog

Click **Analog** in the **User Input** section. The **User Input Analog** window appears.

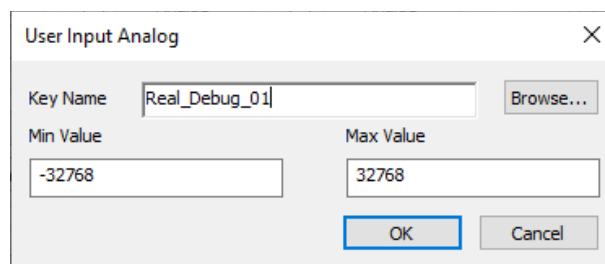


Figure F-119. User Input Analog Configuration Window

Master Properties – User Input – Analog	
Key Name	Keyname of an analog tag.
Min Value	Minimum value in the range of valid input values
Max Value	Maximum value in the range of valid input values
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

Click **OK** to close the **User Input Analog** window.

Click **OK** to save.

Illustration example:

- Design on Grid Designer

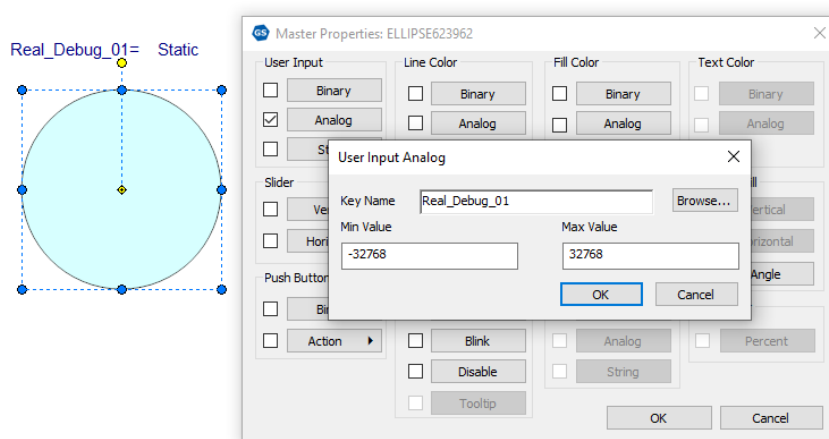
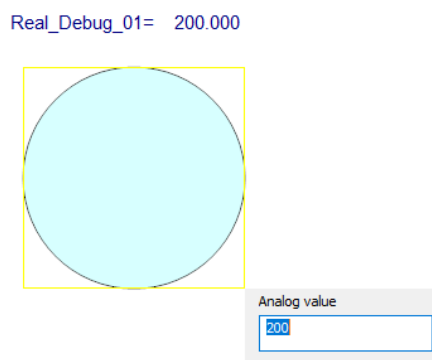


Figure F-120. Example of User Input Analog Configuration

- Effect on Grid Viewer when click to object



F.3.10.4.22 Configuring User Input – String

Open the object's **Master Properties** window.

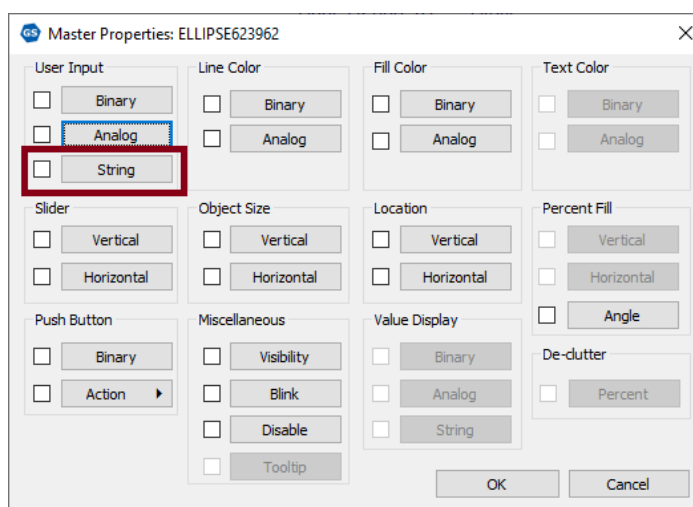


Figure F-121. Master Properties User Input – String

Click **Analog** in the **User Input** section. The **User Input String** window appears.

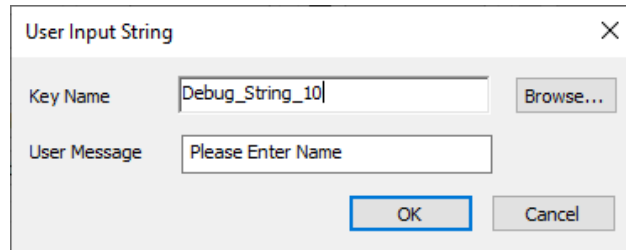


Figure F-122. User Input String Configuration Window

Master Properties – User Input – String	
Key Name	Keyname of a string tag
User Message	Notice in the data input window
Browse	Support tag selection Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

Click **OK** to close the **User Input String** window.

Click **OK** to save.

Illustration example:

- Design on Grid Designer

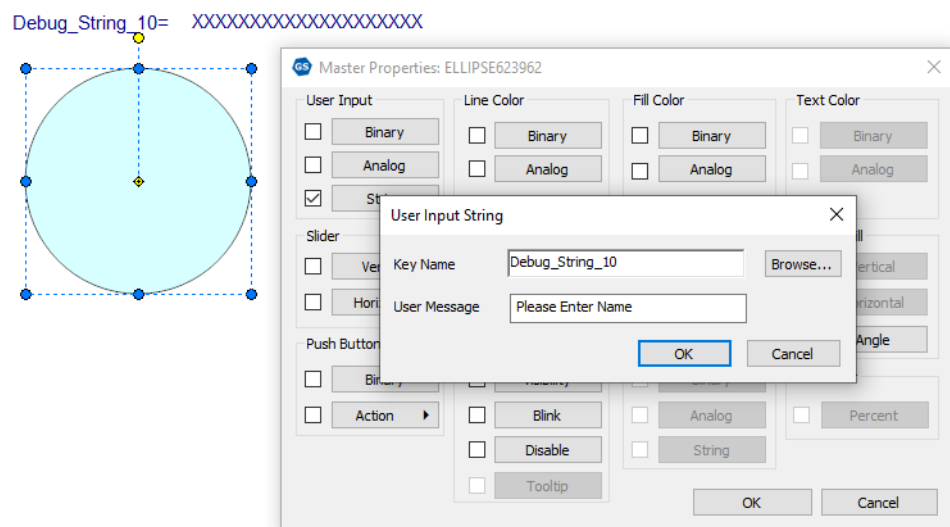
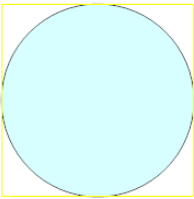


Figure F-123. Example of User Input String Configuration

- Effect on Grid Viewer when click on object

Debug_String_10= "test"



Please Enter Name

F.3.10.4.23 Configuring Slider – Vertical

Open the object's **Master Properties** window.

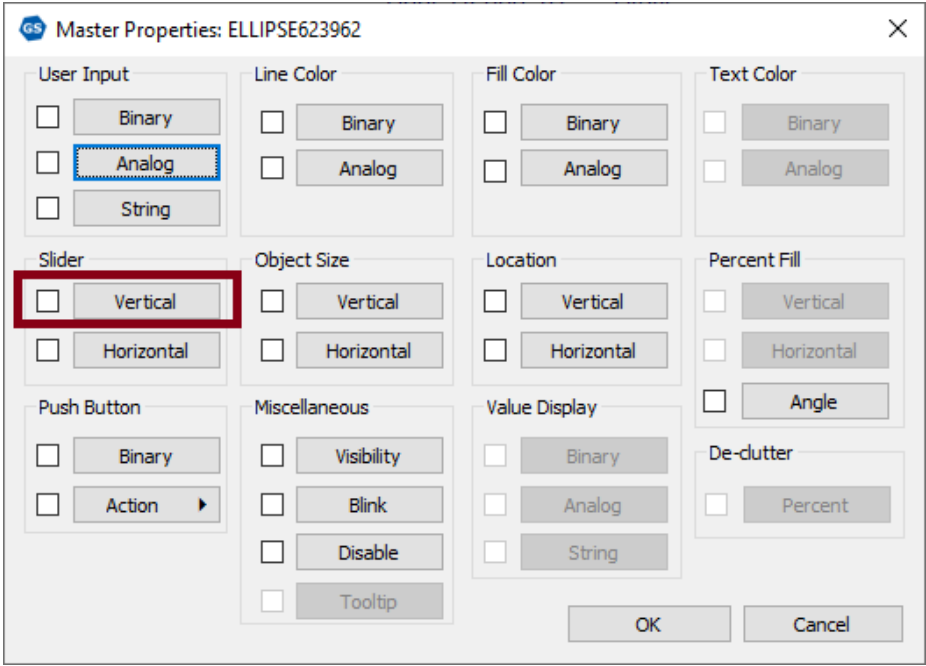


Figure F-124. Master Properties Slider – Vertical

Click **Vertical** in the **Slider** section. The **Slider Vertical** window appears.

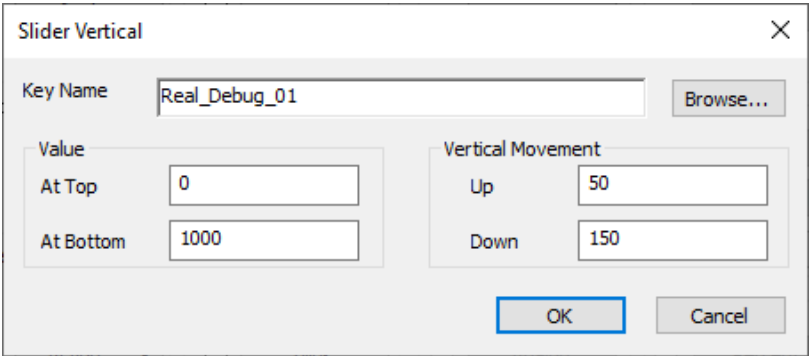


Figure F-125. Slider Vertical Configuration Window

Master Properties – Slider – Vertical

Key Name	Keyname of an analog tag
Value at Top Value at Bottom	The tag's range of values corresponds to the positions at the top and at the bottom.
Vertical Movement Up - Down	Shift coordinates along the Y-axis corresponding to the range of values at the top - at the bottom of the tag.
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

The tag value is calculated from the Y-axis coordinates on the Grid Viewer according to the formula:

$$Y.Viewer = Y.Designer + \frac{\text{Tag value} - \text{at Top}}{\text{at Bottom} - \text{at Top}} \cdot (\text{Down} + \text{Up}) - \text{Up}$$

Click **OK** to close the **Slider Vertical** window.

Click **OK** to save.

Illustration example:

- Design on Grid Designer

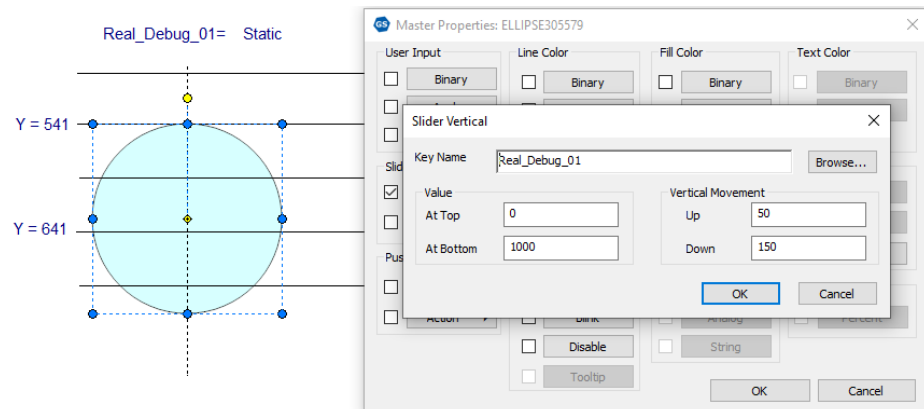
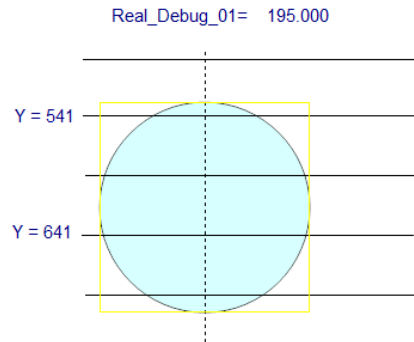


Figure F-126. Example of User Slider Vertical Configuration

- Effect on Grid Viewer when click and move object.



F.3.10.4.24 Configuring Slider – Horizontal

Open the object's **Master Properties** window.

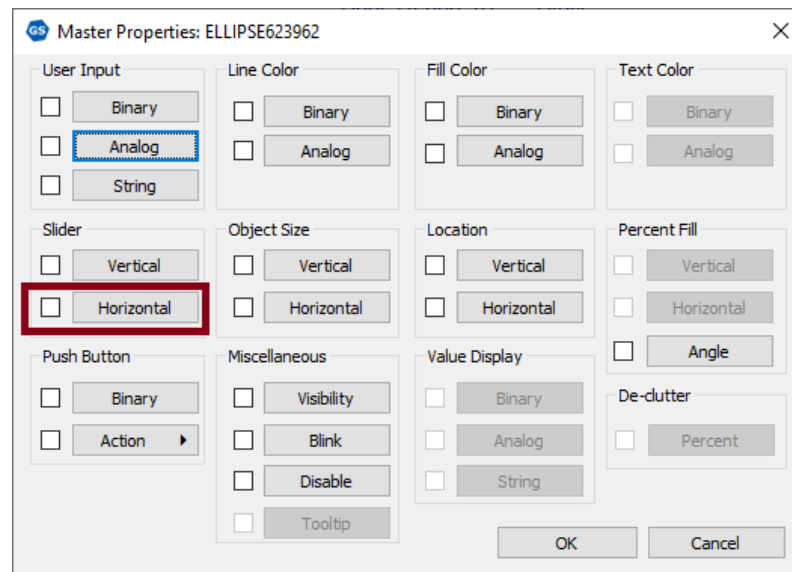


Figure F-127. Master Properties Slider – Horizontal

Click **Vertical** in the **Slider** section. The **Slider Horizontal** window appears.

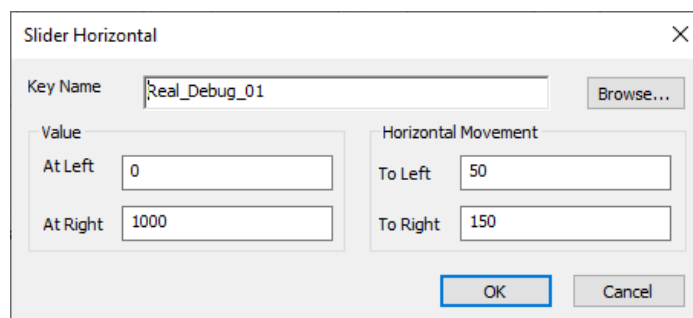


Figure F-128. Slider Horizontal Configuration Window

Master Properties – Slider – Horizontal	
Key Name	Keyname of an analog tag

Value at Left Value at Right	The tag's range of values corresponds to the positions at the top and at the bottom.
Vertical Movement To Left – To Right	Shift coordinates horizontally X axis corresponding to the value range at Left - at Right of the tag.
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

The value of the tag is calculated from the X-axis coordinates on the Grid Viewer according to the formula:

$$X.Viewer = X.Designer + \frac{\text{Expression} - \text{at Left}}{\text{at Right} - \text{at Left}} \cdot (\text{to Right} + \text{to Left}) - \text{to Left}$$

Click **OK** to close **Slider Horizontal**.

Click **OK** to save.

Illustration example:

- Design on Grid Designer

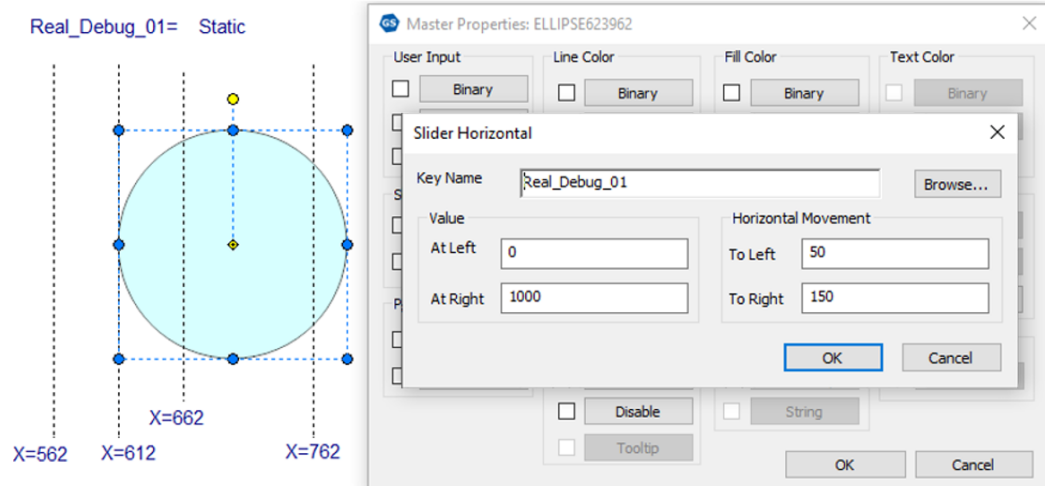
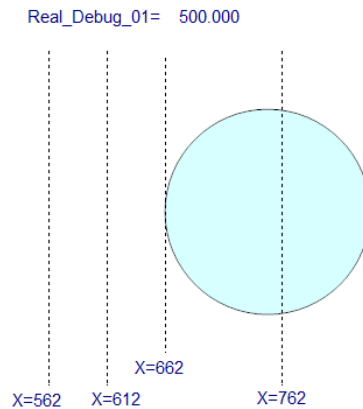


Figure F-129. Example of Slider Horizontal Configuration

- Effect on Grid Viewer



F.3.10.4.25 Configuring Push Button– Binary

Open the object's **Master Properties** window.

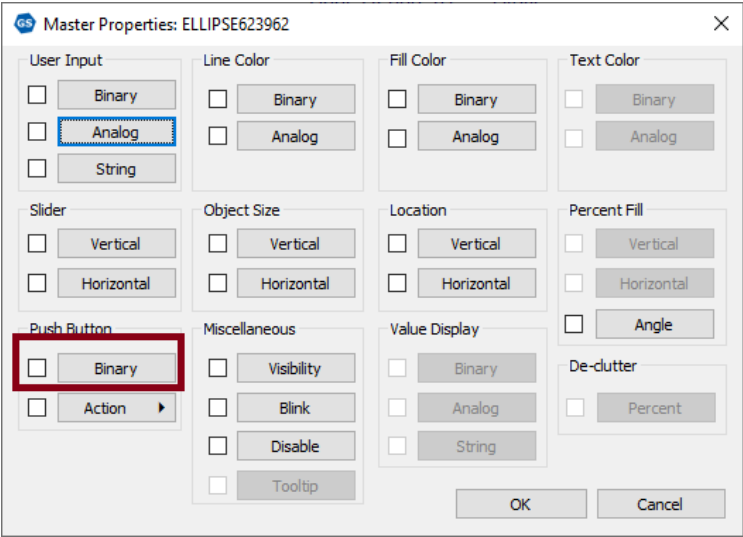


Figure F-130. Master Properties Push Button– Binary

Click **Binary** in the **Push Button**. The **Push Button Binary** window appears.

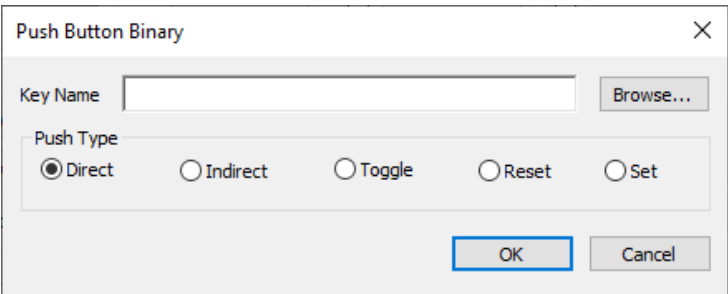


Figure F-131 Slider Horizontal Configuration Window

Master Properties – Push Button – Binary	
Key Name	Keyname of a boolean tag

Push Type - Direct	Press + Hold: tag = true Release: tag = false
Push Type - Indirect	Press + Hold: tag = false Release: tag = true
Push Type - Toggle	Reverse the Tag value (tag = not tag)
Push Type - Set	tag = true
Push Type - Reset	tag = false
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

F.3.10.4.26 Configuring Push Button – Action

Open the object's **Master Properties** window.

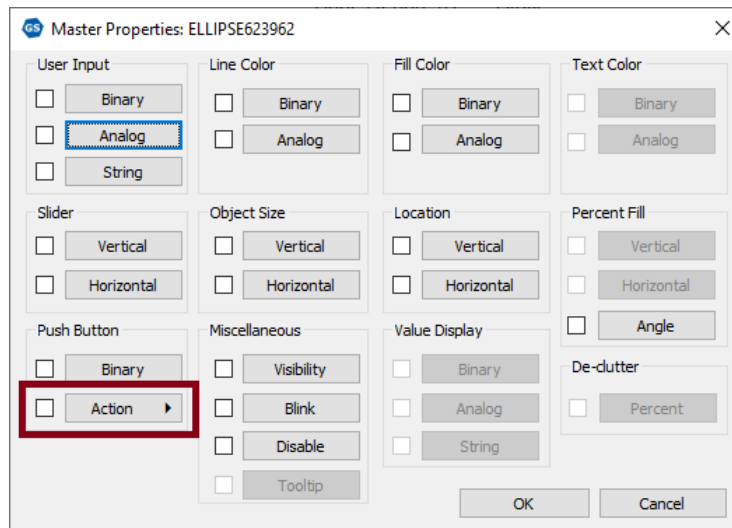


Figure F-132. Master Properties Push Button– Action

Click **Action**→select **Action** in the **Push Button** section.

The **Push Button Action Script** window appears.

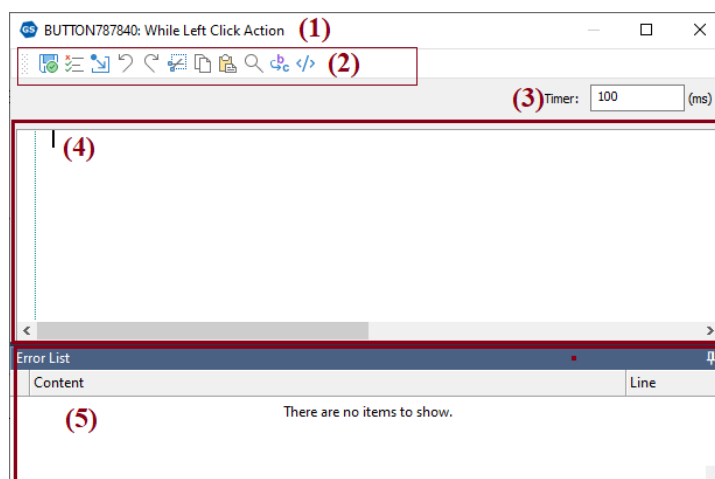


Figure F-133. Push Button Action Configuration Window

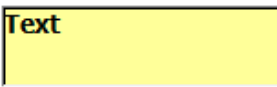
Edit scripts. Refer to F.3.17.2

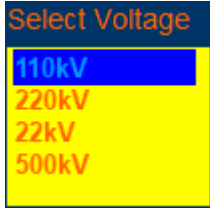
F.3.11. Basic interactive objects

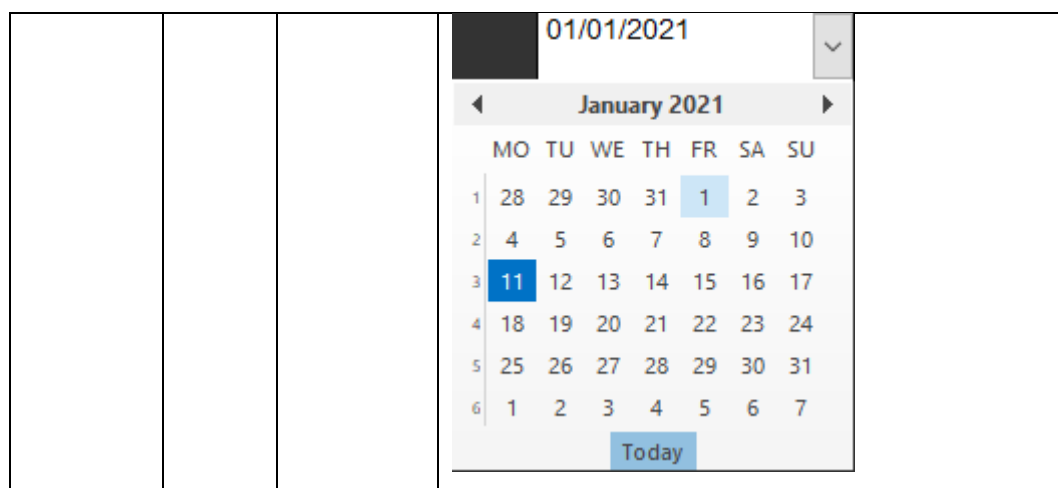


F.3.11.1 List of basic graphics objects

Basic interactive graphics objects are drawn from the Basic Graphics Objects toolbar. Refer to section F.3.1.8

Object	Icon	Data type	Description
Edit box		String Analog	Allows displaying and importing data. The value of the analog/string tag is the input value 
Check Box		Boolean	Allows changing the value of a Boolean tag. The true/false value of the boolean tag corresponds to the Check Box state checked/unchecked. 
List Box		String	Allows selecting an item from the list displayed in the fixed frame.

			The value of the tag is assigned with the caption of the selected item. 
Combo Box			Allows selecting an item from the drop-down list by the button ▾. The value of the tag is assigned with the caption of the selected item. 
Radio Button		Analog	List of options (item), where always one item is selected. The value of the tag is the order of the selected item (order starts at 0). 
Datetime Control		Datetime	Allows you to select the time and save it to a variable of type Datetime.



F.3.11.2 Design support with right-click menu for object

Refer to section [F3.10.2](#)

F.3.11.3 Static Graphic Properties - Designer Properties

Refer to section [F3.10.3](#)

F.3.11.4 Dynamic Graphics Properties – Master Properties

F.3.11.4.1 Edit Box -

The Edit Box object allows displaying and entering data.

The value of the tag (analog/string type) is the input value

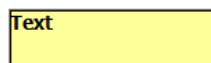


Figure F-134. The Edit Box object

The Edit Box Configuration

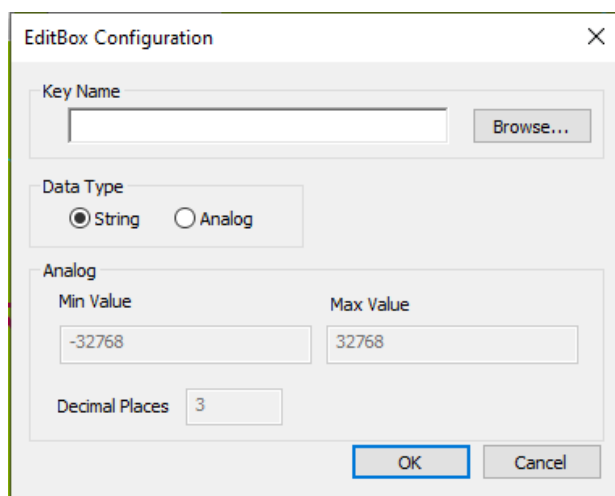


Figure F-135. The Edit Box Configuration window

Configuration – Edit Box	
Properties	Description
Key Name	Tag gets value from Edit Box. The data type corresponds to the Datatype property.
Data Type – String/Analog	Input data type in Edit Box: String/Analog
Analog – Min/Max	Limit to receive input value with Analog data type.
Analog – Decimal Places	Limit the number of decimal places after the decimal point entered with the Analog data type.
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

F.3.11.4.2 Check Box - ☒

Allows changing the value of a Boolean Datapoint.

The true/false value of the tag (boolean type) corresponds to the Check Box state checked/unchecked.



Figure F-136. The Edit Box object

The Check Box Configuration window

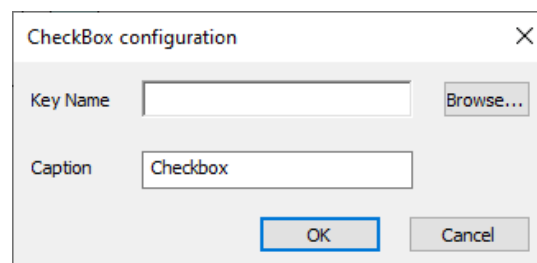


Figure F-137. The Check Box Configuration window

Configuration – Check Box	
Properties	Description

Key Name	Tag gets value from Check Box, type Boolean
Caption	Check box content
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

F.3.11.4.3 List Box -

Allows selecting an item from the list displayed in the fixed frame.

The value of the tag (string type) is assigned the caption of the selected item.

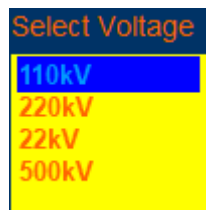


Figure F-138. The List Box object

The **List Box Configuration** window

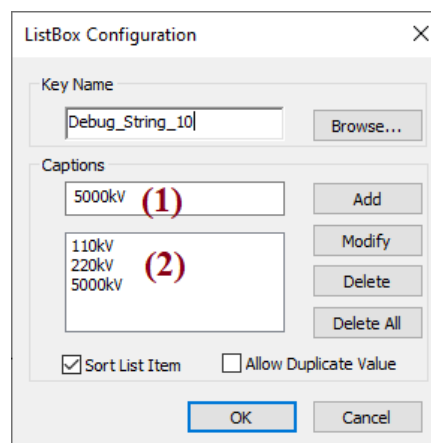


Figure F-139. The List Box Configuration window

Configuration – List Box	
Properties	Description
Key Name	Tag gets value from List Box, type String.

Caption – (1)	Enter new item content or display selected item in Caption (2).
Caption – (2)	List of items in the list box
Add	Add Caption-(1) to the Caption-(2) list
Modify	Edit the contents of the selected Item in Caption-(2) according to Caption-(1).
Delete	Remove the item with Caption-(1) in the Caption-(2) list
Delete all	Remove all items in the Caption-(2) list
Sort List Item	Automatically sort items in Caption-(2) list alphabetically
Allow Duplicate Item	Allow items in Caption-(2) list to have duplicate content
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

F.3.11.4.4 Combo Box -

Allows selecting an item from the drop-down list.

The value of the tag (string type) is assigned the caption of the selected item.

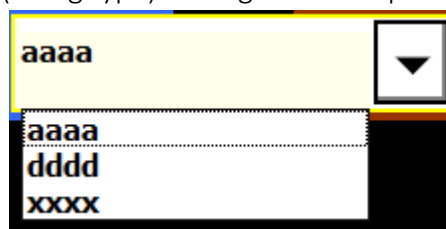


Figure F-140. The Combo Box window

The **Combo Box Configuration** window

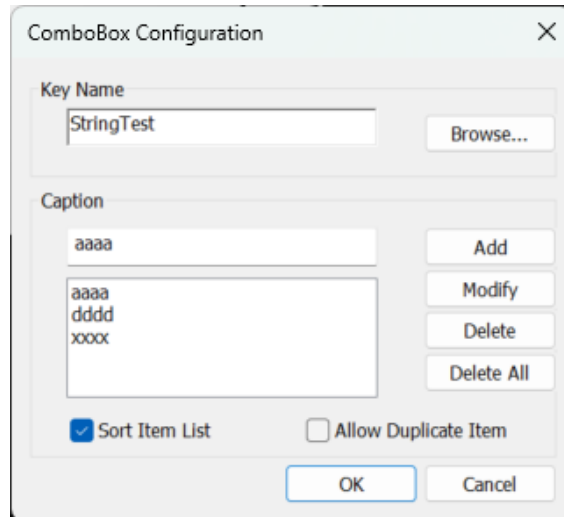


Figure F-141. The Combo Box Configuration window

Configuration – Combo Box	
Properties	Description
Key Name	Tag receives value from Combo Box, type String.
Caption – (1)	Enter new item content or display selected item in Caption (2)
Caption – (2)	List of items in Combo box
Add	Add Caption-(1) to the Caption-(2) list
Modify	Edit the content of Item selected in caption-(2) according to Caption-(1)
Delete	Remove the item with Caption-(1) in the Caption-(2) list
Delete all	Remove all items in the Caption-(2) list
Sort List Item	Automatically sort items in Caption-(2) list alphabetically
Allow Duplicate Item	Allow items in Caption-(2) list to have duplicate content
Browse	Support tag selection.

	Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression
--	---

F.3.11.4.5 Radio Button -

List of options (item), where always one item is selected.

The value of the tag (analog type) is the order of the item selected (order starts at 0).



Figure F-142. The Radio Button object

The **Radio Button Configuration** window

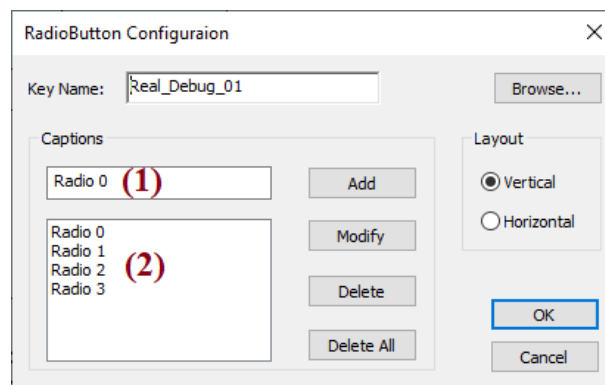


Figure F-143. The Radio Button Configuration window

Configuration – Radio Button	
Properties	Description
Key Name	The tag gets value from Radio Button, analog type.
Caption – (1)	Enter new item content or display selected item in Caption (2).
Caption – (2)	List of items in the list box
Add	Add Caption-(1) to the Caption-(2) list.

Modify	Edit the content of the Item selected in caption-(2) according to Caption-(1).
Delete	Delete the item with Caption-(1) in the Caption-(2) list.
Delete all	Delete all items in the Caption-(2) list.
Layout Vertical/Horizontal	Arrange the items in the Caption-(2) list vertically / horizontally.
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

F.3.11.4.6 Datetime Control -

Allows you to select the time and save it to a variable of type Datetime.

The value of the tag (type datetime) is the selected time.

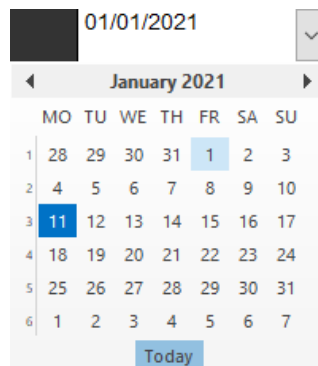


Figure F-144. Datetime / Date Object

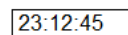


Figure F-145. Datetime / Time Object

The **DateTime Configuration** window

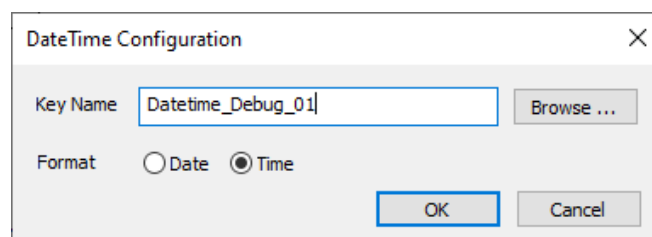


Figure F-146. The Datetime Configuration window

Configuration – Datetime Control	
Properties	Description
Key Name	Tag gets value from Datetime Control, type Datetime.
Format - Date	Choose to enter time as day/month/year. The value of the tag is the date and year entered and the hour minute second is the default value when initializing the tag.
Format - Time	The time input option is hours/minutes/seconds. The value of the tag is the date and year entered by default when the tag is initialized and the hour minute second entered.
Browse	Support tag selection. Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression

Datetime controls are often used in the following contexts:

User uses 2 datetime controls, 1 date config, 1 time config, attached to SAME datetime variable. Then the date configuration will save to the date part of the variable, the time configuration will save the time part of the variable.

F.3.12. Electrical Network Objects

Unlike basic graphic objects, Network objects are closely linked with each other in terms of electrical relationships (voltage, current, ...) and are designed in the data model (Data Model). The links between objects are designed/described under graphically (drawing) form. Electrical Network objects are applied in DMS functions (grid coloring, load flow, ...)



Refer to the general description of objects in the document “A. System Overview” – section “A.5 Electrical Network Objects”.

F.3.12.1 List of Electrical Network objects

STT	Object	Icon	Graphics	Description
1.	Connectivity Node		Point	Electrical connection point of electrical equipment, or power consuming load, or power supply.
2.	Busbar		Straight line	Busbar, the property on the grid is a Connectivity Node.
3.	AC Line		Straight line	Line transmission
4.	Circuit Breaker		Symbol Template designed by users.	Circuit Breaker
5.	Disconnecter			Disconnecter
6.	Ground Disconnecter			Ground Disconnecter
7.	Transformer			Transformer (2 coils, 3 coils)
8.	Series Compensator			Series Compensator
9.	Recloser			Recloser
10.	Load Break Switch			LBS
11.	LBFCO			Load Break Fuse Cutout
12.	FCO			Fuse Cutout
13.	Energy Consumer			Load
14.	Shunt Compensator			Horizontal compensator
15.	Surge Protector			Surge Protector
16.	Generator			Generator
17.	Energy Source			Source
18.	Voltage Transformer			Voltage Transformer
19.	Substation			Substation

20.	Fault Indicator			Fault Indicator
21.	Reactor			Reactor

F.3.12.2 Reference Object and Non-Reference Object

Electrical Network objects are designed with the Reference = Yes/No property. The distinction between these object types comes from the Overview to Detail graphic design style, in which the object carries the electrical connections (topology) described in the overview screen. (Overview, Single Line, ...) and copied + detailed in detail screens (Bay screen).

Non - Reference Object (Reference Symbol Attribute = No)	Reference Object (Reference Symbol Attribute = Yes)
The electrical properties of the object depend on the data and the declared/drawn electrical topologies for the object.	The electrical properties of an object depend on the object (reference ID) referred to.

F.3.12.3 Connectivity Node



Normal Connectivity Node  Normal Connectivity Node is the electrical connection point of electrical devices, which have the form of point graphics.

The **Connectivity Node** represents the connection point, the equipment cable connection point, and represents the source and load point of the grid.

- How to draw from the toolbar: select the **Connectivity Node** icon  → select the voltage level → left click to select the Connectivity Node position on the current screen.
- **(Normal) Connectivity Node** can be drawn with Ctrl + Left -click when you draw the terminal connection of a device Electrical Network. The newly created node has the same voltage level as the previously created node.
- **Connectivity Node – Reference** is a reference object created by Copy/Paste Reference a Connectivity Node object.

The default size of the **Connectivity Node** is configured from the general configuration for the Application **Menu Tool** → **Option** → **Graphic** → **Network – Connectivity Radius**. Refer to section F.3.2.2

Configuration for a Connectivity Node:

Select a Connectivity Node, edit the parameters in the Properties:

Properties	
Control	
Name	472CAMLO/472CLO_53DCL2
Unique ID	afd9997e_d955_47a1_bd1e_047a98983aac
Device Type	Connectivity
Reference Symbol	☐
Voltage Level (kV)	22
Power Source	☐
Viewer Visibility	☐
Appearance	
Coordinate X	1268
Coordinate Y	1558
Rotation Angle	0
Bounding Width	4
Bounding Height	4
Layer	Background

Figure F-147. The Connectivity Node configuration

Properties	
Control	
Name	472CAMLO/472CLO_53DCL2_Copy
Unique ID	ff21c2a8_c74a_4f19_acae_3a83a9c11545
Device Type	Connectivity
Reference Symbol	☒
Refer To Name	472CAMLO/472CLO_53DCL2
Refer To ID	afd9997e_d955_47a1_bd1e_047a98983aac
Voltage Level (kV)	22
Viewer Visibility	☐
Appearance	
Coordinate X	1048
Coordinate Y	1635
Rotation Angle	0
Bounding Width	4
Bounding Height	4
Layer	Background

Figure F-148. The Connectivity Node attribute has a Reference Symbol configuration

Properties	Description
Name	Connectivity Node Name
Unique ID	ID of object

Device Type	<i>Connectivity</i>
Reference Symbol	Reference to another Symbol object by name ID. The charged property of an object will depend on the object being referenced.
Refer To Name	The name of the object is referenced.
Refer To ID	The ID of the object is referenced.
Power Source	Connectivity Node option as source compute point in coloring function.
Power Source State	The Boolean tag represents the live state of the source point.
Viewer Visibility	Connectivity Node display option on Viewer.
Appearance	The design parameters of coordinates, line color, background color, and the display layer of the object are set in the Properties window. Refer to section F.3.1.13

F.3.12.4 Busbar



Busbar  is the electrical connection point of electrical devices in the form of a straight line graphic.

Busbar represents the actual busbar device.

- How to draw from the toolbar: select **Busbar** icon  → select voltage level or Line Container in the Substation panel → left click to select Busbar position on the current screen.
- **Busbar – Reference** is a reference object created by Copy/Paste Reference a Busbar object.

Busbar default size is configured from the general configuration for the application Menu Tool/Option/Graphic/Network – Busbar Width. Refer to section F.3.2.2

Configure Busbar:

Select **Busbar**, adjust the parameters in the Properties section:

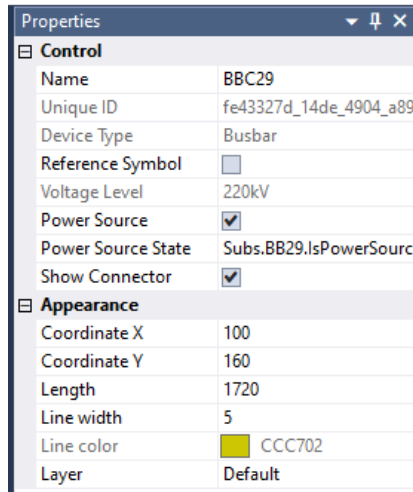


Figure F-149. Busbar properties with source configuration

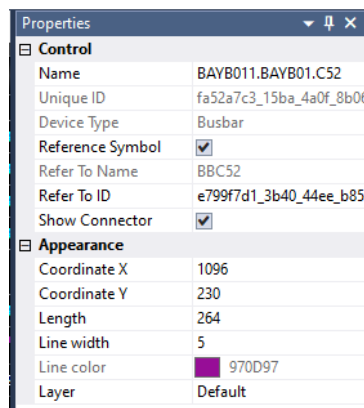


Figure F-150. The Busbar property has a Reference Symbol configuration

Properties	Description
Name	Busbar name
Unique ID	ID of object
Device Type	Busbar
Reference Symbol	Reference to another Symbol object by name ID. The charged property of an object will depend on the object being referenced.
Refer To Name	The name of the object is referenced.

Refer To ID	The ID of the object is referenced.
Power Source State	The Boolean tag represents the state of the source point.
Appearance	The design parameters of coordinates, line color, background colors, and the display layer of the object are set in the Properties window. Refer to section F.3.1.13
Show Connector	Display connection points from devices to the busbar.

F.3.12.5 AC Line

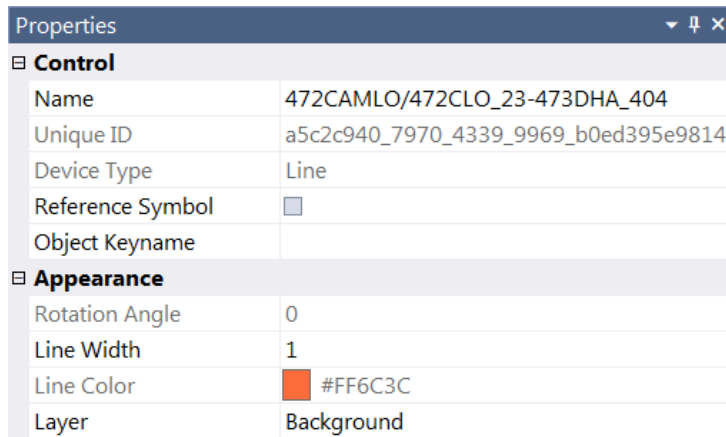
AC Line is a section/line of power transmission.

AC Line model is in fact the busbar segments, connecting devices with busbars, ...

- How to draw from the toolbar: select AC Line  icon → select Line Container from the Substation window → connect **ACLine** terminals to Connectivity Node/Busbar points.

AC Line Configuration

Select **AC Line**, adjust the parameters in the Properties section:



Properties	
Control	
Name	472CAMLO/472CLO_23-473DHA_404
Unique ID	a5c2c940_7970_4339_9969_b0ed395e9814
Device Type	Line
Reference Symbol	
Object Keyname	
Appearance	
Rotation Angle	0
Line Width	1
Line Color	 #FF6C3C
Layer	Background

Figure F-151. AC Line Properties

Properties

Control

Name

CAMLO/C42_Copy-CAMLO/C42

Unique ID

ebf30be3_6778_4a6b_944b_7157787630e4

Device Type

Line

Reference Sym

☒

Refer To Name

473DHACL/473DHA_327-327/4

Refer To ID

e8e7bea8_1b3e_43f5_89d5_693e380a7dff

Object Keyname

Appearance

Rotation Angle

0

Line Width

1

Line Color

 #FF6C3C

Layer

Background

Figure F-152. AC-Line properties have a Reference Symbol configuration

Properties	Description
Name	AC Line name
Unique ID	ID of object
Device Type	Line
Reference Symbol	Reference to another Symbol object by name ID. The charged property of an object will depend on the object being referenced.
Refer To Name	The name of the object is referenced.
Refer To ID	The ID of the object is referenced.
Object Keyname	Key name of the line in data model

Properties of AC Line

STT	Object	Number of terminals	Voltage Properties
1.	AC Line	2	The terminals have the same voltage level. The terminals will have the same electric carrying state.

F.3.12.6 Electrical devices

Electrical devices have custom template graphic properties in **Menu/Symbol→Symbol Template Library**.

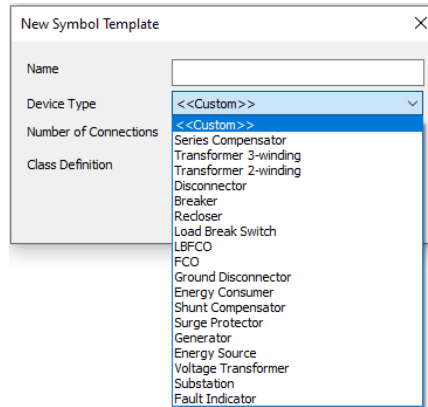


Figure F-153. Electrical Network devices

STT	Object	Number of terminals	Status data	Voltage properties
1	Circuit Breaker	2	Open/Close/Unknow status	The terminals have the same voltage level
2	Disconnector	2		In the closed, the two terminals will have the same electric carrying state.
3	Ground Disconnector	1	Open/Close status	In the closed state & good data (quality good / manual), the terminal has an electric carrying state = "grounded"
4	2-windings Transformer	2		Terminal have the multiple voltage levels.
5	3-windings Transformer	3		The terminals will have the same electric carrying state.
6	Series Compensator	2		

7	Recloser	2		
8	Load Break Switch	2		
9	LBFCO / Load Break Fuse Cutout	2		
10	FCO /Fuse Cutout	2		
11	Energy Consumer	1		
12	Shunt Compensator	1		The terminals will have the same electric carrying state.
13	Surge Protector	1		The terminals will have the same electric carrying state.
14	Generator	1		
15	Energy Source	1		
16	Voltage Transformer	1		
17	Substation	0		
18	Fault Indicator	2		
19	Reactor	2		The terminals will have the same electric carrying state.

An integrated circuit breaker can be modelled as a circuit breaker, where the switching state is the sum of the breaker state and the (in/out) operating position.

F.3.12.7 Connect terminal → Connectivity Node

- When drawing (drag) the Electrical Network object from the toolbar or the Symbol Template Library, the application will ask to draw (drawing terminal connection mode) the connection from the Terminal to the Connectivity Node. If you don't want to connect to the Connectivity Node, when the terminal connection drawing mode is active, use Right-Click to exit this mode and skip the Terminal connection.
- To adjust, redraw the terminal connection → **Connectivity Node**:
Right-Click to **Electrical Network object** → **Redraw terminals (Ctrl+R)**
- To delete terminal connections → **Connectivity Node**:
Right-Click on Electrical Network object → **Remove terminals**

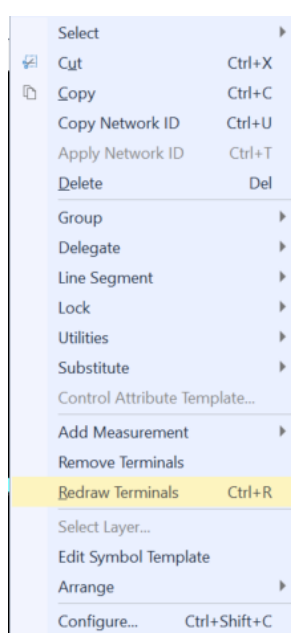


Figure F-154. Redraw/Remove Terminals Menu

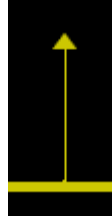
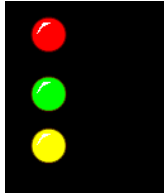
F.3.13. Sample Object



F.3.13.1 List of sample objects

Template graphic objects are drawn from the template graphic objects toolbar. Refer to section F.3.1.11.

Object	Icon	Description
Control Arrow		Control Arrow is a graphic object to draw lines that support the fill feature.

		
LED		LED Object is a graphical object to draw the round LEDs commonly found on device face. 

F.3.13.2 Control Arrow

Control Arrow is a graphic object for drawing transmission lines have a support fill color function on single-line diagrams.

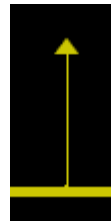


Figure F-155. Control Arrow Object

Control Arrow has properties:

- Connect to only one Connectivity Node/Busbar.
- There are colors on the Grid Viewer that correspond to the electric carrying state of the Connectivity Node/Busbar to which it is connected.

Draw Control Arrow object:

- Select Control Arrow icon  on the toolbar. Move the mouse to select the Connectivity Node that Control Arrow connects to.
- Control Arrow has a drawing similar to a Polyline: Left-click to create folding points, Double Click to create ending points.

Configuring a Control Arrow

- Edit line style, arrows from the Line toolbar. Refer to section F.3.1.6
- Select Control Arrow object, edit Properties of object:

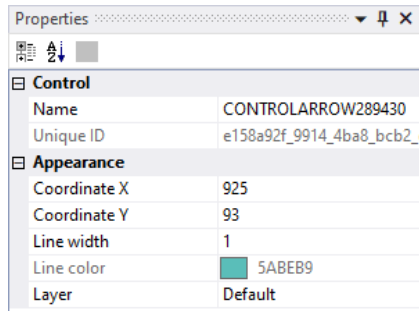


Figure F-156. Control Arrow properties

Properties	Description
Name	Control Arrow name
Unique ID	ID of object
Appearance	The design parameters of coordinates, line and background colors, and the display layer of the object are set in the Properties window. Refer to section F.3.1.13

F.3.13.3 LED Object

LED Object is a graphical object to draw the round LEDs commonly found on device faces.

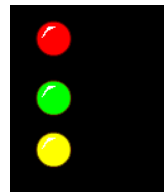


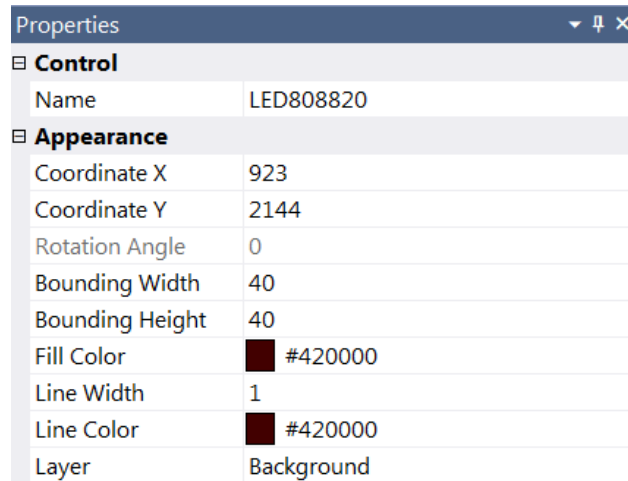
Figure F-157. LED Object

Draw LED Object:

- Select LED Object icon  on the toolbar. Hold Shift while drawing to create a circular LED.

Configure properties for LED Object:

- Edit line style, fill color from Line toolbar. Refer to section F.3.1.6
- Select the LED Object, edit the object's Properties:

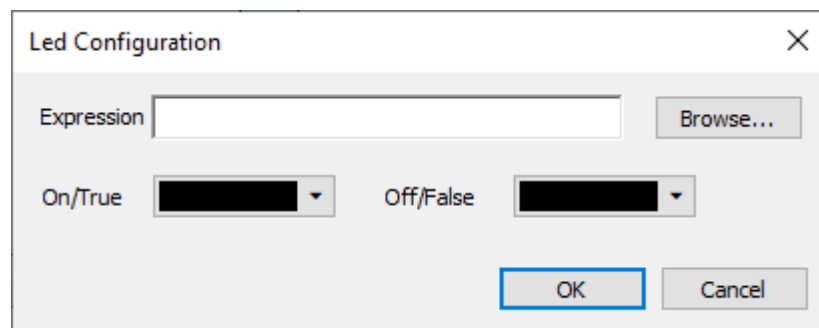


Properties	
Control	
Name	LED808820
Appearance	
Coordinate X	923
Coordinate Y	2144
Rotation Angle	0
Bounding Width	40
Bounding Height	40
Fill Color	 #420000
Line Width	1
Line Color	 #420000
Layer	Background

Figure F-158. LED Object properties

Properties	Description
Name	Led name
Appearance	The design parameters of coordinates, line and background colors, and the display layer of the object are set in the <i>Properties</i> window. Refer to section F.3.1.13

Configuring dynamic properties for LED Object:



Led Configuration

Expression Browse...

On/True  Off/False 

OK Cancel

Figure F-159. The LED Configuration window

Configuration - LED	
Expression	Keyname of a boolean tag or expression with a boolean result
On/True	Color when Expression = TRUE
Off/False	Color when Expression = FALSE
Browse	Support tag selection.

	Open the Data Model View window. Select tag from the list → right click → Copy Key Name → Close Data Model View → Paste to Expression
--	---

F.3.14. Advanced Objects



F.3.14.1 List of Advanced Objects

Advanced graphics objects are drawn from the Advanced Graphics Objects toolbar. Refer to section F.3.1.9

Object	Icon	Description
Event View		Object used to view and manage alerts and events.
Trend View		Object is used to create line graphs with up to 4 data points on a Trend View object.
Data Grid		The object is used to display data from a *.csv file as a table.
Relay Event View		The object is used to quickly retrieve SEL relay event information with the SUM/HIS command through a TCP/IP Telnet connection.
Chart		Objects are used to create integrated graphs of many types (lines, columns), the unlimited number of data points (depending only on computer hardware capacity), support Min max calculation functions, and data export files.
Tabular		The object supports displaying Online data from DataServer, supports Online checking of DataServer data from Grid Viewer environment during operation.
Active X		Integrate Active X objects registered with the Windows operating system.
Search Box		Show list of controls satisfied search pattern by name or caption. On Viewer, user can locate control on screen by Double Click

F.3.14.2 Event View

Event View is used to view and manage alerts and events. For a detailed description of the Alarm/Event and its properties, refer to the OneATS DataServer – Alarm Configuration document.

ACTIVE ALARM						50 event(s)
Timestamp	Tagname	Ack	Operator	Message	Value	
28/01/2022 14:38:...	RP204BCU.ST.Time...	Unack		TIME SYNCH STATUS (B20_4 - S451 BCU)	False	
28/01/2022 14:38:...	RP203BCU.ST.Time...	Unack		TIME SYNCH STATUS (B20_3 - S451 BCU)	False	
28/01/2022 14:38:...	RPR05BCU.ST.Time...	Unack		TIME SYNCH STATUS (BR2_B05 - S451 BCU)	False	
28/01/2022 14:38:...	COMM.M.Status	Unack		HMI/GW1 DISCONNECTED FROM 10.0.0.31	False	
28/01/2022 14:38:...	COMM.H.Status	Unack		HIS/ENG DISCONNECTED FROM 10.0.0.33	False	
28/01/2022 14:38:...	COMM.SCADA_GW1...	Unack		A1 LDPC DISCONNECTED (SCADA-GW1)	0	
28/01/2022 14:38:...	COMM.SCADA_GW1...	Unack		GATEWAY A1 NOT READY (SCADA-GW1)	False	
28/01/2022 14:38:...	COMM.SCADA_GW2...	Unack		GATEWAY A1 NOT READY (SCADA-GW2)	False	
28/01/2022 14:38:...	COMM.SCADA_GW2...	Unack		A0 LDPC DISCONNECTED (SCADA-GW2)	0	
28/01/2022 14:38:...	COMM.SCADA_GW1...	Unack		A0 LDPC DISCONNECTED (SCADA-GW1)	0	
28/01/2022 14:38:...	RP204BCU.ST.SG1	Unack		GROUP 1 (B20_4 - S451 BCU)	False	
28/01/2022 14:38:...	RP204.BAY.McbFVT...	Unack		FVTSM4B MCB FAIL (BAY-582/B20-4)	True	
28/01/2022 14:38:...	B12.XCBR1.F74Coil2C	Unack		CB1/582: F74-2C FAIL	True	
28/01/2022 14:38:...	B12.XCBR1.F74Coil2B	Unack		CB1/582: F74-2B FAIL	True	
28/01/2022 14:38:...	B12.XCBR1.F74Coil2A	Unack		CB1/582: F74-2A FAIL	True	

Figure F-160. Event View

F.3.14.2.1 Create a new Event View object

- Click the **Event View** icon  on the toolbar.
- Move the mouse to the desired position inside the design screen, drag and drop to create a rectangle with the desired size.

F.3.14.2.2 Editing Object Properties

Properties	
Control	
Name	CRITICAL ALARM 4K
Unique ID	bfa01c9_77f8_4b7a...
Appearance	
Coordinate X	0
Coordinate Y	63
Bounding Width	960
Bounding Height	644
Layer	SUMMARY 4K

Figure F-161. Event View Properties

Properties	Description
Name	Trend View object name
Unique ID	ID of object

Appearance	The design parameters of coordinates, line and background colors, and the display layer of the object are set in the <i>Properties</i> window. Refer to section F.3.1.13
------------	--

F.3.14.2.3 Configure the Event View panel display interface

- Open the Event View Configuration by Double Click on Event View or Right Click → Configure or Ctrl + Shift + C



The dialog box 'Event View Configuration' has two tabs: 'General' and 'User Interface'. The 'User Interface' tab is active. It contains a 'Visible columns' section with a table of columns and their widths. To the right of this table are input fields for 'Name' and 'Width' (set to 50), and buttons for 'Update', 'Up', and 'Down'. Below this is an 'Active alarm colors' section with a table of severity levels and their corresponding row, Ack, and UnAck colors. To the right of this table are 'Add' and 'Remove Last' buttons. At the bottom, there are sections for 'Inactive alarm colors' and 'Default active alarm colors', each with color pickers for 'Row', 'Ack', and 'UnAck'. At the very bottom are 'OK', 'Cancel', and 'Apply' buttons.

Visible columns	Column name	Width	Type
☒	Timestamp	150	Device...
☐	Server Time	150	Server...
☐	Name	150	Name
☒	Tagname	150	Source
☒	Ack	50	Ackno...
☐	Active	150	Status
☐	Type	150	Type
☐	Condition Name	150	Condi...
☐	Severity	150	Severity

Active alarm colors	Severity	Row color	Ack text color	UnAck text co...
250				
500				
750				
999				

Figure F-162. Event View Configuration

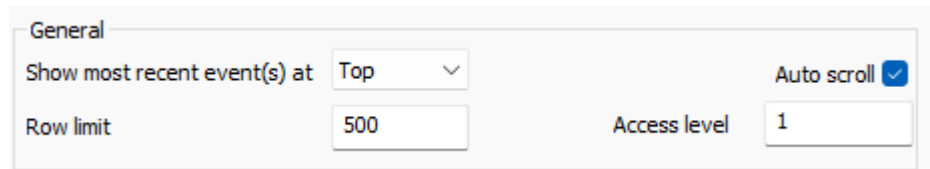
- Edit fonts for Event View from the font toolbar F.3.1.5
- Default value of the Event View
- Edit display title

Title

Properties	Description
Title	Alarm/event table header

- Adjust the display order parameters.

Event View Configuration → General



Properties	Description				
Tab General / General Configuration					
Show most recent event(s) at	Display order of alarms/events by time. <table border="1"> <tr> <td>Top</td><td>In chronological order, alarms/events are arranged from bottom to top. The latest alarm will be displayed at the top.</td></tr> <tr> <td>Bottom</td><td>In chronological order, alarms/events are arranged from top to bottom. The latest alarm will be displayed at the bottom.</td></tr> </table>	Top	In chronological order, alarms/events are arranged from bottom to top. The latest alarm will be displayed at the top.	Bottom	In chronological order, alarms/events are arranged from top to bottom. The latest alarm will be displayed at the bottom.
Top	In chronological order, alarms/events are arranged from bottom to top. The latest alarm will be displayed at the top.				
Bottom	In chronological order, alarms/events are arranged from top to bottom. The latest alarm will be displayed at the bottom.				
Row limit	Maximum number of alarm lines appearing in Event View. In case the Event View table is full of alarms/events (Row Limit is reached), when there are new alarms/events, the oldest alarms/events will be pushed out of the Event View table.				
Auto scroll	Automatically scroll the screen when a new alarm appears.				
Access level	The minimum account access allowed to configure Event View at runtime. Value range 0-255.				

- Double Click the Event View object → **Event View Configuration** → **User Interface**

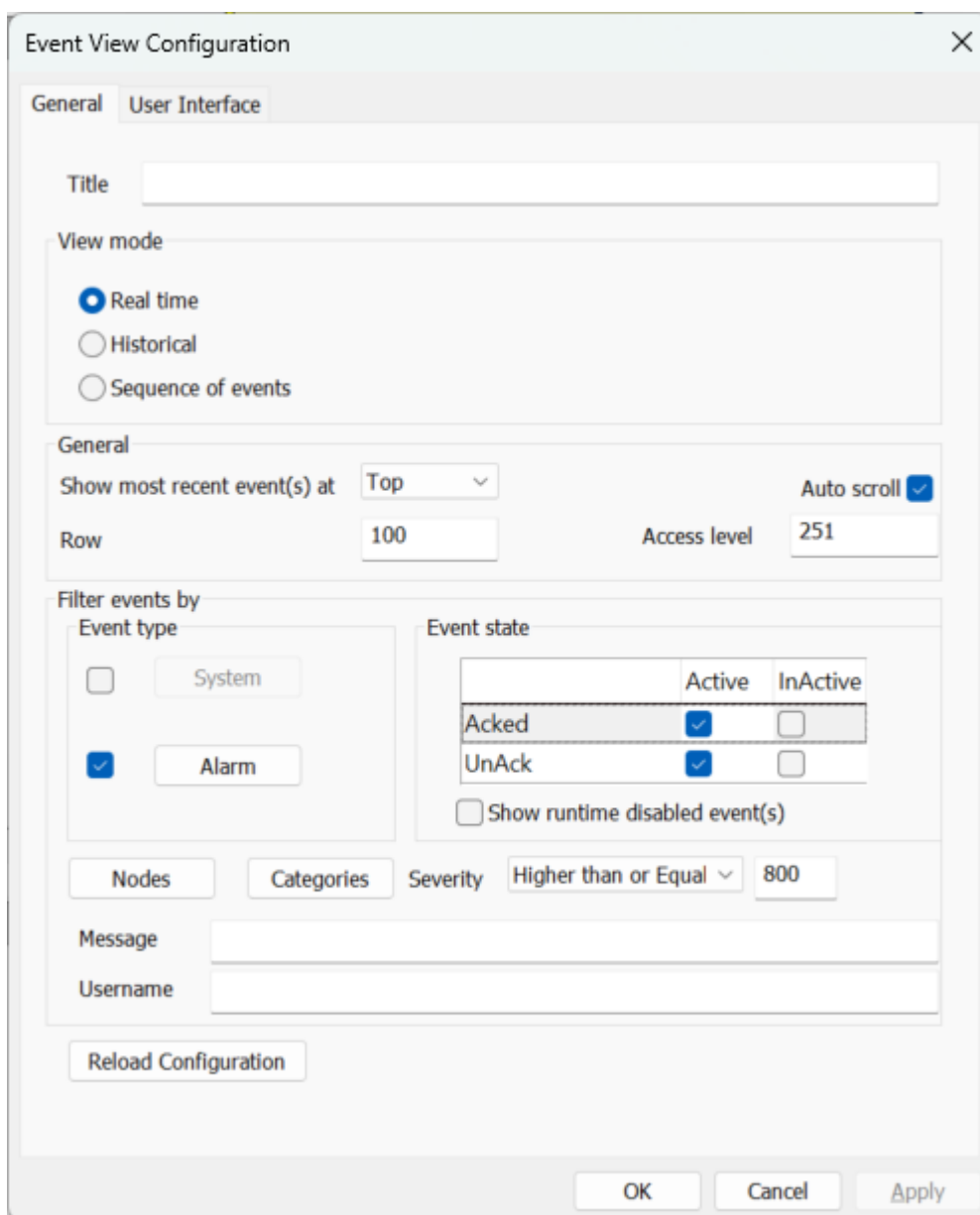
Properties	Description
User Interface / Active Alarm Color:	

Configure text & row colors for alarm messages with Active status with a specified Severity range.	
Severity	Severity range The lowest Severity range starts at 0.
Row Color	Active Event row background color
Ack Text Color	Active Event text color after Acked
Unack Text Color	Active Event text color before Acked
User Interface / Inactive Alarm Color: Configure text & row colors for alarm messages with Inactive status.	
Row Color	Inactive Event row background color
Text Color	Inactive Event text color
User Interface / Default Active Alarm Color: Configure the text & row colors for alarm messages with Active and Severity status outside the range configured in the Active Alarm Color panel.	
Row Color	Default Active Event row background color
Ack Text Color	Default Active Event text color after acked
Unack Text Color	Default Active Event text color after before acked
User Interface / Visible Columns: Configure the information fields displayed in the Event View panel.	
(1st Column)	Check / uncheck the box at the beginning of the row to add / remove the Type column to / from the table.
Column Name	Column headers/information fields
Width	Column width
Type	Corresponding information field
• Update: Update the Name/Width information for the column • Move up, Move down to change the order of columns. • Two type Device Time and Server Time are fixed in the first and second positions.	

- **Server Time** and **Device Time** in the Grid Viewer environment are in the form “**Time Date**”. Configure the formats for **Time** and **Date** in Application General Tab / **Date and Time format** section. Refer to section F.3.2.1

F.3.14.2.4 Configure the data update mode Event View

Double Click the Event View object → **Event View Configuration**.



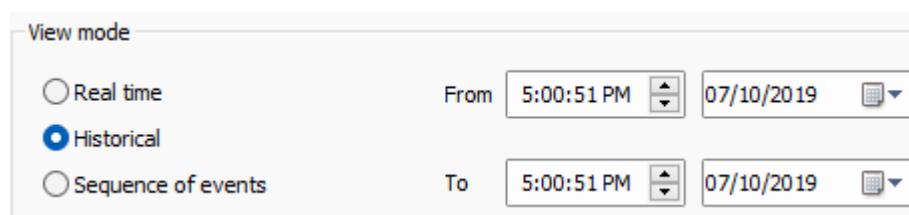
The 'Event View Configuration' dialog box has two tabs: 'General' and 'User Interface'. The 'General' tab is active. It contains the following fields and controls:

- Title:** A text input field.
- View mode:** Three radio buttons: 'Real time' (selected), 'Historical', and 'Sequence of events'.
- General:**
 - Show most recent event(s) at:** A dropdown menu with 'Top' selected.
 - Auto scroll:** A checked checkbox.
 - Row:** A text input field with '100'.
 - Access level:** A text input field with '251'.
- Filter events by:**
 - Event type:** Two checkboxes: 'System' (unchecked) and 'Alarm' (checked).
 - Event state:** A table with columns 'Active' and 'Inactive'.

	Active	InActive
Acked	☒	☐
UnAck	☒	☐
 - Show runtime disabled event(s):** An unchecked checkbox.
- Nodes:** A text input field.
- Categories:** A text input field.
- Severity:** A dropdown menu with 'Higher than or Equal' selected.
- 800:** A text input field.
- Message:** A text input field.
- Username:** A text input field.
- Reload Configuration:** A button.

At the bottom are 'OK', 'Cancel', and 'Apply' buttons.

Figure F-163. Event View Configuration Interface



This section shows the 'View mode' configuration for the 'Historical' mode (selected). It includes the following fields:

- View mode:** Three radio buttons: 'Real time' (unchecked), 'Historical' (selected), and 'Sequence of events' (unchecked).
- From:** A time and date selector showing '5:00:51 PM' and '07/10/2019'.
- To:** A time and date selector showing '5:00:51 PM' and '07/10/2019'.

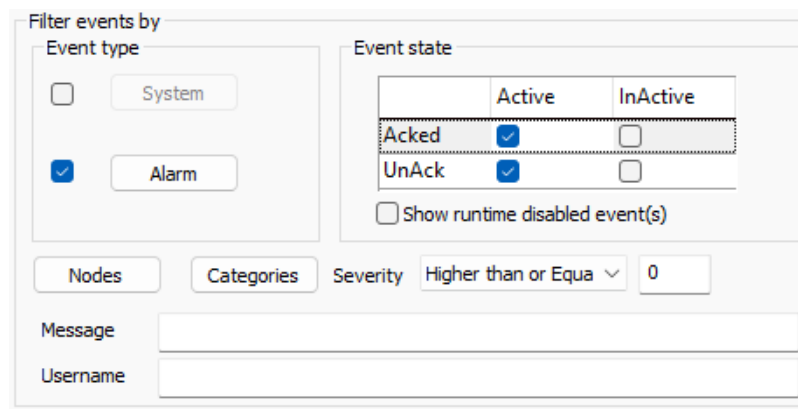
Figure F-164. Event View Configuration - View mode

Properties	Description				
Tab General / View Mode Configuration					
Real time	Alarms/events received from DataServer are automatically updated in real time. Each point (Alarm source) has only 1 line of the latest alarm/event message status on the Event View				
Historical	Alarms/events query from HIS Server <table border="1"> <tr> <td>From</td><td>Time to start querying data.</td></tr> <tr> <td>To</td><td>Time to stop querying data.</td></tr> </table>	From	Time to start querying data.	To	Time to stop querying data.
From	Time to start querying data.				
To	Time to stop querying data.				
Sequence of events	Alarms/events received from DataServer are automatically updated in real time. Event View displays all alarm/event messages without deleting previous alarm/event messages of the same point (alarm source)				

F.3.14.2.5 Configure data filtering displayed in the Event View table

Double Click the Event View object → **Event View Configuration**

Event View support filter conditions alarm/event is flexible according to the combination of different criteria.



The dialog box 'Filter events by' contains the following sections:

- Event type:** Two buttons, 'System' and 'Alarm'. 'Alarm' is selected with a blue checkmark.
- Event state:** A table with columns 'Active' and 'Inactive'.

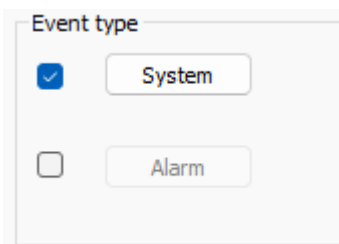
	Active	InActive
Acked	☒	☐
UnAck	☒	☐

Below the table is a checkbox labeled 'Show runtime disabled event(s)' which is currently unchecked.
- Nodes:** A button.
- Categories:** A button.
- Severity:** A dropdown menu showing 'Higher than or Equa' and a text input field with '0'.
- Message:** A text input field.
- Username:** A text input field.

Figure F-165. Event View Configuration – Filter events

Filter criteria	Description
Event type	Filter according to alarm event type (System events, binary events, ...)
Event State	Filter by alarm status (Ack/UnAck, Active/Inactive)
Node	Filter by data Node in the Data Model
Categories	Filter by data grouping
Severity	Filter by importance (Severity) of alarm
Message	Filter by description (alarm message) of alarm
Username	Filter by system login account (operator) at the time the alarm occurs

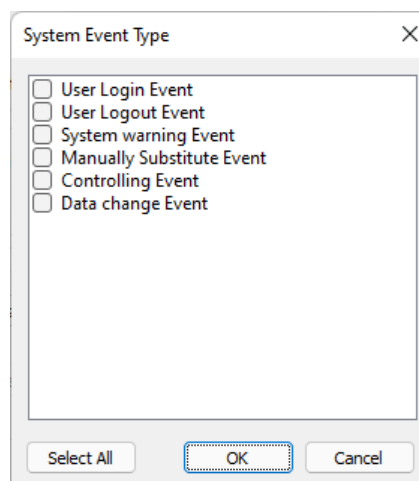
Filter by Event Type



A dialog box titled "Event type" with two options: "System" (selected with a checked checkbox) and "Alarm" (unselected with an unchecked checkbox).

System	System events or data changes
Alarm	Data Alarms

- To display events: Event View Configuration: Filter / **Event Type** → **System**.



A dialog box titled "System Event Type" with a list of events and checkboxes: "User Login Event", "User Logout Event", "System warning Event", "Manually Substitute Event", "Controlling Event", and "Data change Event". At the bottom, there are buttons for "Select All", "OK", and "Cancel".

Figure F-166. Event View Configuration – System event type

Select Events you want to display.

User Login Event	When the User Login to the HMI
User Logout Event	When user performs Logout in HMI
System Warning Event	System warning (*)
Manually Substitute Event	When the user forces a one-point value on DataServer (*)
Controlling Event	When the user calls a controller method on the "Notify event" configured method in the Data Model
Data Change Event	When the one-point value changes for the point that is configured the Data Change Event property in the Data Model

- To display alarms: Event View Configuration: Filter / **Event Type** → **Alarm**.

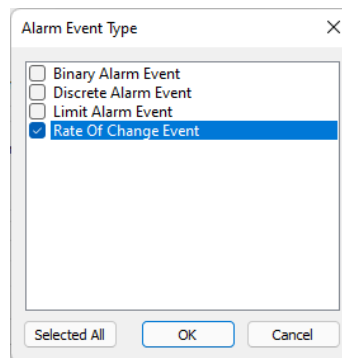


Figure F-167. Event View Configuration – Alarm event type

Select the Alarms you want to display.

Binary Alarm Event	Displays binary alarms.
Discrete Alarm Event	Displays Discrete alarms.
Limit Alarm Event	Displays Limit alarms (Low, High, ...)
Rate of Change Event	Displays Rate of Change alarms

Filter by data node

- Event View Configuration: Filter / **Nodes**. → List of Nodes according to the Data Model (DataServer)

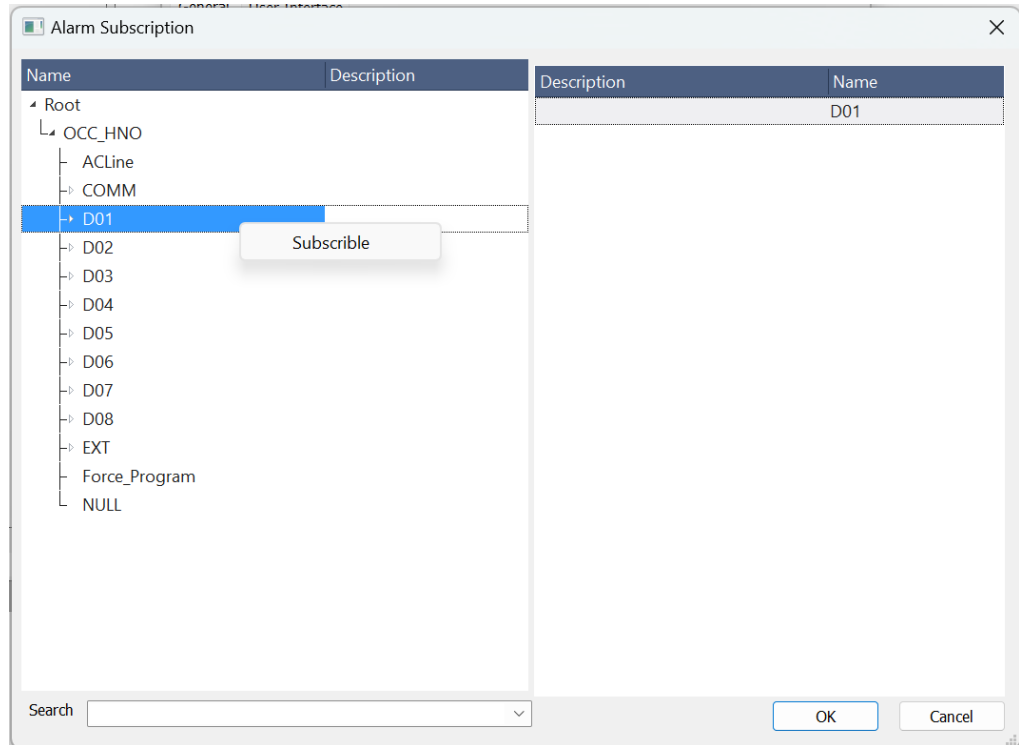


Figure F-168. Event View Configuration – Alarm Subscription

- Click to select Nodes need → Right-Click → Subscribe to display alarm. User can filter Search Box by Name or Description of the node.
- Click OK to save and close the Alarm Nodes Filter window.

Filter by alarm groups

- Event View Configuration: Filter / **Categories**. → List of Alarm Groups (Categories) is configured on the Data Model (DataServer)

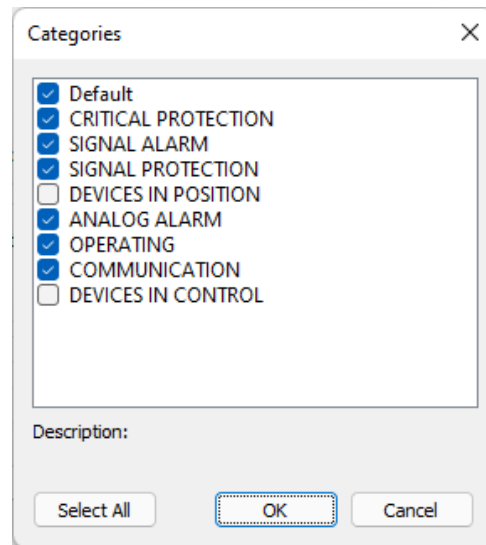


Figure F-169. Event View Configuration – Categories

- Select alarm groups to display. **Select All / Deselect All** option allows to select / deselect all alarm groups.
- Click OK to save and close the Alarm group selection window.

Filter by event status

- Event View Configuration: Filter / **Event States**.

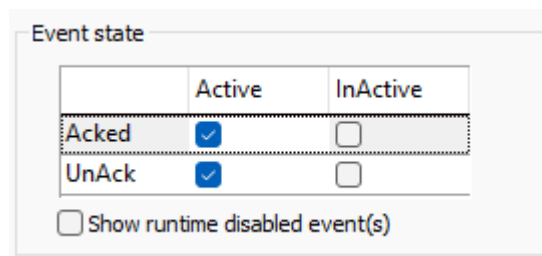
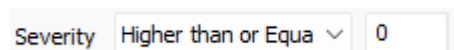


Figure F-170. Event View Configuration – Event State

- Select alarm status to display according to Acked/Unack and Active/Inactive

Filter by priority

- Event View Configuration: Filter / **Severity**.



- Select one of the comparison types and values to compare:

	Comparison Function
--	------------------------

Equals	=
Greater than	>
Less than	<
Greater than or equal	≥
Less than or equal	≤
Between	In the range of values

Filter by the Message

- Event View Configuration: Filter / **Message**.

Message

- Enter content you want to filter to the Message box
- Click OK.

Filter by User name

- Event View Configuration: Filter / **User Name**.

Username

- Enter content you want to filter to the Username box
- Click OK

F.3.14.2.6 Configuration synchronization between Grid Designer & Grid Viewer environments

On a computer, the Grid Designer and Grid Viewer application environments can operate in parallel, and both can change the configuration of the Event View object. To perform manual synchronization of the Even View configuration between the Designer and Viewer environments, Grid Studio provides function buttons.

Reload  Event View configuration from Viewer → Designer

The configuration of Event View objects is saved to the Alarm.cnfg file in the project's data directory.

F.3.14.3 Trend View

Trend View is an object used to build and manage trend graphs.

- Maximum number of data points per object **Trend View: 4**
- Vertical axis: value axis, support 01 vertical axis
- Horizontal axis: time axis, supports 01 horizontal axis
- Grid Studio supports a number of scripts that interact with Trend View objects. See details of Scripts for Trend at [F5.7.7](#)

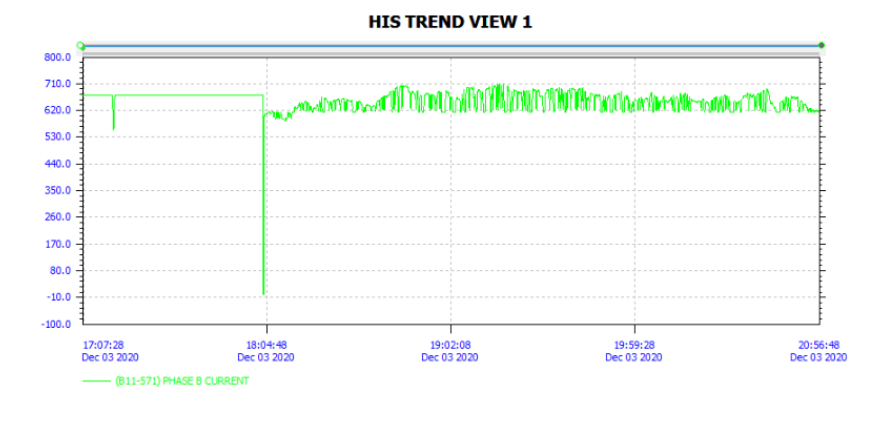


Figure F-171. Trend View

Classify

Type	Description
RealTime Trend	Real-time data trend graph. Data is updated automatically with data source from DataServer.
Historical Trend	Historical data trend graph The data is updated every time the user Updates the configuration with the data source from the HIS Server

F.3.14.3.1 Create a new Trend View object

- Select **Trend View** icon  on the toolbar.
- Move the mouse to the desired position inside the design screen, drag and drop to create a rectangle with the desired size.

F.3.14.3.2 Edit Object Properties

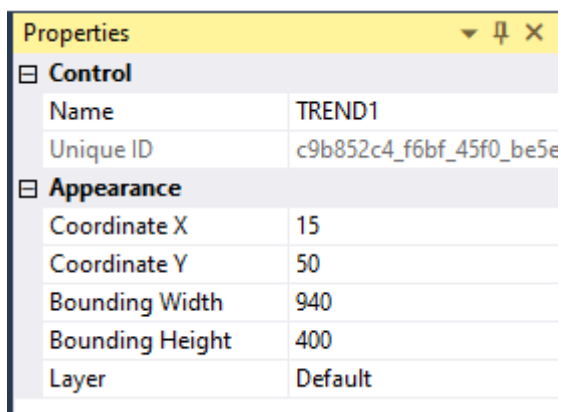


Figure F-172. Trend View Properties

Properties	Description
Name	Trend View object name
Unique ID	ID of object
Appearance	The design parameters of coordinates, line and background colors, and the display layer of the object are set in the <i>Properties</i> window. Refer to section F.3.1.13

F.3.14.3.3 Coordinate axis configuration and display data

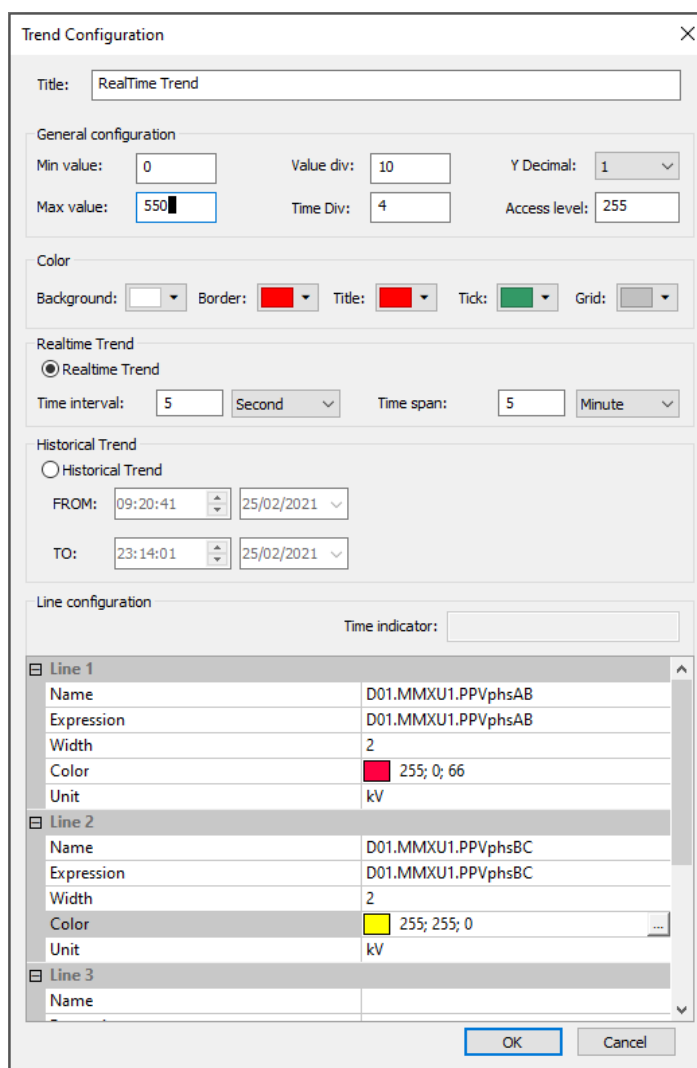


Figure F-173. Trend View object configuration window

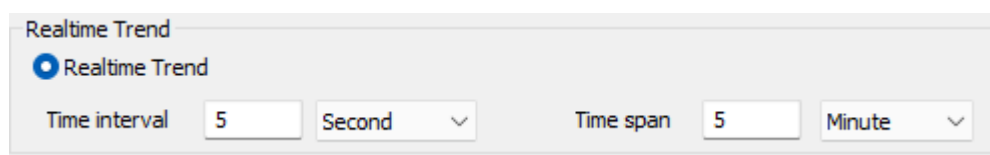
Properties	Description
Title	Title content for graph
Access Level	Minimum level of user access in the Grid Viewer environment to change the data display configuration for the Trend View object. Value range 0-255
General configuration - <i>Display coordinate axis configuration:</i>	
Min/max value	Vertical axis display value range

Value div	Number of divisions (lines) on the vertical axis
Time div	Number of divisions (lines) on the horizontal axis
Y Decimal	Number of decimal places of the reference value on the vertical axis
Color configuration - <i>Configuring display colors</i>	
Background	Object background color
Border	Graph frame border color.
Title	Title color
Tick	Reference value color displayed on vertical and horizontal axis
Grid	Color for grid in graph
Line configuration - <i>Configuring the plot of data points</i>	
Description	Description of the data point
Key name	Key Name of the data point
Width	Thickness for net drawing Integer value from 1-10
Color	Paint color. RGB value - decimal - separated by ";"
Unit	Unit of data point

F.3.14.3.4 Configure data update model

Data Mode configuration - <i>Configure data update mode</i>	
Realtime Trend	The data is automatically updated periodically with the data source from DataServer.
Historical Trend	The data is updated each time the user Updates the configuration with the data source from the HIS Server.

Realtime Trend Mode



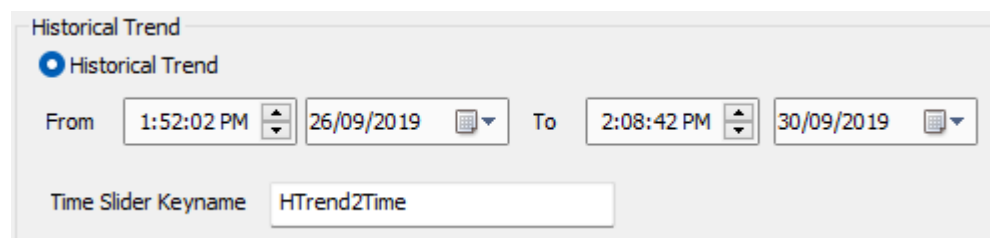
Realtime Trend

☒ Realtime Trend

Time interval Time span

Properties	Description
Time interval	Data update frequency Unit: - Millisecond - Second - Minute - Hour
Time span	The time interval (horizontal axis) of the graph Unit: - Minute - Hour

Historical Trend Mode



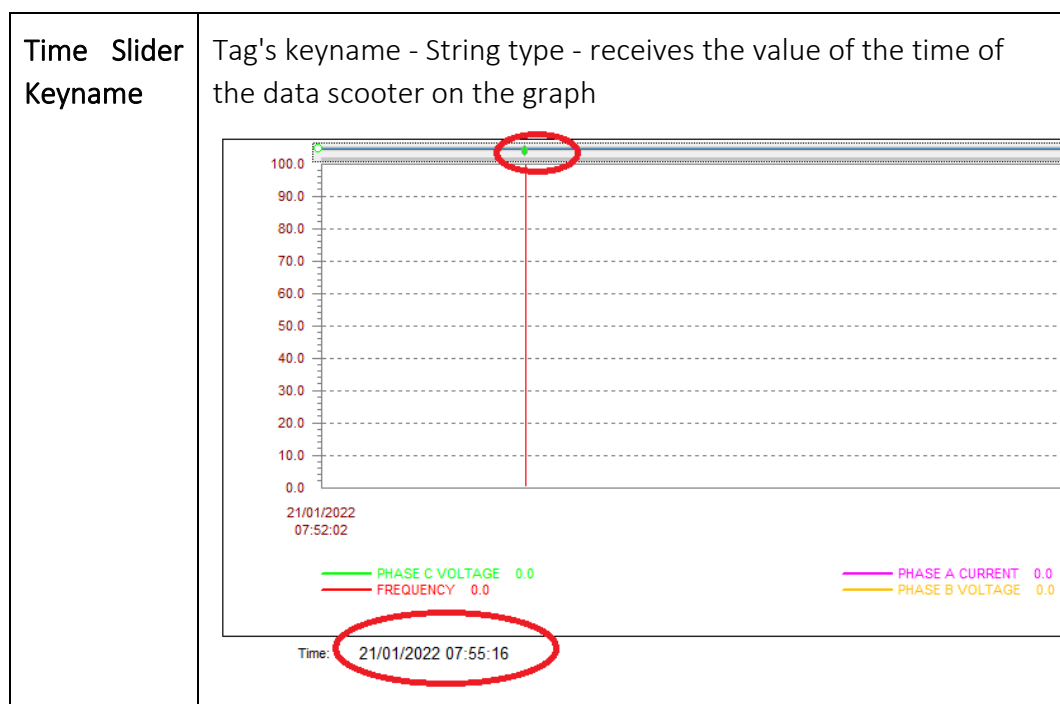
Historical Trend

☒ Historical Trend

From To

Time Slider Keyname

Properties	Description
From	Start time in data query command
To	End time in data query command



F.3.14.4 Data Grid

The DataGrid object is used to display data from a *.csv file as rows and columns.

Operation Description:

- Display the content of .csv files on the **Data Grid**
- Edit the content on the DataGrid table and update it in the corresponding .csv file. The program provides a list of scripts that allow operations on the Data Grid. For details, refer to script for DataGrid at F.5.7.9

F.3.14.4.1 Create a new Data Grid

Click to select Data Grid icon  on the toolbar.

Move the mouse to the desired position inside the design screen, drag and drop to create a rectangle with the desired size. The **Data Grid** object appears on the screen.

F.3.14.4.2 Edit Properties of Object

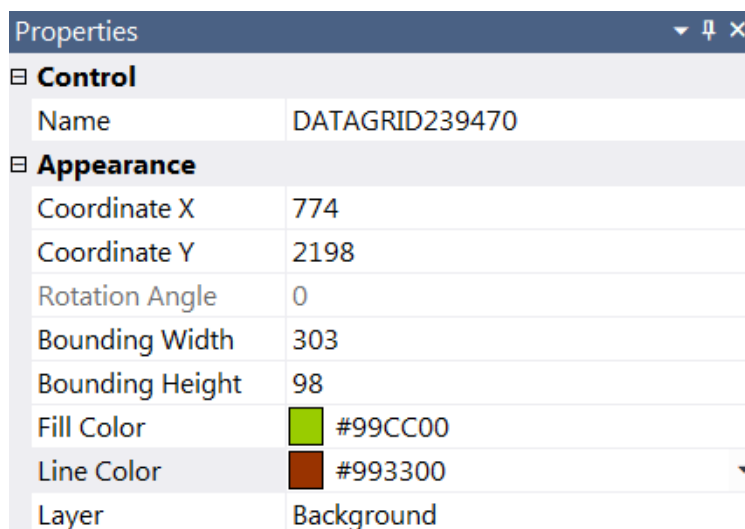


Figure F-174. Data Grid Properties

Properties	Description
Name	Data Grid object name
Unique ID	ID of object
Appearance	The design parameters of coordinates, line and background colors, and the display layer of the object are set in the <i>Properties</i> window. Refer to section F.3.1.13

Edit font, font color from font toolbar F.3.1.5

F.3.14.4.3 Data Grid Configuration at Designer

- ❖ Double Click on the **Data Grid** object you want to configure to open the **DataGrid Configuration** window.

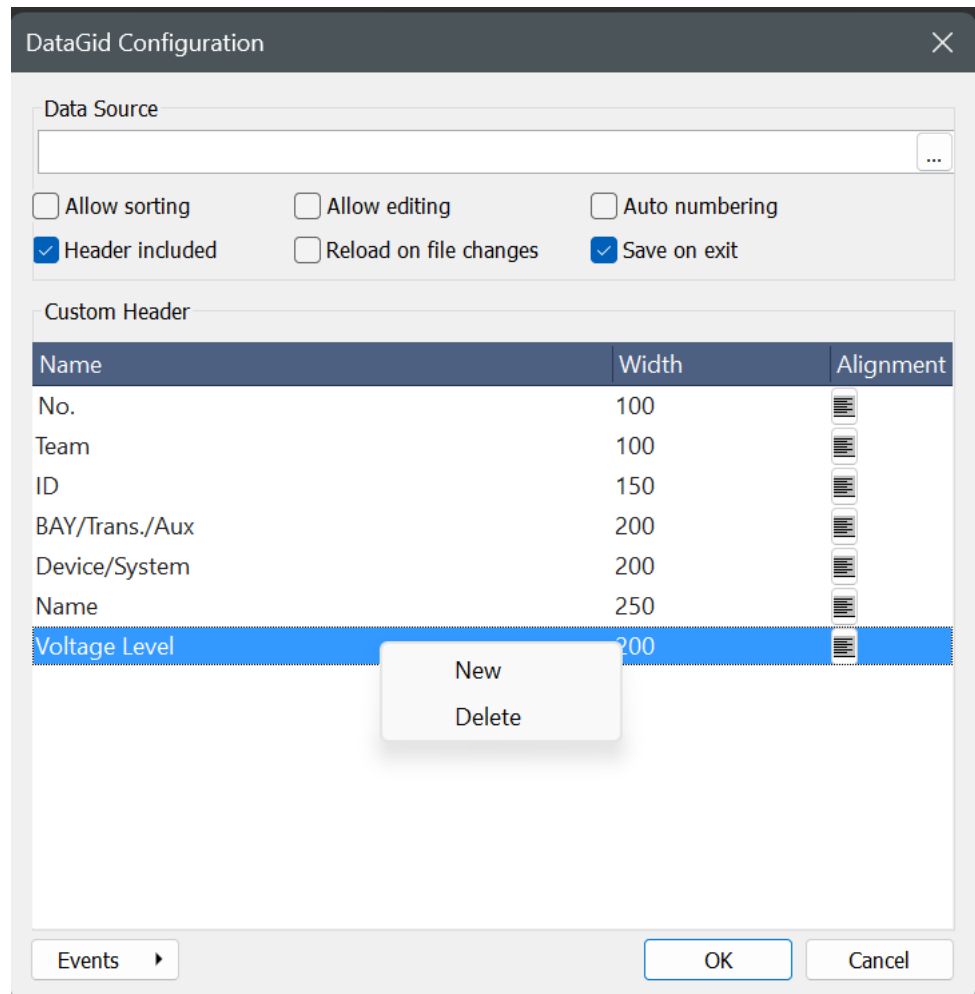
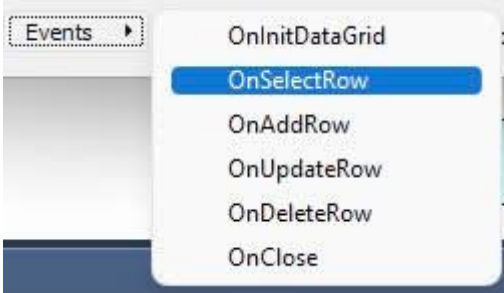


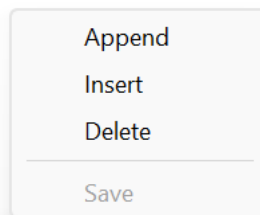
Figure F-175. Configure Data Grid

Properties	Description
Data source	Fill the *.csv file path you want to use on the Data Grid View
Allow sorting	Allow column sorting by column on the Viewer
Allow editing	Allows editing of the contents of the tiles on the Viewer
Auto numbering	Automatically add ordinal number columns
Header included	Datasheet includes header
Reload on file change	Automatically update the data table when the file changes data when running the runtime (even from other software)
Save on exit	Save before exiting Viewer

Config Header	When the Using first row as header option is not selected, this property is used to configure the table headers to include: - Configure information per column <table border="1"> <tr> <td>Name</td><td>Content of column</td></tr> <tr> <td>Width</td><td>Width of column</td></tr> <tr> <td>Alignment</td><td> Align in columns:  : right alignment  : center alignment  : left alignment </td></tr> </table> - To change the number and order of columns: Right-Click in the Config Header area, select: <table border="1"> <tr> <td>New</td><td>Create new columns</td></tr> <tr> <td>Delete</td><td> Delete selected columns. (Can select multiple columns at the same time to delete) </td></tr> </table> - The order of rows in the Config Header is the order of columns displayed on the Grid Viewer from left to right	Name	Content of column	Width	Width of column	Alignment	Align in columns:  : right alignment  : center alignment  : left alignment	New	Create new columns	Delete	Delete selected columns. (Can select multiple columns at the same time to delete)
Name	Content of column										
Width	Width of column										
Alignment	Align in columns:  : right alignment  : center alignment  : left alignment										
New	Create new columns										
Delete	Delete selected columns. (Can select multiple columns at the same time to delete)										
Events	 Scripts to perform desired automatic actions when events occur <table border="1"> <tr> <th>Event type</th><th>Description</th></tr> <tr> <td>OnInitDataGrid</td><td>Script to be executed when Data Grid is activated</td></tr> </table>	Event type	Description	OnInitDataGrid	Script to be executed when Data Grid is activated						
Event type	Description										
OnInitDataGrid	Script to be executed when Data Grid is activated										

	OnSelectRow	Script to be executed when a row in the Data Grid is selected (Left-click on Viewer)
	OnAddRow	Script to be executed when adding a new row on the Data Grid
	OnUpdateRow	Script to be executed when updating a row
	OnDeleteRow	Script to be executed when deleting a row
	OnClose	Script to be executed when the screen containing the Data Grid is closed

- ❖ Append/Insert/Delete row, Save Right-click menu on Viewer



- ❖ Allow fill relative path of Project (Example: .\filename.csv)
- ❖ From the Grid Designer environment, design graphical objects containing scripts for interaction on the Viewer, refer to section script for Data Grid in F.5.7.9

F.3.14.5 Relay Event View

Object used to quickly retrieve SEL relay information with the SUM/HIS command through a TCP/IP Telnet connection.

Operation Description:

- Display **SEL relay** information with **SUM/HIS** command over TCP/IP Telnet connection
- The SUM/HIS command is sent to the relay through the execution of the **RelayEventRefresh** script. The parameters for the script are set through the configuration of the Relay Event View object and the parameters for the scripts. For details, refer to **RelayEventRefresh** at F.5.7.10.1

F.3.14.5.1 Create a new Relay Event View

Click to select the Relay Event View icon  on the toolbar.

Move the mouse to the desired position inside the design screen, drag and drop to create a rectangle with the desired size. The **Relay Event View** object appears on the screen.

F.3.14.5.2 Edit Properties of Object

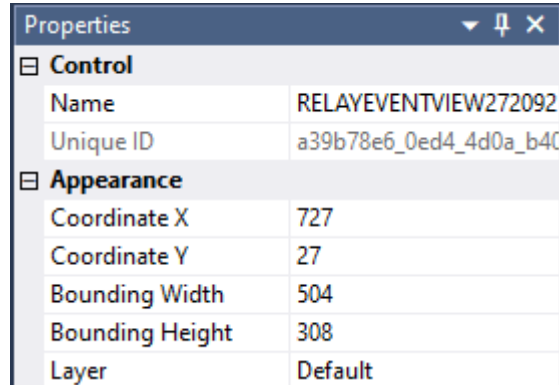


Figure F-176. Relay Event View Properties

Properties	Description
Name	Relay Event View
Unique ID	ID of object
Appearance	The design parameters of the coordinates, the size of the object is set in the Properties window. Refer to section F.3.1.13

Edit font, font color from font toolbar at F.3.1.5

F.3.14.5.3 Configure a Relay Event View at Designer

- ❖ Double Click on the **Relay Event View** icon you want to configure to open **Relay Event View Configuration** window.

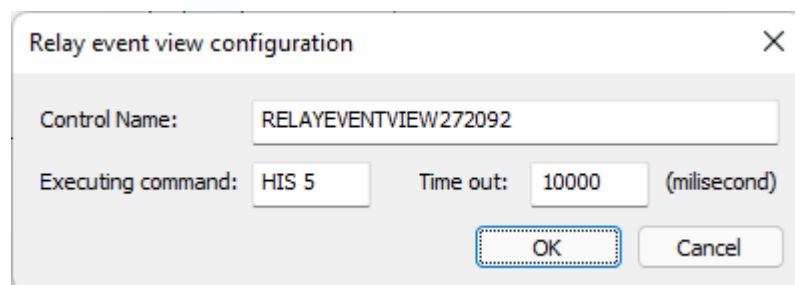


Figure F-177. Relay event view configuration

Properties	Description
------------	-------------

Control Name	Relay Event View object name
Executing Command	ASCII command send to relay SEL. Commonly used commands: - SUM 5 - HIS 5 - EVE 5 (Refer to the relay manual)
Timeout	Timeout time of command sent to relay

F.3.14.6 Chart

Chart View is an object used to build and manage graphs with a variety of representations (columns, lines), and flexible information.



Figure F-178. Chart view

Compare Trend View and Chart

Parameter	Trend View	Chart
Horizontal axis	Time axis, support 1 axis system	
Vertical axis	Value axis, support 1 axis system	Value axis, support multiple axes system

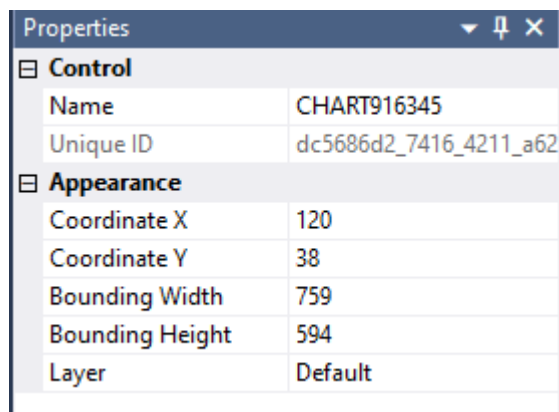
Maximum number of strokes	4	20
Graph form	Support: - Line graph format	Support: - Line graph format - Column graph format
Automatic calculation information	No support	Support values - Max - Min - Average - Interpolate

F.3.14.6.1 Create a new Chart

Select **Chart** icon  on the menu toolbar.

Move the mouse to the desired position inside the design screen, drag and drop to create a rectangle with the desired size.

F.3.14.6.2 Edit Properties of Object



Properties	
Control	
Name	CHART916345
Unique ID	dc5686d2_7416_4211_a62
Appearance	
Coordinate X	120
Coordinate Y	38
Bounding Width	759
Bounding Height	594
Layer	Default

Figure F-179. Chart Properties

Properties	Description
Name	Chart View object name
Unique ID	ID of object

Appearance	The design parameters of the coordinates, the size of the object is set in the Properties window. Refer to section F.3.1.13
------------	---

Edit font, font color from font toolbar at F.3.1.5

F.3.14.6.3 General Configuration & Colors

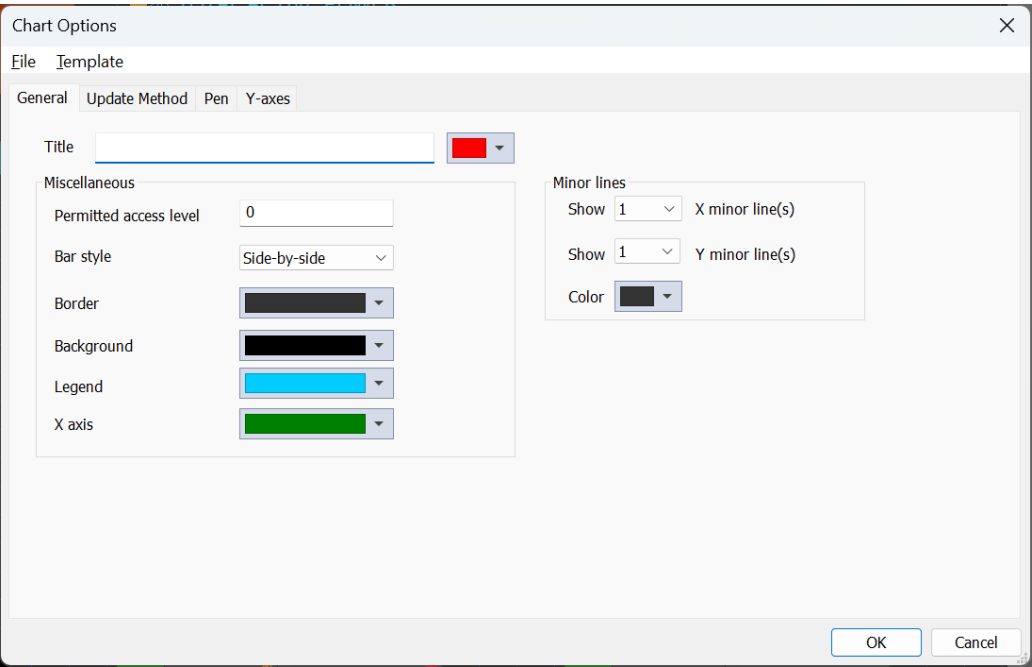


Figure F-180. General Configuration Chart Object

Properties	Description
General configuration	
Miscellaneous	
Bar style (Side-by-side Overlapped)	Style of bar chart
Permitted access level	Minimum level of user access in the Grid Viewer environment to change the data display configuration for the Chart object Value range 0-255

Background	Object background color
Border	Graph frame border color
Legend	Color of notes
X-axis	Color of X-axis
Minor Lines	
Number of X minor lines	The number of lines between the smallest division on the horizontal axis
Number of Y minor lines	The number of lines between the smallest division on the vertical axis
Color	Color for grid in chart

F.3.14.6.4 Configuring data plotting interval (Update Method)

The interval plots the data on the chart

- Start Time
- End Time

Update Method configuration - <i>Configuring the data plotting interval update mode (Start Time & End Time)</i>	
Sliding Window	• End Time: time to initialize/update the configuration Chart. • Start Time: = End Time – Window Duration. End Time and reference values on the horizontal axis (time axis) are automatically updated periodically along with the data update frequency.
History	• Start Time: = Start At • End Time: = End At • Data is updated once user Update configuration in History Mode
Cyclic Update	• Start Time: = Cyclic Update Time + Start After • End Time: = Start Time + Cycle Duration

	Start Time and reference values on the horizontal axis (time axis) are updated at a separate frequency from the data update frequency.
--	--

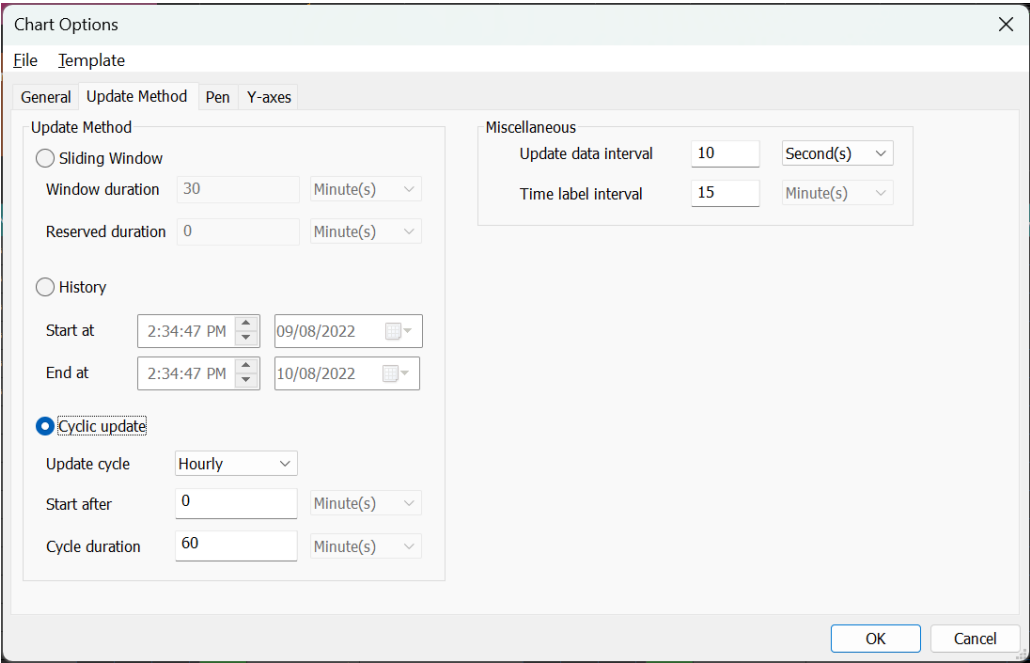


Chart Options

File Template

General Update Method Pen Y-axes

Update Method

☐ Sliding Window

Window duration 30 Minute(s)

Reserved duration 0 Minute(s)

☐ History

Start at 2:34:47 PM 09/08/2022

End at 2:34:47 PM 10/08/2022

☒ Cyclic update

Update cycle Hourly

Start after 0 Minute(s)

Cycle duration 60 Minute(s)

Miscellaneous

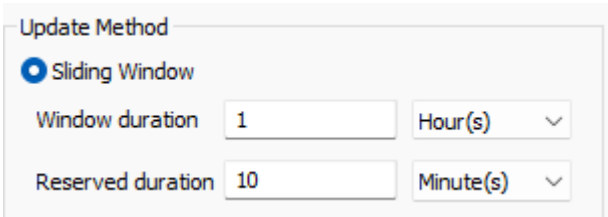
Update data interval 10 Second(s)

Time label interval 15 Minute(s)

OK Cancel

Figure F-181. Configuration window Update Method Chart object

Sliding Window Mode



Update Method

☒ Sliding Window

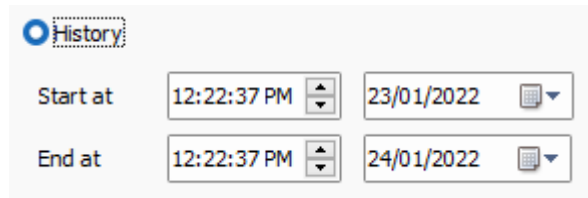
Window duration 1 Hour(s)

Reserved duration 10 Minute(s)

Properties	Description
Sliding Window configuration	
Window duration	Data plot interval. Unit: - Minute(s) - Hour(s) - Day(s) - Week(s) - Month(s)

Reserved duration	Time interval on the graph. Unit: - Minute(s) - Hour(s) - Day(s) - Week(s) - Month(s) Note: total time on the graph = Window duration + Reserved duration.
-------------------	---

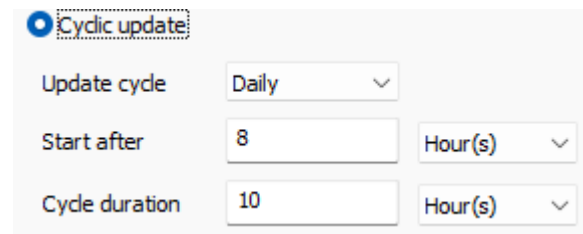
History Mode



The History Mode configuration interface shows a radio button labeled 'History' which is selected. Below it are two rows of date and time pickers. The first row is labeled 'Start at' and has a time picker set to '12:22:37 PM' and a date picker set to '23/01/2022'. The second row is labeled 'End at' and has a time picker set to '12:22:37 PM' and a date picker set to '24/01/2022'.

Properties	Description
History configuration	
Start at	Start time in data query command
End at	End time in data query command

Cyclic Update Mode



The Cyclic Update Mode configuration interface shows a radio button labeled 'Cyclic update' which is selected. Below it are three rows of configuration options. The first row is labeled 'Update cycle' and has a dropdown menu set to 'Daily'. The second row is labeled 'Start after' and has a text input set to '8' and a dropdown menu set to 'Hour(s)'. The third row is labeled 'Cycle duration' and has a text input set to '10' and a dropdown menu set to 'Hour(s)'.

Properties	Description
Cyclic update configuration	
Update cycle	Frequency of updating reference values on the time axis (horizontal axis)

	- Quarter Hourly: every 15 minutes (0, 15, 30, 45) - Hourly: every hour (0 minute 0 second) - Daily: everyday (0 hour 0 minute 0 second) - Weekly: every week (Monday, 0 hour 0 minute 0 second) - Monthly: every month (day 01, 0 hour 0 minute 0 second)
Start after	The time period starts on the time axis (horizontal axis) from the time of Update the reference value on the time axis Unit: - Minute(s) - Hour(s) - Day(s) - Week(s)
Cycle duration	The interval width plots the data on the time (horizontal) axis. Unit: - Minute(s) - Hour(s) - Day(s)

F.3.14.6.5 Configuration of vertical - value axis systems (Y axis)

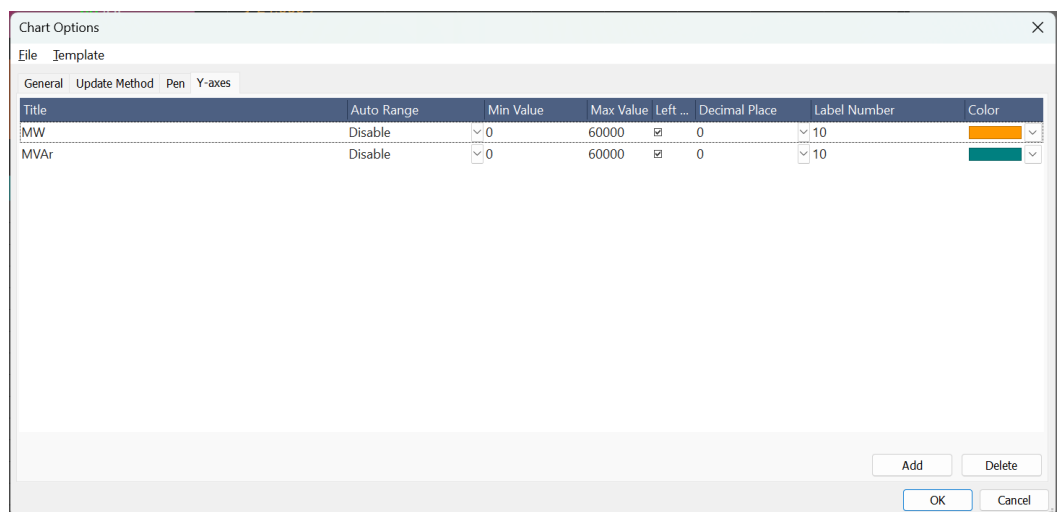


Figure F-182. Configuration window of the Y axis Chart object

- Click **Add** button to add vertical axis
- Select the vertical axis, click the Delete button to delete the vertical axis
- Parameters of the vertical axis

Y axis parameters	
Title	Name of the axis
Auto Range	Automatically update the min – max reference value for the axis based on the data points (pen) of the axis. The reference value update frequency is configured in Tab General / Miscellaneous / Y update Periodical Algorithm: • Vmax, Vmin: the maximum / minimum value in the data points of the same axis • valueRange = Vmax – Vmin • avgRange = Vmax – valueRange/2 • yMax = avgRange + 1.25* valueRange/2 • yMin = avgRange - 1.25* valueRange/2
Min value Max value	Min – Max reference value on vertical axis when initializing Chart object (Open Screen, update Chart)
Left Side	The vertical axis is displayed on the left side of the chart (if this option is not selected, the vertical axis will be displayed on the right side of the chart)
Decimal	Number of decimal places of the reference value on the vertical axis
Label Number	Number of reference values and corresponding alignment (line) on the vertical axis
Color	Color of vertical axis and reference value

General/Minor Lines configuration	
Number of Y minor lines	The number of minor lines between the lines corresponding to the reference values on the vertical axis

F.3.14.6.6 Configuring the pen

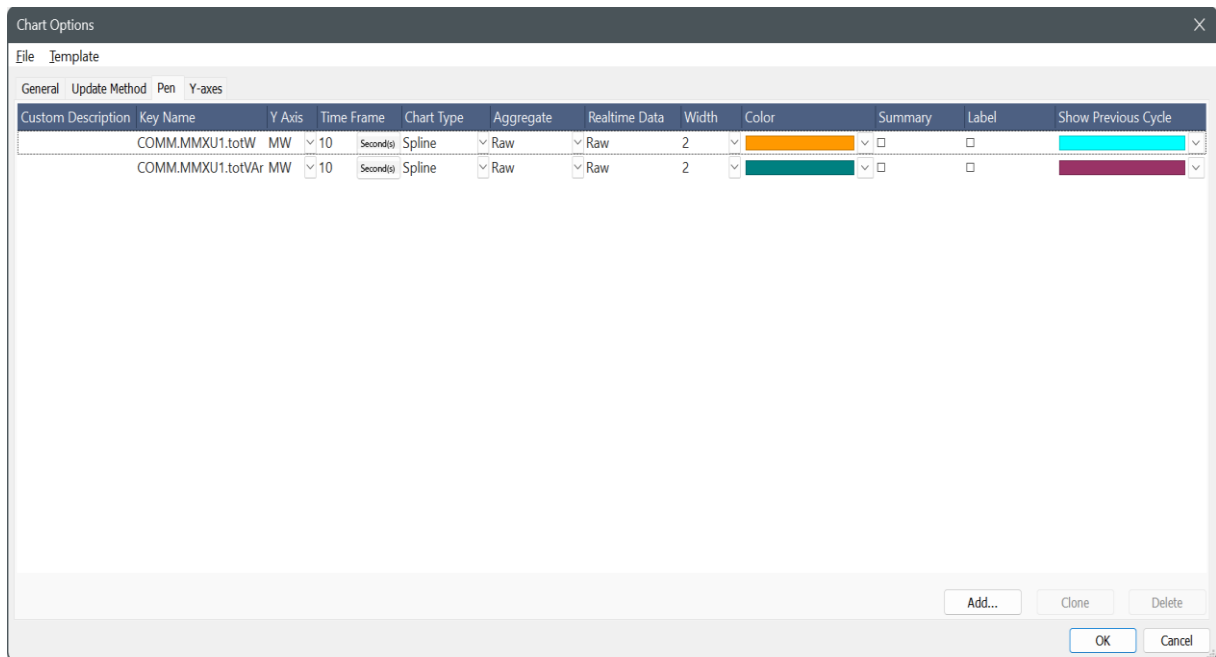


Figure F-183. The pen configuration window for the Chart object

Parameters of pen

Pen parameters	
Custom Description	Description of the drawing.
Key Name	Key Name of the data point. Autofill value according to selected data point.
Y Axis	The system of vertical axes to which the pen belongs. Value: select from a list of vertical axes (Tab Y Axis) declared for the Chart object.
Time Frame	How often data is queried for calculated data types (min, max, average, ...) Unit: • Second(s) • Minute(s) • Hour(s) • Day(s) • Week(s)

	Month(s) Units are automatically scaled according to the Time span (configured on the Update Method tab. The program automatically sets the number of points [min, max] for each line based on the timeframe to avoid the case that the chart has too many points. → application hang.	
Chart Type	Type pen	
	Straight line	Line type connecting points on the chart
	Spline	Curve style connecting points on the chart
	Step Line	The type of ladder line connecting points on the chart
	Bar	Column type
Aggregate	The calculated data type to be queried.	
	Raw	The data points are stored. With HIS Server (Realtime Data = Processed), Cyclic/Sliding mode, this is updated data from Snapshot)
	Interpolate	Interpolated calculated data points with calculation intervals = Time Frame
	Average	The data points are averaged with calculation intervals = Time Frame
	Min	Min calculated data points with calculation intervals = Time Frame
	Max	Max calculated data points with calculation intervals = Time Frame
	Average-abs	Calculated data points average absolute values over calculation intervals = Time Frame

	Min-abs	Calculated data points min absolute values for calculation intervals = Time Frame
	Max-abs	Calculated data points max absolute values for calculation intervals = Time Frame
Realtime Data	The updated data source for pen	
	Raw	Updated data from DataServer. For a DataServer data source, the value points of the data are stored in the buffer memory of the Chart object and periodically (Miscellaneous/Update Data Interval) is updated on the graph.
	Processed	Updated data from HIS Server
Width	Pen thickness.	
Color	Color of pen	
Summary	Option to display the summary information of the pen: • The max value and the time when the max value is reached in the set of value points of that line on the chart • The min value and the time when the min value is reached in the set of value points of that line on the chart • The average value in the set of value points of that line on the chart	
Label	Option to display the values of the data points on the chart	
Show Previous Cycle	Displays a graph of the past cycle of the corresponding pen (4.1.7031 onwards)	

Add new pen

- Click the Add button to add a pen

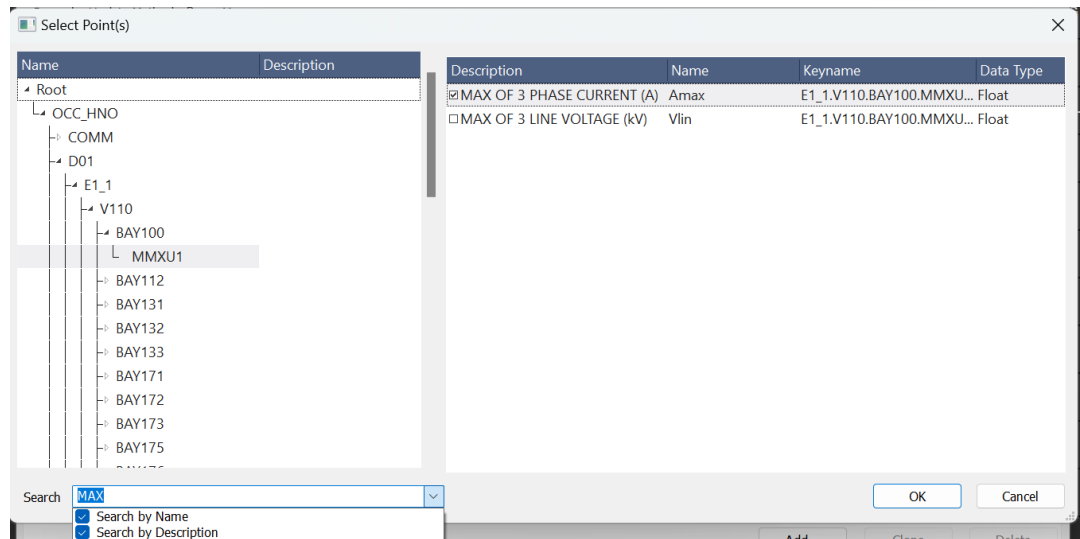


Figure F-184. Select the data point for the Chart object

The application supports data point selection tool from Data Model

- Select data points to add. Can use Search Box to filter points by Name/Description.
- Click OK button.

The new added pen will have the same properties as the previous data point (bottom in the list of pen nets), except the color attribute is randomly generated by the application

Add a pen exactly like the previous one

- Select the pen
- Click **Clone** button to add pen

Delete a pen

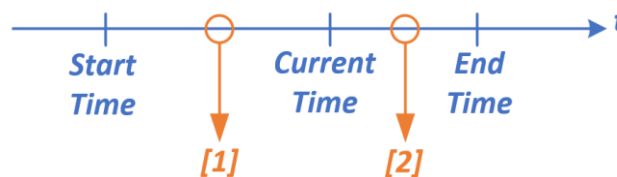
- Select the pen
- Click **Delete** button to delete the pen

F.3.14.6.7 Configure data update frequency

General/Miscellaneous configuration		
Update interval	data	Frequency of sending queries and updating data to the chart Unit: - Second(s) - Minute(s) - Hour(s)

Time label interval	The display time between two-time milestones on the x-axis. Unit: - Minute(s) - Hour(s) - Day(s) - Week(s) - Month(s)
---------------------	---

The data query command parameters (Data Source, Start/End Time, Data Point, Aggregate type, interval, ...) are configured in the Update Method Configuration & Pen Configuration items.



When initializing (Open Screen) or updating the configuration of the Chart object:

- Data belonging to the past time period [1] (Start Time – Current Time) will be queried from the HIS Server. The data query function is configured with the pen/Aggregate parameter.
- Data in the current word period [2] (Current Time – End Time) is updated from DataServer or HIS Server depending on pen/Realtime Data configuration. The data query function is configured with the pen/Aggregate parameter.

F.3.14.6.8 Reload - Synchronize Chart configuration between Designer & Viewer

Configuration of the Chart object can be done simultaneously by the Designer & Viewer. Synchronization is performed by the **Reload** function.

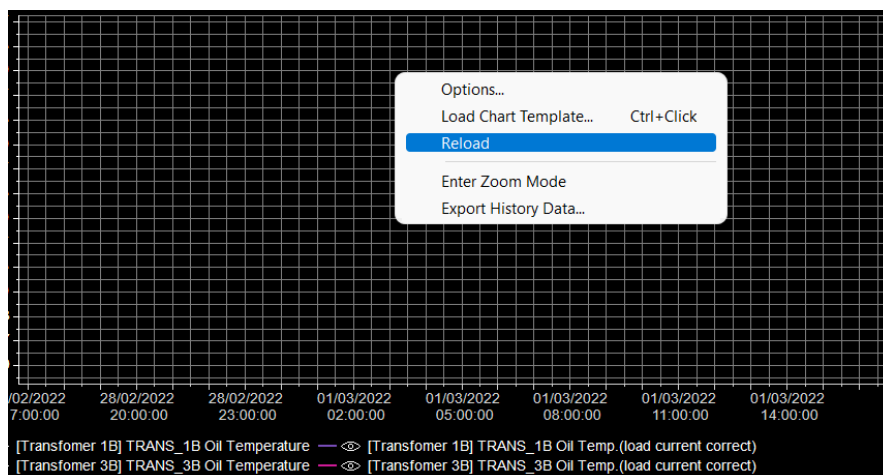
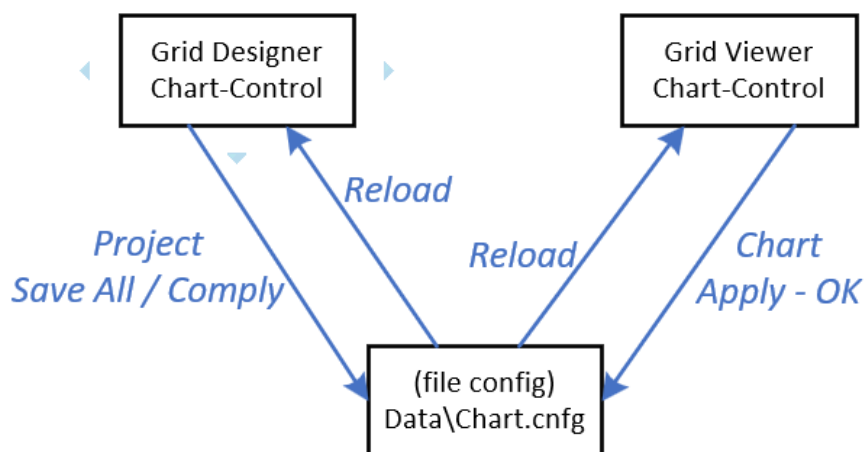


Figure F-185. Reload Chart

Synchronicity principle



F.3.14.6.9 Template

Users can create and save chart configurations as templates.

Create new Template

- After configuring the chart → Double Left-Click to chart → Template → Save Configuration As...

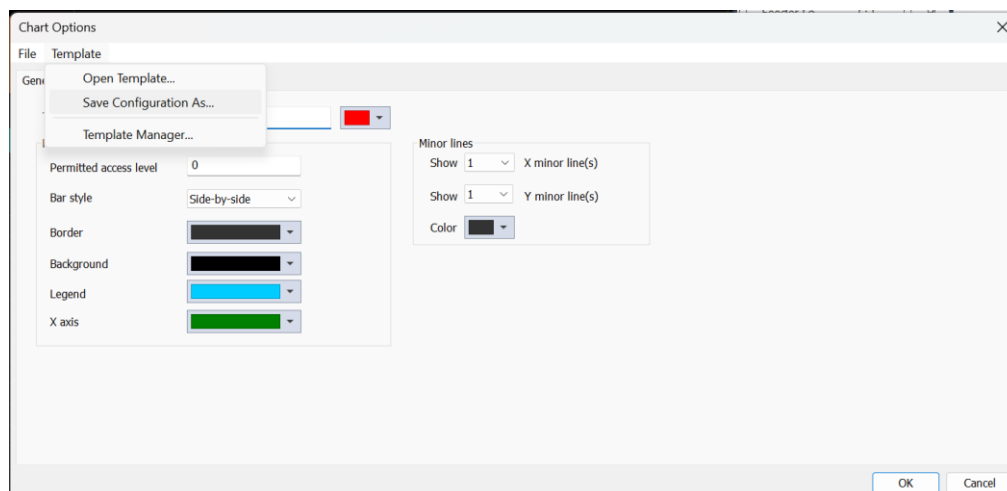


Figure F-186. Save Configuration As

- Select storage at node.

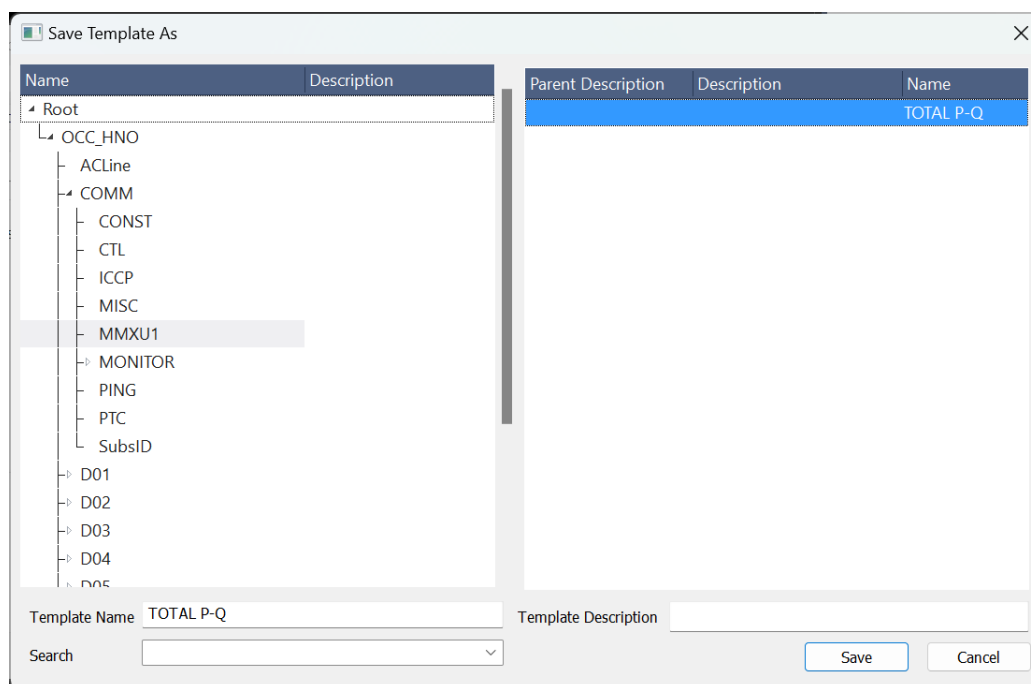


Figure F-187. Browse Save Location

Open Template

- Double Left-Click to chart → Template → Open Template → Browse Template

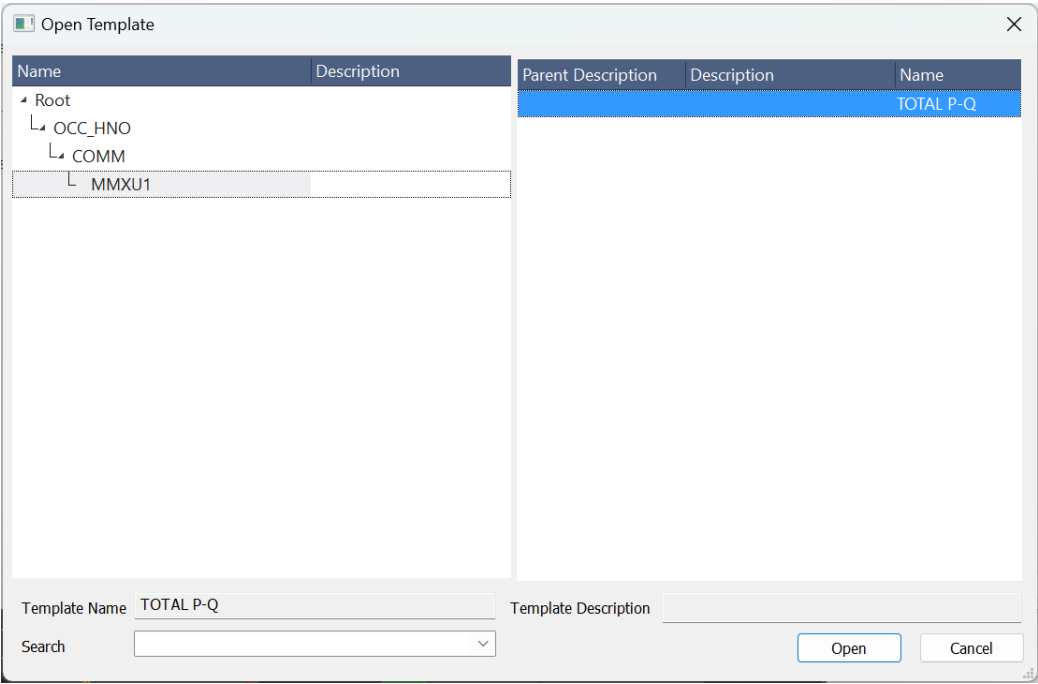


Figure F-188. Browse Template

Template Manager

- Double Left-Click to chart → Template → Template Manager...

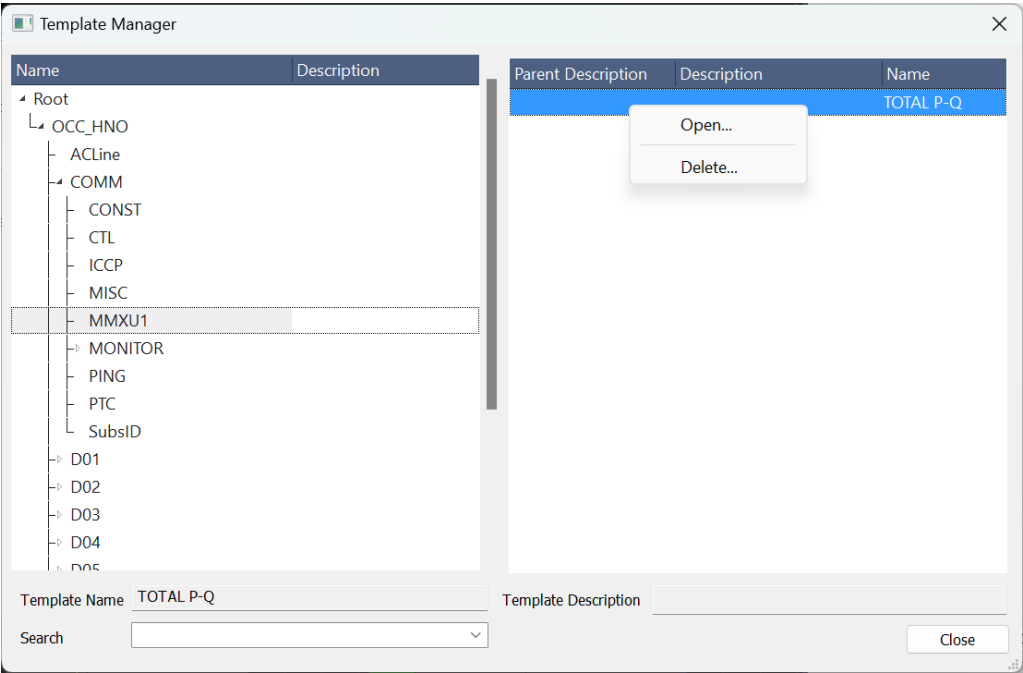


Figure F-189. Template Manager

- Right -Click to template → Choose Open or Delete Template

F.3.14.7 Tabular

The tabular object used to display/write Realtime Remote Data from DataServer as a table. This is a tool to support checking system data in the Grid Viewer environment.

No.	Keyname	Description	Data Type	Value	Quality	Source Timestamp
1	E1_8.V22.BAYTUC44.BAYBB1	CONNECT TO BUSBAR 1	String		Good	08/07/2024 14:39:20
2	E1_8.V22.BAYTUC44.BAYBB2	CONNECT TO BUSBAR 2	String		Good	08/07/2024 14:39:20
3	E1_8.V22.BAYTUC44.BAYBB3	CONNECT TO BUSBAR 3	String		Good	08/07/2024 14:39:20
4	E1_8.V22.BAYTUC44.BAYFeederName		String	E1.8 - TUC44	Good	08/07/2024 14:39:20
5	E1_8.V22.BAYTUC44.BAYFeederName		String		Good	08/07/2024 14:39:20
6	E1_8.V22.BAYTUC44.BAYFuseTrip	FUSE TRIP	Boolean	False	Bad	08/07/2024 14:39:20
7	E1_8.V22.BAYTUC44.BAYGasPresMon	GAS PRESSURE MONITORING	Boolean	False	Bad	08/07/2024 14:39:20
8	E1_8.V22.BAYTUC44.BAYIsLive		Boolean	False	Good	08/07/2024 14:39:20
9	E1_8.V22.BAYTUC44.BAYKey		String	E1_8.V22.BAYTUC44.BAY	Good	08/07/2024 14:39:20
10	E1_8.V22.BAYTUC44.BAYMcb	MCB TRIP/OFF	Boolean	False	Bad	08/07/2024 14:39:20
11	E1_8.V22.BAYTUC44.BAYMcbVTP	VTP MCB TRIP/OFF	Boolean	False	Bad	08/07/2024 14:39:20
12	E1_8.V22.BAYTUC44.BAYRelayBAYSParePanel	RELAY BAY SPARE PANEL	String		Good	08/07/2024 14:39:20
13	E1_8.V22.BAYTUC44.BAYRelayBAYType	RELAY BAY TYPE	String		Good	08/07/2024 14:39:20
14	E1_8.V22.BAYTUC44.CC.DeviceName		String		Good	15/01/2024 14:51:18
15	E1_8.V22.BAYTUC44.CC.IsLive		Boolean	False	Good	08/07/2024 14:39:20
16	E1_8.V22.BAYTUC44.CC.Key		String	E1_8.V22.BAYTUC44.CC	Good	08/07/2024 14:39:20
17	E1_8.V22.BAYTUC44.MMXU1.Amax_AlmLim.Enab.	Enabled	UInt8	0	Good	08/07/2024 14:39:20
18	E1_8.V22.BAYTUC44.MMXU1.Amax_AlmLim.LoLo	LoLo	Double	0.000	Good	08/07/2024 14:39:20
19	E1_8.V22.BAYTUC44.MMXU1.Amax_AlmLim.Lo	Lo	Double	0.000	Good	08/07/2024 14:39:20
20	E1_8.V22.BAYTUC44.MMXU1.Amax_AlmLim.Hi	Hi	Double	0.000	Good	08/07/2024 14:39:20
21	E1_8.V22.BAYTUC44.MMXU1.Amax_AlmLim.HiHi	HiHi	Double	0.000	Good	08/07/2024 14:39:20
22	E1_8.V22.BAYTUC44.MMXU1.Amax_AlmLim.State	State	Int8	0	Good	08/07/2024 14:39:20
23	E1_8.V22.BAYTUC44.MMXU1.Amax	MAX OF 3 PHASE CURRENT (A)	Float	0.000	Bad	08/07/2024 14:39:20
24	E1_8.V22.BAYTUC44.MMXU1.Hz	FREQUENCY (Hz)	Float	0.000	Bad	08/07/2024 14:39:20
25	E1_8.V22.BAYTUC44.MMXU1.Key		String	E1_8.V22.BAYTUC44.MMXU1	Good	08/07/2024 14:39:20
26	E1_8.V22.BAYTUC44.MMXU1.PPvphsAB	LINE AB VOLTAGE (kV)	Float	0.000	Bad	08/07/2024 14:39:20
27	E1_8.V22.BAYTUC44.MMXU1.PPvphsBC	LINE BC VOLTAGE (kV)	Float	0.000	Bad	08/07/2024 14:39:20
28	E1_8.V22.BAYTUC44.MMXU1.PPvphsCA	LINE CA VOLTAGE (kV)	Float	0.000	Bad	08/07/2024 14:39:20
29	E1_8.V22.BAYTUC44.MMXU1.PhVphsA	PHASE A VOLTAGE (kV)	Float	0.000	Bad	08/07/2024 14:39:20
30	E1_8.V22.BAYTUC44.MMXU1.PhVphsB	PHASE B VOLTAGE (kV)	Float	0.000	Bad	08/07/2024 14:39:20
31	E1_8.V22.BAYTUC44.MMXU1.PhVphsC	PHASE C VOLTAGE (kV)	Float	0.000	Bad	08/07/2024 14:39:20
32	E1_8.V22.BAYTUC44.MMXU1.Vin_AlmLim.Enabled	Enabled	UInt8	0	Good	08/07/2024 14:39:20
33	E1_8.V22.BAYTUC44.MMXU1.Vin_AlmLim.LoLo	LoLo	Double	0.000	Good	08/07/2024 14:39:20
34	E1_8.V22.BAYTUC44.MMXU1.Vin_AlmLim.Lo	Lo	Double	0.000	Good	08/07/2024 14:39:20
35	E1_8.V22.BAYTUC44.MMXU1.Vin_AlmLim.Hi	Hi	Double	0.000	Good	08/07/2024 14:39:20
36	E1_8.V22.BAYTUC44.MMXU1.Vin_AlmLim.HiHi	HiHi	Double	0.000	Good	08/07/2024 14:39:20
37	E1_8.V22.BAYTUC44.MMXU1.Vin_AlmLim.State	State	Int8	0	Good	08/07/2024 14:39:20
38	E1_8.V22.BAYTUC44.MMXU1.Vin	MAX OF 3 LINE VOLTAGE (kV)	Float	0.000	Bad	08/07/2024 14:39:20
39	E1_8.V22.BAYTUC44.PRO.F27	F27 TRIP	Boolean	False	Bad	08/07/2024 14:39:20
40	E1_8.V22.BAYTUC44.PRO.F59	F59 TRIP	Boolean	False	Bad	08/07/2024 14:39:20
41	E1_8.V22.BAYTUC44.PRO.F81LV1	F81LV1 TRIP	Boolean	False	Bad	08/07/2024 14:39:20
42	E1_8.V22.BAYTUC44.PRO.F81LV2	F81LV2 TRIP	Boolean	False	Bad	08/07/2024 14:39:20
43	E1_8.V22.BAYTUC44.PRO.F81LV3	F81LV3 TRIP	Boolean	False	Bad	08/07/2024 14:39:20
44	E1_8.V22.BAYTUC44.PRO.F81LV4	F81LV4 TRIP	Boolean	False	Bad	08/07/2024 14:39:20
45	E1_8.V22.BAYTUC44.PRO.GenPickup	GENERAL PICKUP	Boolean	False	Bad	08/07/2024 14:39:20
46	E1_8.V22.BAYTUC44.PRO.GenPickupphsA	PHASE A GENERAL PICKUP	Boolean	False	Bad	08/07/2024 14:39:20
47	E1_8.V22.BAYTUC44.PRO.GenPickupphsB	PHASE B GENERAL PICKUP	Boolean	False	Bad	08/07/2024 14:39:20
48	E1_8.V22.BAYTUC44.PRO.GenPickupphsC	PHASE C GENERAL PICKUP	Boolean	False	Bad	08/07/2024 14:39:20
49	E1_8.V22.BAYTUC44.PRO.GenTrip	GENERAL TRIP	Boolean	False	Bad	08/07/2024 14:39:20
50	E1_8.V22.BAYTUC44.XSWI4.Bik		Int32	0	Good	08/07/2024 14:39:47
51	E1_8.V22.BAYTUC44.XSWI4.BikCB	XSWI in blocking	Boolean	False	Good	08/07/2024 14:39:20
52	E1_8.V22.BAYTUC44.XSWI4.BikDS		Boolean	False	Good	08/07/2024 14:39:20
54	E1_8.V22.BAYTUC44.XSWI4.BikLogic	Show XSWI Logic Interlock Panel	Boolean	False	Good	08/07/2024 14:39:47
55	E1_8.V22.BAYTUC44.XSWI4.DeviceName	EVN name in single line	String	TUC44-4	Good	08/07/2024 14:39:20

Figure F-190. Tabular on Grid Viewer

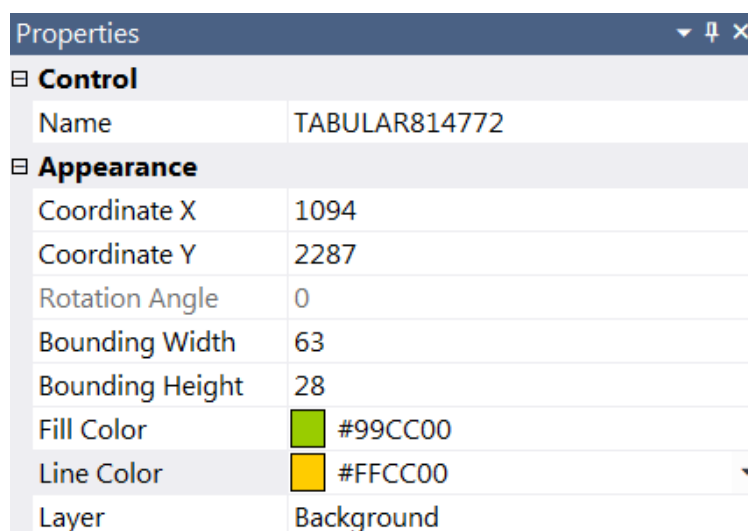
The application supports scripts to interact with Tabular objects. For details, see Tabular scripts in section F.5.7.11

F.3.14.7.1 Create a new Tabular

Click Tabular icon  on the toolbar.

Move the mouse to the desired position inside the design screen, drag and drop to create a rectangle with the desired size. The **Tabular** object appears on the screen.

F.3.14.7.2 Edit Object Properties:



Properties	
Control	
Name	TABULAR814772
Appearance	
Coordinate X	1094
Coordinate Y	2287
Rotation Angle	0
Bounding Width	63
Bounding Height	28
Fill Color	#99CC00
Line Color	#FFCC00
Layer	Background

Figure F-191. Tabular Properties

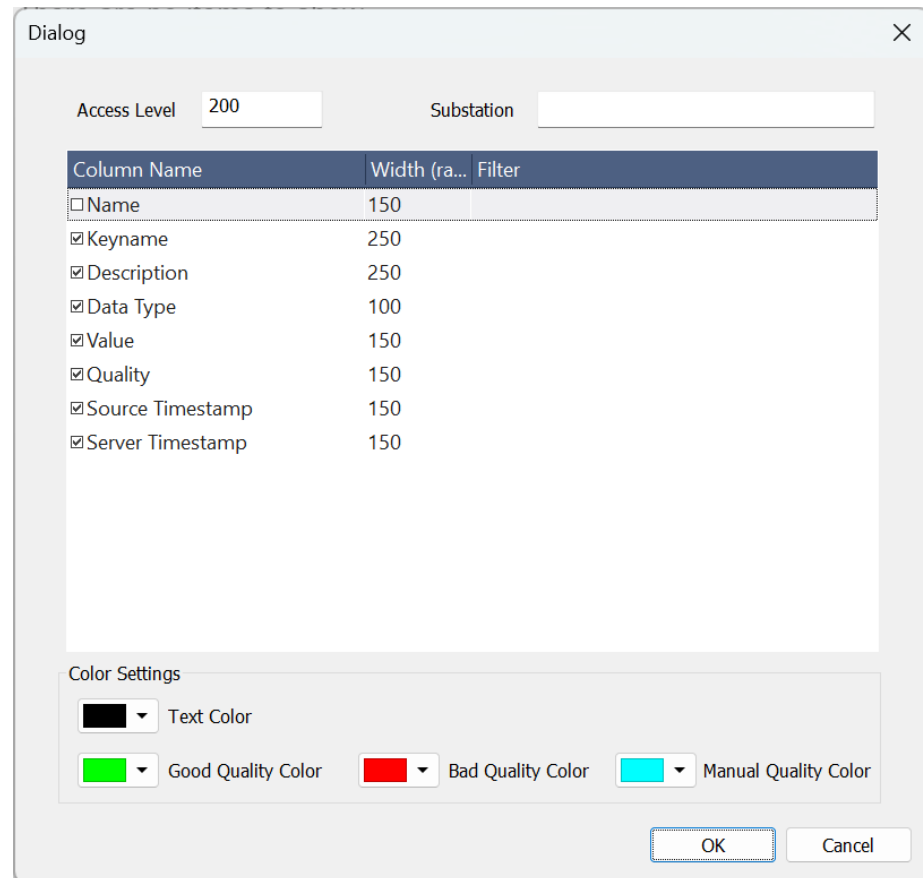
Properties	Description
Name	Tabular Object Name
Unique ID	ID of object
Appearance	The design parameters of the coordinates, the size of the object is set in the Properties window. Refer to section F.3.1.13

Edit the font, from the font toolbar at F.3.1.5.

Adjust the font color according to the value/quality of the data from the Tabular configuration panel.

F.3.14.7.3 Configure Tabular data at Designer

- ❖ Double Click on the **Tabular** object you want to configure to open the **Tabular Configuration** window.



Column Name	Width (ra...	Filter
☐ Name	150	
☒ Keyname	250	
☒ Description	250	
☒ Data Type	100	
☒ Value	150	
☒ Quality	150	
☒ Source Timestamp	150	
☒ Server Timestamp	150	

Color Settings

Text Color Good Quality Color Bad Quality Color Manual Quality Color

OK Cancel

Figure F-192. Configure Tabular

Properties	Description	
Access Level	Minimum level of user access in the Grid Viewer environment to change the data display configuration for Tabular objects Value range 0-255	
Substation	Substation Node on Data Model that Tabular displays the data	
Columns	Configure the display information fields	
	Visible/ Invisible	Option to hide/show information of data
	Column Name	Information field of data
	Width (ratio)	Column width in ratio

	Filter	Filter condition corresponding to the information field of the data
Color Setting – Text Color	Configure text color selection of data fields • Name • Keyname • Description • Data Type • Source Timestamp • Server Timestamp	
Color Setting – Good Quality Color	Configure text color selection of data fields corresponding to the value Quality = Good: • Value • Quality	
Color Setting – Bad Quality Color	Configure the text color selection of the data fields corresponding to the Quality = Bad value: • Value • Quality	
Color Setting – Manual Quality Color	Configure the text color selection of the data fields corresponding to the Quality = Manual: • Value • Quality	

❖ From the Grid Designer environment, design graphic objects containing scripts to interact on the Viewer. For details, refer to Tabular scripts in section F.5.7.11.

F.3.14.8 Active X

The application allows the insertion of Active X objects registered with Windows.

- Click Active X icon  on the toolbar.
- Select ActiveX

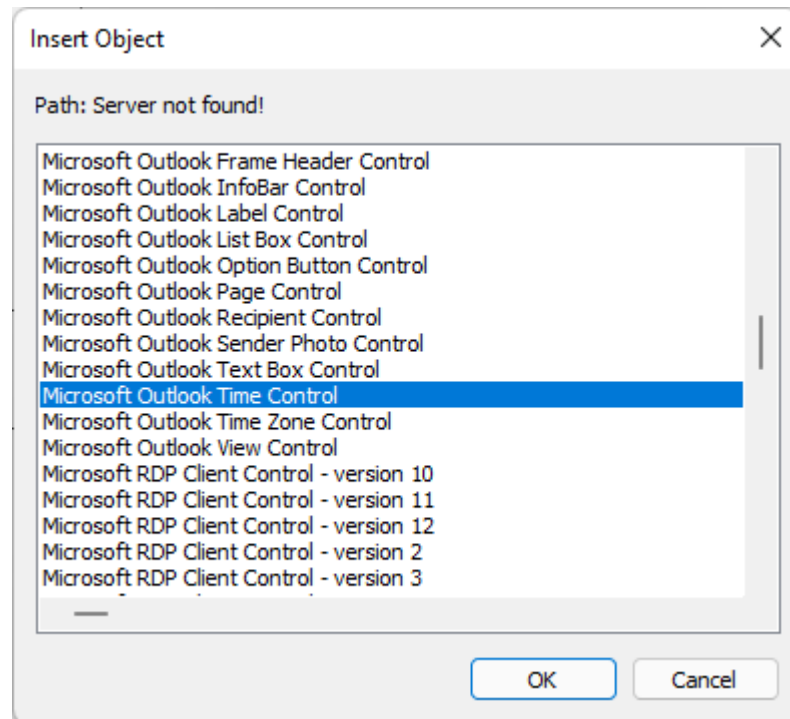


Figure F-193. Insert Active X

- Move the mouse to the desired position inside the design screen, drag and drop to create the desired size. The **Active X** object appears on the screen.

F.3.15. Symbol Template

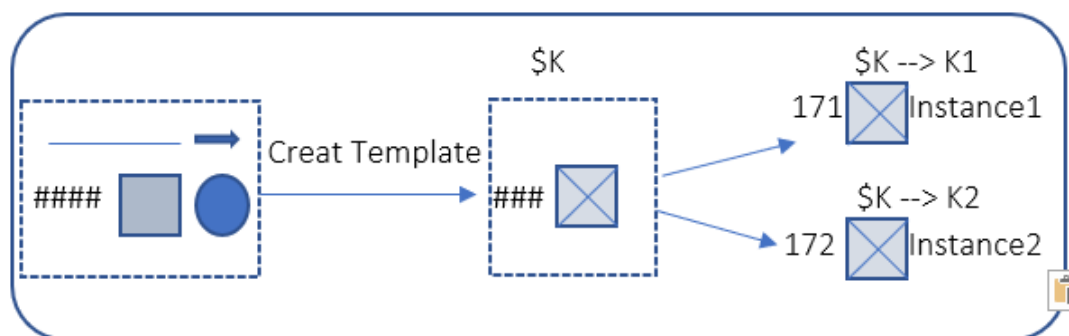


Figure F-194. Using Symbol Template

- Templates are user-designed built-in graphic object templates from basic graphic objects and stored in a library.
- Instance is the actual graphic object created by dragging and dropping the template from the Symbol Library to the screen
 - Has the same graphic properties as the template (Properties, Master Properties)
 - There is a list of distinct real data points of the instance
 - Can be kept the same or rotated relative to the template
- Templates are saved as *.ssl files in the project's Libraries folder.

- Each template only needs to be configured once, then it can be reused many times in the project, increasing the uniformity for objects with the same properties.
- When updating the template, the instances created from the template are automatically updated within the project scope.
- Templates can be imported and exported for use in other projects.
- The template uses the dollar character (\$) to declare input data, the number of input data is unlimited depending on the user's wishes. \$K is the built-in input data available.

Example: All circuit breaker identifiers share the same format: font size 24, Bold, Font Arial; We want the instances built from this Template to be the same in the above 3 formats, only different in the display name (171, 172, ...). So, the Template we build will need an input of \$(Name) to customize the display for each instance.

F.3.15.1 Manage Template

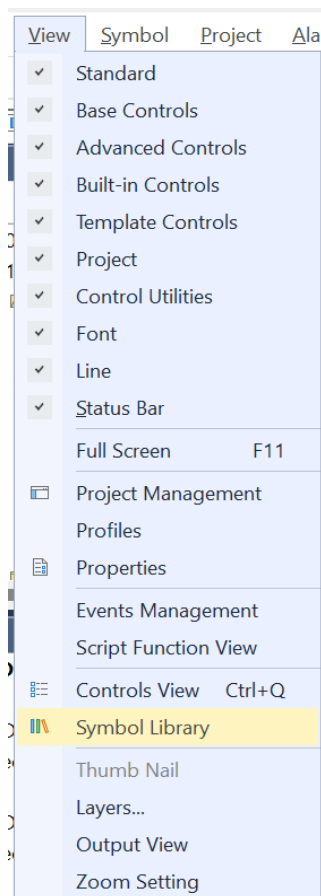


Figure F-195. Menu View – Symbol Library

From the menu bar, selects **View >> Symbol Library** to display to **Symbol Library** window.

Number of Connections	Attributes of electrical equipment. Refer to F.3.12.6	
I/O Status Point	Used to specify the state for the device model. Right Click on the Template design window >> Configuration to open the Symbol Template Configuration window. The information for the I/O Status Point is on the bottom line.	
Class Definition	Serving DMS calculations. <<Not specified>>	
Symbol Update	Keep Size	Changes related to the size of template will not be updated to the instances. Instance will retain the changes in size compared to the original template.
	Keep Rotation	Changes related to template rotation will not be updated to instances. Instance will retain rotation angle changes compared to the original template.

- Click OK to appear the Template design window.

F.3.15.3 Template Design

1. Only one Template design window can be opened at a time.
2. Templates are not nested.

F.3.15.3.1 Template Design Window

The template design window has the properties below. Refer to F.3.8.1 and F.3.8.11

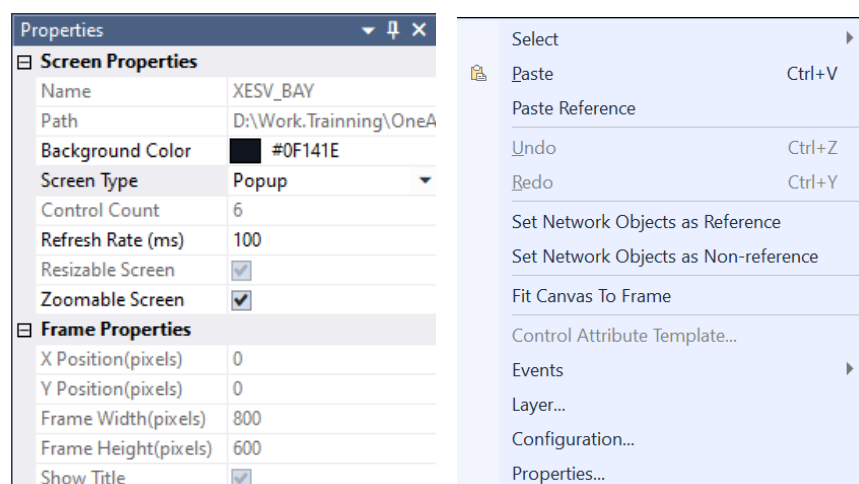


Figure F-198: Template Design properties

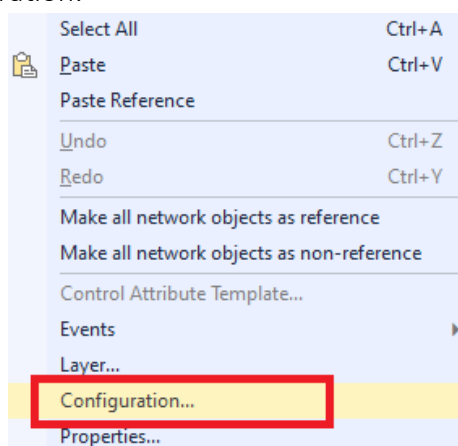
F.3.15.3.2 Graphics Design

Graphic design for templates from basic graphic objects.

- Design reference for graphic objects at F.3.10.
- In expression/script, data prefixed with \$ is used as input data for the Template. These data do not need to be declared in advance and are not real data variables. When using instances from the template, the \$input data needs to be replaced by actual data points

Symbol Variables		
Key Name	Current Value	New Value
\$(BAY)		
\$(XSWI)		
\$K		

For Templates with Device Type of Build-in types (Breaker, Disconnecter, ...), Right-Click menu → Configuration:



Symbol Template Configuration

Device Type

Breaker

Class Definition

<<Not Specified>>

Number of Connections

2

I/O Status Point

ibstation).\$(Voltage).\$(BAY).\$(XCBR).PosSt

Symbol Update

☐ Keep Size
☒ Keep Rotate

OK

Cancel

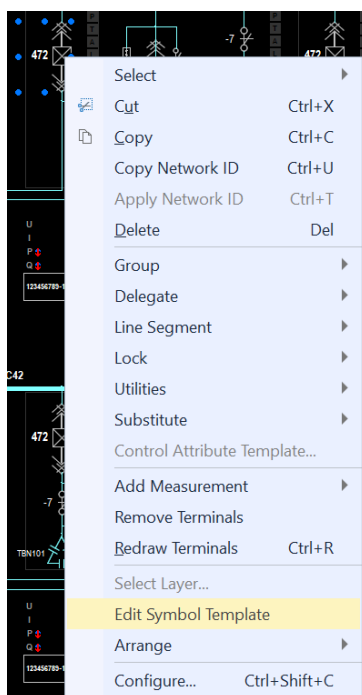
Figure F-199: Configure Connection Point and I/O Status Point

F.3.15.4 Edit Template

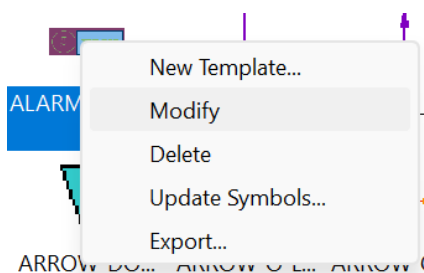
Templates offer bulk design capabilities and great flexibility with the synchronous nature of template instances. All changes in a Template are automatically updated to all instances built from that Template.

There are two ways to edit a symbol template:

- 1 - Right click on the instance designed in the screen, select Template → Symbol → Edit Template.



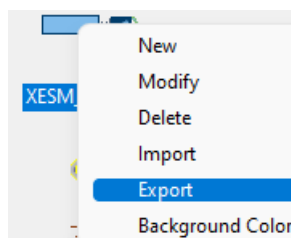
- 2 - In the Symbol Library window, right click on the symbol, select **Modify**.



- Template Design Window appears → Edit Template. Refer to F.3.15.4.
- Press **Ctrl + S** to update all changes from Template to instance or close Template design window to update changes and close **Template** design window.

F.3.15.5 Export Template

- In the Symbol Library window, right click on the symbol, select **Export**



- Select a folder location to save the exported symbol

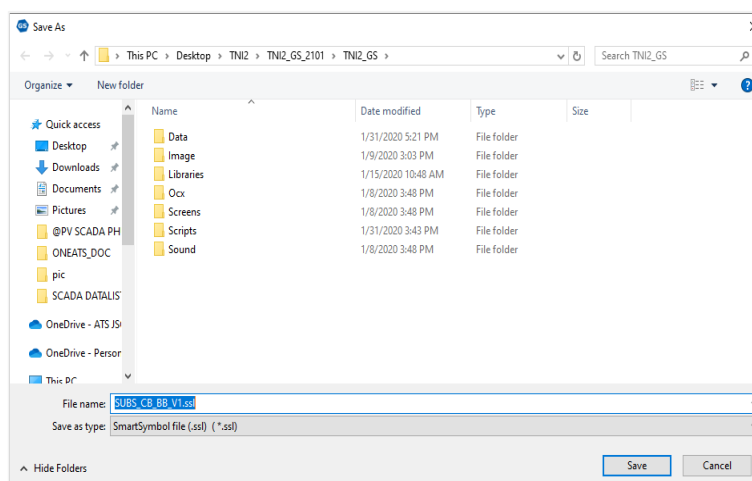
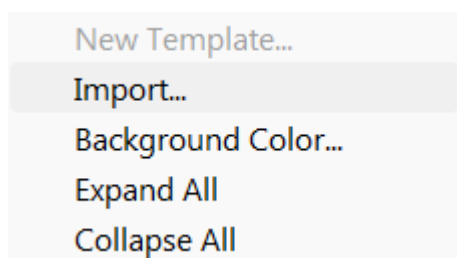


Figure F-200. Export template

The exported *.ssl file has the same format and properties as the *.ssl file in the Libraries folder, so we can use the file in this folder as an export file (convenient in case you want to export many Templates).

F.3.15.6 Import Template

- In the Symbol Library window, right click on the background, select **Import**



- Select the existed file *.ssl to import

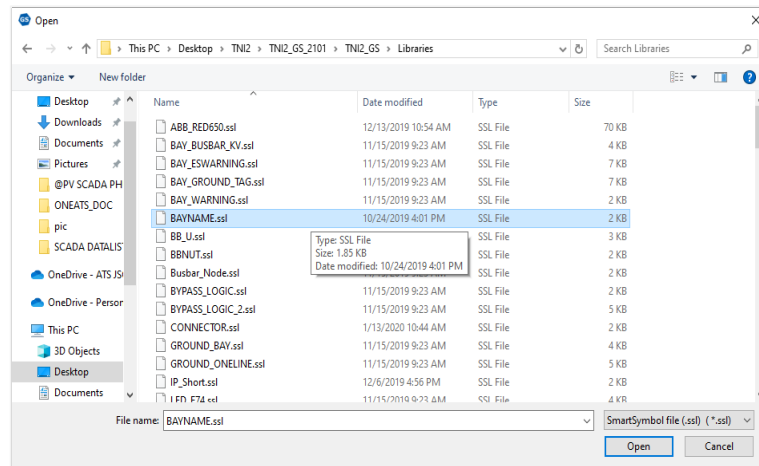


Figure F-201. Import Template

In case you want to import multiple Templates: close the project from Grid Studio Designer, copy the templates (*.ssl) into the Libraries folder, then reopen project.

F.3.15.7 Expand/Collapse All Symbols

In the Symbol Library window, select **Expand All** to view all symbols, select **Collapse All** to group symbols by device type

F.3.15.8 Delete Template

In the Symbol Library window, select the symbol, or right-click on the Template and select **Delete**.

Click Yes in the confirmation window that appears.

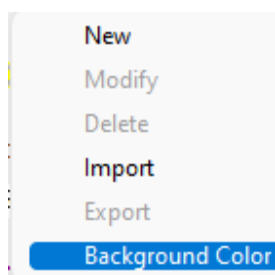
1. Delete Template: When a user deletes a Template from the library, instances built from that Template still exist and are considered "orphaned". → Should consider carefully before deleting Template.
2. Restore the original template for "orphaned" instances: import or create a new Template with the same name as the instance name points to, Right Click the "orphaned" instance and select Edit Symbol, make any changes in the Template and save to update changes.

F.3.15.9 Properties of Template

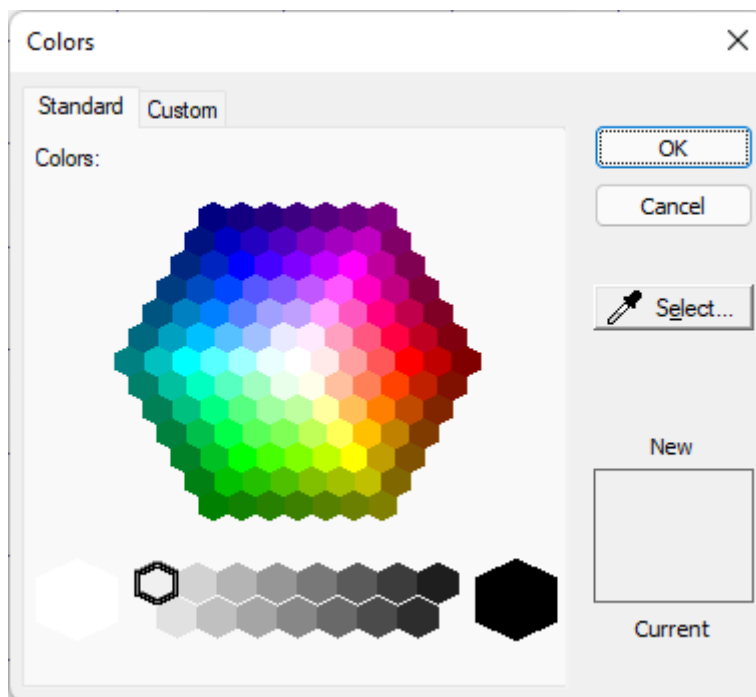
Templates can be replaced tags and text, name as basic objects.

F.3.15.10 Background Color

This function is used to change background color of Template design window.



Right-click on the template, select **Background Color** → Choose the color.



F.3.15.11 Using Template

Using a **Template** is creating instances from a **Template**.

Create and configure information for instance

Open the **Symbol Library** window, select Template, drag and drop it into screen. For Templates with Build-in type, we need to select the Area, Connectivity Node to connect to Connection Points. Connectivity Nodes connect depending on the Area selected.

Example: The Area part for the selected device is D01 (under the 220kV voltage level in the Model tree), the Connection Point section will only allow connection to the **Connectivity Nodes** of the 220kV voltage level.

The object that appears on the screen is an instance created from Template. You can set the parameters for the instance:

- **Name:** Symbol Name

- **Reference Symbol:** Symbol can refer to another Symbol.
- **Layer:** select feature layer in case of more than 1 layer

Select the instance, press Ctrl + E to open the configuration window for input parameters:

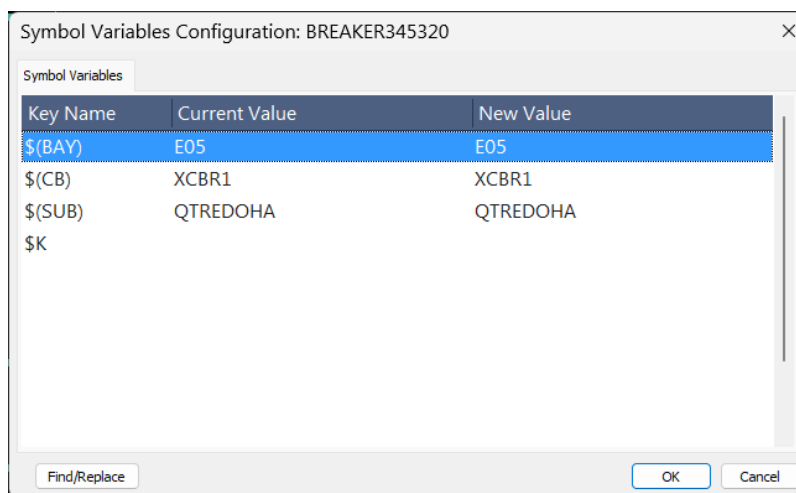


Figure F-202: Configure input parameters for Template

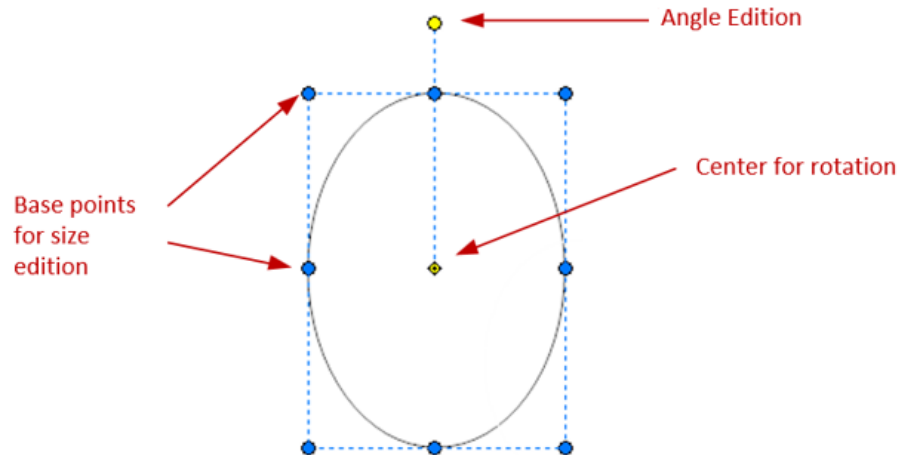
F.3.16. Editing tools in Screen

F.3.16.1 Drawing Graphic Objects

Graphic object is drawn by mouse actions:

- [1] Click to select the object on the toolbar
- [2] Click and drag to draw object on the screen
- [3] Adjust the size using the **Properties** window or interactively at the resizing landmarks on the graphic
- [4] Adjust the center of rotation by clicking the center point, dragging and dropping it to a new position

- [5] Adjust the rotation angle using the **Properties** window or rotating object directly on screen by using mouse pointer.



- [6] Move an object by holding down the left mouse button and dragging it to a new position or enter the coordinates for the object from the Properties panel.
- [7] Flip an object by selecting the object and click flip button on the toolbar
- Support Retain drawing control (b7343 onwards): This mode allows you to continue drawing operations without exiting the current command. You can perform multiple consecutive drawing actions, such as drawing lines, shapes, or details, without having to repeatedly select the tool or command. To exit the current command or drawing mode, press **Esc** or **Right-click** on the screen.

F.3.16.2 Configure Master Properties (Ctrl + Shift + C)

Select the object (ungrouped object), press [Ctrl+Shift+C] key to open Master Properties dialog/window, for example the properties of a rectangle as below:

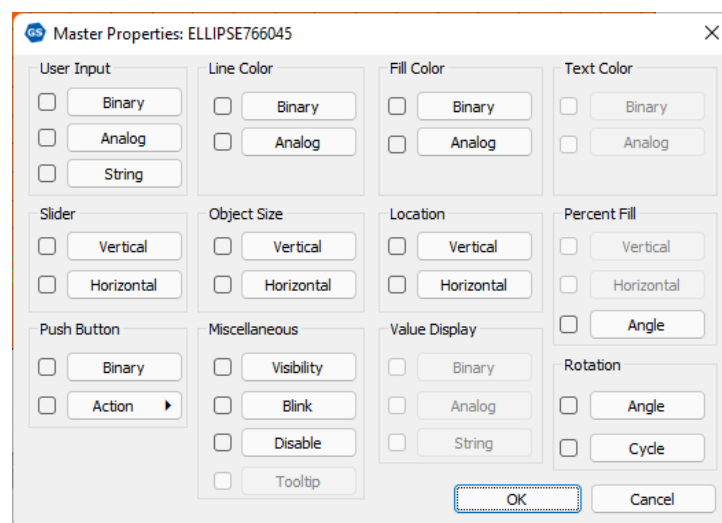


Figure F-203. Attribute of an object

F.3.16.3 Replace Tag (Ctrl+E)

To replace the data/template assigned to an object, select the object and press **Ctrl + E**, the Symbol Variables Configuration dialog appears:

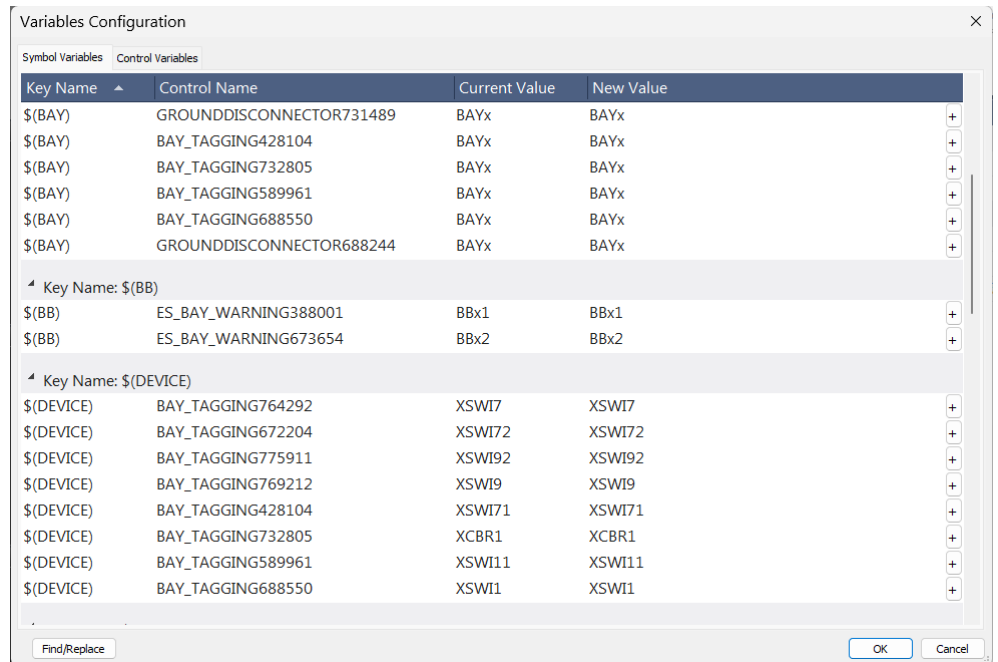


Figure F-204. Replace object's tag

Enter the replacement tag in the **New Value** column and click **Apply/OK** to submit the changes. The **Find/Replace** function is used for searching variables or quick replacement.

F.3.16.4 Replace Text (Ctrl+L)

Press **Ctrl+L** to open Texts Configuration dialog allows to rename the object and replace content inside the object.

At the **Names** tab, click into the row of object name in the New Name column to replace/rename, then click **Apply/OK** button to submit the changes:

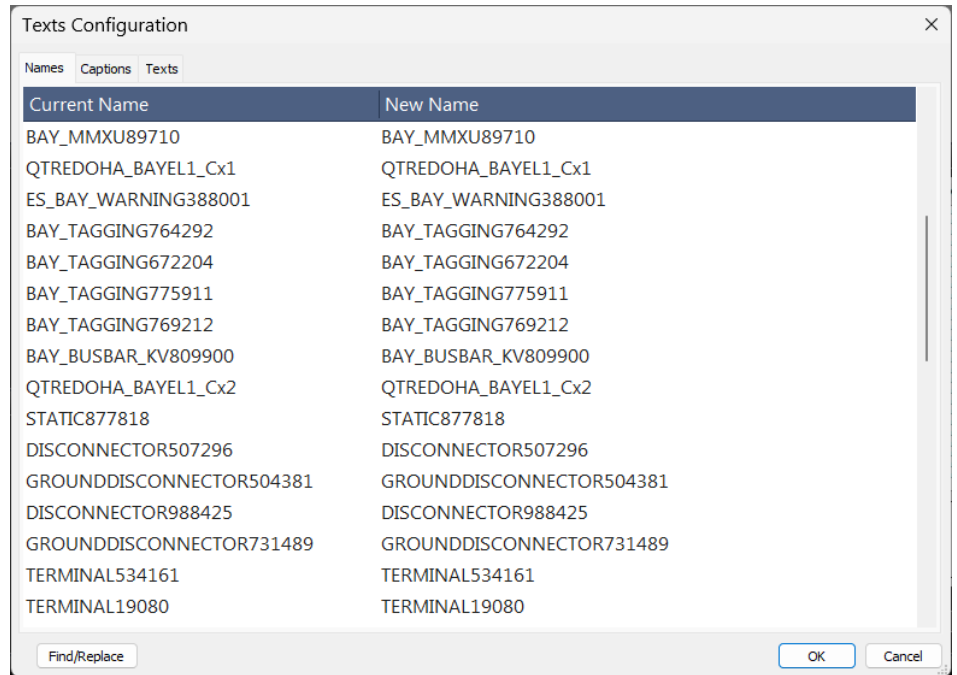


Figure F-205. Replace object name

At the **Captions** tab, current caption of object can be replaced by a new one in New Caption column.

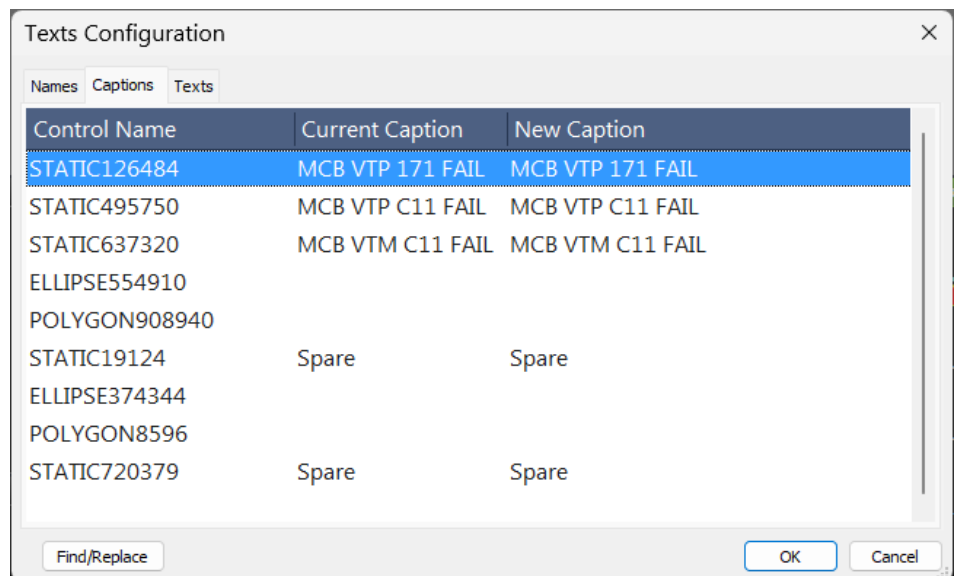


Figure F-206. Replace the caption of the object

The Find/Replace function is useful for searching and replacing large amount of content.

F.3.16.5 Copy/Paste

There are two ways to copy/paste objects:

1: Use keyboard and mouse

- Select the object
- Press the key **Ctrl + C** or right click on the object, then select Copy, or click Copy button  on toolbar to copy
- Left click on a new coordinate
- Press the key Ctrl + V, or right click on the object, then select Paste or click Paste button  on the toolbar to create the new object.

The new object has the same properties as the original object, exclude the electric carrying properties, and its coordinates are the position of left mouse click

2: : Use mouse action

- Press Ctrl key + left mouse click
- Drag and drop in a new coordinates/position on screen to clone a new object
- The new object has the same properties as the original object, excluding the electric carrying properties, and its coordinates are the position of left mouse click

F.3.16.6 Copy electric carrying properties (Copy/Paste Reference)

- With the objects which have electrical properties:
- Select the object
- Press the key **Ctrl + C** or **right click** on the object, then select Copy, or click Copy button  on toolbar to copy
- **Right click and select Paste Reference** to create new object.
- The new object has the same properties as the original object, including the electric carrying properties, and its coordinates are the position of left mouse click

F.3.16.7 Zoom/Pan/Full Screen

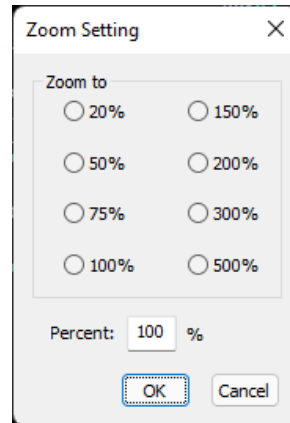


Figure F-207. Zoom/Pan Toolbar

Zoom in, zoom out:

There are 4 ways to zoom in/out:

- 1: use the functions on the toolbar: **Zoom In**  / **Zoom Out**  / **Zoom Rect** 
- 2: use mouse action: hold CTRL + mouse wheel
- 3: Menu **View** → **Zoom Setting**



- 4: use the Status bar
The aspect ratio is displayed on the Status bar in the lower right corner of the screen. User can also drag the slider or tap +, - to change the aspect ratio.



Figure F-208. Change aspect ratio on status bar

HMI provides shortcut keys Ctrl + 3, Ctrl + 2, Ctrl + 1 to zoom in the respective ratios 300%, 200%, 100%.

Moving in the current screen:

- 1: use the function on the toolbar: **Pan** .
- 2: use mouse action: hold the mouse wheel + move the cursor position

Full Screen View mode: hide toolbar and windows title

- Using Menu **View** → **Full Screen** or shortcut key **F11** to view/exit full screen view.

F.3.16.8 Group (Ctrl+G) / Ungroup (Ctrl+Shift+G)

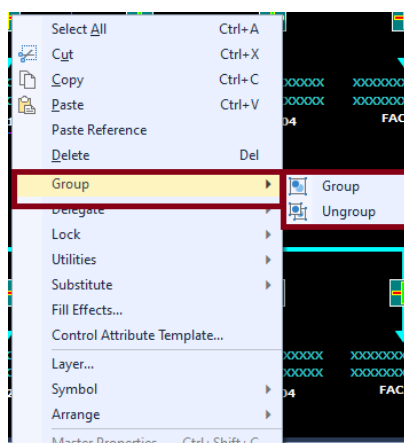
The position ratio of object remains after grouping

To group/ungroup objects:

- Select at least two objects
- Right-Click → **Group** → **Group/Ungroup**.

Or using the shortcut key

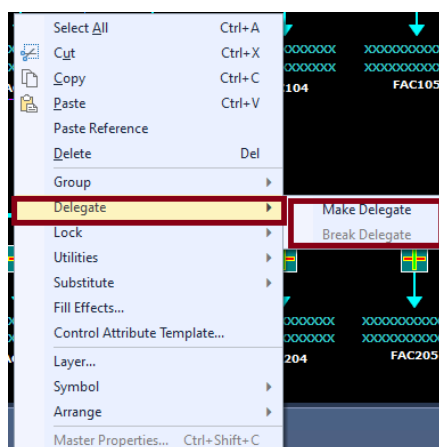
Ctrl+G (=group) / Ctrl+Shift+G (= ungroup)



F.3.16.9 Make/ Break Delegate

Objects remain the position ratio and same master properties after delegating.

To Make/ Break Delegate objects:

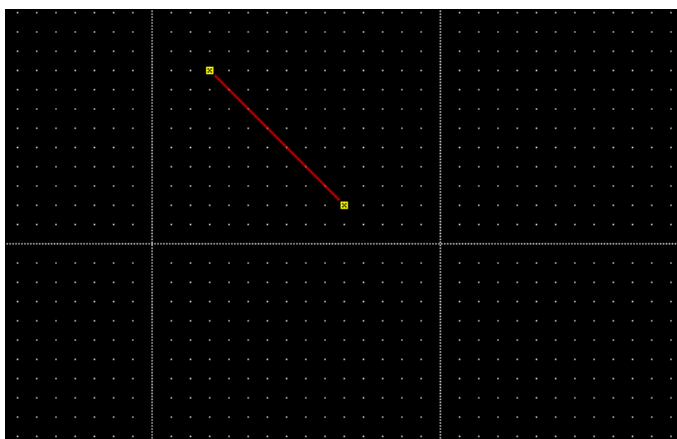


- Select at least two objects
- Right-Click → Delegate → Make Delegate/Break Delegate

F.3.16.10 Fast & accurate drawing techniques

OneATS Grid Studio supports a high speed and accurate drawing technique by using mouse and keyboard, achieving high efficiency in drawing diagrams such as grid diagrams, information diagrams, equipment diagrams, ...

F.3.16.10.1 Grid display and snap



Turn on Grid and Snap mode in the lower right corner of Status bar.. Refer to F.3.1.7

F.3.16.10.2 Move object perpendicularly or 45°

To move the object in perpendicular or 45° directions, hold down the SHIFT button while dragging the mouse to move the object.

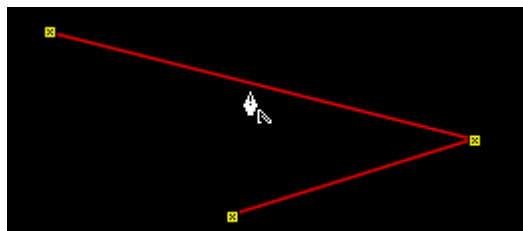
F.3.16.10.3 Draw a 45° or perpendicular line

To draw a line (line, Polyline, AC Line, Control Arrow, Terminal Connection, ...) vertical (90°) or horizontal (0°) or 45° inclined, hold down the Shift button while drawing

F.3.16.10.4 Create Polyline from line segment

With a drawn line segment, you can "broken line" the line segment with mouse operation:

- Click to select the line you want to edit
- Hold Ctrl
- Move the mouse closer to the point where you need to "broken line"
- When the pointer changes to a pen shape 

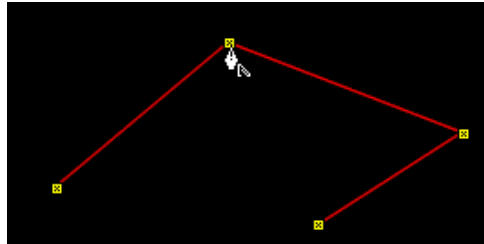


- Click & hold + drag the left mouse button to "broken line"

F.3.16.10.5 Edit Polyline

- Click to select the polyline to be edited. The broken points show up as yellow dots

- Move the mouse close to the point where you need to "broken line" to adjust
- When the pointer changes to a pen shape 



- Click & hold + drag left mouse to edit
- Hold Ctrl + Left Click to delete the break point

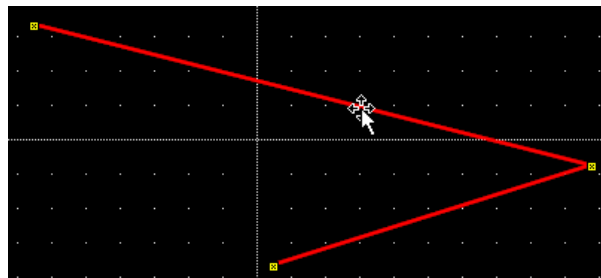
F.3.16.10.6 Automatic Connectivity Node Creation.

When drawing (drag) an Electrical Network object from Symbol Template Library, the application will require to draw a connection from the Terminal to the Connectivity Node. If the Connectivity Node has not been created, the Connectivity Node can be created automatically in the Terminal connection drawing mode by holding the Ctrl + Left-Click button.

F.3.16.10.7 Duplicate (Copy-paste) objects with keyboard + mouse operation

To clone an object, key + mouse operation can be used

- Left-Click to select the object
- Move the mouse pointer over the object, the mouse icon changes to an



- Hold Ctrl + Left-Click, hold + drag to duplicate the object

F.3.16.10.8 Align coordinates of objects

- There are two ways to align coordinates of object:
- 1:
 - Selection of objects
 - Enter X-Y coordinates in the Properties window
- 2: use the alignment tool on the Control Utilities toolbar. Refer to F.3.1.4

F.3.16.11 Replace Tag (Ctrl+E)

To replace the data/template assigned to an object, the user selects the object and presses **Ctrl + E**, a replacement dialog box appears:

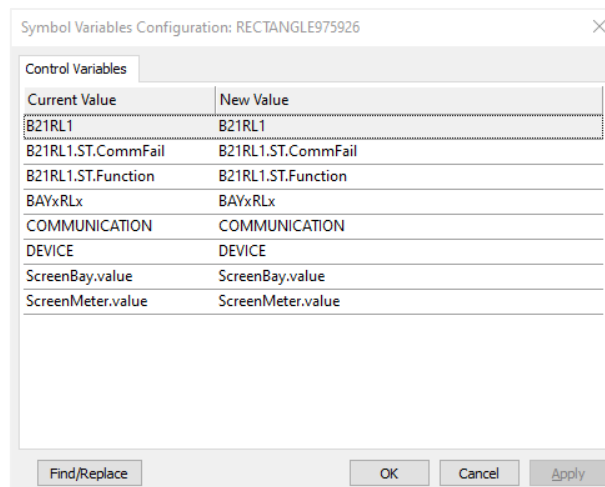


Figure F-209. Replace object's tag

- The user enters the replacement tag in the **New Value** column and clicks **Apply/OK** to confirm. The **Find/Replace** function allows searching for and replacing a string throughout the entire Control Variable table.

F.3.16.12 Searching for graphic objects.

When you need to inspect the list, search for graphical objects, commonly encountered in debugging situations, check the operation of the coloring function, use the Control View tool window. Refer to F.3.1.14. This window provides tools

- List of Network objects with the Power Source property
- List of graphic objects
- Find tools by object's name/ID

F.3.17. Script

Script is a set of command lines that perform a certain purpose in a project.

Script includes the properties of the programming program.

- Script execution conditions.
- Data variables, including parameters, local variables, global variables
- Programming language
- The ability to program “function within function”

For details on scripting languages, refer to section F.5

F.3.17.1 Classification of scripts

OneATS GridStudio includes two Script groups.

- Script Function** is a function that is executed when called by another script.

- **Script Event:** is a function that is executed when an event occurs

Event Script Groups:

- **(Application) Project Events:** events are activated when the **Viewer** module is active.
- **Screen Events:** events are activated when the Screen is displayed (note that the Screen can be displayed at a coordinate that is outside the display range of the physical screen).
- **Key Events:** events are activated when a key or key shortcut is pressed or released.
- **Condition Events:** events are activated when a conditional function (Expression) is true, false, or changes the value.
- **Action Events** are activated when a mouse click occurs on an object (including basic graphics objects, advanced graphics objects, ...)

F.3.17.2 Script Editor

For details on scripting languages, refer to section F.5.

F.3.17.2.1 Script editor windows

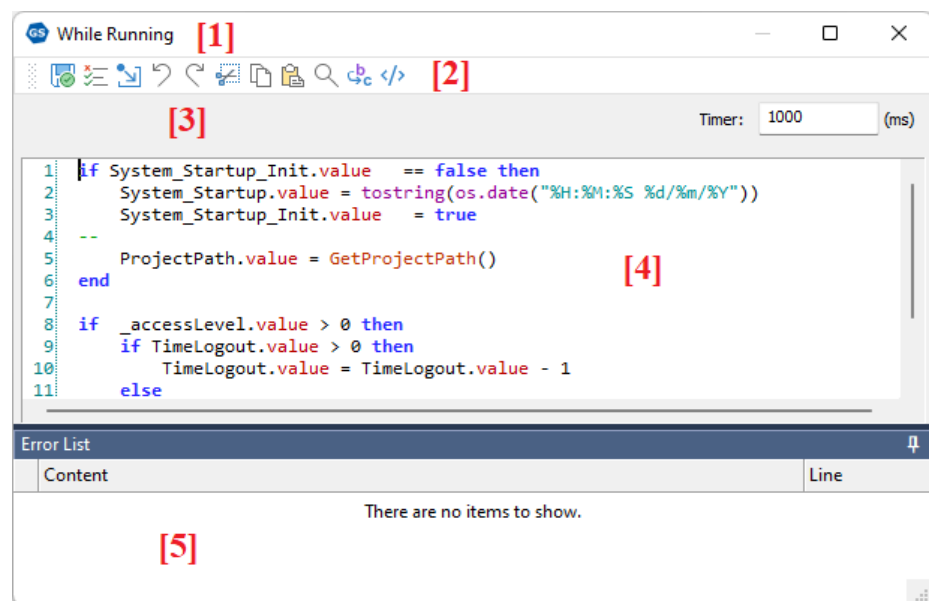


Figure F-210. Script editor windows

- [1]: Title bar
 - Event/ Script name
 - Close 
 - Zoom in  / Zoom out 
- [2]: Script Editor Toolbar
- [3]: Area of input parameters, event type parameters of script
- [4]: Script Content
- [5]: Error List area – script error message when compiling

F.3.17.2.2 Character color in script

```

1 Output(DeviceName)
2 CTRLx.Refer(DeviceName)
3 ScreenConfirm.value = ScreenBay.value
4 --Message(CTRLx.RefName(),0)
5 Output("CTRLx.Refer --> "..CTRLx.RefName().."\\n")
6 HideWindows()
7 ShowScreenAt("CONTROL_PANEL", (ScreenBay.value -1)*1920+760,389)
8 x = math.abs(25)

```

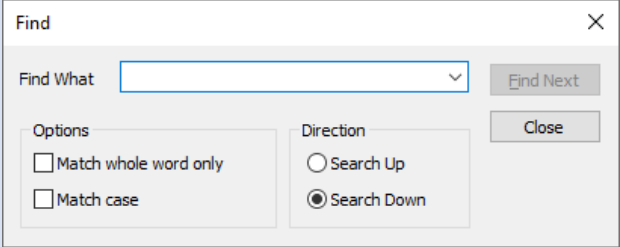
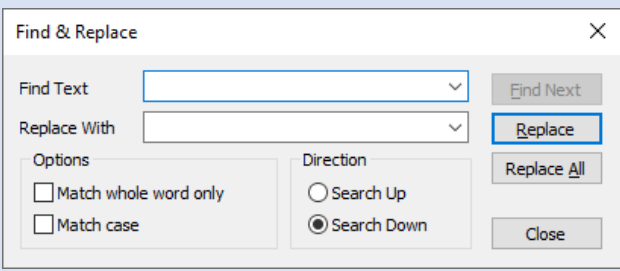
Script editor window supports automatic character coloring according to classification

Color	Character type	Example
Red (Starts with a capital letter)	Grid Studio's built-in function	Output ShowScreenAt ...
Red (Starts with a lowercase letter)	Grid Studio's data properties field	value quality
Green	Comment	//comment
Blue	Key word of Lua programming language	and break do else elseif end ...
Violet	Lua's Built-in Functions	math. abs
Light blue	String format	"Hello world"

F.3.17.2.3 Script Editor Toolbar



Figure F-211. Scripts Editor Toolbar

	Ctrl + S	Save script
	Ctrl + P	Verify / Check script syntax
		Insert Data Point: Open the Data Model View window. Select tag → right -click → Copy key name Paste (Ctrl +V) in the script editor window
		Undo
		Redo
		Cut
		Copy
		Paste
		Find / Find text in the script 
		Replace / Relace text in the script 
		Insert Defined Function

F.3.17.2.4 Copy / Cut / Paste / Find

The operations are similar to those in Notepad's editing window, including tools supported from the toolbar/shortcut/right-click menu.

Function	Toolbar	Shortcut keys	Description
Copy		Ctrl + C	Copy
Cut		Ctrl + X	Cut
Paste		Ctrl + V	Paste
Find		Ctrl + F	Search

F.3.17.2.5 Show suggestions for built-in scripts (Ctrl+Space)

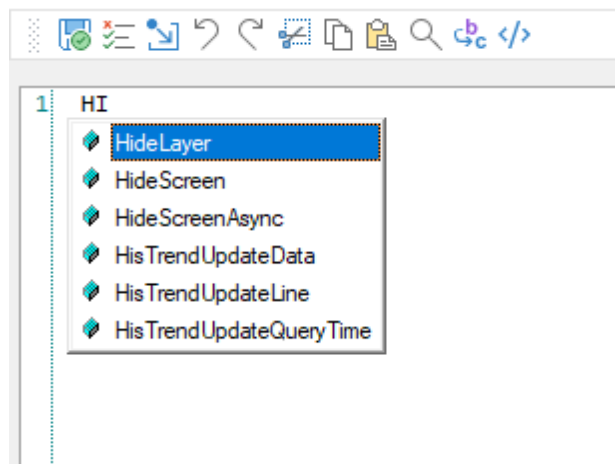
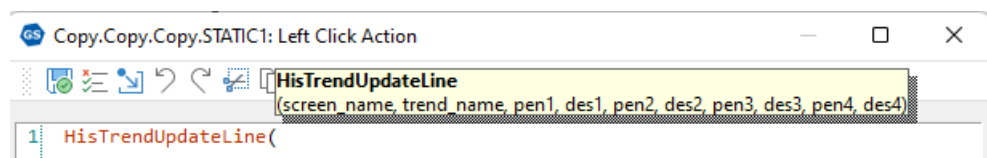


Figure F-212. Show suggestions for built-in scripts

In the script editor window, users can use key shortcuts to edit scripts:

- Use **Ctrl+Space** to display a suggested list of built-in scripts.
- Use arrow keys to select script
- Type **Space** to autofill selected built-in script.
- Continue typing "(" according to the syntax, the application will show a suggested list of script parameters.



F.3.17.2.6 Verify script

- Press the **Verify** button  on the Script toolbar window.
- The system will check the syntax and compile the script. The result will be displayed in the **Error List** window.

F.3.17.2.7 Save or discard changes with a Script

When you work in the Script editor window, or at the end of writing the Script, the user can save or discard the changes.

- Click **Save (Ctrl+S)**  to save the changes and close the Script editor window.
- Click **Cancel**  to ignore the changes and close the Script editor window.

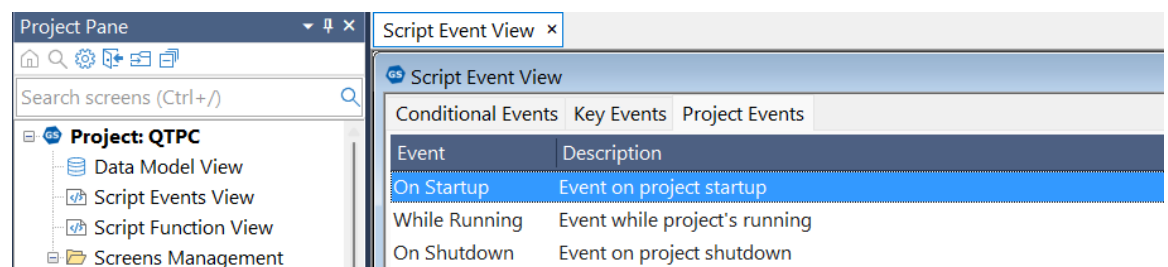
F.3.17.3 (Application) Project Events script

Application Events script has scope in the Grid Viewer. Project Events script type:

APPLICATION SCRIPT	DESCRIPTION
On Start Up	This script is executed once at Viewer initialization.
While Running	This script is executed continuously with a specified frequency (timer) while the Viewer is running.
On Shutdown	This script is only executed once when the Viewer stops working

F.3.17.3.1 Managing Application scripts

- In the Project Pane → **Script Event View** or Menu View → **Event Management** → **Project events**



- Double Click to open (Application) Project script (On Startup, While Running, On Shutdown)

F.3.17.3.2 Configure Application script

- Open the (Application) Project Events script.
- With While Running script, set **timer** parameter – script execution frequency.
- Editor script. Refer to F.3.17.2

F.3.17.3.3 Delete Application script

- Open the Application Project Events script.
- Select the entire Script content → Click delete button to delete.

- Press Save (Ctrl+S)  to save the blank content and close the editor window.

F.3.17.4 Screen Events script

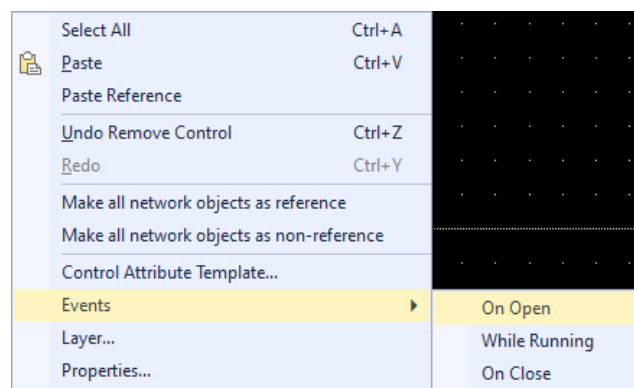
Screen Events scripts are tied to a specific screen, including types:

SCREEN EVENTS SCRIPT	DESCRIPTION
On Open	This script is executed once when the screen is opened
While Running	This script is executed continuously with a specified frequency (timer) while the screen is displayed. (Screen can be displayed at coordinates beyond the limits of the physical screen)
On Close	This script is executed once when the screen is closed.

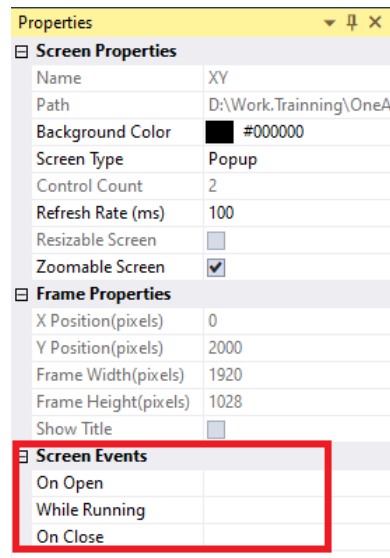
F.3.17.4.1 Managing screen scripts

There are two ways to manage screen events:

- 1: Open the Screen → Right Click on the area that does not contain the object → **Events** → Screen scripts (On Open, While Running, On Close).



- 2: Open the Screen → the Screen Properties window → Double Click on the Screen scripts (On Open, While Running, On Close).



F.3.17.4.2 Screen script Configuration

- Open the Screen Script.
- With Script **While Running**, set **timer** value – script execution frequency.
- Script editor. Refer to F.3.17.2

F.3.17.4.3 Delete Screen script

- Open the Screen Script.
- Select the entire Script content → Press **Delete** key to delete.
- Press Save (Ctrl+S)  to save the blank content and close the editor window.

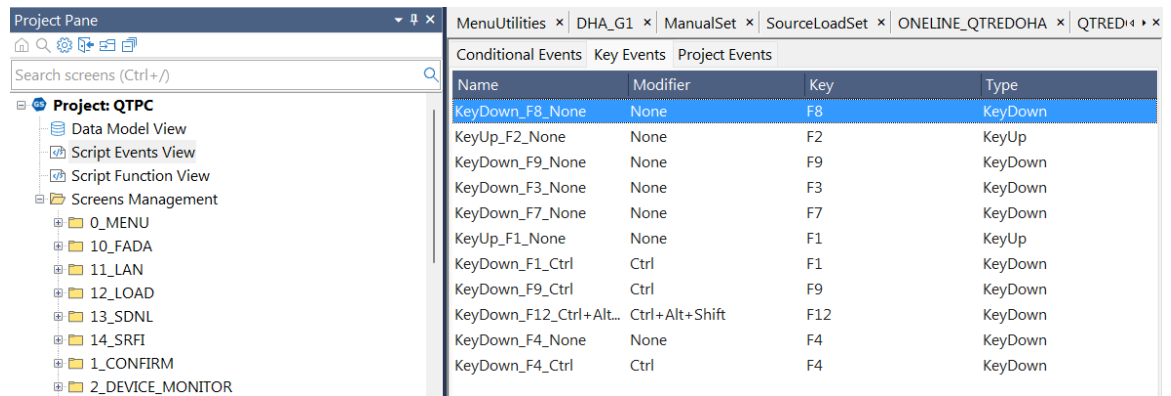
F.3.17.5 Key Events script

Key Events script is executed when there is a keypress effect, including:

KEY SCRIPT	DESCRIPTION
KeyDown	Script to be executed at the time the key is pressed down
KeyUp	Script to be executed when the button is released.

F.3.17.5.1 Managing key scripts

- In the Project Pane → **Script Event View** or Menu View → **Event Management**
- → **Key events**
- Double Click to open the Key script.



F.3.17.5.2 Create a new key event script

- Open key event script management window
- Right-Click in the Key Events window→ select Add

F.3.17.5.3 Configuring a key event script

- Open the key event script.
- Event type parameter configuration.

Modifier: Type: Key:

Properties	Description	
Modifier	Additional key pressed at the same time as the main key	
	None	No simultaneous key press
	Ctrl	Press the Ctrl key at the same time.
	Ctrl+Alt	Press the Ctrl and Alt keys simultaneously.
	Ctrl+Shift	Press the Ctrl and SHIFT keys simultaneously.
	Ctrl+Alt+Shift	Press the Ctrl and Shift and Alt keys simultaneously
Type	Keypad style	
	Key Up	Script to be executed when the button is released.
	Key Down	Script to be executed at the time the key is pressed down

	—
Key	Main key Select one of the keys from F1, F2, ... F12

- Script Editor. Refer to F.3.17.2

F.3.17.5.4 Editing Key Script

- Open key-scripts management window
- Double Click on the script, or Right Click on the script→**Update**
- Editing Scripts. Refer to F.3.17.2

F.3.17.5.5 Delete Key Script

- Open key-scripts management window
- Right Click on the script → **Delete**.

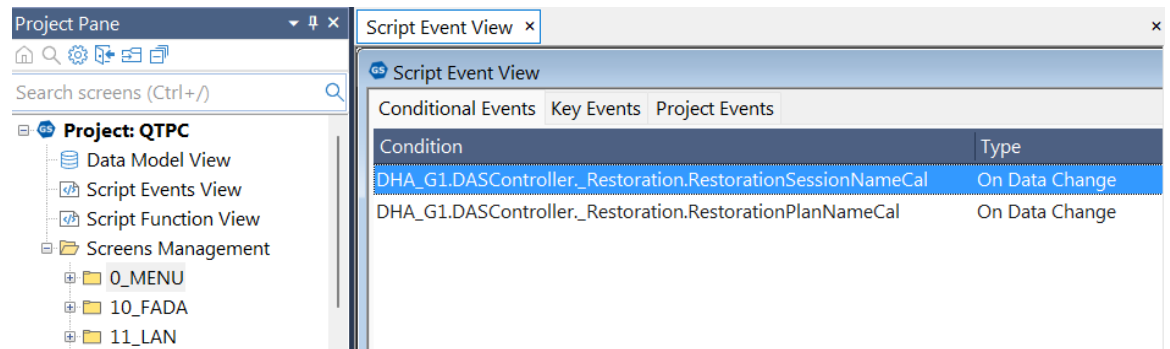
F.3.17.6 Condition Events script

Condition Events script is executed when the **Expression** condition is satisfied:

CONDITION SCRIPT	DESCRIPTION
On True	The script is executed once when the Expression value changes from False to True (Rising Edge).
On False	Script to be executed once when the Expression value changes from True to False (Falling Edge)
While False	This script is executed continuously at a specified period (timer) when the Expression value is False.
While True	This script is executed continuously for a specified period (timer) when the Expression value is True.
On Data Change	The script is executed once when the Expression value changes.

F.3.17.6.1 Managing the Conditional Events scripts

- In the Project Pane → **Script Event View** or Menu View → **Event Management**
- → Conditional Events.



- Double Click to open the **Conditional Events** script.

F.3.17.6.2 Create new **Conditional Events** -scripts

- Open the **Conditional Events** -scripts management window
- Right-Click in the **Conditional Events** window→ select Add

F.3.17.6.3 **Conditional Events** Script Configuration

- Open the Conditional Events script.
- Configure the event type parameter.

Type: Timer: (ms)

Expression

Properties	Description	
Type	Event type	
	On False	Script to be executed once when the Expression value changes from True to False (Falling Edge)
	On True	The script is executed once when the Expression value changes from False to True (Rising Edge).
	While False	This script is executed continuously at a specified period (timer) when the Expression value is False.
	While True	This script is executed continuously for a specified period (timer) when the Expression value is True.
	On Data Change	Script to be executed once when the data point value changes.
	—	

Timer	Script execution frequency.
Expression	Condition expression. Refer to F.3.10.4.2

- Editing Scripts. Refer to F.3.17.2

F.3.17.6.4 Editing Conditional Events Script

- Open the **Conditional Events** -scripts management window
- Double Click on the script, or Right Click on the script→**Update**
- Editing Scripts. Refer to F.3.17.2

F.3.17.6.5 Delete Conditional Events Script

- Open the **Conditional Events** -scripts management window
- Right Click on the script → **Delete**.

F.3.17.7 Action script

Action Script is executed once or cyclically when the user has a mouse action on a specific object.

- Types of action scripts with basic graphics objects: refer to F.3.10.4.1
- Types of action scripts with advanced objects (DataGrid, Chart)

F.3.17.7.1 Open Action script

- With the basic graphic object: in the Master Properties/Push Button - Action properties section of that object. Refer to section F.3.10.4.26
- With advanced graphics objects (Event View, DataGrid, ActiveX, ...): in the object's event script configuration.

F.3.17.7.2 Configuring an Action script

- Open Action -scripts
- Double Click on the script
- Editing Scripts. Refer to F.3.17.2

F.3.17.7.3 Delete Action script

- Open Action Script.
- Select the entire Script content → Click delete button to delete.
- Press Save (Ctrl+S)  to save the blank content and close the editor window.

F.3.17.8 Function Scripts

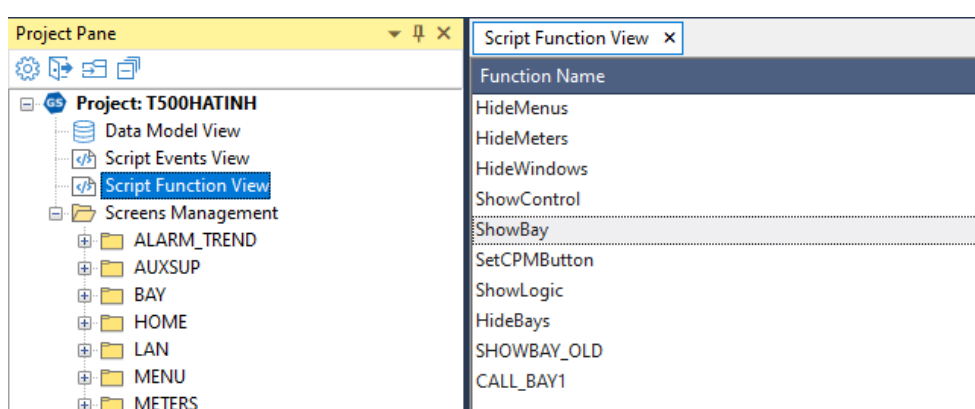
Function script are user-built functions for a specific purpose. These functions can be called from other scripts in the program. Building functions reduces code duplication when editing scripts.

Function script have 3 basic parts:

- Name
- List of parameters
- Script content

F.3.17.8.1 Managing function scripts

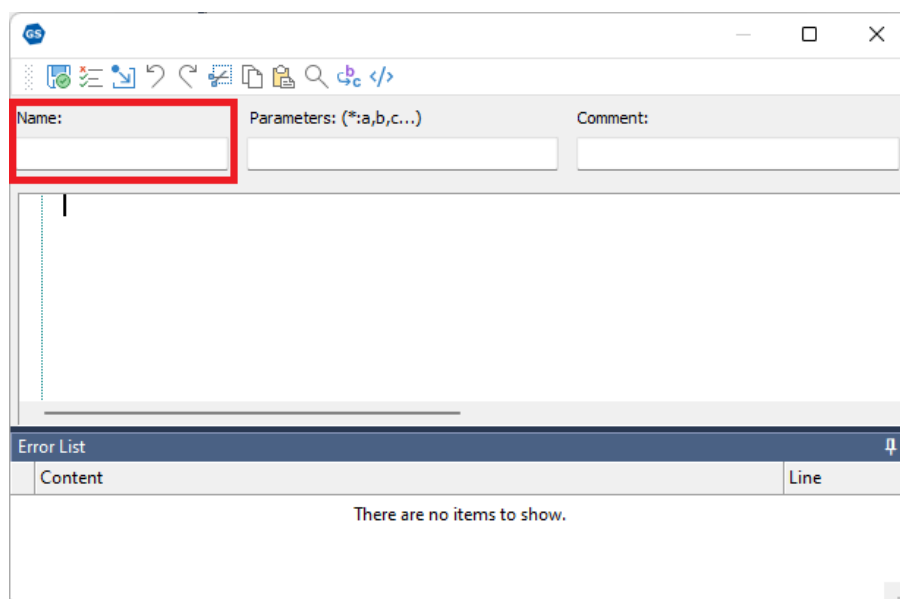
- Project Pane → **Script Function View** or Menu View → **Script Function View**



- Double Click to open the **Function script**

F.3.17.8.2 Create a new function script

- Open the function-scripts management window
- Right-Click in the Function Script window → select Add
- Enter a name for the function scripts



F.3.17.8.3 Function script configuration

Name:	Parameters: (*:a,b,c...)	Comment:

- Open function-scripts
- Configure parameters

Properties	Description
Name	Function script name
Parameters	Parameter of the function script. Parameters are separated by “,” Parameters: (*:a,b,c...) BAYxNAME, ScreenIndex
Comment	Description content of the function script.

- Editing Scripts. Refer to F.3.17.2

F.3.17.8.4 Editing the Function Script

- Open the **Function** -scripts management window
- Double Click on the script, or Right Click on the script→**Update**
- Editing Scripts. Refer to F.3.17.2

F.3.17.8.5 Delete Function Script

- Open the **Function** -scripts management window
- Right Click on the script → **Delete**.

F.3.18. Tagging

During operation, when a piece of equipment is being repaired, maintained, or shut down, warnings should be given to the operator to recognize and limit certain actions. The tagging function provides a list of sign types with different levels.



Rework (4.1.6905 onwards)

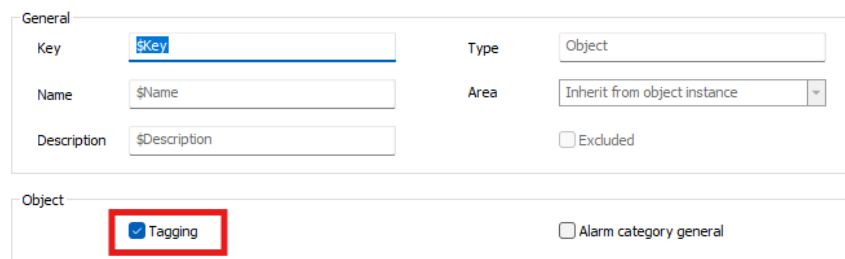
There are 16 types of tagging as shown in the table below, ranked in ascending order of severity:

Tagging Type	Description
Blue Tag /blue (Info) tagging 	Often used for operation instruction purposes Tagging value: 1 Severity: 0
Pink Tagging / (Check) tagging 	Commonly used for directive purposes Tagging value: 2 Severity: 1
Yellow Tagging / (Hot_Line) tagging 	Dangerous Warning Often used for hot-line jobs Tagging value: 3 Severity: 2
Red Tagging	Dangerous Warning

Tagging Type	Description
/ (Prohibit) tagging 	Disable device closure Tagging value: 4 Severity: 3
Note Tagging 	Often used for note-taking purposes Tagging value: 5 Severity: 4
Work Tagging /Work_here tagging 	Often used for the purposes of indicating the working position Tagging value: 6 Severity: 5
Working Tagging 	Usually used for the purposes of instructing the work being carried out Tagging value: 7 Severity: 6
Survey Tagging 	Often used for the purposes of instructing the implementation of surveys Tagging value: 8 Severity: 7
Warning Tagging 	Common warning Tagging value: 9 Severity: 8
Grounded Tagging 	Usually used for the purposes of grounded indications Tagging value: 10 Severity: 9
Maintenance Tagging 	Usually used for the purpose of instructing under maintenance Tagging value: 11 Severity: 10

Tagging Type	Description
Hi-voltage Tagging 	Commonly used for high-voltage indication purposes Tagging value: 12 Severity: 11
Alarm-dis Tagging 	Commonly used for indication purposes with alarm turned off Tagging value: 13 Severity: 12
Not-connected Tagging 	Often used for the purposes of the disconnection indication Tagging value: 14 Severity: 13
Error Tagging 	Often used for the purposes of faulty directives Tagging value: 15 Severity: 14
Fault Tagging 	Often used for the purposes of directives with serious errors Tagging value: 16 Severity: 15

A device can have multiple taggings, each of which can be activated or deactivated independently. The tagging list is a device properties defined in the **Data Model** on the **DataServer** module. To suspend or lower a tagging, the device must have the Tagging attribute enabled.



The screenshot shows the 'Data Model' configuration window. The 'General' tab is active, displaying fields for 'Key' (containing '\$Key'), 'Name' (containing '\$Name'), and 'Description' (containing '\$Description'). The 'Type' is set to 'Object' and the 'Area' is set to 'Inherit from object instance'. The 'Excluded' checkbox is unchecked. The 'Object' tab is also visible, showing the 'Tagging' checkbox checked and highlighted with a red box, and the 'Alarm category general' checkbox unchecked.

Figure F-213. Enable Tagging properties for device in the Data Model

The signage/tagging is stored and synchronized by the DataServer module.

Tagging management operations (viewing, hanging signs, lowering signs) are executed on Grid Viewer through built-in windows (Tagging Manager, Tagging

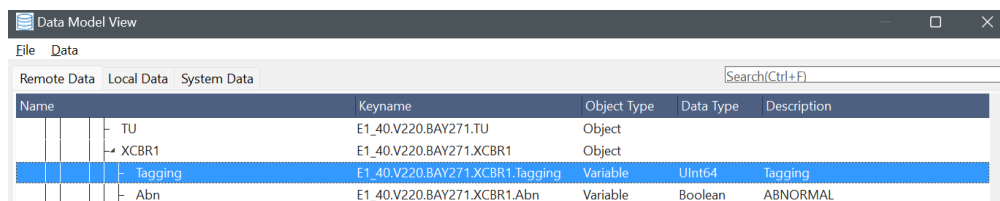
Device) in the application. These windows are displayed by scripts (see details at [Scripts for Tagging](#)).

F.3.18.1 Attributes of a tagging

Properties	Description
Device	The name of the device is tagged with a control.
Device Description	Description of the device.
Tagging Template	Tagging type.
Tagging Template Description	Description of the type of tagging.
Tagging Template Severity	The severity of the type of tagging.
Activated Operator	The name of the operator set tagging.
Activated Time	The time to set tagging.
Activated Comments	Notes when set tagging.
Deactivated Operator	Name of the operator removing the tagging.
Deactivated Time	Time to remove the tagging.
Deactivated Comments	Notes when removing the tagging.
Work Order	Working order code.
Text Filter	Filter the tags by device name.
Tagging Type	Filter by tagging type.
Time Filter	Filter by set time and remove tagging

F.3.18.2 System variables ".Tagging" of the device

Each device on the Data Model with the Tagging option selected will have the Tagging system variable. This variable is used to serve the display of warning signs on the HMI (Grid Studio).

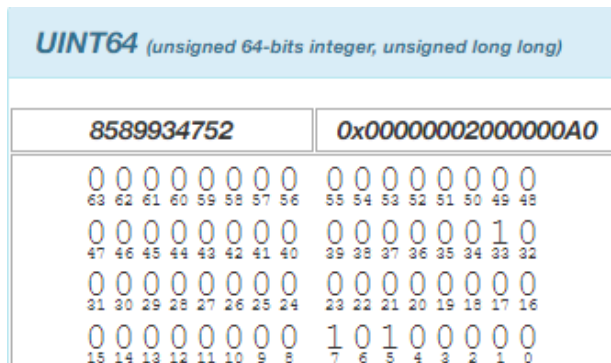


Name	Keyname	Object Type	Data Type	Description
TU	E1_40.V220.BAY271.TU	Object		
XCBR1	E1_40.V220.BAY271.XCBR1	Object		
Tagging	E1_40.V220.BAY271.XCBR1.Tagging	Variable	UInt64	Tagging
Abn	E1_40.V220.BAY271.XCBR1.Abn	Variable	Boolean	ABNORMAL

Figure F-214. System variables - Tagging of the device

Parent node's ".Tagging" variable combines the tagging severity and the amount of tagging child nodes.

The properties of the Object.Tagging

Object.Tagging	
Object Type	Variable
Data Type	uint64
Value	Combination of tagging severity and the amount of tagging child nodes. Severity indicated by the first 32 bits: only the first 16 bits are used to indicate the corresponding mount/remove status for 16 types of tags The number of tagging attached is combined by the last 32 bits For example: The CB is set to 2 tagging with severities are separately 5 and 7, the ".Tagging" variable has a decimal value of 8589934752. 

F.3.18.3 Tagging Manager

Grid Studio supports scripts to interact with Tagging (Manage, Set, Remove). For detail, refer to F.5.7.8

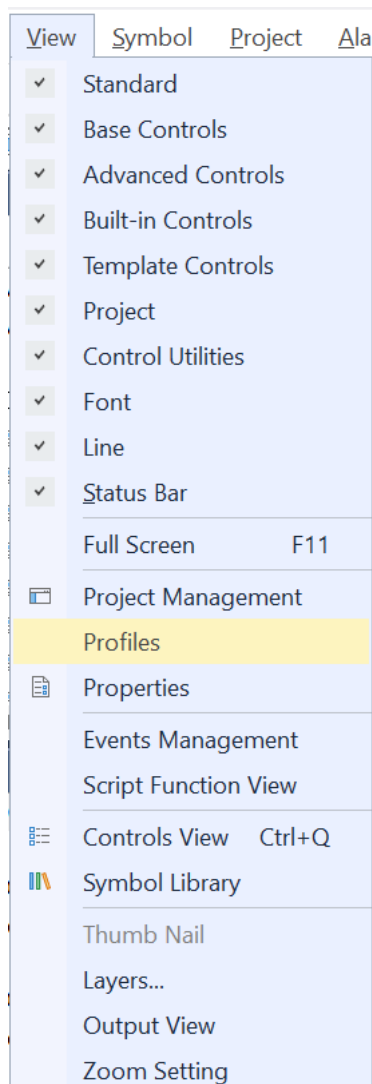
F.3.19. Collaboration

Unavailable

F.3.20. Profile

Profile includes a Data Model partition registered to the configured AOR group partition and a list of screens that can be displayed on the Viewer. Users can launch each profile separately.

F.3.20.1 Manage Profile



From the menubar, selects **View >> Profile** to display to **Profile** window.

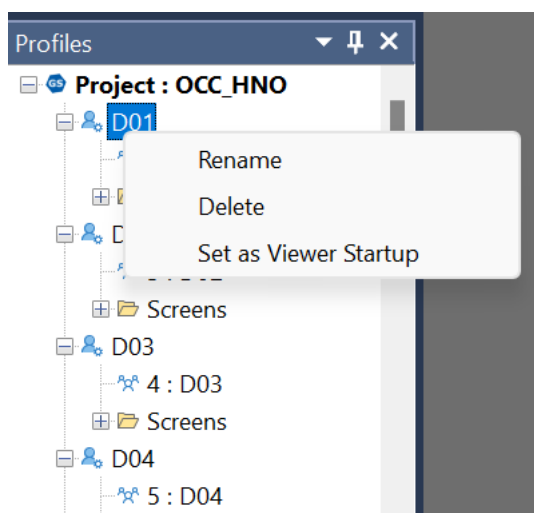


Figure F-215: Profile window

F.3.20.2 Create Profile

- Right Click on **Project** (Default Profile) in profile window→ **New Profile**.

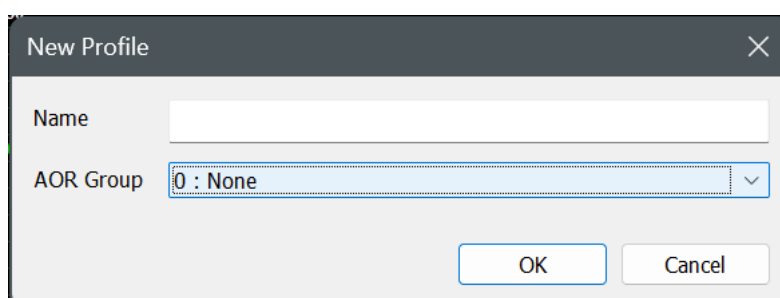


Figure F-216: New Profile creation window

Parameter	Description
Name	Profile name
AOR Group	Select AOR Group created on Data Model

- Click OK to appear the New Profile window.

F.3.20.3 Select Screen

- Right Click on **Screen folder**→ **Select**.

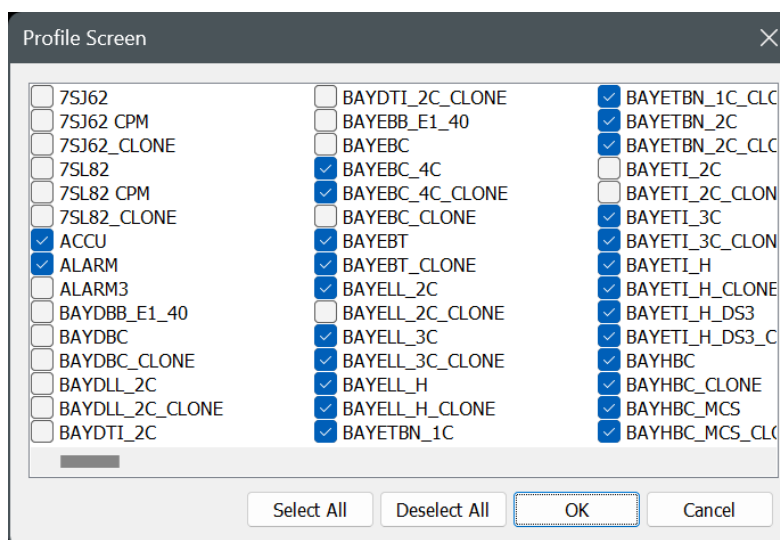


Figure F-217: Profile screen window

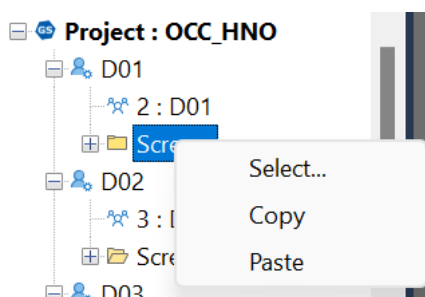
- Click OK displays on Profile screen window.

When deleting a screen in Screens Manager, the screen in screen list of Profile is also deleted

F.3.20.4 Copy/Paste Screen List

List of screens are allowed to be copied into other profiles

- Right Click on **Screen** folder→ **Copy**.

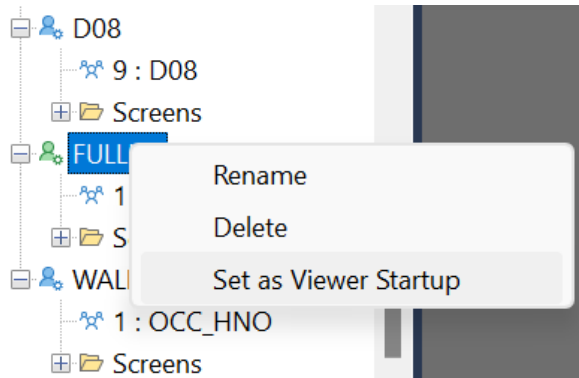


- Right Click on **Screen** folder of other profile→ **Paste**.
→ The screens are copied to Screen folder of other profile.

F.3.20.5 Configure startup Profile

Configure the profile to run when starting the Viewer

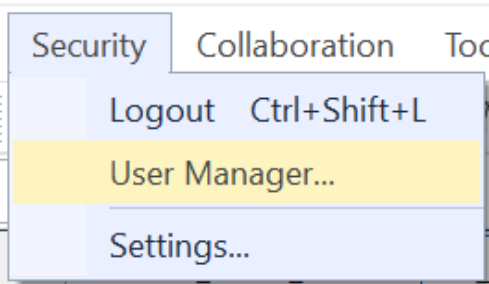
- Right Click on **Profile name** → **Set as Viewer Startup**.



F.3.21. User Management

F.3.21.1 Add new users

From **Toolbar** → **User Manager**.



The **User Manager** window appears. **File** → **New user**.

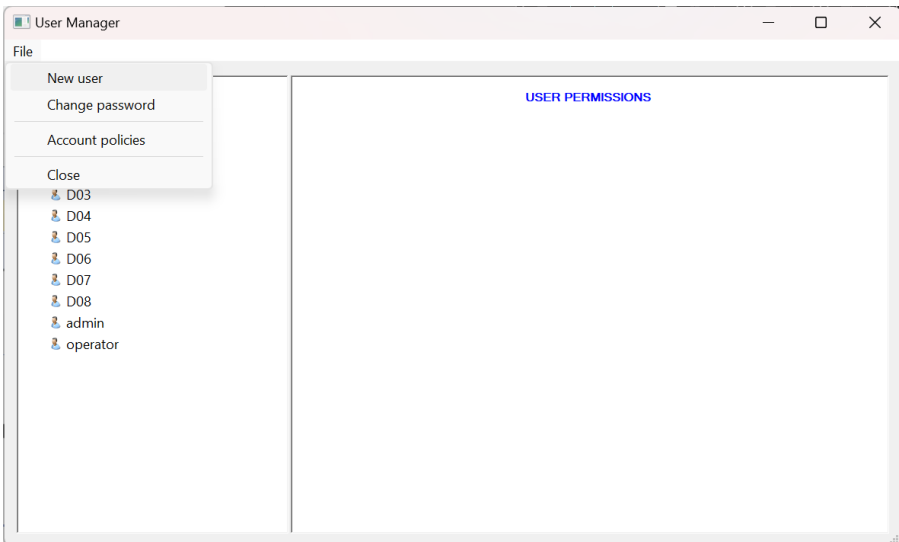


Figure F-218. User Manager Dialog

The **New User** window appears.

Fill in information related to the user:

- **User name.**
- **Password:** password to access.
- **Confirm password:** confirm password match.
- **User type.**
- **Account is disabled:** disable user (database cannot be retrieved).
- **User must change password at next logon:** ask the user to change the password on the next login.

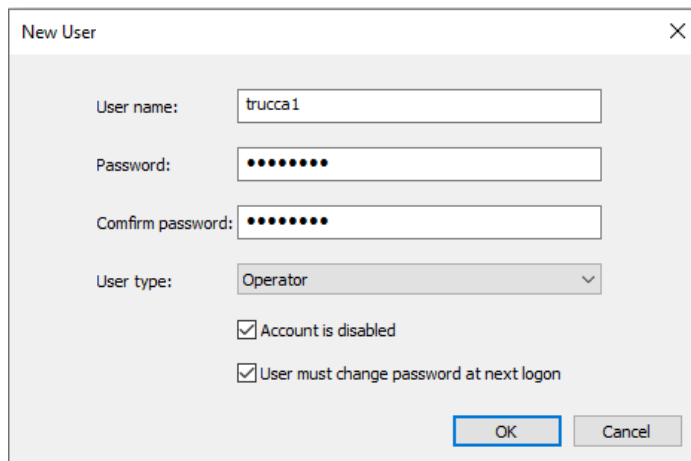


Figure F-219. Create user information

Can use script to open User Management window on Viewer. Refer to F.5.7.1.3

F.3.21.2 Decentralization

Open the window **User Manager**, click on the user who wants to decentralize.

Disable user: Yes/No - ban/allow users.

Access Level: fill integers in ranges 0 – 255.

DataEditor: decentralize users to DataEditor (**View Only/View & Write**)

HMI Designer: *Unavailable.*

Access to FEP: decentralize users to FEP (**YES/NO**)

The table below helps decentralize control to users in AOR (Area of Responsibility Groups. Refer to *ChapterC.OneATS DE-Project Management – AOR management*)

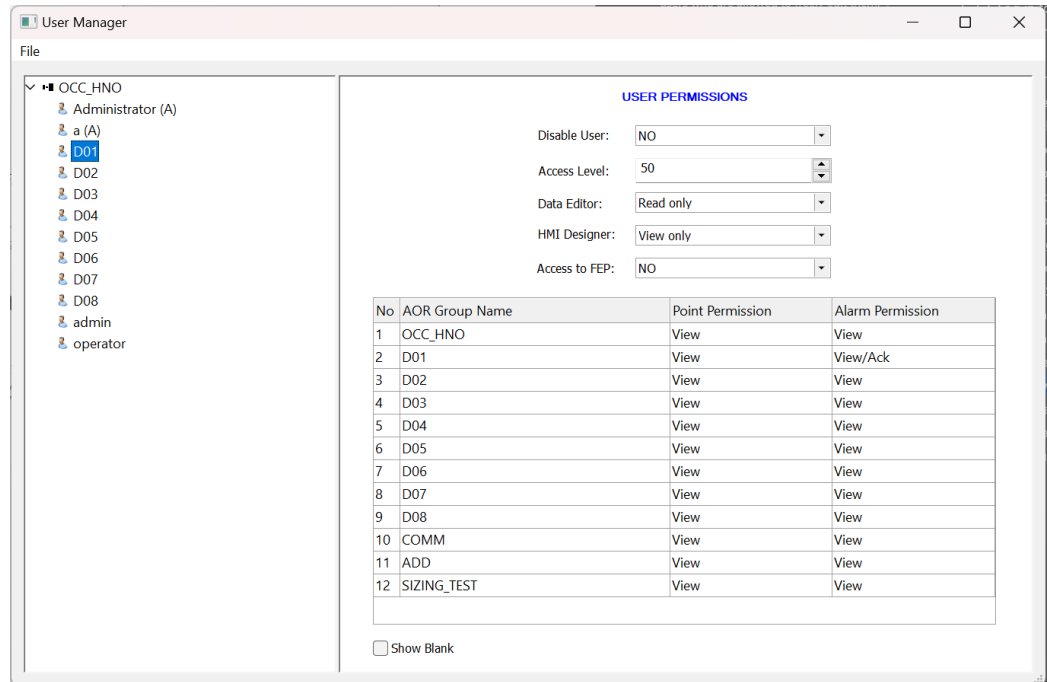


Figure F-220. User decentralization

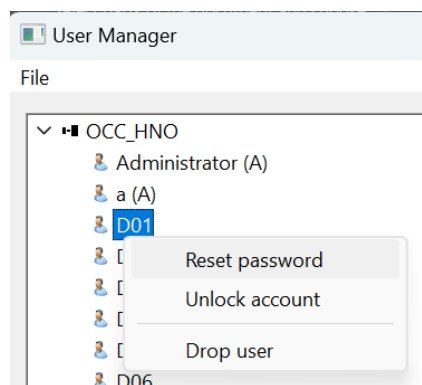
AOR Group Name: List AOR group

Point Permission: user's permission to the data point in AOR (**View/Control**).

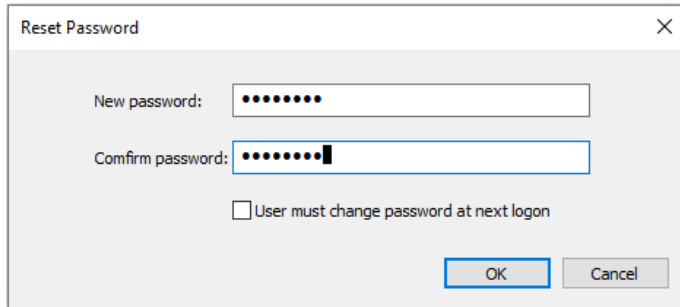
Alarm Permission: user's permission to the alarm in AOR (**View/Ack/Delete**).

F.3.21.3 Reset password

Open the window **User Manager**, click Right Click select the username, select **Reset password**.



Enter a password and confirm a new password.

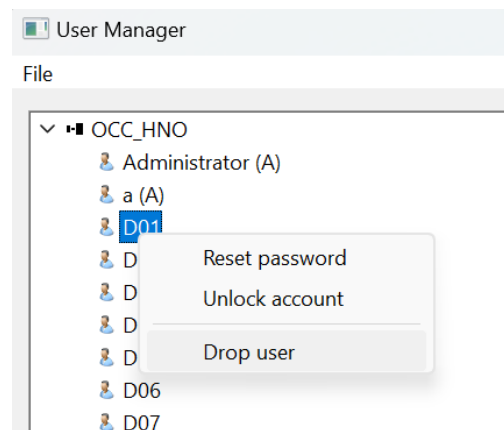


Reset Password dialog box. It contains two password input fields: "New password:" and "Confirm password:". Below the fields is a checkbox labeled "User must change password at next login". At the bottom right are "OK" and "Cancel" buttons.

Figure F-221. Reset password

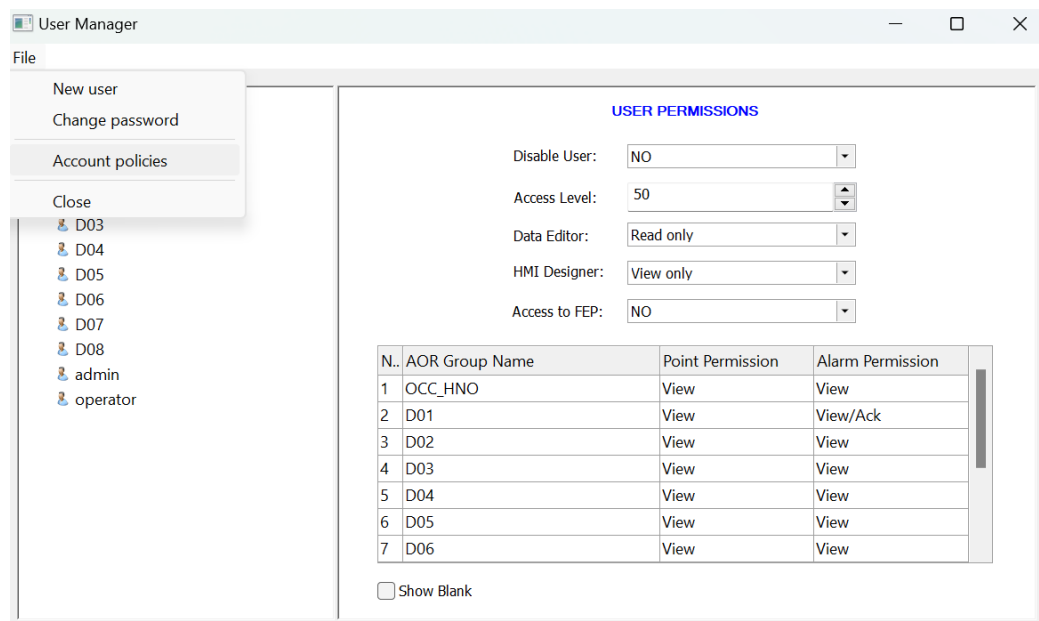
F.3.21.4 Delete users

Open the window **User Manager**, click Right Click select the username, select **Drop user**.



F.3.21.5 Account policies

The **User Manager** window appears. **File** → **Account Policies**.



User Manager window. The "File" menu is open, showing options: "New user", "Change password", "Account policies", and "Close". The "Account policies" option is selected. The main area displays "USER PERMISSIONS" settings and a table of user permissions.

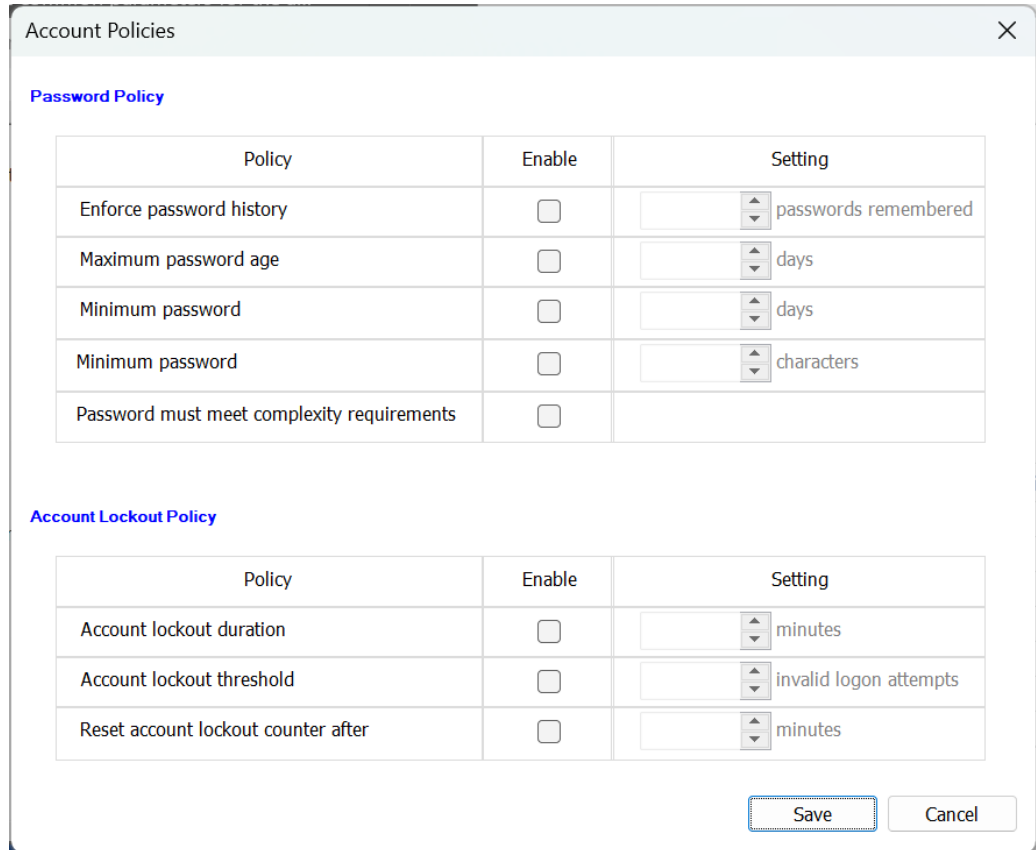
USER PERMISSIONS

Disable User: NO
Access Level: 50
Data Editor: Read only
HMI Designer: View only
Access to FEP: NO

N.	AOR Group Name	Point Permission	Alarm Permission
1	OCC_HNO	View	View
2	D01	View	View/Ack
3	D02	View	View
4	D03	View	View
5	D04	View	View
6	D05	View	View
7	D06	View	View

☐ Show Blank

Figure F-222. User Manager Dialog



Account Policies

Password Policy

Policy	Enable	Setting
Enforce password history	☐	passwords remembered
Maximum password age	☐	days
Minimum password	☐	days
Minimum password	☐	characters
Password must meet complexity requirements	☐	

Account Lockout Policy

Policy	Enable	Setting
Account lockout duration	☐	minutes
Account lockout threshold	☐	invalid logon attempts
Reset account lockout counter after	☐	minutes

Save **Cancel**

Figure F-223. Account Policies Dialog

Enforce password history: Avoid users using the same password more than once. For example, when set to 5 times, the first 5 times to change the password cannot be the same as the original password.

Maximum password age (MAPA): The maximum time in days a password can be used before the system asks the user to change it, range 0-999. Set = 0 password never expires.

Minimum password age (MIPA): The minimum time in days a password can be used before the system asks the user to change it, range 0-998. MIPA < MAPA unless MAPA = 0.

Minimum password length: Minimum password length.

Password must meet complexity requirements: Requires a strong level of password security
Account lockout duration: the time to recover locked accounts after a failed login exceeds a certain amount

Account lockout threshold: Number of failed login attempts --- > Temporary account lockout.

Reset account lockout counter after: Reset account lockout counter after a period of time.

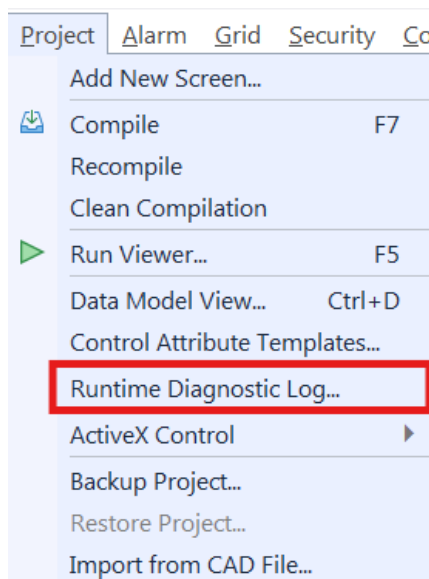
F.4. OneATS Grid Viewer

F.4.1. Configuring and Deploying Viewer

OneATS Grid Viewer is a module that runs as a Desktop application

F.4.1.1 Configure Error Log in OneATS Grid Viewer

From the Designer environment/menu bar, select Project, in the user drop-down list there are options:



Runtime Diagnostic Log: warnings, error messages during runtime. Users can not allow log to the file by unchecking Enable Diagnostic Log or allow logging to the file with levels: Debug, Information, Warning, Error, Critical

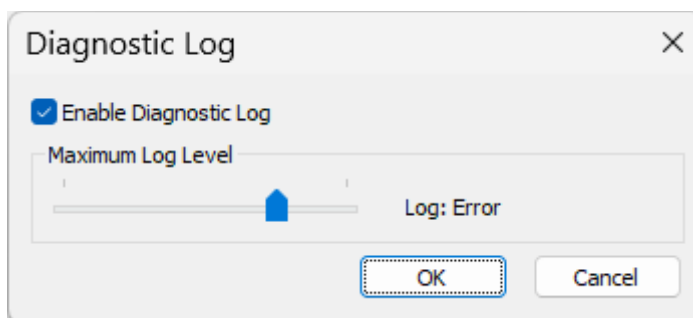


Figure F-224. Diagnostic Log Configuration Window

The runtime log file named project_name.dlog is stored in the project directory.

F.4.1.2 Run OneATS Grid Viewer

F.4.1.2.1 Run OneATS Grid Viewer from project file

Left Click the *.hrt file of the project to run the OneATS Grid Viewer module

F.4.1.2.2 Run OneATS Grid Viewer with Windows startup

In the Project Properties Generals window, launch Grid Viewer on start-up and select the parameters at startup. Reference F.3.3.7.1

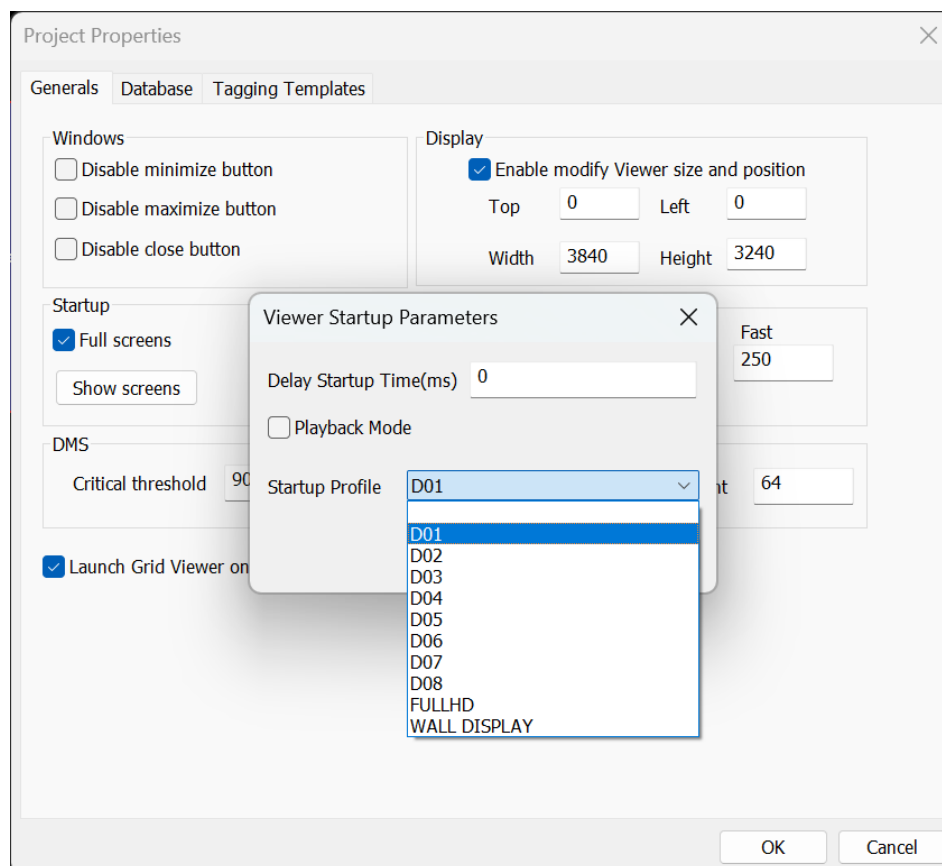


Figure F-225. Startup Parameters

F.4.1.2.3 Run OneATS Grid Viewer from Designer

To switch from Designer mode to Viewer mode, on the menubar select **Project / Run Viewer**; or use the keyboard shortcut **F5**.

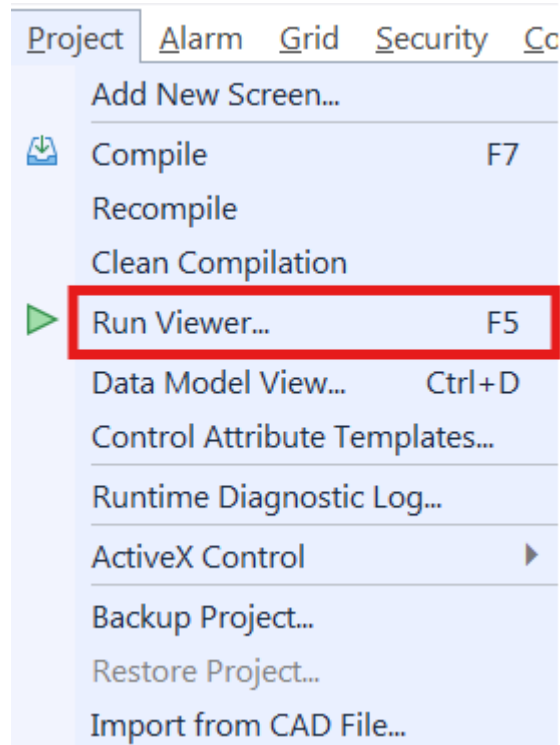
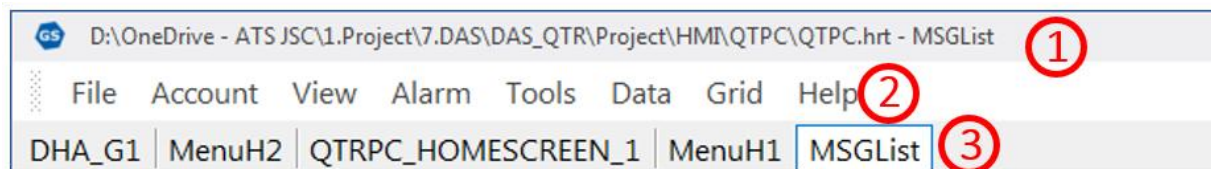


Figure F-226. Run OneATS Grid Viewer

F.4.2. Function menu of OneATS Grid Viewer



Press F11 key to change Menubar display/hide mode for OneATS Grid Viewer

[1]	Title bar
[2]	Menu bar
[3]	Screen

F.4.2.1 Title bar



- File Path of the Project - Name of the Active Screen
- Button: Minimize / Hide task bar -

- Button: Maximize / Enlarge to full screen - 
- Button: Restore Down / Zoom out - 
- Button: Close application- 

These buttons can be set to Enable/disable via Project Properties in Grid Designer

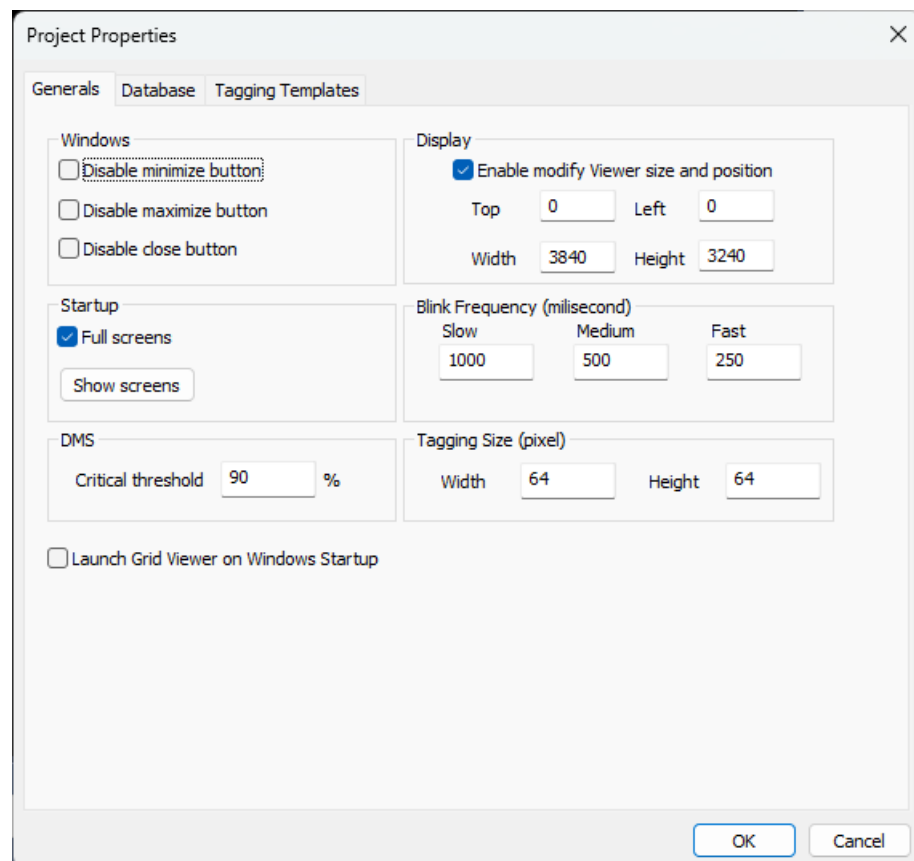
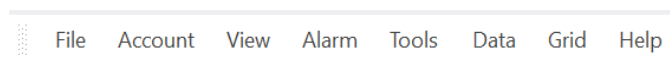


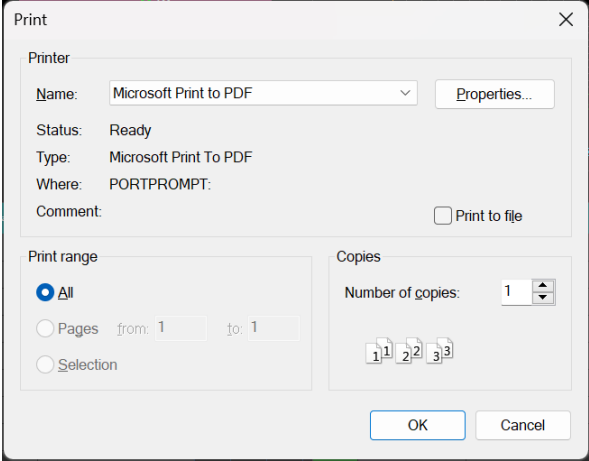
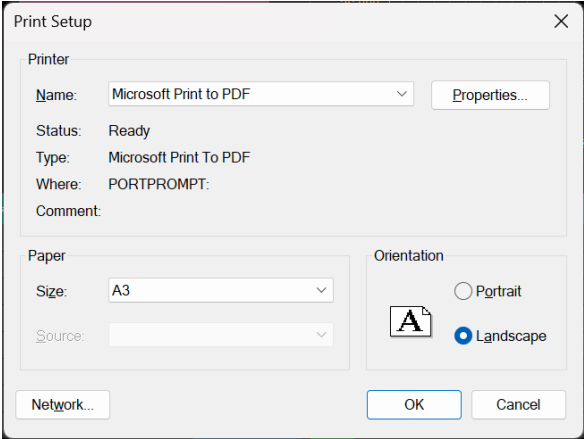
Figure F-227. Project Properties

F.4.2.2 Menu bar

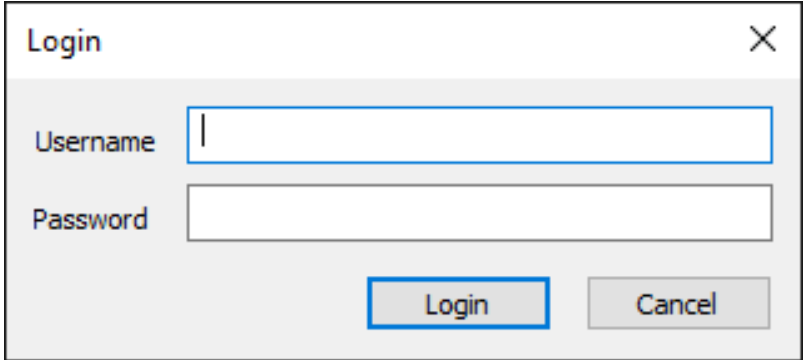


F.4.2.2.1 Menu – File

Sub Menu	Key shortcut	Description
Print	Ctrl+P	Print as setting

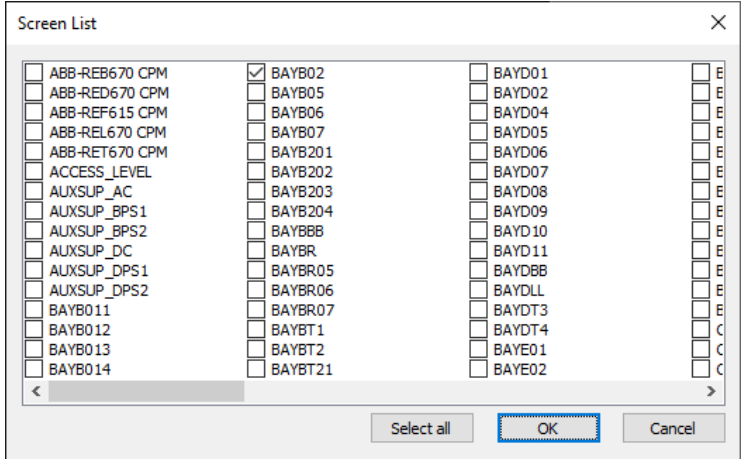
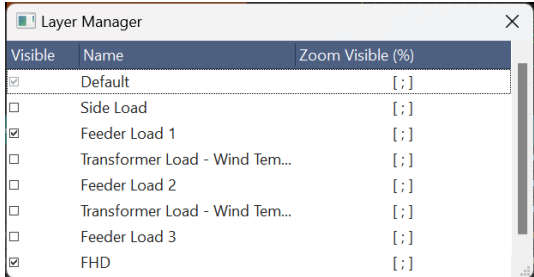
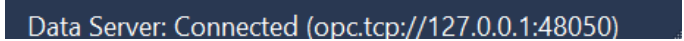
		
Print Setup		Setting Printer, Paper, Orientation 
Exit		Close the application

F.4.2.2.2 Menu – Account

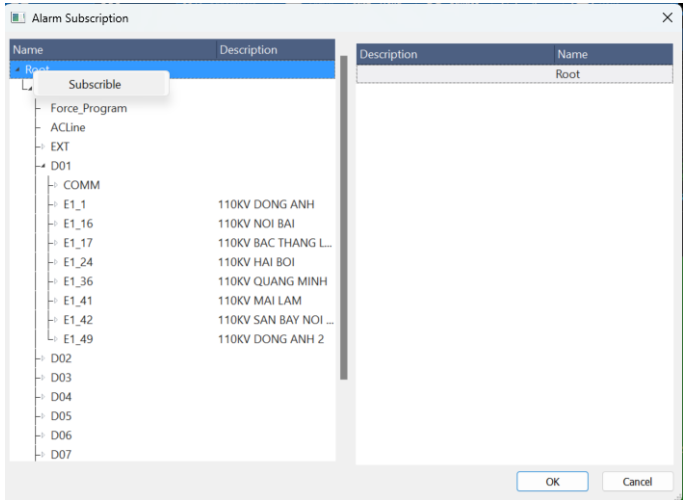
Sub Menu	Key shortcut	Description
Log in	Ctrl+L	Login 

Log out	Ctrl+L	Logout
---------	--------	--------

F.4.2.2.3 Menu – View

Sub Menu	Key shortcut	Description
Screen List	Ctrl+O	Open list of screens.  Open the screen: select screen → Click OK
Full Screen	F11	Switch full screen mode, hide title bar, Menu, Screen Tabs
Layer...	F4	Show/Hide layer in active screen 
Status bar		Display the connection status bar between the Viewer and the DataServer. 

F.4.2.2.1 Menu – Alarm

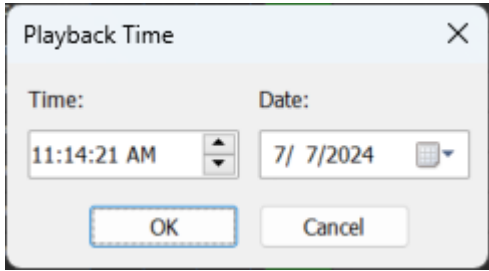
Sub Menu	Key shortcut	Description
Alarm → Subscription Node		List of nodes on DataModel that Grid Viewer subscribes to receive Alarm. 
Alarm → Sound Configuration		View Sound Alarm configuration. Enable/Disable alarm sound.
Alarm Popup		View Alarm, Popup configuration.

F.4.2.2.2 Menu – Tools

Sub Menu	Key shortcut	Description
Debug Tool	Ctrl + Shift + D	Open the debug window. Refer to F.4.3

F.4.2.2.3 Menu – Data

Sub Menu	Key shortcut	Description
Stop Realtime		Switch to past data mode (HIS Server data connection).

Playback Time...		Send command to query data in Playback Mode at a set time. 
------------------	--	--

F.4.2.2.4 Menu – Grid

Sub Menu	Key shortcut	Description
Coloring → Setting		Configure colors in the coloring function. See more details in F.6
Coloring → Enable		Enable/Disable coloring function.
Calculation		<i>Unavailable</i>
Results		<i>Unavailable</i>
Export Network Model		<i>Unavailable</i>

F.4.2.2.5 Menu – Help

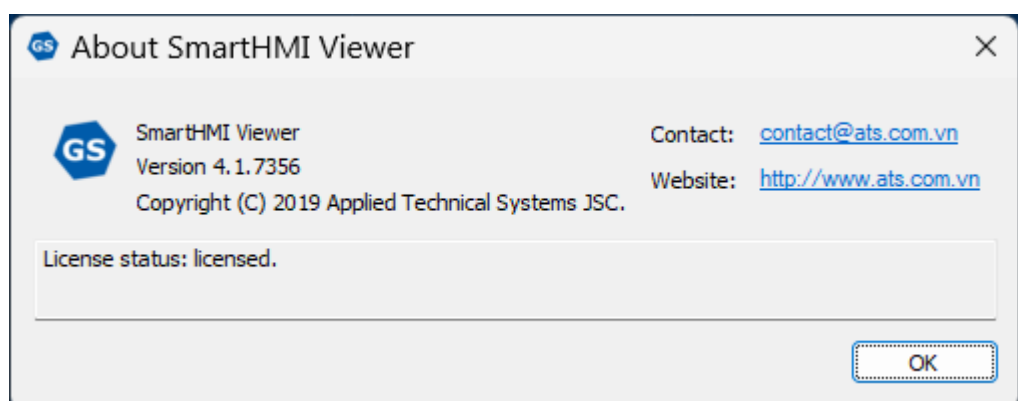
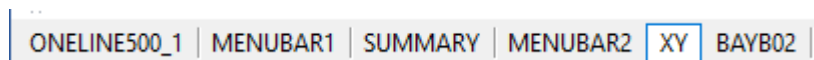


Figure F-228. About Grid Studio Viewer

Information about:

- Version: software version
- License Status
- Other information about the software

F.4.2.3 Screen bar



- Click to switch the Screen

The screen can be displayed at coordinates outside the screen's viewport according to the design in the Grid Designer.

F.4.3. Login on the Grid Viewer environment

In the Grid Viewer environment, login can be done through the Account menu, or the buttons with the login script installed are designed in the Designer environment.

The login operation from the **Viewer** environment is controlled by the user/password via the security function of the **DataServer** module and the **Database MongoDB** module.

F.4.4. Check data with Debug Tool

Grid Viewer supports Debug Tool to check the value of tags, expressions, test scripts.

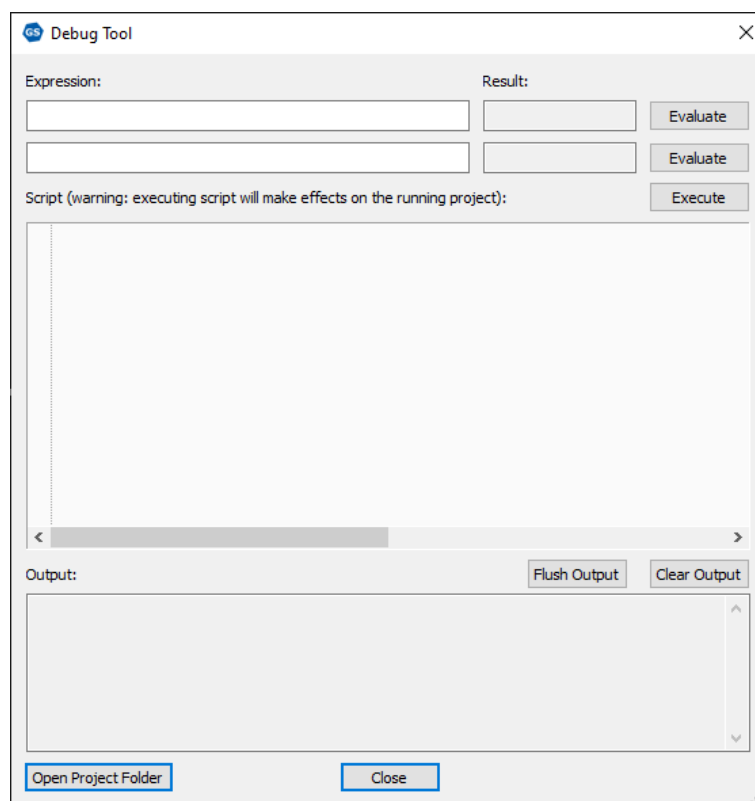


Figure F-229: The Debug Tool window

Open the Debug Tool: Menu → Tool → Debug Tool

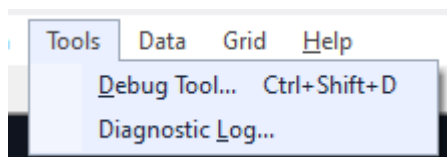
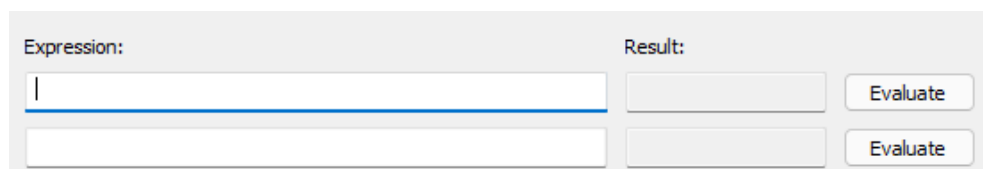


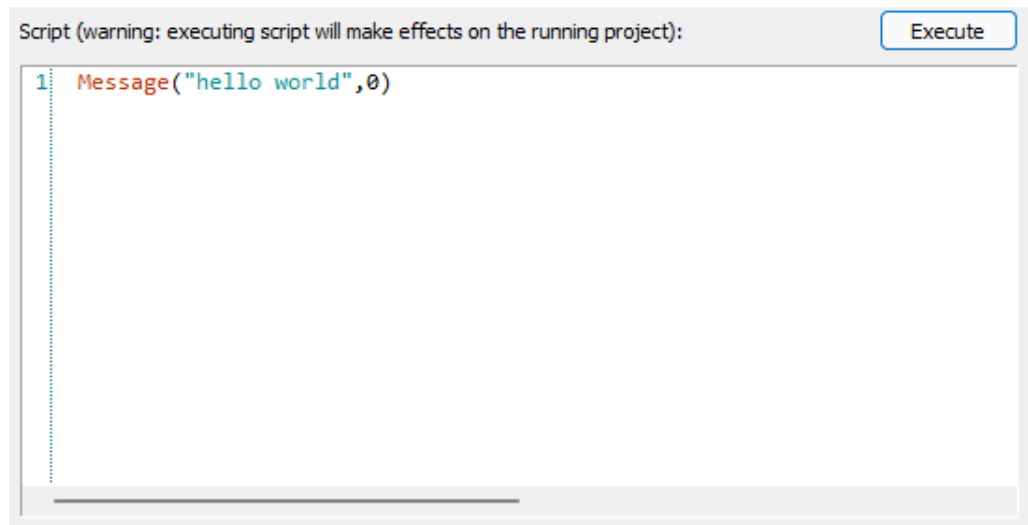
Figure F-230: Open the Debug Tool

F.4.4.1 Check the Expression value



- Enter expression values. Syntax expression refer to F.3.10.4.2
- Click Evaluate button
- The results are displayed in the Result box

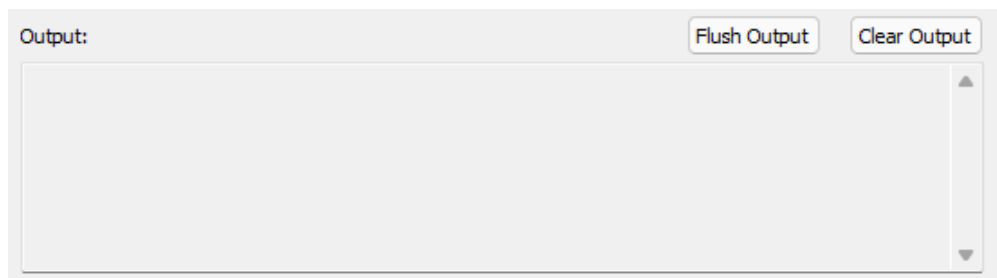
F.4.4.2 Run script



Scripts executed from the Debug Tool have the same effect as scripts designed in the project.

- Log in to Grid Viewer with an account of type Administrator.
- Enter Script content. Script syntax refer to scripting language F.5
- Click the Execute button
- The result of running the script is similar to the scripts designed in the project.

F.4.4.3 Check the log on the Output window



Logs during operation, Output() scripting functions are displayed on the Debug Tool / Output area.

- The function button:

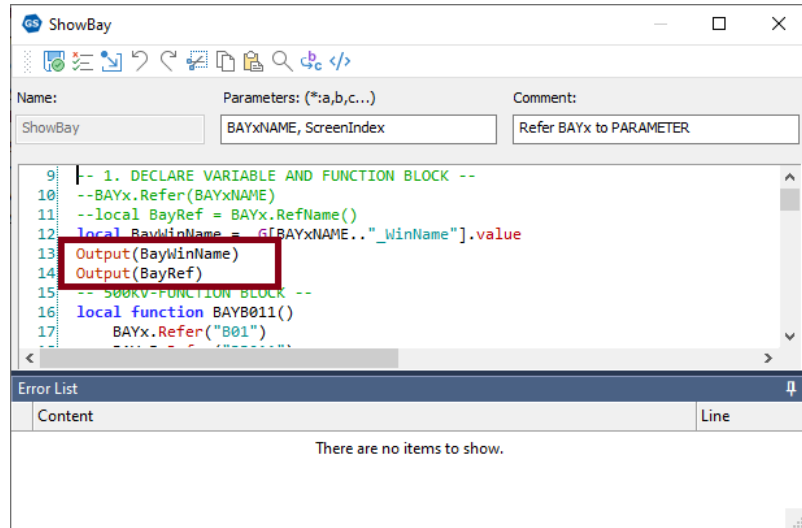
Clear Output	Delete content in the Output window
--------------	-------------------------------------

- In the Output window, you can edit (select, copy) the contents of the log messages.

...

F.4.5. Check working with scripts in Designer

To check the operation of scripts in the Viewer environment, you can combine and insert the commands "Output".



Output commands will output the above value in the Debug Tool/Output window of the OneATS Grid Viewer.

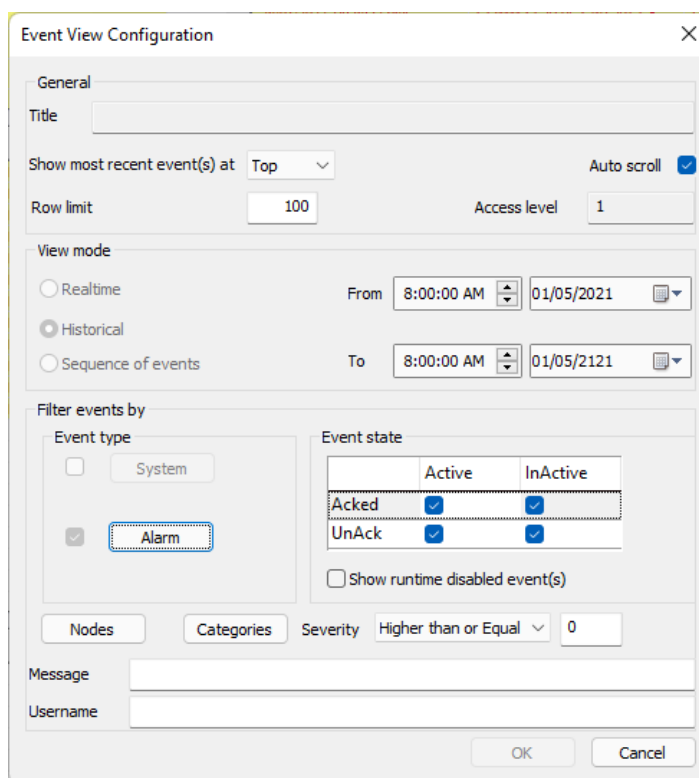
F.4.6. Interact with Event View in Grid Viewer

F.4.6.1 Event View Configuration

- Log in with an account with an authorized Access Level.
- Double Click on the alarm object on the **Viewer** to configure the **Event View** object.
 - For History Mode: allows to change the data query interval (**From – To**)
 - Configure the general parameters of the display order. Refer to F.3.14.2.3
 - Configure data filter conditions. Refer to F.3.14.2.5.

In the Grid View environment, parameters cannot be changed:

- Title
- Data Update Mode View Mode (Realtime Mode / Historical /Sequence of events)
- Access Level
- Event Type – System/Alarm. (Only allows changing options in Event Type System/Alarm)
- Display interface parameters (Tab User Interface)



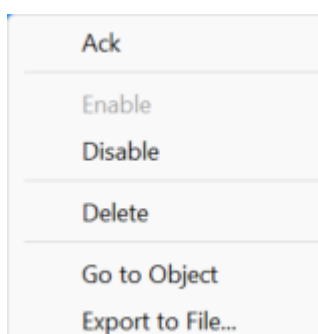
The 'Event View Configuration' dialog box contains the following sections:

- General:** Title (text field), Show most recent event(s) at (dropdown: Top), Auto scroll (checkbox: checked), Row limit (text field: 100), Access level (text field: 1).
- View mode:** Realtime (radio), Historical (radio: selected), Sequence of events (radio). Time range: From (8:00:00 AM, 01/05/2021) to To (8:00:00 AM, 01/05/2121).
- Filter events by:**
 - Event type:** System (checkbox: unchecked), Alarm (checkbox: checked).
 - Event state:** Table with columns Active and InActive. Rows: Acked (Active: checked, InActive: checked), UnAck (Active: checked, InActive: checked). Show runtime disabled event(s) (checkbox: unchecked).
- Buttons:** Nodes, Categories, Severity (Higher than or Equal), 0.
- Message:** (text field)
- Username:** (text field)
- OK** and **Cancel** buttons at the bottom right.

- Click OK to save and close the **Event View Configuration** window.

F.4.6.2 Operations with Realtime alarm

On the list of alarms from the Event View / Mode Realtime Alarm object → Right Click on an alarm → A list of options:



The context menu displays the following options:

- Ack
- Enable
- Disable
- Delete
- Go to Object
- Export to File...

Function	Description
Ack	Confirm alarm
Enable	On DataServer/Data Model, set Alarm Enable properties of point/tag = true → calculate events/alarms with point/tag Display alarm on alarm list. (Only setup via Scripts)

Disable	On DataServer/Data Model, set Alarm Enable properties of point/tag = false → do not calculate events/alarms with point/tag Hide alarm on alarm list.
Delete	Remove alarm event from event view
Go to Object	—
Export to File	Export List of alarms to file *.csv

F.4.6.3 View number of alarm/event

At the corner of each Event View object, the number of alarms/events in the table is displayed

ACTIVE ALARM						50 event(s)
Timestamp	Tagname	Ack	Operator	Message	Value	
28/01/2022 14:38:...	RP204BCU.ST.Time...	Unack		TIME SYNCH STATUS (B20_4 - S451 BCU)	False	
28/01/2022 14:38:...	RP203BCU.ST.Time...	Unack		TIME SYNCH STATUS (B20_3 - S451 BCU)	False	
28/01/2022 14:38:...	RPR05BCU.ST.Time...	Unack		TIME SYNCH STATUS (BR2_B05 - S451 BCU)	False	
28/01/2022 14:38:...	COMM.M.Status	Unack		HMI/GW1 DISCONNECTED FROM 10.0.0.31	False	

F.4.6.4 Alarm Popup

Users can modify the Alarm Popup configuration in both Designer and Viewer.
Reference F.3.6.5

F.4.7. Interact with Trend View in Grid Viewer

F.4.7.1 Configure data display parameters

In Grid Viewer environment, parameters cannot be changed:

- Realtime Mode / Historical Mode
- Access Level
- Time Slider Key Name

- Log in with an account with an authorized Access Level
- Double click Trend View to configure display parameters, parameters related to data updates (frequency, time, ...) similar to Grid Designer, refer to section F.3.14.3.3, F.3.14.3.4

F.4.7.2 View data value

- Move the time pointer (Time scooter) to see the corresponding data value of the data points (pen)

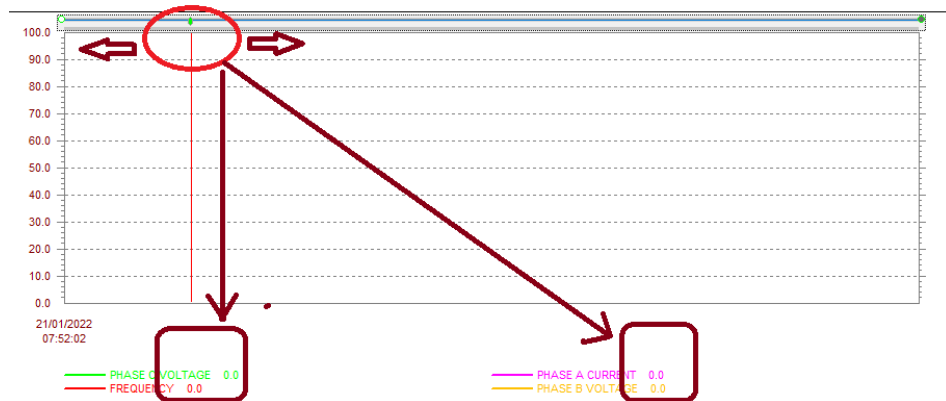


Figure F-231. Interact with Trend View

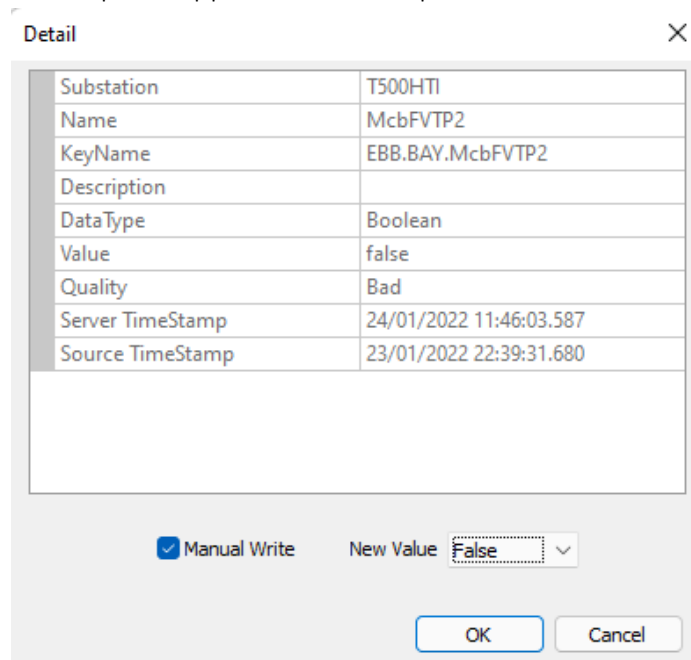
F.4.8. Interact with Data Grid in Grid Viewer

On the Data Grid object, Double Left-Click to select lines. The corresponding events are designed via scripts from the Grid Designer environment.

F.4.9. Interact with Tabular in Grid Viewer

F.4.9.1 Force value to variable

- Log in with an account with an authorized Access Level
- Double Left-Click on the data point
- Data information panel appears with the option to force data



Property	Value
Substation	T500HTI
Name	McbFVTP2
KeyName	EBB.BAY.McbFVTP2
Description	
DataType	Boolean
Value	false
Quality	Bad
Server TimeStamp	24/01/2022 11:46:03.587
Source TimeStamp	23/01/2022 22:39:31.680

☒ Manual Write New Value: False ▼

OK Cancel

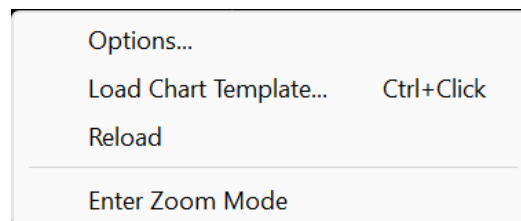
Figure F-232. Tabular Properties

Properties	Description
------------	-------------

Detail	Values of information fields of tag
Manual Write	Options to assign (force) data
New Value	Value to force data

F.4.10. Interact with Chart in Grid Viewer

F.4.10.1 Chart Configuration



- **Right-click** on the Chart object → **Option** to configure the parameters of the same Chart object in the Grid Designer, more detail refers to section F.3.14.6

F.4.10.2 View data value

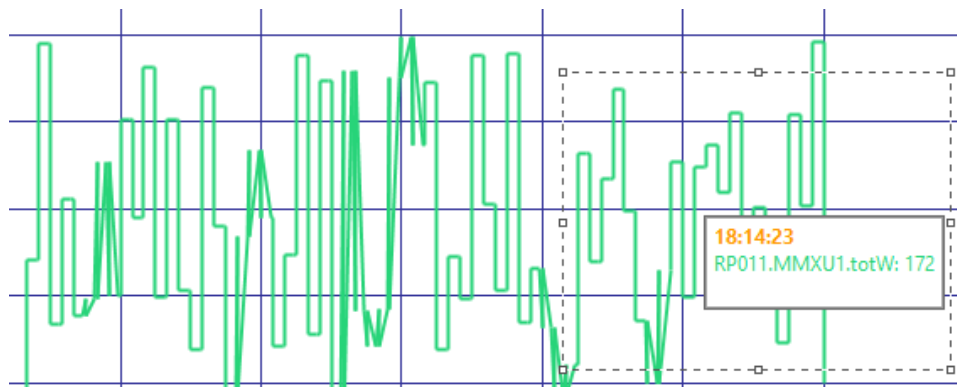


Figure F-233. View data value on chart

- Move the mouse to the data point, the time and value of the data point are displayed on the graph

F.4.10.3 Zoom/Pan on the chart

- **Right-click** on the Chart object → **Zoom Mode** to access Zoom/Pan view.
- Use mouse roller to Zoom.
- Hold + drag left mouse to perform Pan
- **Right-click** on the Chart object → **Zoom Mode** to exit Zoom Mode.

F.4.10.4 Hide lines on charts

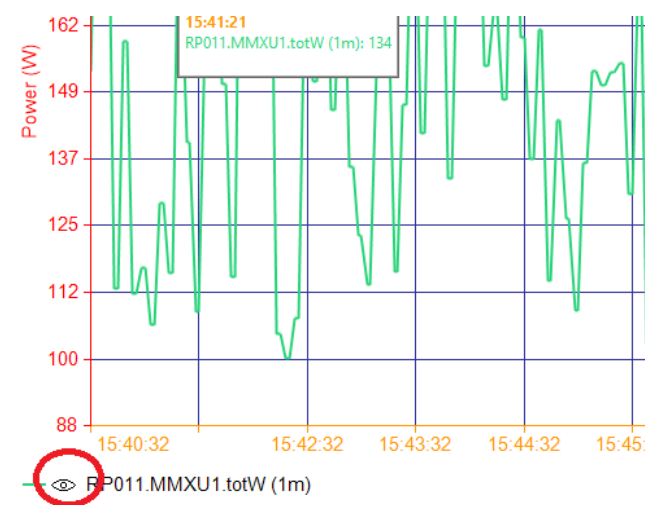


Figure F-234. Hide lines on chart

- **Left-click** on Visible icon  to hide/show the line on the chart

F.4.10.5 Export data as a file

The Chart allows exporting data in History mode to a .csv file:

- Right-click on the Chart → Select **Export History Data**.
- Enter a name and select a location to save the .csv file.

F.4.10.6 Reload - Synchronize Chart configuration between Designer & Viewer

Similar configuration on Grid Designer. Refer to F.3.14.6.8

F.4.11. Interact with Tagging in Grid Viewer

Tagging interaction windows including Tagging Manager, Tagging Device, Set Tagging are called out from Tagging scripts designed in Grid Designer.

Rework (4.1.6905 onwards)

F.4.11.1 Tagging Manager

Tagging Manager manages the tagging for all devices in the system.

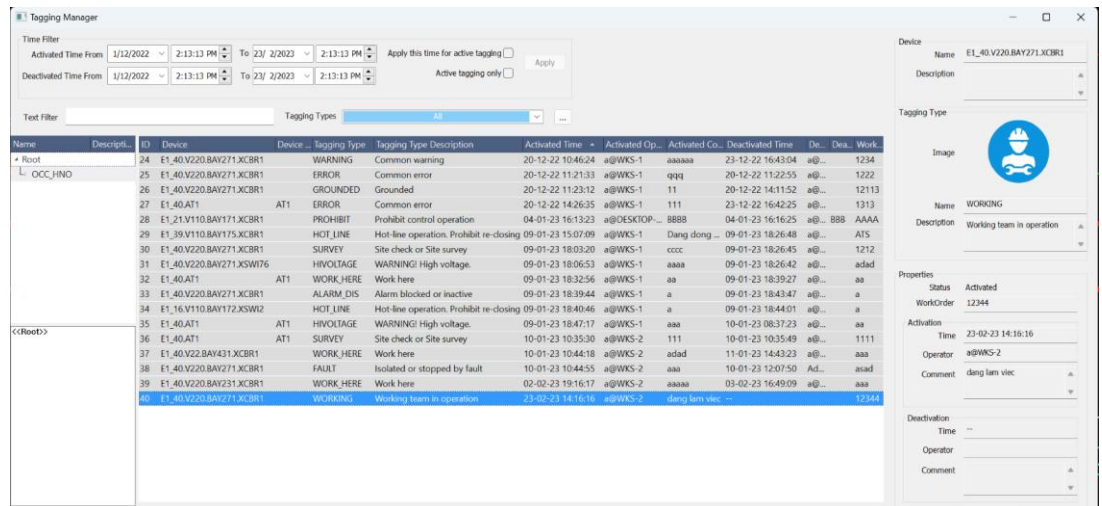


Figure F-235. Manage device tagging

Include: Refer to [F.3.18.1](#)

F.4.11.1.1 Filter by device name

Text Filter

In the Tagging Manager window.

- Enter the device name in the *Device Name* box.

F.4.11.1.2 Filter by Start Tagging of history tagging

Remove, use Filter by Time instead (4.1.6905 onwards)

F.4.11.1.3 Filter by End Tagging of history tagging

Remove, use Filter by Time instead (4.1.6905 onwards)

F.4.11.1.4 Filter by Time

Time Filter

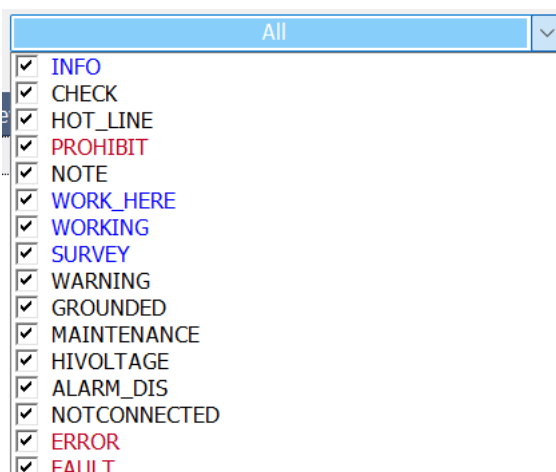
Activated Time From 2/ 8/2023 3:54:33 PM To 3/ 8/2023 3:54:33 PM Apply this time for active tagging ☒ Apply

Deactivated Time From 2/ 8/2023 3:54:33 PM To 3/ 8/2023 3:54:33 PM Active tagging only ☐

In the Tagging Manager window.

- Select Enable of Apply this time for active tagging or Active tagging only.
- Choose a time period Activated/Deactivated From – To.
- Click Apply filter.

F.4.11.1.5 Filter by Tagging type



In the Tagging Manager window.

- Choose Type according to tagging type table at F.3.18

F.4.11.2 Tagging Device

Manage the types of signs that are still in effect (hanging) of the device.

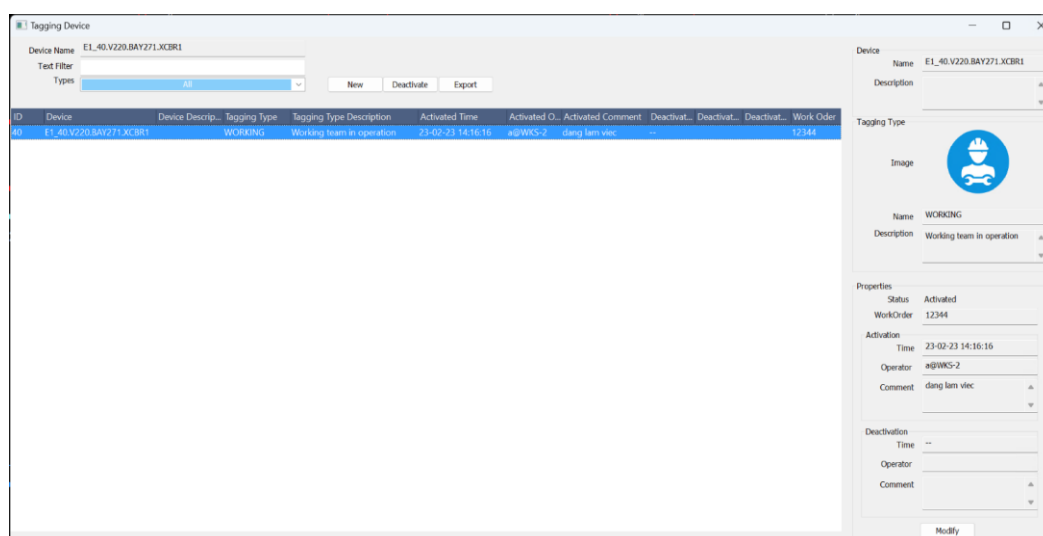


Figure F-236. Manage device tagging

Filter by Tagging type

Like the filtering by sign type in the Tagging Manager window, except time filter.

F.4.11.3 Hanging signs

F.4.11.3.1 Hang a sign from the Tagging Device window

In the **Tagging Device**, press **New** button.

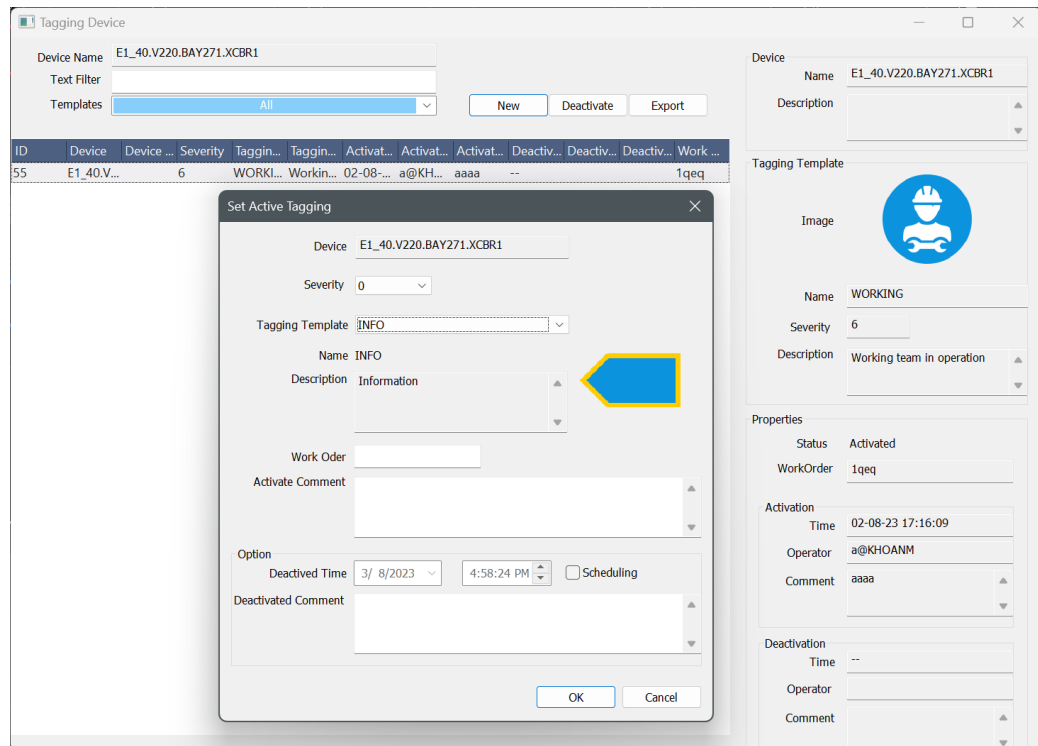


Figure F-237. Hanging sign

The **Set Active Tagging** window displays allowing you to enter content with signs.

Device	The device is hung with the sign.
Severity	The severity of sign.
Tagging Template	Corresponding signage type by severity
Name	Name of sign.
Description	Description of sign
Work Order	Work order code
Active Comment	Signs content.
Scheduling	<i>Set a schedule to lower sign.</i> <i>Deactivated time: time lower sign</i> <i>Deactivated Comment: content when lower sign</i>

Choose **OK** button.

The newly created Tagging will have a valid status (**Current**), and the **Tagging Device** and **Tagging Manager-Tab Tagging** windows will appear.

F.4.11.3.2 Set Tagging from the Set Active Tagging window.

Grid Studio supports minimalistic as **TaggingSet...** scripts for signage/tagging.

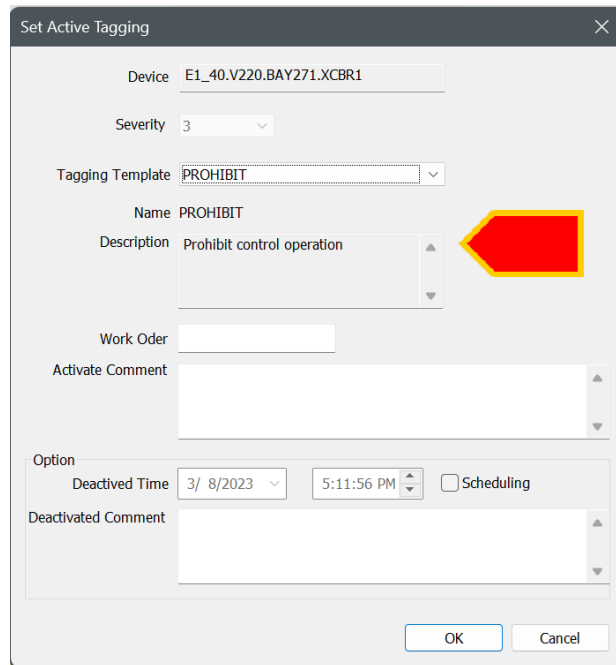


Figure F-238. Set Tagging from Set Active Tagging

Device Name	The name of the main unit is assigned a sign. <i>Passed into the TaggingSet...</i>	
Type	Signage type. <i>According to the script is called:</i>	
	Script	Tagging Type
	TaggingSetRed	Red (PROHIBIT)
	TaggingSetYellow	Yellow (HOT_LINE)
	TaggingSetPink	Pink (CHECK)
	TaggingSetBlue	Blue (INFO)

Press **OK** button.

F.4.11.4 Remove Tagging

Confirm the Tagging is out of date.

From the **Tagging Device** window, follow these steps:

ID	Device	Devic...	Sever...	Taggi...	Taggi...	Activ...	Activ...	Activ...	Deact...	Deact...	Deact...	Work...
55	E1_40...		6	WOR...	Worki...	02-0...	a@K...	aaaa	--			1qeq
56	E1_40...		10	MAIN...								1112...

Deactivate Tagging

Export Tagging List

Figure F-239. Remove signs

- Select one or more signs.
- Right click → **Deactivate**.
- In the **Confirm delete** window → **Yes** to confirm delete item.

Grid Viewer automatically updates properties for tagging

Deactivate time	<i>Time to lower the sign</i> <i>Automatically update according to the system time</i>
Deactivate Operator	<i>User/Operator lowering the sign</i> <i>Automatically update according to the currently logged in user</i>

F.5. Programmable language script

F.5.1. Script language

OneATS Grid Studio's scripts support Lua Language syntax and function libraries that combine application built-in functions.

F.5.2. Variables, parameters in script

Global variable:

- Include **tags (Remote Data, Local Data, System Data)** that are project-wide, used in all scripts without needing to be defined in the script body.

Local variable:

- Are variables that have an effect only in the scope of the script in which the variable is declared.
- Local variables are declared in the script body with the keyword **local**.

Parameter:

- A variable whose value is passed in when calling the script.
- Parameters are only valid within the scope of the **script** in which the parameter is defined.
- Parameters separated by commas “,”

F.5.2.1 Rules for naming variables and parameters

- May not begin with a number or underscore. The following examples are not accepted: `_abc.def` or `5abc.def`
- May not contain spaces or special characters: `` ~ ! @ # $ % ^ & * () - = + ' " ; : , . / < > ?`
- May not contain Unicode control characters.
- May not contain wildcard characters.
- Can't be more than 64 characters.

F.5.2.2 Data type

Data Type	Description	Value range
Boolean		true / false; 0 / 1
Int8	8-bit signed integer	-128 → 127
Int16	16-bit signed integer	-32,768 → 32,767
Int32	32-bit signed integer	-2,147,483,648 → 2,147,483,647

Int64	64-bit signed integer	-9,223,372,036,854,775,808 → 9,223,372,036,854,775,807
UInt8	8-bit unsigned integer	0 → 255
UInt16	16-bit unsigned integer	0 → 65,535
UInt32	32-bit unsigned integer	0 → 4,294,967,295
UInt64	64-bit unsigned integer	0 → 18,446,744,073,709,551,615
Float	32-bit floating-point number	$1,5 \cdot 10^{-45} \rightarrow 3,4 \cdot 10^{38}$
Double	64-bit floating-point number	$5,0 \cdot 10^{-345} \rightarrow 1,7 \cdot 10^{308}$
String	strings	
time_t	(Int64) number of seconds from 00:00:00 1/1/1970 UTC to the specified time.	Apply: - Script: os.time() - System Tag: _time - Expression: TimeFormat(<i>time_t</i>)
DateTime	Time type variable The variable has a value equivalent to the number of seconds from 00:00:00 1/1/1970 UTC to the specified time. Variables of type "DateTime" are also automatically converted to the corresponding time_t when used in the script.	Apply: ○ Declare Local Data ○ Value Display string property. (Application automatically converts to string format) ○ Datetime control. ○ Expression ○ HisTrend script
int64_t	(Int64) number of seconds from 07:00:00 1/1/1970 UTC to the specified time. When used in scripts, int64_t will be converted "intact" to time_t.	Apply: - Script: Chart scripts

F.5.2.3 Dotfield of tag

In script language, Data consists of 3 fields:

Dot field	Data type	Description
.value	(According to the data type defined for Remote Data)	The value of the tag .
.timestamp	UInt64	<i>(not implemented on Grid Studio yet)</i>
.quality	Int64	Confidence level of the value of the data assigned to the tag . There are three main types of quality: Good, Uncertain, Bad . Table of values: refer to F.5.2.4

In the script, the value assignment statement must specify the **tag's .value** field

F.5.2.4 Table of quality values according to standards OPC UA:

Good Quality Values

Code	Description	Code
00000000	Good	
002D0000	GoodSubscriptionTransferred	
002E0000	GoodCompletesAsynchronously	
002F0000	GoodOverload	
00300000	GoodClamped	
00960000	GoodLocalOverride	
00A20000	GoodEntryInserted	
00A30000	GoodEntryReplaced	
00A50000	GoodNoData	
00A60000	GoodMoreData	
00A70000	GoodCommunicationEvent	
00A80000	GoodShutdownEvent	
00A90000	GoodCallAgain	
00AA0000	GoodNonCriticalTimeout	
00BA0000	GoodResultsMayBeIncomplete	

Uncertain Quality Values

Code Description	Code
Uncertain	40000000
UncertainReferenceOutOfServer	406C0000
UncertainNoCommunicationLastUsableValue	408F0000
UncertainLastUsableValue	40900000
UncertainSubstituteValue	40910000
UncertainInitialValue	40920000
UncertainSensorNotAccurate	40930000
UncertainEngineeringUnitsExceeded	40940000
UncertainSubNormal	40950000
UncertainDataSubNormal	40A40000
UncertainReferenceNotDeleted	40BC0000
UncertainNotAllNodesAvailable	40C00000

Bad Quality Values

Bad	80000000
BadUnexpectedError	80010000
BadInternalError	80020000
BadOutOfMemory	80030000
BadResourceUnavailable	80040000
BadCommunicationError	80050000
BadEncodingError	80060000
BadDecodingError	80070000
BadEncodingLimitsExceeded	80080000
BadUnknownResponse	80090000
BadTimeout	800A0000
BadServiceUnsupported	800B0000
BadShutdown	800C0000
BadServerNotConnected	800D0000
BadServerHalted	800E0000
BadNothingToDo	800F0000
BadTooManyOperations	80100000
BadDataTypeIdUnknown	80110000
BadCertificateInvalid	80120000
BadSecurityChecksFailed	80130000
BadCertificateTimeInvalid	80140000
BadCertificateIssuerTimeInvalid	80150000
BadCertificateHostNameInvalid	80160000
BadCertificateUriInvalid	80170000
BadCertificateUseNotAllowed	80180000
BadCertificateIssuerUseNotAllowed	80190000

BadCertificateUntrusted	801A0000
BadCertificateRevocationUnknown	801B0000
BadCertificateIssuerRevocationUnknown	801C0000
BadCertificateRevoked	801D0000
BadCertificateIssuerRevoked	801E0000
BadUserAccessDenied	801F0000
BadIdentityTokenInvalid	80200000
BadIdentityTokenRejected	80210000
BadSecureChannelIdInvalid	80220000
BadInvalidTimestamp	80230000
BadNonceInvalid	80240000
BadSessionIdInvalid	80250000
BadSessionClosed	80260000
BadSessionNotActivated	80270000
BadSubscriptionIdInvalid	80280000
BadRequestHeaderInvalid	802A0000
BadTimestampsToReturnInvalid	802B0000
BadRequestCancelledByClient	802C0000
BadNoCommunication	80310000
BadWaitingForInitialData	80320000
BadNodeIdInvalid	80330000
BadNodeIdUnknown	80340000
BadAttributeIdInvalid	80350000
BadIndexRangeInvalid	80360000
BadIndexRangeNoData	80370000
BadDataEncodingInvalid	80380000
BadDataEncodingUnsupported	80390000
BadNotReadable	803A0000
BadNotWritable	803B0000
BadOutOfRange	803C0000
BadNotSupported	803D0000
BadNotFound	803E0000
BadObjectDeleted	803F0000
BadNotImplemented	80400000
BadMonitoringModelInvalid	80410000
BadMonitoredItemIdInvalid	80420000
BadMonitoredItemFilterInvalid	80430000
BadMonitoredItemFilterUnsupported	80440000
BadFilterNotAllowed	80450000
BadStructureMissing	80460000
BadEventFilterInvalid	80470000
BadContentFilterInvalid	80480000

BadFilterOperandInvalid	80490000
BadContinuationPointInvalid	804A0000
BadNoContinuationPoints	804B0000
BadReferenceTypeInvalid	804C0000
BadBrowseDirectionInvalid	804D0000
BadNodeNotInView	804E0000
BadServerUriInvalid	804F0000
BadServerNameMissing	80500000
BadDiscoveryUriMissing	80510000
BadSempahoreFileMissing	80520000
BadRequestTypeInvalid	80530000
BadSecurityModeRejected	80540000
BadSecurityPolicyRejected	80550000
BadTooManySessions	80560000
BadUserSignatureInvalid	80570000
BadApplicationSignatureInvalid	80580000
BadNoValidCertificates	80590000
BadRequestCancelledByRequest	805A0000
BadParentNodeIdInvalid	805B0000
BadReferenceNotAllowed	805C0000
BadNodeIdRejected	805D0000
BadNodeIdExists	805E0000
BadNodeClassInvalid	805F0000
BadBrowseNameInvalid	80600000
BadBrowseNameDuplicated	80610000
BadNodeAttributesInvalid	80620000
BadTypeDefinitionInvalid	80630000
BadSourceNodeIdInvalid	80640000

F.5.1. Notes in the code

Comment for a statement or a block

Line-by-line notes: use `--`

Example:

```
-- ham start app
```

Note the whole block: use `-- [[ -- ]]`

Example:

```
--[[tagstr=tag1.Name.."Hello World"
```



```
ref1:Refer(tagstr.Name)
--]]
```

F.5.2. Value assignment command

Syntax: assignment command used with dot field tags and the operator “=”

```
tag.value = 1
```

Example:

Nahnf.value = true—*value assignment command.*

F.5.3. Call function

Functions in the program are divided into three types:

- Built-in functions (including built-in functions of OneATS Grid Studio and of LUA Language). See details at F.5.6
- User-defined functions (**function script**)
- Method in **Remote Data**.

F.5.3.1 Call Built-in function

FunctionName(Parameters) -- Calling a function with no return value.

Datapoint.value = FunctionName(Parameters)-- the return value of the function assigned to a Datapoint.

- *When calling using the built-in function in the script, the program will provide a list of supported functions corresponding to the characters entered via **Ctrl + Space**. The detailed list of these functions is given in the **_LUA.schclass** file in the software installation directory.*

F.5.3.2 Call User-defined functions

FunctionName(Parameters)

Example: Showbay(“D01”,1)

F.5.3.3 Call method – CallMethod

CallMethod(MethodName,Parameters)

Example: CallMethod(“E01_XCBR1_PosCtl”,true)

F.5.4. Structure Commands (if-then-else, loop)

For details, see the LUA language documentation, chapter 4, section 4.3 – **Control Structures**

F.5.5. Operators

For details see LUA language documentation, chapter 3- **Expression**

F.5.6. Introduce some commonly used Lua functions

F.5.6.1 Mathematical calculation function

Use *Lua.math* library

F.5.6.1.1 Functions to floor and change signs

math.abs

Definition: calculates the absolute value of a value or an expression.

Syntax:

```
double math.abs(double x)
```

Example: `x=math.abs(2-3)`

The returned result is `x=1`.

math.floor

Definition: returns the smallest integer value by flooring down if necessary.

Syntax:

```
double math.floor(double x)
```

Example: `x=math.floor(2.56) y=math.floor(-2.56)`

The returned results are `x=2` and `y=-3`.

math.ceil

Definition: returns the next largest integer value by rounding up if necessary

Syntax:

```
double math.ceil(double x)
```

Example: `x=math.ceil(2.56) y=math.ceil(-2.56)`

The returned results are `x=3` and `y=-2`.

math_Trunc

Definition: Returns the integer part, no rounding required.

Syntax:

```
double math_Trunc(double x)
```

Example: $x=\text{math_Trunc}(2.42)$ $y=\text{math_Trunc}(-2.42)$

The returned results are $x=2$ and $y=-2$.

math_Sign

Definition: determine the sign of the number

If number $< 0 \rightarrow$ function returns -1

If number $=0 \rightarrow$ function returns 0

If number $>0 \rightarrow$ function returns 1

Syntax:

```
double math_Sign(double x)
```

Example: $x=\text{math_Sign}(-5)$ $y=\text{math_Sign}(0)$ $z=\text{math_Sign}(5);$

The results are $x=-1$, $y=0$ and $z=1$.

F.5.6.1.2 Trigonometry function

Note: The parameter is in radians.

math.sin

Syntax:

```
double math.sin(double x)
```

Example: $x=\text{math.sin}(1)$

math.cos

Syntax:

```
double math.cos(double x)
```

Example: $x=\text{math.cos}(1)$

math.tan

Syntax:

```
double math.tan(double x)
```

Example: $x=\text{math.tan}(1)$

math.asin

Syntax:

```
double math.asin(double x)
```

Example: `x=math.asin(1)`

math.acos

Syntax:

```
double math.acos(double x)
```

Example: `x=math.acos(1)`

math.atan

Syntax:

```
double math.atan(double x)
```

Example: `x=math.atan(1)`

math.sinh

Syntax:

```
double math.sinh(double x)
```

Example: `x=math.sinh(1)`

math.cosh

Syntax:

```
double math.cosh(double x)
```

Example: `x=math.cosh(1)`

math.tanh

Syntax:

```
double math.tanh(double x)
```

Example: `x=math.tanh(1)`

F.5.6.1.3 Functions that return pi, infinity, convert degrees - radians

math.pi

Definition: return the numeric value PI

Syntax:

```
double math.pi
```

Example: `x=math.pi`

Return `x=3.14159265358979`

math.huge

Definition: is a constant, returns the positive value infinity.

Syntax:

```
double math.huge
```

Example:

`math.huge` -> return inf

math.deg

Definition: Convert to degrees

Syntax:

```
double math.deg(double x)
```

Example: `x=math.deg(math.pi)` `y=math.deg(math.pi/2)`

Return `x=180` and `y=90`

math.rad

Definition: convert to radian

Syntax:

```
double math.rad(double x)
```

Example: `x=math.rad(180)`

Return `x=3.1415926535898`

F.5.6.1.4 Logarit functions

math.exp

Syntax:

```
double math.exp(double x)
```

Example: `x=math.exp(2)`

math.log

Syntax:

```
double math.log(double x, double y)
```

Example: `x=math.log(10,10)`

math.log10

Syntax:

```
double math.log10(double x)
```

Example: `x=math.log10(10)`

F.5.6.1.5 Square root functions, exponents

math.sqrt

Definition: returns the square root

Syntax:

```
double math.sqrt(double x)
```

Example: `x=math.sqrt(9)`

Return `x=3`

math.pow

Definition: Returns the exponent of a number

Syntax:

```
double math.sqr(double x, double y)
```

Example: `x=math.pow(9,2);`

Return `x=81`

F.5.6.1.6 Functions to find the largest and the smallest number, generate random numbers

math.max

Syntax:

```
double math.max(double x, double y)
```

Example: `x=math.max(5,7)`

Return `x=7`

math.min

Syntax:

```
double math.min(double x, double y)
```

Example: `x=math.min(5,7)`

Return `x=5`

math.random

Syntax:

```
double math.random()
```

Example: `x=math.random()`

F.5.6.2 String Functions

Using the Lua.strings library

F.5.6.2.1 tostring – Convert from number to string

Definition: This function is used to convert integers, real numbers, binary numbers or tag names with discrete, analog values to string.

Syntax:

```
tostring(double x)
```

Example: `Message(ToString(6.7),0)` -> The viewer will display a message box that returns a string value of "6.7".

F.5.6.2.2 Trim – The function removes whitespace characters at the beginning and end of a string

Definition: The function removes whitespace characters at the beginning and end of a string.

Syntax:

```
Trim(s)
```

`s`: is a string.

Example: `Trim(" Hello World ")` -> return result is string "Hello World"

F.5.6.2.3 string.sub – String slicing function

Definition: This function is used for string trimming.

Syntax:

```
string.sub(s,i[,j])
```

`s`: input string.

i,j: cut string s starting from position i to position j.

i,j can be negative. If you don't enter j, it will cut to the end of the string length.

Example:

Cut a string of characters from position x (calculate the landmark from the beginning of the string) to the end of the string (cut the right string)

```
string.sub("Hello Lua user", 7)
```

The system will cut from the 7th character to the end of the string, and return the string "Lua user"

Cut the character string from position x to position y

```
string.sub("Hello Lua user", 7,9)
```

Return the string "Lua"

Cut the string from position x (calculate the landmark from the end of the string) to the end of the string (cut the right string)

```
string.sub("Hello Lua user", -8)
```

Return the string "Lua user"

Cut the string from the x position (calculate the landmark from the end of the string) to the y position (calculate the landmark from the beginning of the string)

```
string.sub("Hello Lua user", -8,9)
```

Return the string "Lua"

Cut the string from position x (calculate the landmark from the end of the string) to position y (calculate the landmark from the end of the string)

```
string.sub("Hello Lua user", -8,-6)
```

Return the string "Lua"

F.5.6.2.4 string.lower – string.upper – Function to change uppercase to lowercase

Definition: Returns the value of a string all lowercase characters.

Syntax:

```
string.lower(x)
```

x: is the input string.

Example: x=string.lower("Hello World")

Result: x="hello world"

F.5.6.2.5 *string.upper* – Function to convert lowercase to uppercase

Definition: Returns the value of a string all uppercase.

Syntax:

```
string.upper(x)
```

x: is the input string.

Example: x=string.upper("Hello World")

Result: x="HELLO WORLD"

F.5.6.2.6 *string.find* – Find function

Definition: searches for a character phrase and returns the position of the character found in the string. If not found, return is nil

Syntax:

```
string.find("string","kitucantim", n)
```

Example: string.find("ATS SMART HMI","ATS", n)

F.5.6.2.7 *string.gsub* – Find and replace function

Definition: searches for a character phrase and replaces the character phrase found in the string with the new character phrase.

Syntax:

```
string.gsub("string","kitucantim", "kituthaythe")
```

Example:

```
s = string.gsub("Lua is cute", "cute", "great")
```

Result: s="Lua is great"

F.5.6.3 *Functions interact with the operating system*

Use the library Os supported by *Lua Script*

F.5.6.3.1 *os.clock()*

Definition: Returns the CPU time since Lua started in seconds

Syntax:

```
os.clock()
```

Example: var x = os.clock()

CPU time since Lua started is 1000, so returns x = 1000

F.5.6.3.2 *os.date()*

Definition: return a date string in the format, or table of timestamps, which resembles the format of the strftime () function in C

Syntax:

```
os.date('*t')  
os.date('!*t')  
os.date()
```

Example: var x = os.date()

F.5.6.3.3 *os.difftime()*

Definition: calculate the number of seconds between time t1 and t2

Syntax:

```
os.difftime(t1,t2)
```

Example: var x = os.difftime(10,5)

F.5.6.3.4 *os.execute()*

Definition: execute command of the operating system

Syntax:

```
os.execute([command])
```

Example: os.execute("echo hello")

F.5.6.3.5 *os.exit()*

Definition: call the exit function of C, with optional code, and terminate the host program.

Syntax:

```
os.exit([code])
```

Example: os.exit(0) end Lua command

F.5.6.3.6 *os.remove()*

Definition: file delete function. If it fails, return "nil" and the error description string

Syntax:

```
os.remove(filename)
```

Example: `os.remove("banana.txt")`

F.5.6.3.7 `os.rename()`

Definition: file rename function. If the function fails, return "nil" and the error description string

Syntax:

```
os.rename(oldname, newname)
```

Example: `os.rename("banana.txt", "apple.txt")`

F.5.6.3.8 `os.time()`

Definition: returns the system time in seconds, data type `time_t` (Int64)

Syntax:

```
os.time()
```

Example: `var x = os.time`

The current system time is 200, so `x = 200`

F.5.6.4 Functions for File Management

Refer Lua Language

Example:

- Read the csv file and print out the content of this file, including 3 columns timestamp, value, quality:

```
local contents, headers = csv.parse("hisData2021_04_16.csv", ",")
for index, data in ipairs(contents) do
    Output(data.timestamp, data.value, data.quality)
end
```

- Write the content from the `contents` variable (table format) to the `csv` . file:

```
local fileOutput = csv.encode(contents, ",")
local file = io.open("hisOutput.csv", "w")
file:write(fileOutput)
file:close()
```

F.5.7. Grid Studio built-in function

F.5.7.1 System scripts

F.5.7.1.1 Login

Definition: Call the login form at position x,y so that the user can login to the application.

Syntax:

```
Login(x,y)
```

Example: Login(10,10) → When the application runs the login form appears at coordinates (10,10)

F.5.7.1.2 Logout

Definition: This function allows the user to logout from the application.

Syntax:

```
Logout()
```

F.5.7.1.3 UserManager

Definition: Displays the user manager dialog, the user can add, edit, and delete users (if they have enough permissions)

Syntax:

```
UserManager()
```

F.5.7.1.4 ExecuteFile

Definition: Run exe. file

Syntax:

```
ExecuteFile(file_path, [optional] params)
```

Example:

```
ExecuteFile("C:\\ATS\\PrintScreen\\PrintScreen.exe","-capture 0 0 1920  
1080 D:/capture")
```

F.5.7.1.5 Exit

Exit()

Definition: close Viewer

Syntax:

Exit()

F.5.7.2 Scripts for Object/Tag

F.5.7.2.1 Refer

Definition: Reference for pointer.

Support Fallback variable (4.1.6905 onwards)

Syntax:

```
ref_Object.Refer(KeyName_Object,[optional]Fallback_var)  
ref_Variable:Refer(KeyName_Variable)
```

ref_Object: object pointer.

ref_Variable: variable pointer.

KeyName_Object: string type representing the Keyname of the referenced object.

Fallback_var: Variables that do not exist in KeyName_Object but exist in ref_Object will be referenced to Fallback_var.

KeyName_Variable: string variable representing the **Keyname** of the referenced variable.

Example:

Keyname	Description
BAYx	Object type pointer
BAYxWinName	String deformed pointer
D01	Object Key name
D01_WinName	String variable

Correct syntax:

BAYx.Refer("D01")

After this command is executed, all places where the **BAYx** variable is used will be replaced with D01.

BAYxWinName:Refer("D01_WinName")

After the script executes, all places where the **BAYxWinName** variable is used will be replaced with **D01_WinName**.

The pointer and the referenced object must be of the same type.

F.5.7.2.2 ReferFromTo

Definition: Reference from refName1 to refName2 (object or variable).

Syntax:

```
ReferFromTo(refName1, refName1)
```

refName1, refName1: pointers of type Object / Variable.

F.5.7.2.3 SymbolRefer

Definition: Attribute reference of **Coloring** one symbol to another symbol.

Syntax:

```
SymbolRefer(Symbol1_Name, Symbol2_Name)
```

Symbol1_Name: String format representing the name of the Symbol to be referenced.

Symbol2_Name: String form representing the name of the referenced Symbol.

Example:

```
SymbolRefer("BAYELL.CB", "ONELINE.CB")
```

Symbol1, Symbol2 can be Network, Coloring or Template objects. If it is a Template, they must be the same in Device Type.

Symbol1 must be a Reference Symbol.

F.5.7.2.4 UnRefer

Definition: Clears all reference state of the referenced variable or object, returning to the default state (like when initializing the Viewer).

Syntax:

```
ref_Object:UnRefer()
```

ref_Object: pointer of type object/variable.

F.5.7.2.5 KeyName

Removed (4.1.6869 onwards)

F.5.7.2.6 *DisplayName*

Removed (4.1.6869 onwards)

F.5.7.2.7 *RefName*

Definition: Returns the String KeyName of the Object referred to.

Syntax:

```
refObject:RefName()
```

Example: BAYx.Refer("D01")

BAYx.Name:RefName() → "D01.Name"

F.5.7.2.8 *GetDescription*

Definition: This function returns the description of the KeyName node on the DataModel.

Syntax:

```
string GetDescription(string keyName)
```

Example: GetDescription("E1_1.V110.BAY100") → "BAYBC"

F.5.7.2.9 *GetValue*

Removed (4.1.6869 onwards)

F.5.7.2.10 *SetValue*

Removed (4.1.6869 onwards)

F.5.7.2.11 *description*

Definition: Returns the description of the Variable.

Syntax:

```
Varriable.description
```

Example: D01.MMXU1.Amax.desription → "MAX OF 3 PHASE CURRENT (A)"

F.5.7.2.12 *descriptionExt*

Definition: Returns the extended description of the Variable in fomate "[DescriptionParent1.DescriptionParent2...] description".

Support descriptionExt (4.1.6905 onwards)

Syntax:

Varriable.descriptionExt

Example: D01.MMXU1.Amax.desriptionExt → “[220KV TAY HO.DLL] MAX OF 3 PHASE CURRENT (A)”

F.5.7.2.13 ResetNoRef

Definition: to reference unreferenced variables of the same type in ObjectRef to the fallbackVar variable. If the fallbackVar variable is a string type, only string type variables will be referenced. If the fallbackVar variable is not of type string, it will only reference variables that are not of type string.

Support ResetNoRef() (4.1.6964 onwards)

Syntax:

ref_Object.ResetNoRef(Fallback_var)

ref_Object: object pointer.

ref_Variable: variable pointer.

Fallback_var: Variables that do not exist on *the* referenced KeyName_Object & are existing on *the ref_Object* will be referenced to *Fallback_var*.

Example:

Keyname	Description
BAYx	Object type pointer
BAYxWinName	String deformed pointer
D01	Object Key name
D01_WinName	String variable

Correct syntax:

BAYx.Refer("D01")

After this command is executed, all places where the BAYx variable is used will be replaced with D01.

BAYx.Refer("D02")

After this command is executed, all places where the BAYx variable is used will be replaced with D02.

BAYx:ResetNoRef(Var)

After the script executes, the variables (tags) that have been referenced that time (D01) if they do not exist in this reference (D02) will be referenced to the *Var variables*.

F.5.7.2.14 GetDataInfo

Definition: Return data type, description of object/variable (tag)

Attributes: type (*int* type value), types (*string* type value), subtype(*int* subtype value), subtypes (*string* subtype value), description (*string*).

Support GetDataInfo() (4.1.6964 onwards)

Syntax:

```
GetDataInfo(key_name)
```

Example: Get data type of tag "E1_40.V220.BAY271.MMXU1.Amax":

```
GetDataInfo("E1_40.V220.BAY271.MMXU1.Amax").type → 6
```

```
GetDataInfo("E1_40.V220.BAY271.MMXU1.Amax").types → "Variable"
```

```
GetDataInfo("E1_40.V220.BAY271.MMXU1.Amax").subtype → 10
```

```
GetDataInfo("E1_40.V220.BAY271.MMXU1.Amax").subtypes → Float
```

F.5.7.2.15 GetBusbarState

Definition: Return energized state of a busbar or connectivity node

0 – deenergized, 1- energized, 2 - grounded

Syntax:

```
GetBusbarState (screen_name, busbar_  
connectivitynode _name)
```

Example: local x = GetBusbarState("ONELINE_QTREDHA","QTREDHA_C41")

F.5.7.3 Scripts to query past data

F.5.7.3.1 SwitchMode – Realtime conversion – History Mode

Syntax:

SwitchMode()

Description:

- Switch data source connection mode from DataServer → HIS Server and vice versa:
 - If GridView is connecting data to DataServer → Script to convert data connection to HIS Server
 - If GridView is connecting data to HIS Server → Script executes data connection to DataServer
- Connection status is reflected by the system variable **_mode.value**

F.5.7.3.2 QueryHisPlayback

Syntax:

QueryHisPlayback()

Description:

- Displays a window to choose when to query data.
- The user chooses the time.
- Click OK ® Send data query command to HIS Server.
- The status of the process querying data from the HIS Server is reflected by the system variable **_queryHisState**
 - **_queryHisState = true**: Waiting for data from Server
 - **_queryHisState = false**: Data query complete or timeout

F.5.7.3.3 PlayBack

Support Playback() (4.1.6987 onwards)

Syntax:

Playback(screen_name, x, y)

Description: Open PlayBack menu bar in the screen at location (x, y)

F.5.7.3.4 PlaybackMultiScreens

Support PlaybackMultiScreens() (4.1.7031 onwards)

Syntax:

```
PlaybackMultiScreens (x, y, screen_name1,  
screen_name1,...)
```

*Description: Open PlayBack menu bar for many screens at location (x, y).
All icons of the selected screens will display Playback data.*

F.5.7.4 Scripts for interacting with Screens

F.5.7.4.1 ShowScreen

ShowScreen()

Definition: open a screen

Syntax:

```
ShowScreen("ScreenName")
```

Example: ShowScreen("ThotNot")

When the application runs, the **ThotNot** screen is automatically opened.

F.5.7.4.2 ShowScreenAt

ShowScreenAt()

Definition: open a screen, move the screen to a specific coordinate.

Syntax:

```
ShowScreenAt("ScreenName",x,y)
```

Example: ShowScreenAt("ThotNot",10,10)

When the application runs, the ThotNot screen is automatically opened at coordinates (10,10).

F.5.7.4.3 ShowScreenModal

Definition: Displays a screen (popup or overlay) as a dialog and allows the user to manipulate the screen only while it is visible.

(Unlike ShowScreen for popup, it still allows users to manipulate another screen when there is a popup)

Syntax:

```
ShowScreenModal("ScreenName")
```

F.5.7.4.4 ShowScreenModalAt

Definition: open a screen, move the screen to a specific coordinate.

Syntax:

```
ShowScreenModalAt("ScreenName", x, y)
```

F.5.7.4.5 HideScreen

Definition: Command to close one screen from a graphic/script object on another screen.

Support Hide Multi Screens (4.1.6894 onwards)

Syntax:

```
HideScreen("ScreenName_1", "ScreenName_2", ...  
          , "ScreenName_n")
```

Example: HideScreen ("ThotNot1", "ThotNot2") → Close "ThotNot1", "ThotNot2" screen

F.5.7.4.6 HideScreenAsync

Definition: Command to close a screen from a graphics/script object on another screen or the screen itself.

Syntax:

```
HideScreenAsync("ScreenName")
```

F.5.7.4.7 GetScreenXCoordinate

Removed

F.5.7.4.8 GetScreenYCoordinate

Removed

F.5.7.4.9 ZoomScreen

Definition: zoom screen up %

Syntax:

```
ZoomScreen("screenName", scale_percent)
```

Example: ZoomScreen("test", 90)

→ When the application runs, the screen test is zoomed at 90%.

F.5.7.4.10 ZoomToFitScreen

Removed

F.5.7.4.11 OpenApp

OpenApp()

Support OpenApp() (4.1.7031 onwards)

Definition: open an App in a screen

Syntax:

```
OpenApp(file_path, screen_name, [optional]  
window_class)
```

Example: OpenApp("C://Program
Files//ATS//SmartHis//His.Admin.App.exe", "ThotNot ")

When the application runs, the **SmartHis Administrator** App is automatically opened in **ThotNot** screen.

F.5.7.4.12 CloseApp

CloseApp()

Support CloseApp() (4.1.7031 onwards)

Definition: close App in a screen

Syntax:

```
CloseApp(screen_name)
```

Example: CloseApp("ThotNot ")

When the application runs, the App is automatically closed in **ThotNot** screen.

F.5.7.4.13 IsShowApp

IsShowApp ()

Support IsShowApp() (4.1.7031 onwards)

App support only: SmartHis Administrator, OneATS DataEditor, notepad2

Definition: check any App open in a screen

Syntax:

```
IsShowApp(screen_name)
```

Example: `IsShowApp("ThotNot ")`

Check App open status in **ThotNot** screen. If any App is already open, return **true** else return **false**.

F.5.7.4.14 ScaleScreen

ScaleScreen()

Support ScaleScreen() (4.1.6932 onwards)

Definition: Scale size of screen in percentage

Syntax:

```
ScaleScreen(screen_name, percent)
```

Example: `ScaleScreen("ThotNot ",200)`

Scale up **ThotNot** screen to 200%.

F.5.7.4.15 GetScreenInfo

GetScreenInfo()

Support GetScreenInfo() (4.1.6932 onwards)

Additional attributes: scale (4.1.6982 onwards)

Additional attributes: bottom, right, centerX (center X-axis), centerY (center Y-axis), isVisible (Visible) (4.1.6987 onwards)

Definition: Get informations of a screen: top, bottom, left, right, width, height, scale, centerX (center X-axis), centerY (center Y-axis), isVisible (Visible)

Syntax:

```
GetScreenInfo(screen_name)
```

Example: `local top = GetScreenInfo("ThotNot ").top`

Get top location of **ThotNot** screen.

F.5.7.4.16 OpenWebScreen

OpenWebScreen ()

Support OpenWebScreen() (4.1.7051 onwards)

Definition: open a Website using url in a screen

Syntax:

OpenWebScreen(screen_name, url,x,y)

Example: OpenWebScreen("ThotNot
","https://www.youtube.com/watch?v=uuBETyA_yc",0,0)

When the application runs, the **Web App** automatically open a website using url in **ThotNot** screen.

F.5.7.4.17 CloseWebScreen

CloseWebScreen()

Support CloseWebScreen() (4.1.7051 onwards)

Definition: close Web App in a screen

Syntax:

CloseWebScreen(screen_name)

Example: CloseWebScreen ("ThotNot ")

When the application runs, the Web App is automatically closed in **ThotNot** screen.

F.5.7.4.18 GetScreenList

GetScreenList()

Support GetScreenList () (4.1.6987 onwards)

Definition: Get screen name list, return a table containing screen name list

Syntax:

GetScreenList(pattern)

Parameter	Data type	Description
pattern	string	Returns a list of screen names containing the specified pattern. If no pattern is provided, the function will return to a list of all screen names in the project.

Example: local List = GetScreenList("BAY*"): Return List contains all BAY screen

F.5.7.5 Scripts for Layer

F.5.7.5.1 ShowLayer

Description: Display Layer.

Syntax:

```
ShowLayer(screen_name, layer_name)
```

Example: ShowLayer("AT1", "SEL751_1"): show layer SEL751_1 in the AT1 screen.

F.5.7.5.2 HideLayer

Description: Hide Layer.

Support Hide Layers (4.1.6894 onwards)

Syntax:

```
HideLayer(screen_name, layer_name1, layer_name2...)
```

Example: HideLayer("AT1", "SEL751_1", "SEL751_2"): hide layers SEL751_1, SEL751_2 in the AT1 screen.

F.5.7.5.3 HideAllLayerExcept

Description: Hide all Layers except one.

Support Hide all Layers except (4.1.6894 onwards)

Syntax:

```
HideAllLayersExcept (screen_name, layer_name)
```

Example: HideAllLayersExcept("AT1", "SEL751_1"): hide all layers except SEL751_1 in the AT1 screen.

F.5.7.6 Scripts for Alarm – Event View

F.5.7.6.1 Ack

Script: Ack

Description:

- Change the state of a tag from **Unack** to **Ack**.

Syntax:

```
Ack(Keyname)
```

Parameters:

Parameter	Data type	Description
key_name	string	Key name of the tag/point

Example: Ack("D01.XCBR1.SF6Lok")

F.5.7.6.2 AckAll

Script: **AckAll**

Description:

- Change the state of all tags from **Unack** to **Ack**

Syntax:

```
AckAll()
```

F.5.7.6.3 EnableAlarm

Script: **EnableAlarm**

Description:

- Enable/disable the alarm of a variable

Syntax:

```
EnableAlarm(key_name, is_enable)
```

Parameters:

Parameter	Data type	Description
key_name	string	Key name of the tag/point
is_enable	bool	Value = true: allow an alarm to occur. Value = false: alarm is prohibited.

Example: EnableAlarm("D01.XCBR1.RemEnaSt",false) – alarm is prohibited with variable D01.XCBR1.RemEnaSt.

F.5.7.6.4 AlarmAddSubstation

Script: **AlarmAddSubstation**

Description:

- Add a Substation (Substation Data Type of Equipment Container in the Data Model) to the Alarm Subscription

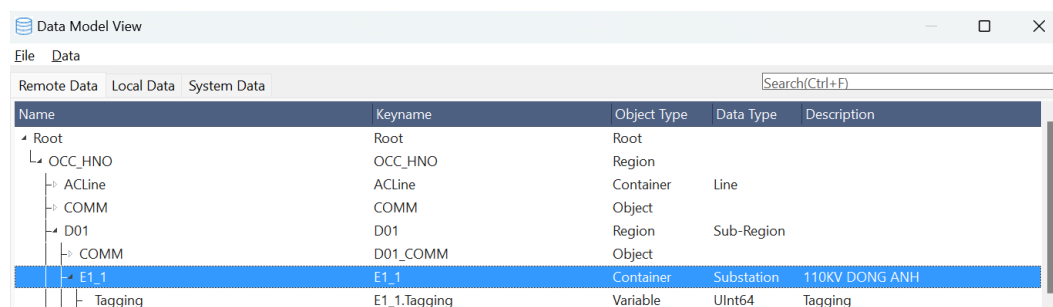
Syntax:

```
AlarmAddSubstation(substation_name)
```

Parameters:

Parameter	Data type	Description
substation_name	string	Substation name in Data Model

Example: AlarmAddSubstation("E1_1"): Subscribe alarm in E1_1 Substation (*Open Alarm Subscription windows to check result*) as location and information of E1_1 in Data Model



Name	Keyname	Object Type	Data Type	Description
Root	Root	Root		
OCC_HNO	OCC_HNO	Region		
ACLine	ACLine	Container	Line	
COMM	COMM	Object		
D01	D01	Region	Sub-Region	
COMM	D01_COMM	Object		
E1_1	E1_1	Container	Substation	110KV DONG ANH
Tagging	E1_1.Tagging	Variable	UInt64	Tagging

F.5.7.6.5 AlarmDeleteSubstation

Script: AlarmDeleteSubstation

Description:

- Remove a Substation (Substation Data Type of Equipment Container in the Data Model) from the Alarm Subscription.

Syntax:

```
AlarmDeleteSubstation(substation_name);
```

Parameters:

Parameter	Type	Description
substation_name	string	Substation name in Data Model

Example: AlarmDeleteSubstation ("E1_1") - Unsubscribe alarm in E1_1 Substation

F.5.7.6.6 AlarmShowSubscription

Script: **AlarmShowSubscription**

Description:

- Open the Alarm Subscription window at a specified location.

Syntax:

```
AlarmShowSubscription(x, y);
```

Parameters:

Parameter	Data type	Description
x	int	Coordinates along the X-axis
Y	int	Coordinates along the Y axis

F.5.7.6.7 AlarmSoundConfig

Script: **AlarmSoundConfig**

Description:

- Open the Alarm Sound Configuration window at a specified location.

Syntax:

```
AlarmSoundConfig(x, y);
```

Parameters:

Parameter	Data type	Description
x	int	X-axis coordinates
Y	int	Y-axis coordinates

F.5.7.6.8 AlarmSoundEnable

Script: **AlarmSoundEnable**

Description:

- Enable Alarm Sound.

Syntax:

```
AlarmSoundEnable(value);
```

Parameters:

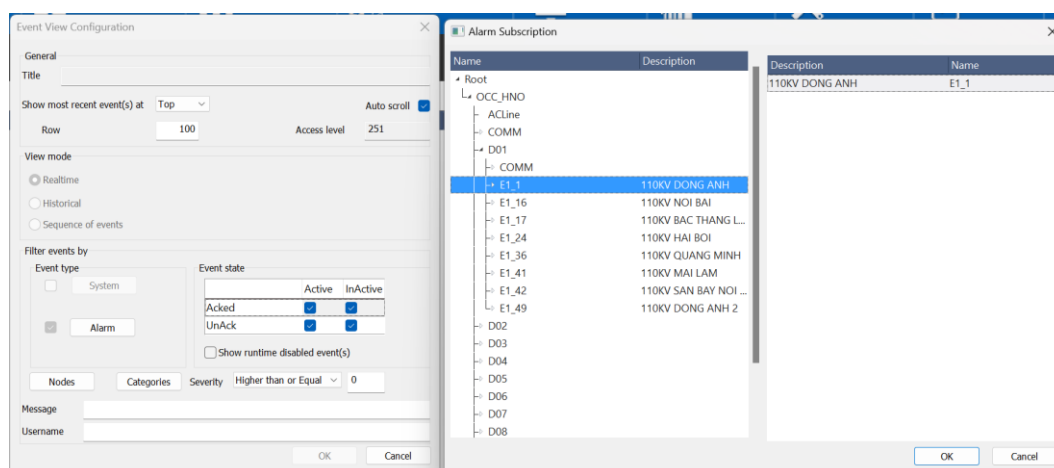
Parameter	Data type	Description
value	int	Value = 0: Disable Alarm Sound Value = 1: Enable Alarm Sound

F.5.7.6.9 EventViewFilterByNodes

Script: EventViewFilterByNodes

Description:

- Update Node Alarm subscription for Event View



Syntax:

```
EventViewFilterByNodes(screen_name,  
eventView_name, nodes);
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Event View object
eventView_name	string	Name of the Event View object

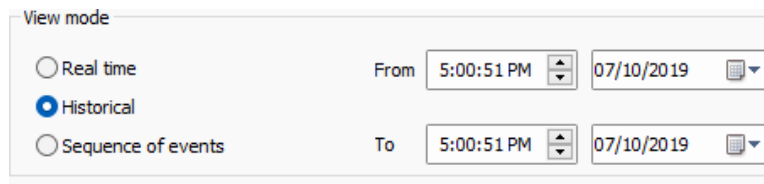
nodes	string	Node name, nodes are separated by ",."
-------	--------	---

F.5.7.6.10 EventViewUpdateHisTime

Script: EventViewUpdateHisTime

Description:

- Update data query interval in Historical mode



Syntax:

```
EventViewUpdateHisTime(screen_name,  
eventView_name, time_begin, time_end);
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Event View object
eventView_name	string	Name of the Event View object
time_begin	DateTime	Query start time
time_end	DateTime	Query end time

F.5.7.7 Scripts interact with Trend View

F.5.7.7.1 HisTrendUpdateLine

Script: HisTrendUpdateLine

Description:

Update configuration data points for Trend View object in Historical Trend mode.

Line configurations	
Line 1	
Description	E02.MMXU1.Hz
Keyname	E02.MMXU1.Hz

Syntax:

```
HisTrendUpdateLine(screen_name, trend_Name, pen1,
des1, pen2, des2, pen3, des3, pen4, des4);
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Trend View object
trend_name	string	Name of the Trend View object
pen1	string	Key name of line 1
des1	string	Description of line 1
pen2	string	Key name of line 2
des2	string	Description of line 2
pen3	string	Key name of line 3
des3	string	Description of line 3
pen4	string	Key name of line 4
des4	string	Description of line 4

F.5.7.7.2 HisTrendUpdateQueryTime

Script: HisTrendUpdateQueryTime

Description:

Update data (send data query command) for Trend View object in Historical Trend mode with changing parameters over time.

From	7:52:02 AM	21/01/2022	To	8:08:42 AM	21/01/2022
------	------------	------------	----	------------	------------

Syntax:

```
HisTrendUpdateQueryTime(screen_name, trend_Name,
time_from, time_to);
```

Parameters:

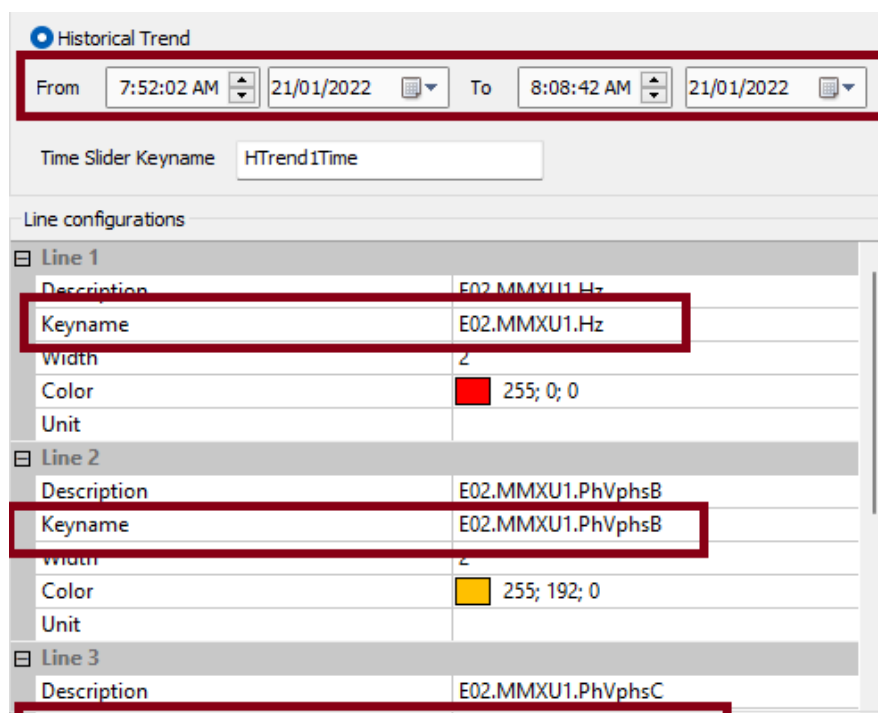
Parameter	Data type	Description
screen_name	string	The name of the screen containing the Trend View object
trend_name	string	Name of the Trend View object
time_from	time_t	The start time of the data query command
time_to	time_t	The end time of the data query command.

F.5.7.7.3 HisTrendUpdateData

Script: HisTrendUpdateData

Description:

Update data (send data query command) for Trend View object in Historical Trend mode with changing parameters of time and data point.



Syntax:

```
HisTrendUpdateData(screen_name, trend_Name, pen1,
pen2, pen3, pen4, time_from, time_to);
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Trend View object
trend_name	string	Name of the Trend View object
pen1	string	Key name of line 1
pen2	string	Key name of line 2
pen3	string	Key name of line 3
pen4	string	Key name of line 4
time_from	time_t	The start time of the data query command
time_to	time_t	The end time of the data query command.

F.5.7.8 Scripts for Tagging

Rework (4.1.6905 onwards)

F.5.7.8.1 TaggingShowManager()

Definition: This function displays the management form Tagging Manager.

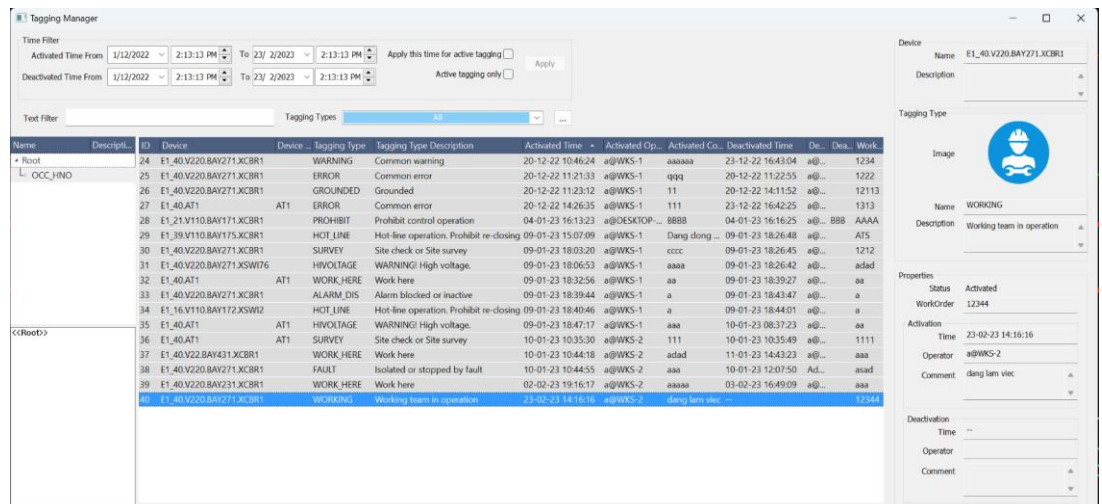


Figure F-240. Form tagging manager

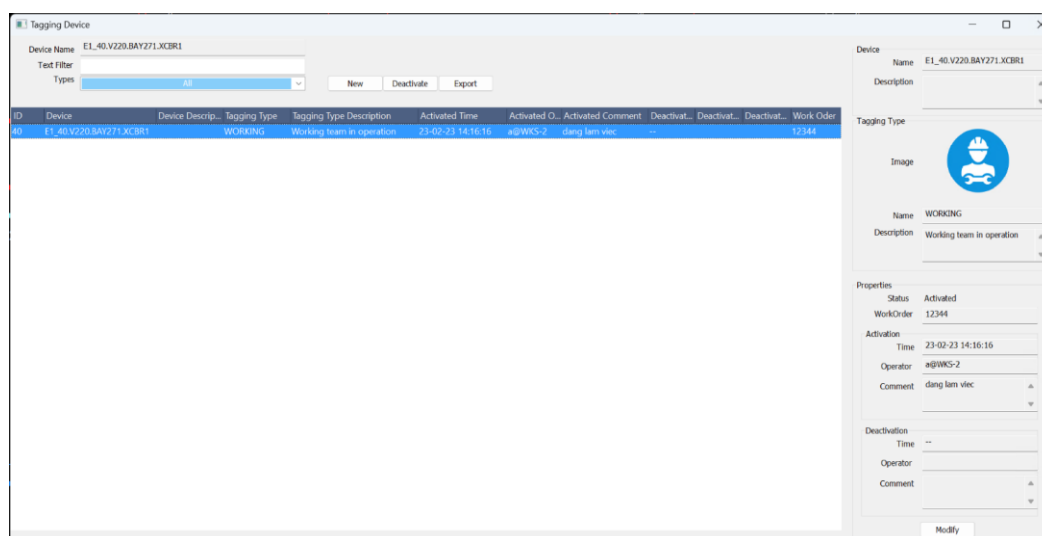
Syntax:

```
TaggingShowManager(x,y)
```

x,y: coordinate position (x,y) , default (-1,-1)

F.5.7.8.2 TaggingShowDevice()

Definition: This function displays the Tagging Device form at coordinates (x,y) to set the status of hanging/lowering the sign for a device. You can hang multiple signs for one device.



ID	Device	Device Description	Tagging Type	Tagging Type Description	Activated Time	Activated O...	Activated Comment	Deactivat...	Deactivat...	Deactivat...	Work Order
40	E1_40.V220.BAY271.XCBR1		WORKING	Working team in operation	23-02-23 14:16:16	aspWMS-2	dang lam vlc	--			12344

Figure F-241. Form Tagging Device

Syntax:

```
TaggingShowDevice(string device_name, int x, int y)
```

Device_name: Device name

x,y: coordinate position (x,y)

Example: ShowTaggingDeive("E1_40.V220.BAY271.XCBR1",200,300)

In the viewer when calling this script, the system will display the Tagging Device form at coordinates (200,300), allowing the user to hang signs, lower the sign and search for the status of the device "device1".

F.5.7.8.3 TaggingSetDevice()

Definition: This function will display the Set Tagging form at the coordinate position (x,y). In this form, the user enters Comment information for Tagging, other parameters of Tagging are automatically updated according to the system and parameters sent to the script.

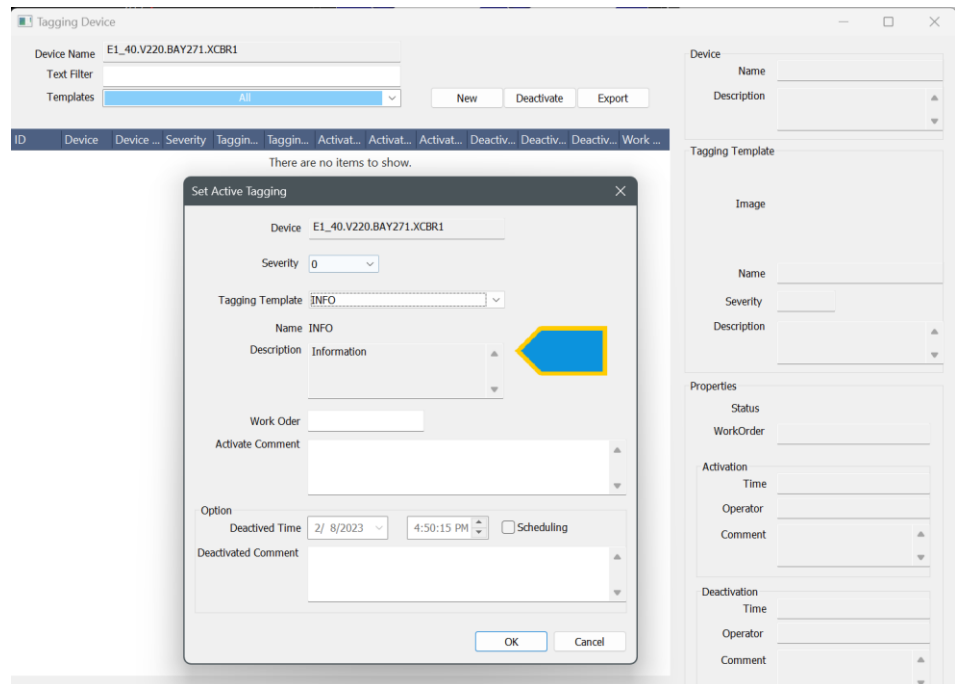


Figure F-242. The set tagging device window

Syntax:

```
TaggingSetDevice(string device_name, int tagging_type,
int x, int y)
```

device_name: Device name.

tagging_type : type of tagging (value from 0 to15)

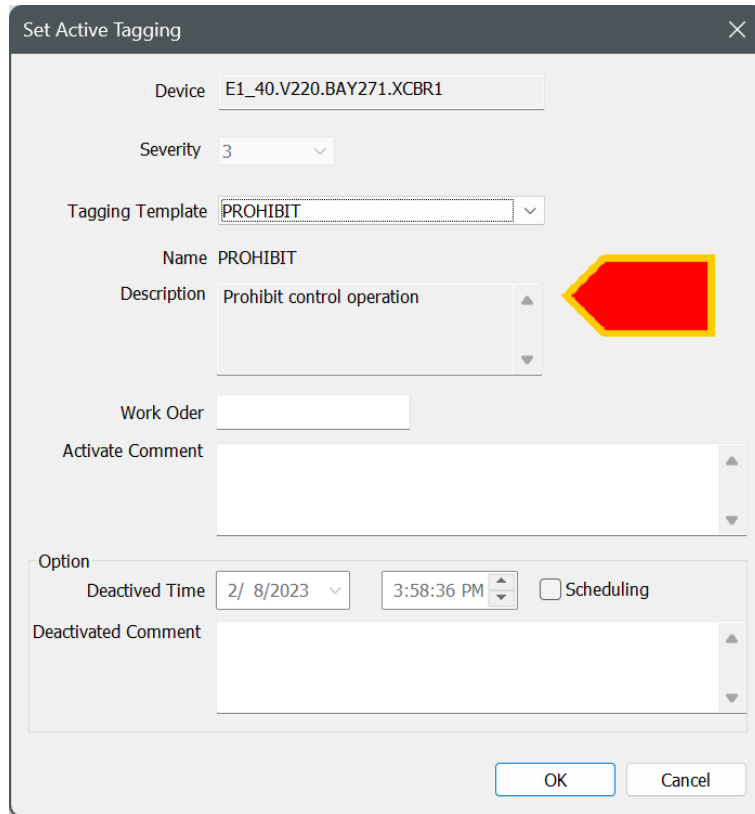
x,y: coordinate position (x,y)

Example: SetTaggingDevice ("E1_40.V220.BAY271.XCBR1", 1, 200,300)

When running the script, the system will display the Tagging Device window form with type =1 at coordinates (200,300). This form allows manager Tagging with E1_40.V220.BAY271.XCBR1.

F.5.7.8.4 TaggingSetRed()

Definition: This function will display the Set Red Tagging form (red sign) at the coordinate position (x,y). In this form, the user enters Comment information for Tagging, other parameters of Tagging are automatically updated according to the system and parameters passed to the script.



The 'Set Active Tagging' dialog box contains the following fields and controls:

- Device:** Text input field with value 'E1_40.V220.BAY271.XCBR1'.
- Severity:** Dropdown menu with value '3'.
- Tagging Template:** Dropdown menu with value 'PROHIBIT'.
- Name:** Text input field with value 'PROHIBIT'.
- Description:** Text input field with value 'Prohibit control operation'. To the right of this field is a red arrow-shaped tag with a yellow border.
- Work Oder:** Text input field.
- Activate Comment:** Text input field.
- Option:** A section containing:
 - Deactivated Time:** A date and time selector showing '2/ 8/2023' and '3:58:36 PM'.
 - Scheduling:** A checkbox that is currently unchecked.
- Deactivated Comment:** Text input field.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom right.

Figure F-243. The set Red tagging window

Syntax:

```
TaggingSetRed(string device_name, int x, int y)
```

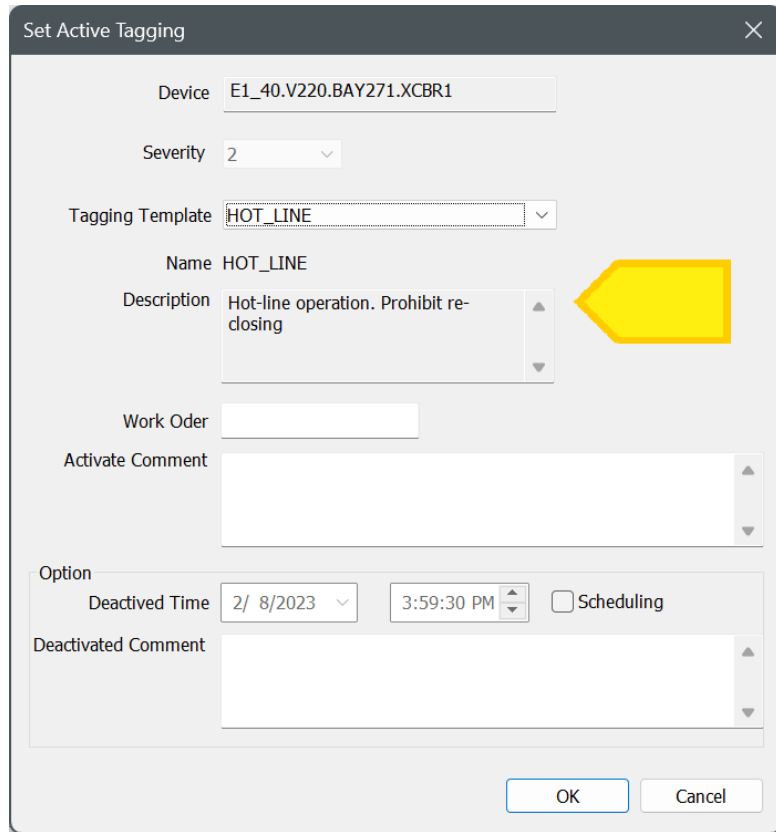
Device_name: Device name.

Example: TaggingSetRed ("E1_40.V220.BAY271.XCBR1", 200,300)

At viewer, when running the script, the system will display the Set Tagging form at coordinates (200,300). This form allows you to hang a red sign with E1_40.V220.BAY271.XCBR1.

F.5.7.8.5 TaggingSetYellow()

Definition: This function will display the Set Yellow Tagging form (sea yellow leopard) at coordinate position (x,y). In this form, the user enters Comment information for Tagging, other parameters of Tagging are automatically updated according to the system and parameters passed to the script.



The 'Set Active Tagging' dialog box contains the following fields and controls:

- Device:** Text field with value 'E1_40.V220.BAY271.XCBR1'.
- Severity:** Dropdown menu with value '2'.
- Tagging Template:** Dropdown menu with value 'HOT_LINE'.
- Name:** Text field with value 'HOT_LINE'.
- Description:** Text area with value 'Hot-line operation. Prohibit re-closing'. To the right of this field is a yellow arrow-shaped tag icon.
- Work Order:** Text field.
- Activate Comment:** Text area.
- Option:** A sub-section containing:
 - Deactivated Time:** A date dropdown (2/ 8/2023) and a time spinner (3:59:30 PM).
 - Scheduling:** A checkbox.
- Deactivated Comment:** Text area.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom right.

Figure F-244. The set yellow tagging window

Syntax:

```
TaggingSetYellow(string device_name, int x, int y)
```

Device_name: device name.

Example: TaggingSetYellow ("E1_40.V220.BAY271.XCBR1", 200,300)

At viewer, when running the script, the system will display the Set Tagging form at coordinates (200,300). This form allows you to hang a yellow sign with E1_40.V220.BAY271.XCBR1.

F.5.7.8.6 TaggingSetPink()

Definition: This function will display the Set Pink Tagging form at the coordinate position (x,y). In this form, the user enters Comment information for Tagging, other parameters of Tagging are automatically updated according to the system and parameters passed to the script.

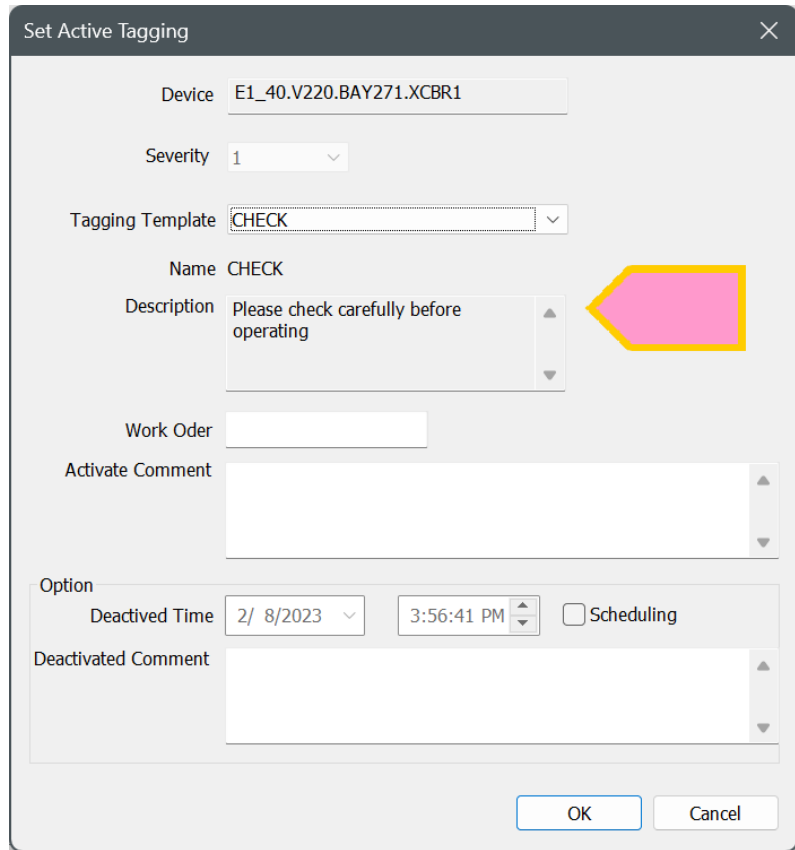


Figure F-245. The set Pink tagging window

Syntax:

```
TaggingSetPink(string device_name, int x, int y)
```

Device_name: device name.

Example: TaggingSetPink ("E1_40.V220.BAY271.XCBR1", 200,300)

At viewer, when running the script, the system will display the Set Tagging form at coordinates (200,300). This form allows you to hang a pink sign with E1_40.V220.BAY271.XCBR1.

F.5.7.8.7 TaggingSetBlue()

Definition: This function will display the Set Blue Tagging form at the coordinate position (x,y). In this form, the user enters Comment information for Tagging, other parameters of Tagging are automatically updated according to the system and parameters passed to the script.

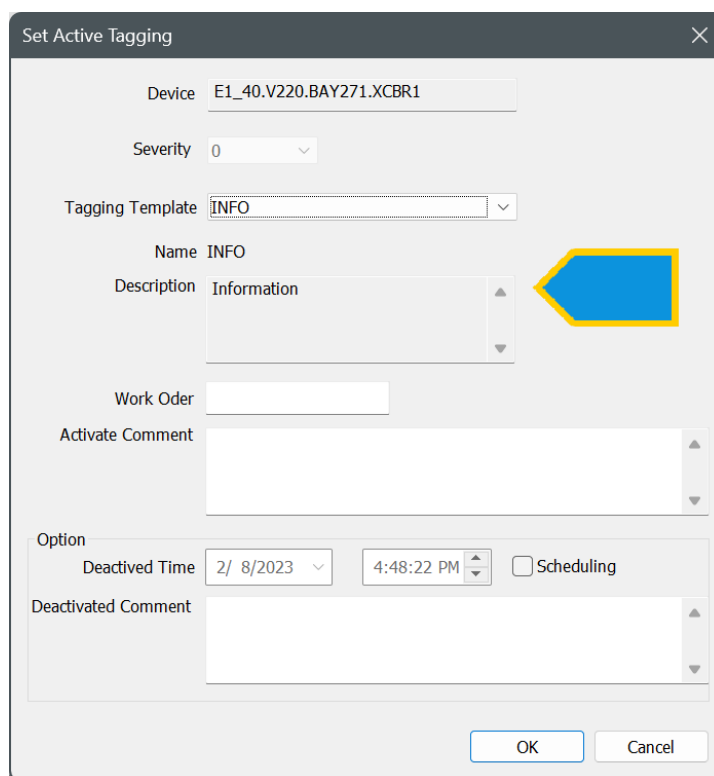


Figure F-246. The set Blue tagging window

Syntax:

```
TaggingSetBlue(string device_name, int x, int y)
```

Device_name: device name.

Example: TaggingSetBlue ("E1_40.V220.BAY271.XCBR1", 200,300)

At viewer, when running the script, the system will display the Set Tagging form at coordinates (200,300). This form allows you to hang a blue sign with E1_40.V220.BAY271.XCBR1.

F.5.7.8.8 TaggingAddNodeToView()

Definition: This function will display the Tagging manager and display the signage of the node and within the node [Optional].

Support TaggingAddNodeToView() (4.1.6982 onwards)

Syntax:

```
TaggingAddNodeToView(string device_name, bool  
isIncludeChild, bool isClearView)
```

Device_name: device name.

isIncludeChild: include tagging of child node in Tagging Manager View.

isClearView: clear Tagging Manager View before add device to view.

Example: TaggingAddNodeToView("E1_40.V220.BAY271",true,true)

At viewer, when running the script, the system will display the Tagging Manager View. All of tags in node E1_40.V220.BAY271 and child will be displayed.

F.5.7.9 Scripts for Data Grid

Rework (4.1.7031 onwards)

F.5.7.9.1 DataGridGetItem

Script: DataGridGetItem

Description:

- The function gets the value of the item at the position with row **x**-column **y** coordinates.
- The function returns the string item value.

Syntax:

```
DataGridGetItem(screen_name, dataGrid_name, row_x,  
col_y)
```

Parameters:

Paremeter	Type	Description
screen_name	string	The screen name contains the Data Grid
dataGrid_name	string	Data Grid object name
row_x, col_y	int	The position (order) of the row - column of the item to get the value. The rows are in ascending order from top to bottom, the first row is in order 0. Columns are in ascending order from left to right, the first column is in order 0.

Example: DataGridGetItem("GRID","DataGrid1",0,0)

F.5.7.9.2 DataGridAddRow

Script: DataGridAddRow

Description:

- Allows adding a new row to *the bottom row* of the DataGrid.
- Boolean return result:
 - true – if more successful
 - false – if add failed.

Syntax:

```
DataGridAddRow(screen_name, dataGrid_name,  
para1;para2; ...)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The screen name contains the Data Grid
dataGrid_name	string	Data Grid object name
para1; para2; ...	string	values for the columns you want to add, columns separated by semicolons ";"

Example: DataGridGetItem("GRID","DataGrid1","171;172")

F.5.7.9.3 DataGridInsertRow

Remove (4.1.7031 onwards)

F.5.7.9.4 DataGridSetItem

Script: DataGridSetItem

Description:

- The function enters a string value for the item at the position with row x-column y coordinates.

Syntax:

```
DataGridSetItem(screen_name, dataGrid_name, row_x,  
col_y, value)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The screen name contains the Data Grid
dataGrid_name	string	Data Grid object name
row_x, col_y	int	Position (order) row - column of the item to be entered value. The rows are in ascending order from top to bottom, the first row is in order 0. Columns are in ascending order from left to right, the first column is in order 0.
value	string	The value of the item

Example: DataGridSetItem("GRID", "DataGrid1", 0, 0, "171") – set the value at row 0 column 0 equal to 171.

F.5.7.9.5 DataGridRemoveRow

Script: **DataGridRemoveRow**

Description:

- Delete a row of DataGrid.

Syntax:

```
DataGridRemoveRow(screen_name, dataGrid_name, row_x)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The screen name contains the Data Grid
dataGrid_name	string	Data Grid object name
row_x	int	The position (order) of the row from which the line was deleted.

		The rows are in ascending order from top to bottom, the first row is in order 0.
--	--	--

Example: `DataGridRemoveRow("GRID ", "DataGrid1", 2)`

F.5.7.9.6 DataGridRemoveAll

Script: **DataGridRemoveAll**

Description:

- Delete all row of DataGrid.

Syntax:

```
DataGridRemoveAll(screen_name, dataGrid_name)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The screen name contains the Data Grid
dataGrid_name	string	Data Grid object name

F.5.7.9.7 DataGridFind

Script: **DataGridFind**

Description:

- Find value in column.
- Results returned: First row index found containing the input string (returns -1 if not found)

Syntax:

```
DataGridFind(screen_name, dataGrid_name, col, input)
```

Parameters:

Parameter	Data type	Description
-----------	-----------	-------------

screen_name	string	The screen name contains the Data Grid
dataGrid_name	string	Data Grid object name
col	int	column you want to search
input	string	value you want to find.

F.5.7.9.8 DataGridFindNoCase

Script: DataGridFindNoCase

Description:

- Find value in column.
- Results returned: First row index found cells containing the input string (not case sensitive and returns -1 if not found).

Syntax:

```
DataGridFindNoCase(screen_name, dataGrid_name,  
col, input)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The screen name contains the Data Grid
dataGrid_name	string	Data Grid object name
col	int	column users want to search.
input	string	value users want to search

F.5.7.9.9 DataGridSelectedRow

Script: DataGridSelectedRow

Description:

- Return index of selected Row in Data Grid

Syntax:

```
DataGridSelectedRow(screen_name, dataGrid_name,  
row_x)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The screen name contains the Data Grid
dataGrid_name	string	Data Grid object name
row_x	int	Column should be selected. The rows are in ascending order from top to bottom, the first row is in order 0.

F.5.7.9.10 DataGridGetRowCount

Script: DataGridGetRowCount

Description:

- Returns the total number of rows of the DataGrid, integer type.

Syntax:

```
DataGridGetRowCount(screen_name, dataGrid_name)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The screen name contains the Data Grid
dataGrid_name	string	Data Grid object name

F.5.7.9.11 DataGridLoadCSV

Script: DataGridLoadCSV

Description:

- Import content from *.csv file onto Data Grid.

Syntax:

```
DataGridLoadCSV(screen_name, dataGrid_name,  
file_path)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The screen name contains the Data Grid
dataGrid_name	string	Data Grid object name
file_path	string	The path to where the *.csv file is saved. Note in the path use “\\” instead of “\”

[F.5.7.9.12 DataGridSaveCSV](#)

Script: DataGridSaveCSV

Description:

- Save the content from the Data Grid into a *.csv file.

Syntax:

```
DataGridSaveCSV(screen_name, dataGrid_name,  
file_path)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The screen name contains the Data Grid
dataGrid_name	string	Data Grid object name
file_path	string	The path to where the *.csv file is saved.

		Note in the path use “\\” instead of “\”
--	--	--

F.5.7.9.13 DataGridReplace

Script: DataGridReplace

Description:

- Within a column, search and replace any string on the Data Grid with another string. The function will replace the first string found.

Syntax:

```
DataGridReplace(screen_name, dataGrid_name, col,
old_string, new_string)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The screen name contains the Data Grid
dataGrid_name	string	Data Grid object name
col	int	Column users want to replace
old_string	string	String users want to replace
new_string	string	New string after replacement

F.5.7.9.14 DataGridReplaceAll

Script: DataGridReplaceAll

Description:

- In all columns, search and replace any string on the Data Grid with another string. The function will replace the first string found..

Syntax:

```
DataGridReplaceAll(screen_name, dataGrid_name,
old_string, new_string)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The screen name contains the Data Grid
dataGrid_name	string	Data Grid object name
old_string	string	String users want to replace
new_string	string	New string after replacement

F.5.7.9.15 GetText

Script: **GetText**

Description:

- Return cell value.

Syntax:

```
local Data_Grid = DataGrid("screen_name", "DataGrid_name")
Data_Grid.GetText(row_index, column_index)
```

Parameters:

Parameter	Data type	Description
row_index	int	Row index of cell
column_index	int	Column index of cell

F.5.7.9.16 SetText

Script: **SetText**

Description:

- Set value to cell.

Syntax:

```
local Data_Grid = DataGrid("screen_name", "DataGrid_name")
Data_Grid.SetText(row_index, column_index,
"value")
```

Parameters:

Parameter	Data type	Description
row_index	int	Row index of cell
column_index	int	Column index of cell
value	string	Value

F.5.7.9.17 SetTextColor

Script: **SetTextColor**

Description:

- Set cell text color.

Syntax:

```
local Data_Grid = DataGrid("screen_name", "DataGrid_name")
Data_Grid:SetTextColor(row_index,
column_index,RGB(r,g,b))
```

Parameters:

Parameter	Data type	Description
row_index	int	Row index of cell
column_index	int	Column index of cell
RGB	int	RGB Color code

F.5.7.9.18 AddRow

Script: **AddRow**

Description:

- Add row.

Syntax:

```
local Data_Grid = DataGrid("screen_name", "DataGrid_name")
Data_Grid:AddRow(para1)
Data_Grid:AddRow(row_index, para1;para2; ...)
```

Parameters:

Parameter	Data type	Description
row_index	int	Row index of cell (add 1 row at row_index). Without this parameter, a new row will be added to the end of the table and the data will be filled in the first column of the row.
para1;para2;...	string	values for the columns you want to add, columns separated by semicolons ";"

F.5.7.9.19 UpdateRow

Script: **UpdateRow**

Description:

- Update 1 row.

Syntax:

```
local Data_Grid = DataGrid("screen_name", "DataGrid_name")
Data_Grid:UpdateRow(row_index, para1;para2;...)
```

Parameters:

Parameter	Data type	Description
row_index	int	Row index of cell
para1;para2;...	string	values for the columns you want to update, columns separated by semicolons ";"

F.5.7.9.20 RemoveRow

Script: **RemoveRow**

Description:

- Remove 1 row.

Syntax:

```
local Data_Grid = DataGrid("screen_name", "DataGrid_name")
```

Data_Grid:RemoveRow(row_index)

Parameters:

Parameter	Data type	Description
row_index	int	Row index of cell

F.5.7.9.21 Reset

Script: **Reset**

Description:

- Clear all content of Data Grid.

Syntax:

```
local Data_Grid = DataGrid("screen_name", "DataGrid_name")
Data_Grid:Reset()
Data_Grid:Reset(clear_header = false)
```

Parameters:

Parameter	Data type	Description
clear_header	bool	Flag clear header (true : clear header)

F.5.7.9.22 SelectRow

Script: **SelectRow**

Description:

- Return row index.

Syntax:

```
local Data_Grid = DataGrid("screen_name", "DataGrid_name")
Data_Grid:SelectRow()
```

F.5.7.9.23 RowCount

Script: **RowCount**

Description:

- Returns the total number of rows in Data Grid.

Syntax:

```
local Data_Grid = DataGrid("screen_name", "DataGrid_name")  
Data_Grid:RowCount()
```

F.5.7.9.24 Load

Script: **Load**

Description:

- Load Data to Data Grid.

Syntax:

```
local Data_Grid = DataGrid("screen_name", "DataGrid_name")  
Data_Grid:Load(file_name)  
Data_Grid:Load(contents, headers)
```

Parameters:

Parameter	Data type	Description
file_name	string	File path
contents, headers		Content, header in Table (CSV Parse)

Example 1: Data_Grid:Load("E:\\HMI\\OCC_HNO\\Grid.csv")

Example 2: Data_Grid:Load({{'a','b'},{'c','d'}},{'header1','header2'})

F.5.7.9.25 Save

Script: **Save**

Description:

- Save Data Grid to file.

Syntax:

```
local Data_Grid = DataGrid("screen_name", "DataGrid_name")  
Data_Grid:Save()  
Data_Grid:Save(file_name)
```

Parameters:

Parameter	Data type	Description
-----------	-----------	-------------

file_name	string	File path. If this parameter is not specified, it will save to the path configured in the Data Source (Data Grid Configuration).– Refer to F.3.14.4.3)
------------------	--------	--

F.5.7.9.26 Filter

Script: **Filter**

Description:

- Filter according to callback (callback returns true to retain the row_index line, the opposite is to delete).

Syntax:

```
local Data_Grid = DataGrid("screen_name", "DataGrid_name")
Data_Grid: Filter(callback)
-- as function callback(Data_Grid,row_index)
```

Parameters:

Parameter	Data type	Description
callback	function	Callback funtion
row_index	int	Row index of cell

Example:

```
-- PointFilter: Input string Local variable
local col = 3    --column
--define filter callback function
local function filter(Data_Grid,row)
    local key = Data_Grid:GetText(row,col)
    if string.find(key,PointFilter.value) then
        return true
    else
        return false
    end
end
end
```

```
-- Filter  
if PointFilter.value ~= "" then  
    Data_Grid:Filter(filter)  
end
```

F.5.7.9.27 ResizeColumn

Script: **ResizeColumn**

Description:

- Resize Column.

Syntax:

```
local Data_Grid = DataGrid("screen_name", "DataGrid_name")  
Data_Grid:ResizeColumn(column_index,width)  
Data_Grid:ResizeColumn(size_array)
```

Parameters:

Parameter	Data type	Description
column_index	int	Column index of cell
width	int	Column width
size_array		Size array in Table (CSV Parse)

F.5.7.9.28 SetVisibleColumn

Script: **SetVisibleColumn**

Description:

- Show/hide column.

Syntax:

```
local Data_Grid = DataGrid("screen_name", "DataGrid_name")  
Data_Grid:SetVisibleColumn(column_index,visible)  
Data_Grid:SetVisibleColumn(visible_array)
```

Parameters:

Parameter	Data type	Description
-----------	-----------	-------------

column_index	int	Column index of cell
visible	bool	true : visible, false : invisible
visible_array		visible array in Table (CSV Parse)

F.5.7.10 Scripts for Relay Event View

F.5.7.10.1 RelayEventRefresh

Script: **RelayEventRefresh**

Description:

Send SUM/HIS command to SEL relay via TCP telnet connection:

- Relay connection address parameters: configured in Parameter command scripts.
- Syntax ASCII commands: configured by the Execute Command properties of the Relay Event View object.

Syntax:

```
RelayEventRefresh(screen_name, relayView_name,
relay_ip, relay_port, pwd_level1, pwd_level2)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Relay Event View
relayView_name	string	Name of the Relay Event View object
relay_ip	string	IP of relay
relay_port	string	Port telnet of relay
pwd_level1	string	Password level 1 – relay SEL
pwd_level2	string	Password level 2 – relay SEL

F.5.7.11 Scripts for Tabular

F.5.7.11.1 TabularSetFilterDataType

Script: **TabularSetFilterDataType**

Description:

Filter the display list of data on Tabular by Data Type.

Syntax:

```
TabularSetFilterDataType(screen_name,  
tabular_name, data_type)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Tabular
tabular_name	string	The name of the Tabular object
data_type	string	Condition Data type is filtered

F.5.7.11.2 TabularSetFilterDescription

Script: **TabularSetFilterDescription**

Description:

Filter the display list of data on Tabular by Description.

Syntax:

```
TabularSetFilterDescription( screen_name,  
tabular_name, description)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Tabular

tabular_name	string	The name of the Tabular object
description	string	Description condition of the filtered data

F.5.7.11.3 *TabularSetFilterKeyName*

Script: **TabularSetFilterKeyName**

Description:

Filter the display list of data on Tabular by KeyName.

Syntax:

```
TabularSetFilterKeyName( screen_name,  
tabular_name, key_name)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Tabular
tabular_name	string	The name of the Tabular object
key_name	string	Condition Key Name of the filtered data.

F.5.7.11.4 *TabularSetFilterName*

Script: **TabularSetFilterName**

Description:

Filter the display list of data on Tabular by Name

Syntax:

```
TabularSetFilterName( screen_name, tabular_name,  
name)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Tabular
tabular_name	string	The name of the Tabular object
name	string	Condition Name of the filtered data

F.5.7.11.5 TabularSetSubstation

Script: **TabularSetSubstation**

Description:

Filter the display list of data on Tabular by Substation node on DataModel.

Syntax:

```
TabularSetSubstation( screen_name, tabular_name,
substation);
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Tabular
tabular_name	string	The name of the Tabular object
substation	string	Substation node condition of filtered data

F.5.7.11.6 TabularSetAllFilters

Script: **TabularSetAllFilters**

Description:

Filter the display list of data on Tabular by Substation, Name, Keyname, Description, and Data Type

Syntax:

```
TabularSetAllFilters(screen_name, tabular_name,
substation, filter name, keyname, description,
data_type)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Tabular
tabular_name	string	The name of the Tabular object
substation	string	Substation node condition of filtered data
name	string	Condition Name of the filtered data
key_name	string	Condition Key Name of the filtered data.
description	string	Description condition of the filtered data
data_type	string	Condition Data type is filtered

F.5.7.12 Scripts for Chart

F.5.7.12.1 ChartAddLine

Script: **ChartAddLine**

Description:

Add a pen line to the chart. The newly added line will have the same properties as the previous data point (bottom in the list of pen nets), except the color attribute is randomly generated by the application.

Syntax:

```
ChartAddLine( screen_name, chart_name, key_name);
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Chart

chart_name	string	The name of the Chart object.
key_name	string	Parameter pen/key_name

F.5.7.12.2 ChartAddLineEx

Script: ChartAddLineEx

Description:

Add a pen line to the chart with full Parameters.

Syntax:

```
ChartAddLineEx(screen_name, chart_name, keyname,
custom_description, color, aggregate_type,
line_type, realtime_source, pen_width, time_frame,
y_axis_index, is_show_summary, is_show_label);
```

Parameters:

Parameter	Data type	Description		
screen_name	string	The name of the screen containing the Chart		
chart_name	string	The name of the Chart object.		
key_name	string	The parameter pen/key_name		
custom_description	string	Custom description of pen/keyname		
Color	string	Parameter pen/color Value range: RGB - hex code with Data type as string, excluding # character For example, the code #FF0000 will only be passed as “FF0000”		
aggregate_type	int	Parameter pen/aggregate Value range: <table><tr><td>Value</td><td>type</td></tr></table>	Value	type
Value	type			

		0	Raw										
		2341	Interpolate										
		2342	Average										
		2346	Min										
		2347	Max										
		11285	Average-Abs										
		11286	Min-Abs										
		11287	Max-Abs										
line_type	int	Parameter pen/Line_type Value range: <table><tr><td>Value</td><td>type</td></tr><tr><td>0</td><td>Straight Line</td></tr><tr><td>1</td><td>Spline</td></tr><tr><td>2</td><td>Step Line</td></tr><tr><td>3</td><td>Bar</td></tr></table>		Value	type	0	Straight Line	1	Spline	2	Step Line	3	Bar
Value	type												
0	Straight Line												
1	Spline												
2	Step Line												
3	Bar												
realtime_source		Parameter pen/Realtime Data Value range: <table><tr><td>Value</td><td>type</td></tr><tr><td>0</td><td>DataSource (Raw)</td></tr><tr><td>1</td><td>His Server (Processed)</td></tr></table>		Value	type	0	DataSource (Raw)	1	His Server (Processed)				
Value	type												
0	DataSource (Raw)												
1	His Server (Processed)												
pen_width	int	Parameter pen/width											
time_frame	Int64_t	Parameter pen/Time_frame Unit: seconds											
y_axis_id	int	Parameter pen/Y_axis											

		The value is the ID of the declared Y-axis. The ID of the vertical axis (Y-axis) is calculated from top to bottom in the table of Y-axis, starting from 0.
is_show_summary	bool	Parameter pen/Summary
is_show_label	bool	Parameter pen/Label

F.5.7.12.3 ChartAddTemplate

Script: **ChartAddTemplate**

Description:

Add Template to Chart.

Syntax:

```
ChartAddTemplate( screen_name, chart_name,
template_name);
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Chart
chart_name	string	The name of the Chart object
template_name	string	The name of the template

F.5.7.12.4 ChartApply

Script: **ChartApply**

Description:

Update the configuration for the chart (after using other scripts to change the configuration)

Syntax:

```
ChartApply( screen_name, chart_name);
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Chart
chart_name	string	The name of the Chart object

F.5.7.12.5 ChartRemoveAllLine

Script: **ChartRemoveAllLine**

Description:

Remove all lines from the chart.

Syntax:

```
ChartRemoveAllLine( screen_name, chart_name);
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Chart
chart_name	string	The name of the Chart object

F.5.7.12.6 ChartRemoveLine

Script: **ChartRemoveLine**

Description:

Remove 1 line (key_name) from the chart.

Syntax:

```
ChartRemoveLine( screen_name, chart_name,  
key_name);
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Chart
chart_name	string	The name of the Chart object
key_name	string	The parameter pen/key_name

F.5.7.12.7 ChartSetCyclicMode

Script: **ChartSetCyclicMode**

Description:

Cyclic mode setting for Chart.

Syntax:

```
ChartSetCyclicMode( screen_name, chart_name,
start_after, duration, cyclic_interval,
update_time_interval);
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Chart
chart_name	string	The name of the Chart object
start_after	int64_t	Parameter Cyclic Mode/Start After Unit: seconds
duration	int64_t	Parameter Cyclic Mode/Duration Unit: seconds
cyclic_interval	int	Parameter Cyclic/Update Cycle <i>Value range:</i>

		<table><tr><td>Value</td><td>type</td></tr><tr><td>0</td><td>Quarter Hourly</td></tr><tr><td>1</td><td>Hourly</td></tr><tr><td>2</td><td>Daily</td></tr><tr><td>3</td><td>Weekly</td></tr><tr><td>4</td><td>Monthly</td></tr></table>	Value	type	0	Quarter Hourly	1	Hourly	2	Daily	3	Weekly	4	Monthly
		Value	type											
		0	Quarter Hourly											
		1	Hourly											
		2	Daily											
		3	Weekly											
4	Monthly													
	—													
update_time_interval	int64_t	Parameter Update Method/Micellaneous/Update Data Interval Unit: seconds												

F.5.7.12.8 ChartSetHistoricalMode

Script: ChartSetHistoricalMode

Description:

Setting History mode for Chart.

Syntax:

```
ChartSetHistoricalMode( screen_name, chart_name,
start_time, end_time, update_time_interval);
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Chart
chart_name	string	The name of the Chart object
start_time	int64_t	Parameter Historical/Start At Value:

		(Int64) number of seconds from 07:00:00 1/1/1970 UTC to the specified time
end_time	int64_t	Parameter Historical/End At Value: (Int64) number of seconds from 07:00:00 1/1/1970 UTC to the specified time
update_time_interval	int64_t	Parameter Update Method/Micellaneous/Update Data Interval Unit: seconds

F.5.7.12.9 ChartSetSlidingMode

Script: ChartSetSlidingMode

Description:

Setting the Sliding window mode for Chart.

Syntax:

```
ChartSetSlidingMode( screen_name, chart_name,
duration, reserve, update_time_interval);
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Chart
chart_name	string	The name of the Chart object
duration	int64_t	Parameter Sliding Window/Window Duration Unit: seconds
reserve	int64_t	Parameter Sliding Window/Reserve Duration

		Unit: seconds
update_time_interval	int64_t	Parameter Update Method/Micellaneous/Update Data Interval Unit: seconds

F.5.7.12.10 ChartSetTimeLabelInterval

Script: ChartSetTimeLabelInterval

Description:

Set the Parameter Miscellaneous/Time Label Interval for the chart.

Syntax:

```
ChartSetTimeLabelInterval( screen_name,
chart_name, interval);
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Chart
chart_name	string	The name of the Chart object
interval	int64_t	Parameter Miscellaneous/Time Label Interval Unit: _

F.5.7.13 Scripts for Combo Box

F.5.7.13.1 ComboBoxAddItem

Script: ComboBoxAddItem

Description:

Add itemText to Combo Box control

Syntax:

```
ComboBoxAddItem(screen_name, control_name,
itemText)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Combo Box
control_name	string	Name of Combo Box object
itemText	string	Item

F.5.7.13.2 ComboboxClear

Script: **ComboboxClear**

Description:

Delete the entire item list of the Combo Box control.

Syntax:

```
ComboboxClear(screen_name, control_name)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Combo Box
control_name	string	Name of Combo Box object

F.5.7.13.3 ComboBoxFindItem

Script: **ComboBoxFindItem**

Description:

Search items by keyword

Syntax:

```
ComboBoxFindItem(screen_name, control_name,  
keyword, searchType)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Combo Box
control_name	string	Name of Combo Box object
keyword	string	Search content.
searchType	int	searchType = 1: Search item = keyword searchType = 0: Search for items similar to keyword

F.5.7.13.4 ComboboxGetCurItem

Script: **ComboboxGetCurItem**

Description:

Get the item value (text) that the user is selecting

Syntax:

```
ComboboxGetCurItem(screen_name, control_name)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Combo Box
control_name	string	Name of Combo Box object

F.5.7.13.5 ComboboxGetItem

Script: **ComboboxGetItem**

Description:

Get the item (text) value by position in the control's item list.

Syntax:

```
ComboboxGetItem(screen_name, control_name,  
position)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Combo Box
control_name	string	Name of Combo Box object
position	int	Item's position (order) - Index 0 (first item has position 0)

F.5.7.13.6 ComboboxRemoveCurItem

Script: **ComboboxRemoveCurItem**

Description:

Delete the item that the user is currently selecting

Syntax:

```
ComboboxRemoveCurItem(screen_name, control_name)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Combo Box
control_name	string	Name of Combo Box object

F.5.7.13.7 ComboBoxRemoveItem

Script: **ComboBoxRemoveItem**

Description:

Remove itemText of control

Syntax:

```
ComboboxRemoveItem(screen_name, control_name,  
itemText)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Combo Box
control_name	string	Name of Combo Box object
itemText	string	Item content to be deleted

F.5.7.13.8 *ComboBoxRemoveItemAt*

Script: **ComboBoxRemoveItemAt**

Description:

Delete item by position in control

Syntax:

```
ComboboxRemoveItemAt(screen_name, control_name,  
position)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Combo Box
control_name	string	Name of Combo Box object
position	int	Item's position (order) - Index 0 (first item has position 0)

F.5.7.13.9 *ComboBoxSortItem*

Script: **ComboBoxSortItem**

Description:

Sort the list of items in the control in order

Syntax:

```
ComboBoxSortItem(screen_name, control_name, order)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the Combo Box
control_name	string	Name of Combo Box object
order	string	Order = "Des": Sort by list of values descending Order = "Acs": Sort by list of ascending values

F.5.7.14 Scripts for List Box

F.5.7.14.1 ListBoxAddItem

Script: **ListBoxAddItem**

Description:

Add itemText to the List Box control

Syntax:

```
ListBoxAddItem(screen_name, control_name,  
itemText)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the List Box
control_name	string	Name of the List Box object
itemText	string	Item

F.5.7.14.2 ListboxClear

Script: **ListboxClear**

Description:

Delete the entire list of items of the List Box control

Syntax:

```
ListboxClear(screen_name, control_name)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the List Box
control_name	string	Name of the List Box object

F.5.7.14.3 ListboxFindItem

Script: **ListboxFindItem**

Description:

Search items by keyword

Syntax:

```
ListboxFindItem(screen_name, control_name,  
keyword, searchType)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the List Box
control_name	string	Name of the List Box object
keyword	string	Search content
searchType	int	searchType = 1: Search item = keyword

		searchType = 0: Search for items similar to keyword
--	--	---

F.5.7.14.4 ListboxGetCurItem

Script: **ListboxGetCurItem**

Description:

Get the item value (text) that the user is selecting

Syntax:

```
ListboxGetCurItem(screen_name, control_name)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the List Box
control_name	string	Name of the List Box object

F.5.7.14.5 ListboxGetItem

Script: **ListboxGetItem**

Description:

Get the item (text) value by position in the control's item list

Syntax:

```
ListboxGetItem(screen_name, control_name, position)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the List Box
control_name	string	Name of the List Box object

position	int	Item's position (order) - Index 0 (first item has position 0)
-----------------	-----	--

F.5.7.14.6 ListboxRemoveCurItem

Script: **ListboxRemoveCurItem**

Description:

Delete the item that the user is currently selecting

Syntax:

```
ListboxRemoveCurItem(screen_name, control_name)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the List Box
control_name	string	Name of the List Box object

F.5.7.14.7 ListBoxRemoveItem

Script: **ListBoxRemoveItem**

Description:

Remove itemText of control

Syntax:

```
ListboxRemoveItem(screen_name, control_name, itemText)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the List Box
control_name	string	Name of the List Box object

itemText	string	Item content to be deleted
-----------------	--------	----------------------------

F.5.7.14.8 *ListBoxRemoveItemAt*

Script: **ListBoxRemoveItemAt**

Description:

Delete item by position in control

Syntax:

```
ListboxRemoveItemAt(screen_name, control_name, position)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the List Box
control_name	string	Name of the List Box object
position	int	Item's position (order) - Index 0 (first item has position 0)

F.5.7.14.9 *ListboxSortItem*

Script: **ListboxSortItem**

Description:

Sort the list of items in the control in order

Syntax:

```
ListboxSortItem(screen_name, control_name, order)
```

Parameters:

Parameter	Data type	Description
screen_name	string	The name of the screen containing the List Box

control_name	string	Name of the List Box object
order	string	Order = "Des": Sort by list of values descending Order = "Acs": Sort by list of ascending values

F.5.7.15 Other Scripts

F.5.7.15.1 Message

Definition: show message box notification at viewer

Syntax:

Message(str, nType)

nType = 0: display message with OK button and return value.

nType = 1: display message with 2 buttons OK, Cancel and return value.

nType = 2: Display message with 3 buttons OK, No, Cancel and return value.

If you press Yes: return 1.

If you press Cancel: return 0

If you press No: return -1

Example:

Message("Hello World",0)

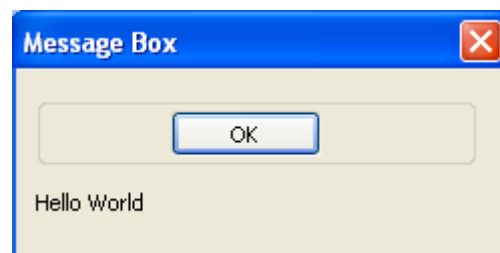


Figure F-247. Message "Hello World" with OK button

Message("Hello World",1):

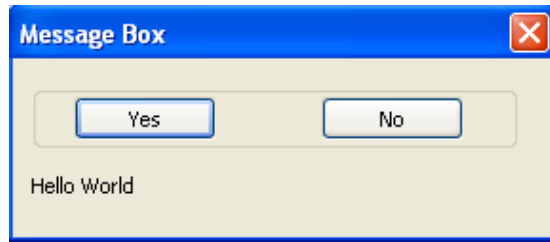


Figure F-248. Message "Hello World" with Yes, No button

Message("Hello World",2):

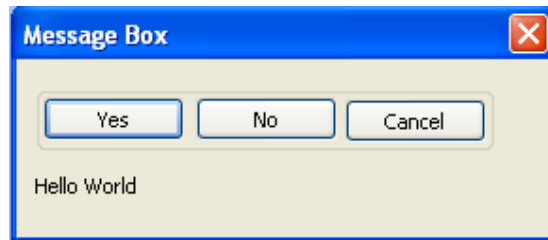


Figure F-249. Message "Hello World" with Yes, No, Cancel button

F.5.7.15.2 *GetProjectPath*

Definition: return the path containing the Project. Returns a value of type string.

Syntax:

```
GetProjectPath()
```

F.5.7.15.3 *Output*

Definition: Output the message to the Debug Tool/Output window

Syntax:

```
Output(message str)
```

Example:

```
Output("BayWinName")
```

F.5.7.15.4 *local function*

Definition: The keyword "local function" defines a function that is scoped for use within the script. Syntax calls a function similar to a regular function script.

"local function" is used to create explicit code structure in a script with many if-then-else statements, loops with a large number of statements executed in the body.

Syntax:

```
local function FUNTION_NAME([parameter])
```

```
command1  
command2  
command3  
...  
end
```

Example:

```
local function BAYB011()  
    BAYx.Refer("B01")  
    BAYx5.Refer("RP011")  
    BAYxBCU.Refer("RP011BCU")  
    BAYxBF.Refer("RP011BF")  
end  
  
BAYB011() -- call local function BAYB011
```

F.5.7.15.5 _G

Definition:

_G is a table containing all object, variable, method, and program data of the project located in both Remote Data, Local Data, and System Data.

_G[] is used as a global variable allowing access to the properties of a global variable with real data, with Parameter being the (string) name of the global variable with real data.

Syntax:

```
_G[var_name str]
```

Where, "var_name" (string type) is the name of global variable *var_name*

$_G["var_name"] \equiv var_name$

_G is used in "meta code" situations, when it is necessary to perform the same tasks with multiple global variables with similar names.

Example:

There are 10 global variables: var1, var2, ..., var10

Requirements:

```
var1 = 1  
var2 = 2  
...  
var10 = 10
```

Code:

```
for i = 1,10 do  
    _G["var"..i] = i  
end
```

--E10_WinName: is a variable (global) Local Data

E10_WinName.value = "BAYELL"

--BAYxNAME: is a Parameter passed to the function, with the value "E10"

Code function script

local BayWinName = _G[BAYxNAME.."_WinName"].value

Explain:

- BayWinName is a local script
- BAYxNAME.."_WinName" → "E10_WinName"
- _G["E10_WinName"].value ≡ E10_WinName.value
- → BayWinName = "BAYELL"

F.5.7.15.6 CheckBit

Script: CheckBit

Description:

Returns the n(zero-index) bit value in value

Syntax:

CheckBit(value, n)

Parameters:

Parameter	Data type	Description
value	int	

n	int	Returns the nth bit value (zero-index) in value (0/1)
---	-----	---

F.5.7.15.7 ClearBit

Script: **ClearBit**

Description:

Set the nth bit value (zero index) in value to 0 and return value.

Syntax:

```
ClearBit(value, n)
```

Parameters:

Parameter	Data type	Description
value	int	Set the nth bit value (zero index) in value to 0 and return value.
n	int	

F.5.7.15.8 ToggleBit

Script: **ToggleBit**

Description:

Inverse the nth bit value in value and return value.

Syntax:

```
ToggleBit(value, n)
```

Parameters:

Parameter	Data type	Description
value	int	Inverse the nth bit value in value and return value.
n	int	

F.5.7.15.9 SetBit

Script: **SetBit**

Description:

Set the nth (zero-index) bit in value to 1 and return value.

Syntax:

```
int SetBit(value, n)
```

Parameters:

Parameter	Data type	Description
value	int	Set the nth (zero-index) bit in value to 1 and return value.
n	int	

F.5.7.15.10 Producer Support Scripts

Not Available

F.5.7.15.11 Log

Definition: Output the message to the file runtime dlog

Support Log() (4.1.7007 onwards)

Syntax:

```
Log(message str)
```

Example: Log("Error code: 1234")

F.5.7.15.12 GetUserToken

Definition: Return Token string of User

Support GetUserToken() (4.1.6987 onwards)

Syntax:

```
GetUserToken()
```

Example:

```
ExecuteFile("C:/Program  
Files/ATS/OAFepProcessMonitor/OAFepProcessMonitorApp.exe  
"..GetUserToken())
```

Open FEP_Process_Monitor App with user token.

F.5.7.15.13 NetworkColorSetting

Definition: Show network color setting dialog

Support NetworkColorSetting() (4.1.7299 onwards)

Syntax:

NetworkColorSetting()

F.5.7.15.14 EnableColoring

Definition: Show Enable Coloring config dialog

Support EnableColoring() (4.1.7299 onwards)

Syntax:

EnableColoring()

F.6. Grid coloring function

Coloring the grid is the most basic application of the objects (Electrical) Network, based on the principle of the voltage relationship of the elements, graphically representing the voltage states of the elements on the grid electricity by color:

- ❖ Carrying electric state, color discrimination at voltage levels (500/220/110, ...)
- ❖ Power failure and no ground
- ❖ Grounding
- ❖ Unknown status



Figure F-250. Network Coloring

In the Designer environment, the primary devices in each bay are declared and connected (drawn) in the correct order.

In a run-time environment, Grid Studio Viewer automatically calculates and colorizes objects based on:

- Carrying electric state of source points
- Status of the switching devices
- Connecting devices (topology) designed on Designer.

The process of opening and closing the primary device, leading to changes in the range of power, and no electricity, ... is shown through the connection order, status, and color on the Network graphic objects.

Some concepts related to coloring function.

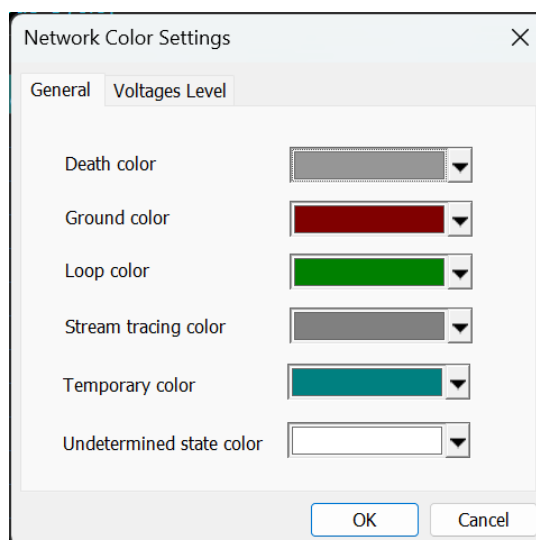


Figure F-251. Network Color Settings

- *Loop Color: Unavailable.*
- *Stream tracing Color: Unavailable*
- *Temporary Color: Unavailable*
- *Undetermined state color: the area isolated from the incoming power supply and at least one device with an unknown state separates this area from the live zone.*
- *Color priority order: Earthing > having power > power failure.*
- *Device Status (IO Status Point):*
 - 0 and 3: unknow.
 - 1: open status.
 - 2: close status



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